

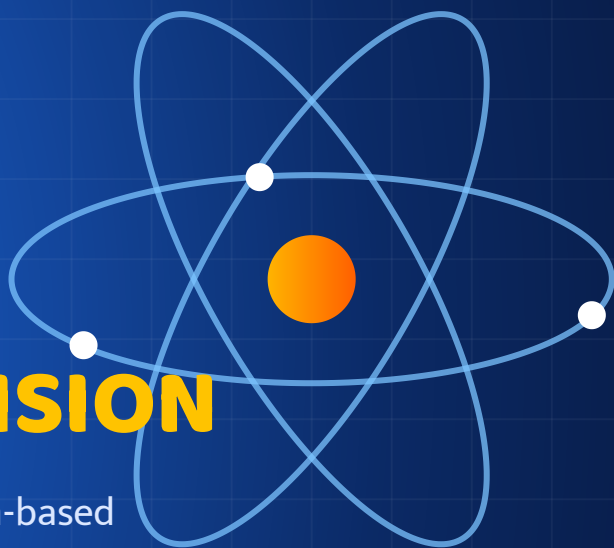
CLASS 11 · NCERT ALIGNED · NEET + JEE MAIN

$$v^2 = u^2 + 2as$$

NEET + JEE PHYSICS

30-DAY RAPID REVISION

Concepts · Formulas · Diagrams · Traps · Pattern-based
practice from NEET & JEE Main previous-year trends



$$E = \frac{1}{2}mv^2$$

$$\tau = I\alpha$$

14 NCERT Chapters

300+ MCQs with Solutions

Full Mock Test

100+ Diagrams & Formula Boxes

$T =$



VS SIR

Vijay Kumar Sharma
20+ years · NEET/JEE Physics · Kota

NEET & JEE Main Physics: 30-Day Rapid Revision (Class 11)

Class 11 Physics · NCERT-aligned · For NEET & JEE Main

by VS Sir (Vijay Kumar Sharma)

AI-Enabled Educator · 20+ years teaching NEET/JEE Physics, Kota

First edition: 2026 · Language: English

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About the questions: All practice questions in this book are original, pattern-based questions written by the author and modelled on the trends of NEET and JEE Main previous-year papers. They are not reproductions of official papers. Syllabus follows the rationalised NCERT Class 11 textbook and the NMC NEET-UG syllabus.

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- Author of "Practical Guide to AI for Educators", "Electrostatics" and "Automation Mastery"
- Known for turning long chapters into one-page concept maps and formula boxes

A message from VS Sir

Dear student,

In the last month before NEET, you do not need another 800-page book. You need a **clear, short, reliable revision** of everything NCERT says, the formulas that actually get used, and the traps where toppers lose marks.

This book is built exactly for that. One chapter per day or two, a formula box you can revise in five minutes, diagrams that make the idea obvious, and practice questions in the pattern NEET and JEE Main have followed year after year – each with a quick solution.

Follow the 30-day plan honestly, tick each box, and keep an error log. Consistency beats intensity. You can do this!

— VS Sir

How to use this book

Every chapter follows the same rhythm so your brain knows exactly where to look.

1. **Read** the chapter-at-a-glance map (2 min).
2. **Revise** concepts, diagrams and formula boxes (40–60 min).
3. **Solve** the 20 practice questions with a timer – 1 min each.
4. **Check** with the answer key; read the solution of every wrong/guessed question.
5. **Log** each mistake in your error log (page at the end). Re-solve on Day 29.

Colour code of the boxes

 Key Concept Core idea explained simply	 Formula Box Formulas to revise in 5 minutes
 Common Trap Where students lose marks	 VS Sir's Exam Tip Exam strategy & shortcuts
 NCERT Line to Remember Exact NCERT facts asked in NEET	 Memory Trick Tricks to remember
 JEE Main Extension Extra depth for JEE Main	

Question tags

NCERT based direct from textbook lines **NEET pattern** typical NEET level **JEE Main pattern** one step deeper
Assertion–Reason

 Honest note on PYQs The questions are original and written in the style of NEET and JEE Main previous-year questions (PYQ trends): same concepts, same difficulty, same traps. Solve your official PYQ papers too – this book prepares you to crack them.

Your 30-Day Revision Plan

Daily target: 2–3 hours for Physics. Tick the box when a day is complete. Weekly tests are built into Days 8 and 17.

Day 1 Ch 1 Units, SF, errors, dimensions	Day 2 Ch 2 Motion in a straight line	Day 3 Ch 3 Vectors & projectile motion	Day 4 Ch 3 Relative velocity, circular motion	Day 5 Ch 4 Newton's laws, FBDs
Day 6 Ch 4 Friction, circular dynamics	Day 7 Ch 5 Work, energy, power	Day 8 Ch 5 Collisions + Test (Ch 1–5)	Day 9 Ch 6 Centre of mass, torque	Day 10 Ch 6 Moment of inertia, rolling
Day 11 Ch 7 Kepler's laws, g variation	Day 12 Ch 7 Satellites, escape speed	Day 13 Ch 8 Elasticity of solids	Day 14 Ch 9 Pressure, Pascal, buoyancy	Day 15 Ch 9 Bernoulli, viscosity, surface tension
Day 16 Ch 10 Expansion, calorimetry	Day 17 Ch 10 Heat transfer + Test (Ch 6–10)	Day 18 Ch 11 Laws of thermodynamics	Day 19 Ch 11 Processes, engines, Carnot	Day 20 Ch 12 Kinetic theory of gases
Day 21 Ch 13 SHM basics & energy	Day 22 Ch 13 Spring & pendulum	Day 23 Ch 14 Wave motion, wave speed	Day 24 Ch 14 Standing waves, pipes, beats	Day 25 Final Formula sheet: Mechanics
Day 26 Final Formula sheet: Rest of Class 11	Day 27 Final FULL MOCK TEST (45 Q, 60 min)	Day 28 Final Mock analysis + error log	Day 29 Final Re-attempt all wrong questions	Day 30 Final Traps list + exam-day plan

Golden rules

- Never skip the practice set – revision without questions is only reading.
- Sleep 7 hours. Memory is stored while you sleep.

If you fall behind

Do the formula box + 10 questions of the missed day, then continue with today. Use Days 28–29 as buffer.

Contents

1	Units and Measurement SI units · Significant figures · Errors · Least count	Day 1	7
2	Motion in a Straight Line Displacement vs distance · Speed & velocity · Acceleration · x - t , v - t , a - t graphs	Day 2	14
3	Motion in a Plane Scalars & vectors · Vector addition · Dot & cross product · Projectile motion	Day 3–4	22
4	Laws of Motion Newton's laws · Momentum & impulse · Free-body diagrams · Apparent weight	Day 5–6	31
5	Work, Energy and Power Work · Work–energy theorem · Potential energy · Conservation of energy	Day 7–8	41
6	System of Particles and Rotational Motion Centre of mass · Torque & angular momentum · Moment of inertia · Axis theorems	Day 9–10	50
7	Gravitation Kepler's laws · Law of gravitation · Variation of g · Potential & PE	Day 11–12	61
8	Mechanical Properties of Solids Stress & strain · Hooke's law · Stress–strain curve · Y , B , G moduli	Day 13	70
9	Mechanical Properties of Fluids Pressure & Pascal · Continuity · Bernoulli's principle · Viscosity & Stokes	Day 14–15	78
10	Thermal Properties of Matter Temperature scales · Thermal expansion · Calorimetry · Latent heat	Day 16–17	87
11	Thermodynamics Zeroth & first law · C_p , C_v , γ · Isothermal & adiabatic · Work = area on PV	Day 18–19	96
12	Kinetic Theory Ideal gas equation · Pressure of a gas · KE & temperature · rms speed	Day 20	105
13	Oscillations Periodic motion · SHM equation · Velocity & acceleration · Energy in SHM	Day 21–22	113
14	Waves Transverse & longitudinal · Wave equation · Speed of waves · Superposition & reflection	Day 23–24	123
	Full Mock Test 45 questions · 60 minutes · NEET marking	Day 27–29	133
	Last-Week Toolkit: Formula Sheets, Traps & Exam Strategy 14 formula sheets · 60 traps · 10 key graphs · exam-day plan · error log	Day 25–30	141

CHAPTER

1

Units and Measurement

17

Day 1

NEET weightage ★★☆☆

NCERT Class 11

SI units

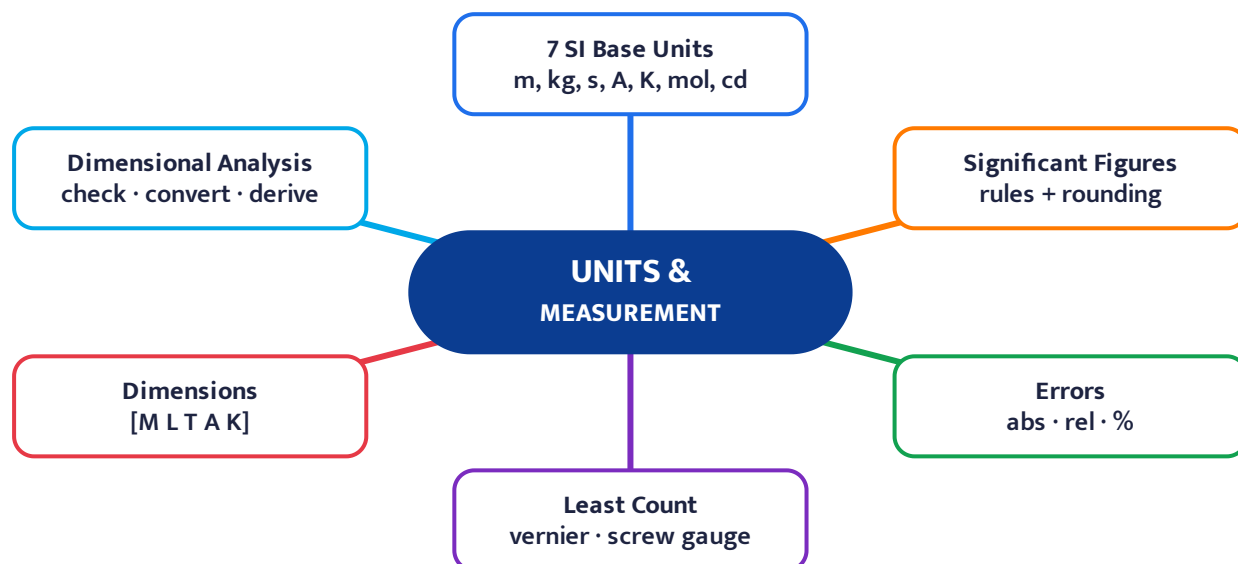
Significant figures

Errors

Least count

Dimensions

Dimensional analysis



Chapter at a glance

1.1 The SI System – 7 Base Quantities

Every physical quantity is expressed as **(numerical value) × (unit)**. The International System (SI) builds all units from **seven base units**; every other unit is a **derived unit**.

Base quantity	SI unit	Symbol	Dimension
Length	metre	m	[L]
Mass	kilogram	kg	[M]
Time	second	s	[T]
Electric current	ampere	A	[A]
Thermodynamic temperature	kelvin	K	[K]
Amount of substance	mole	mol	[mol]
Luminous intensity	candela	cd	[cd]

 NCERT Line to Remember

Plane angle (radian, rad) and solid angle (steradian, sr) are **dimensionless**. $d\theta = \frac{ds}{r}$ and $d\Omega = \frac{dA}{r^2}$. A full sphere subtends 4π sr.

 Memory Trick

Prefixes you must know: **p** 10^{-12} , **n** 10^{-9} , **μ** 10^{-6} , **m** 10^{-3} , **c** 10^{-2} , **k** 10^3 , **M** 10^6 , **G** 10^9 , **T** 10^{12} .

 Handy practical units

- 1 light year = 9.46×10^{15} m
- 1 AU = 1.496×10^{11} m
- 1 parsec = 3.08×10^{16} m $\approx$ 3.26 ly
- 1 Å = 10^{-10} m, 1 fermi = 10^{-15} m
- 1 u = 1.66×10^{-27} kg

1.2 Significant Figures

Significant figures (SF) are the **reliable digits plus the first uncertain digit** of a measurement. They tell you how precise a reading is.

Rule	Example	SF
All non-zero digits are significant	2.345	4
Zeros between non-zero digits count	20.05	4
Leading zeros (before first non-zero) do NOT count	0.00450	3
Trailing zeros after decimal point count	4.500	4
Trailing zeros without decimal: ambiguous – use scientific notation	4.50×10^3	3
Powers of 10 are never counted	6.0×10^{23}	2

 Arithmetic with SF

- **× and ÷** : answer has as many SF as the quantity with the **least SF**.
- **+** and **−** : answer has as many **decimal places** as the quantity with the **fewest decimal places**.

 Rounding off

- Digit dropped $> 5 \rightarrow$ raise; $< 5 \rightarrow$ keep.
- Exactly 5: make preceding digit **even**. 2.745 $\rightarrow$ 2.74, 2.735 $\rightarrow$ 2.74.

 Solved Example

Find 4.237×2.51 and $12.11 + 18.0 + 1.012$ to correct significant figures.

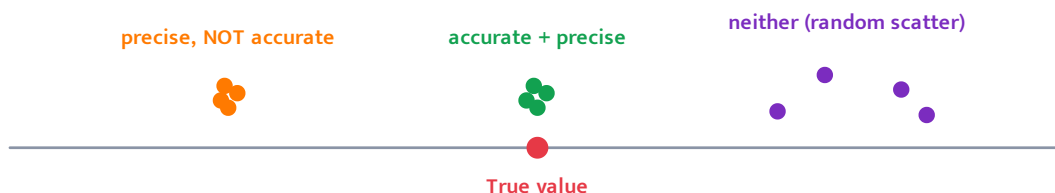
Solution: Product = 10.63487 $\rightarrow$ least SF is 3 (2.51) $\rightarrow$ **10.6**. Sum = 31.122 $\rightarrow$ fewest decimal places is 1 (18.0) $\rightarrow$ **31.1**.

 Common Trap

- Changing units does **not** change SF: 2.30 m = 230 cm = 2.30×10^2 cm (3 SF each).
- 0.0500 has **3** SF, not 1 – the trailing zeros after the decimal count.
- Exact numbers (like 2 in $2\pi r$) have infinite SF – ignore them.

1.3 Errors in Measurement

Accuracy = closeness to the true value. **Precision** = closeness of repeated readings to each other (resolution). Errors are **systematic** (instrument, method, personal – same sign every time) or **random** (irregular, reduced by taking many readings).



Accuracy vs precision – the favourite NCERT picture question

Error formulas

$$\bar{a} = \frac{a_1 + a_2 + \dots + a_n}{n}$$

Mean (best) value

$$\Delta a_i = a_i - \bar{a}$$

Absolute error of each reading

$$\bar{\Delta a} = \frac{\sum |\Delta a_i|}{n}$$

Mean absolute error

$$\delta a = \frac{\bar{\Delta a}}{\bar{a}}$$

Relative (fractional) error

$$\frac{\bar{\Delta a}}{\bar{a}} \times 100\%$$

Percentage error

Combination of errors (always ADD the errors)

Propagation of errors

$$Z = A \pm B \Rightarrow \Delta Z = \Delta A + \Delta B$$

Sum or difference: absolute errors add

$$Z = AB \text{ or } \frac{A}{B} \Rightarrow \frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$$

Product or quotient: relative errors add

$$Z = \frac{A^p B^q}{C^r} \Rightarrow \frac{\Delta Z}{Z} = p \frac{\Delta A}{A} + q \frac{\Delta B}{B} + r \frac{\Delta C}{C}$$

Powers multiply the relative error

Solved Example

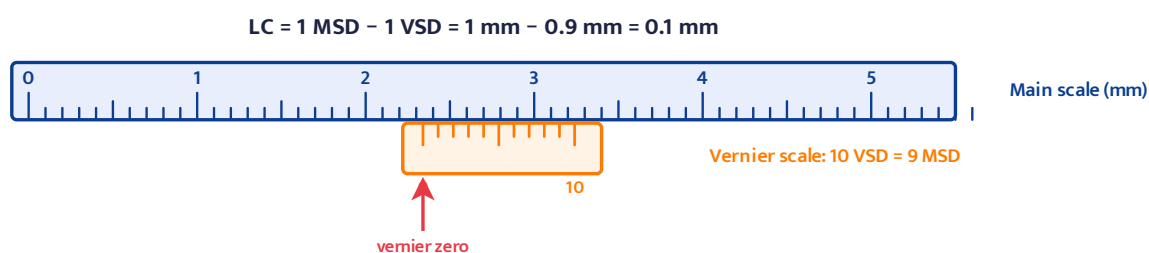
The period of a pendulum is $T = 2\pi\sqrt{l/g}$. If l is measured with 2% error and T with 1% error, find the % error in g .

Solution: $g = 4\pi^2 l / T^2 \Rightarrow \frac{\Delta g}{g} = \frac{\Delta l}{l} + 2 \frac{\Delta T}{T} = 2\% + 2(1\%) = 4\%$.

VS Sir's Exam Tip

In NEET, the quantity with the **highest power** contributes most to the error – measure it most carefully. Signs in the formula never reduce error: $\frac{\Delta Z}{Z}$ for $Z = A / B$ is still a **sum**.

1.4 Least Count – Vernier Calipers & Screw Gauge



Vernier calipers: 10 vernier divisions coincide with 9 main-scale divisions

Least-count formulas

$$LC_{\text{vernier}} = 1 \text{ MSD} - 1 \text{ VSD} \quad \text{Least count of vernier calipers}$$

$$\text{Reading} = \text{MSR} + (\text{VSR} \times LC) - \text{zero error} \quad \text{Vernier reading}$$

$$LC_{\text{screw}} = \frac{\text{pitch}}{\text{no. of circular divisions}} \quad \text{Screw gauge least count (e.g. } 0.5 \text{ mm}/50 = 0.01 \text{ mm)}$$

Common Trap

Zero error is **subtracted with its sign**: positive zero error (+0.02 cm) → subtract 0.02; negative zero error (−0.03 cm) → add 0.03.

1.5 Dimensions & Dimensional Analysis

The **dimensions** of a quantity are the powers to which the base quantities are raised to represent it. Example: force $F = ma \Rightarrow [F] = [M^1 L^1 T^{-2}]$.

Quantity	Dimension	Quantity	Dimension
Velocity	$[LT^{-1}]$	Pressure, stress, Y	$[ML^{-1}T^{-2}]$
Force	$[MLT^{-2}]$	Surface tension, spring k	$[MT^{-2}]$
Work, energy, torque	$[ML^2T^{-2}]$	Viscosity η	$[ML^{-1}T^{-1}]$
Power	$[ML^2T^{-3}]$	Gravitational constant G	$[M^{-1}L^3T^{-2}]$
Momentum, impulse	$[MLT^{-1}]$	Planck constant h, angular momentum	$[ML^2T^{-1}]$
Moment of inertia	$[ML^2]$	Specific heat c	$[L^2T^{-2}K^{-1}]$
Strain, angle, refractive index	dimensionless	Gas constant R	$[ML^2T^{-2}K^{-1}mol^{-1}]$

 3 uses

- **Check** correctness of an equation (principle of homogeneity).
- **Convert** units: $n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$
- **Derive** a relation between quantities (up to a constant).

 Limitations

- Cannot find dimensionless constants (2π , $\frac{1}{2}$).
- Fails for sin, log, exp relations and sums of terms.
- A dimensionally correct equation may still be wrong.
- Fails if more than 3 base quantities are involved (M, L, T).

 Solved Example

Using dimensions, show that the time period of a simple pendulum is $T = k\sqrt{L/g}$.

Solution: Let $T = k m^a l^b g^c$. Then $[T] = [M]^a [L]^b [LT^{-2}]^c$. Equate powers: M: $a = 0$; L: $b + c = 0$; T: $-2c = 1 \Rightarrow c = -\frac{1}{2}, b = \frac{1}{2}$. So $T = k\sqrt{L/g}$ (experiment gives $k = 2\pi$).

 Solved Example

Convert 1 newton into dyne using dimensions.

Solution: $[F] = [MLT^{-2}]$. $n_2 = 1 \times \left(\frac{1\text{ kg}}{1\text{ g}} \right)^1 \left(\frac{1\text{ m}}{1\text{ cm}} \right)^1 \left(\frac{1\text{ s}}{1\text{ s}} \right)^{-2} = 10^3 \times 10^2 = 10^5$. So **1 N = 10^5 dyne**.

 Memory Trick

Quick pairs with the SAME dimensions: work–torque–energy; pressure–stress–Young's modulus–energy density; impulse–momentum; Planck constant–angular momentum; frequency–angular velocity–velocity gradient.

 Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1** **NCERT based** The number of significant figures in 0.06900 is
 (a) 3 (b) 4 (c) 5 (d) 2
- 2** **NEET pattern** Which pair has the same dimensions?
 (a) Momentum and energy (b) Torque and energy
 (c) Work and power (d) Force and pressure
- 3** **NEET pattern** The dimensions of Planck's constant are the same as those of
 (a) energy (b) linear momentum
 (c) angular momentum (d) power
- 4** **NEET pattern** A physical quantity $P = \frac{a^3 b^2}{\sqrt{c} d}$. Percentage errors in a, b, c, d are 1%, 3%, 4% and 2%. The % error in P is
 (a) 7% (b) 10% (c) 13% (d) 14%

- 5 **NCERT based** The dimensional formula of the coefficient of viscosity is
 (a) $[MLT^{-1}]$ (b) $[M^0L^0T^0]$ (c) $[ML^{-1}T^{-1}]$ (d) $[ML^{-2}T^{-2}]$
- 6 **JEE Main pattern** In a vernier calipers, 20 vernier divisions coincide with 19 main scale divisions of 1 mm each. Its least count is
 (a) 0.02 mm (b) 0.05 mm (c) 0.1 mm (d) 0.5 mm
- 7 **NCERT based** The sum of 2.745, 1.32 and 0.004 with correct significant figures is
 (a) 4.069 (b) 4.0 (c) 4.1 (d) 4.07
- 8 **NEET pattern** If force F , velocity v and time T are taken as fundamental quantities, the dimensions of mass are
 (a) $[Fv^{-1}T]$ (b) $[FvT^{-1}]$ (c) $[F^{-1}vT]$ (d) $[FvT]$
- 9 **JEE Main pattern** A screw gauge has pitch 0.5 mm and 50 circular divisions. Main scale reading 2.5 mm, 20th circular division coincides, zero error -0.01 mm. The reading is
 (a) 2.70 mm (b) 2.71 mm (c) 2.69 mm (d) 2.20 mm
- 10 **NCERT based** Which of the following is dimensionless?
 (a) Strain (b) Surface tension
 (c) Stress (d) Angular velocity
- 11 **NEET pattern** The radius of a sphere is measured as (2.0 ± 0.1) cm. The percentage error in its volume is
 (a) 5% (b) 10% (c) 20% (d) 15%
- 12 **NEET pattern** In $y = A\sin(\omega t - kx)$, the dimensions of ω / k are
 (a) $[LT^{-1}]$ (b) $[L^{-1}]$ (c) $[L^{-1}T]$ (d) $[T^{-1}]$
- 13 **NCERT based** The dimensional formula of the universal gravitational constant G is
 (a) $[M^{-1}L^2T^{-2}]$ (b) $[ML^3T^{-2}]$ (c) $[ML^2T^{-2}]$ (d) $[M^{-1}L^3T^{-2}]$
- 14 **JEE Main pattern** If the unit of length is doubled and unit of mass is halved (time unchanged), the numerical value of a given force becomes
 (a) same (b) half (c) double (d) four times
- 15 **Assertion-Reason** Assertion: A dimensionally correct equation must be physically correct. Reason: Dimensional analysis cannot detect missing dimensionless constants.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 16 **NCERT based** Two resistors $R_1 = (100 \pm 3) \Omega$ and $R_2 = (200 \pm 4) \Omega$ are in series. The equivalent resistance is
 (a) $(300 \pm 5) \Omega$ (b) $(300 \pm 1) \Omega$ (c) $(300 \pm 7) \Omega$ (d) $(300 \pm 12) \Omega$

17. **NEET pattern** The dimensions of $\frac{1}{2}\epsilon_0 E^2$ (ϵ_0 = permittivity, E = electric field) are
 (a) $[ML^2T^{-2}]$ (b) $[ML^{-1}T^{-2}]$ (c) $[MLT^{-1}]$ (d) $[ML^2T^{-1}]$
18. **NCERT based** Light year is a unit of
 (a) distance (b) speed of light
 (c) intensity (d) time
19. **JEE Main pattern** In the relation $P = \frac{\alpha}{\beta} e^{-\alpha Z / (k\theta)}$ (P pressure, Z distance, k Boltzmann constant, θ temperature), the dimensions of β are
 (a) $[ML^2T]$ (b) $[M^0L^0T^0]$ (c) $[M^0L^2T^0]$ (d) $[M^{-1}L^1T^2]$
20. **NCERT based** The mean of readings 2.63, 2.56, 2.42, 2.71, 2.80 s is 2.62 s. The mean absolute error is
 (a) 0.01 s (b) 0.05 s (c) 0.21 s (d) 0.11 s

✓ Answer Key & Quick Solutions

- 1(b) 2(b) 3(c) 4(c) 5(c) 6(b) 7(d) 8(a) 9(b) 10(a) 11(d) 12(a) 13(d) 14(a) 15(c)
 16(c) 17(b) 18(a) 19(c) 20(d)

1. (b) Leading zeros don't count; trailing zeros after the decimal do: 6, 9, 0, 0 $\rightarrow$ 4.
2. (b) Both torque and energy are $[ML^2T^{-2}]$.
3. (c) $h = E / \nu \Rightarrow [ML^2T^{-2}] / [T^{-1}] = [ML^2T^{-1}]$ = angular momentum.
4. (c) $3(1) + 2(3) + \frac{1}{2}(4) + 1(2) = 3 + 6 + 2 + 2 = 13\%$.
5. (c) $F = \eta A \frac{dv}{dx} \Rightarrow \eta = \frac{[MLT^{-2}]}{[L^2][T^{-1}]} = [ML^{-1}T^{-1}]$.
6. (b) 1 VSD = 19/20 mm = 0.95 mm; LC = 1 - 0.95 = 0.05 mm.
7. (d) Sum = 4.069; least decimal places = 2 (1.32) $\rightarrow$ 4.07.
8. (a) $F = mv / T \Rightarrow m = FT / v = [Fv^{-1}T]$.
9. (b) LC = 0.01 mm. Reading = 2.5 + 20(0.01) - (-0.01) = 2.71 mm.
10. (a) Strain = $\Delta L / L$, a ratio of lengths.
11. (d) $\Delta r / r = 5\%$; $V \propto r^3 \Rightarrow 3 \times 5\% = 15\%$.
12. (a) ωt and kx are dimensionless $\Rightarrow [\omega] = T^{-1}$, $[k] = L^{-1} \Rightarrow$ ratio = velocity.
13. (d) $G = Fr^2 / m^2 = [MLT^{-2}][L^2] / [M^2]$.
14. (a) $n_2 = n_1(M_1 / M_2)(L_1 / L_2) = n_1(2)(\frac{1}{2}) = n_1$.
15. (c) $s = ut + at^2$ is dimensionally correct but physically wrong (needs $\frac{1}{2}$). So A is false; R is true.
16. (c) In a sum, absolute errors add: $3 + 4 = 7 \Omega$.
17. (b) It is energy density = energy/volume = $[ML^2T^{-2}] / [L^3]$.
18. (a) Distance light travels in one year $\approx 9.46 \times 10^{15}$ m.
19. (c) Exponent dimensionless $\Rightarrow [\alpha] = [k\theta] / [Z] = [ML^2T^{-2}] / [L] = [MLT^{-2}]$. $\beta = \alpha / P = [MLT^{-2}] / [ML^{-1}T^{-2}] = [L^2]$.
20. (d) $|\Delta| = 0.01, 0.06, 0.20, 0.09, 0.18$; sum = 0.54; mean = $0.108 \approx 0.11$ s.

CHAPTER

2

Motion in a Straight Line

Day 2

17

NEET weightage ★★☆☆

NCERT Class 11

Displacement vs distance

Speed & velocity

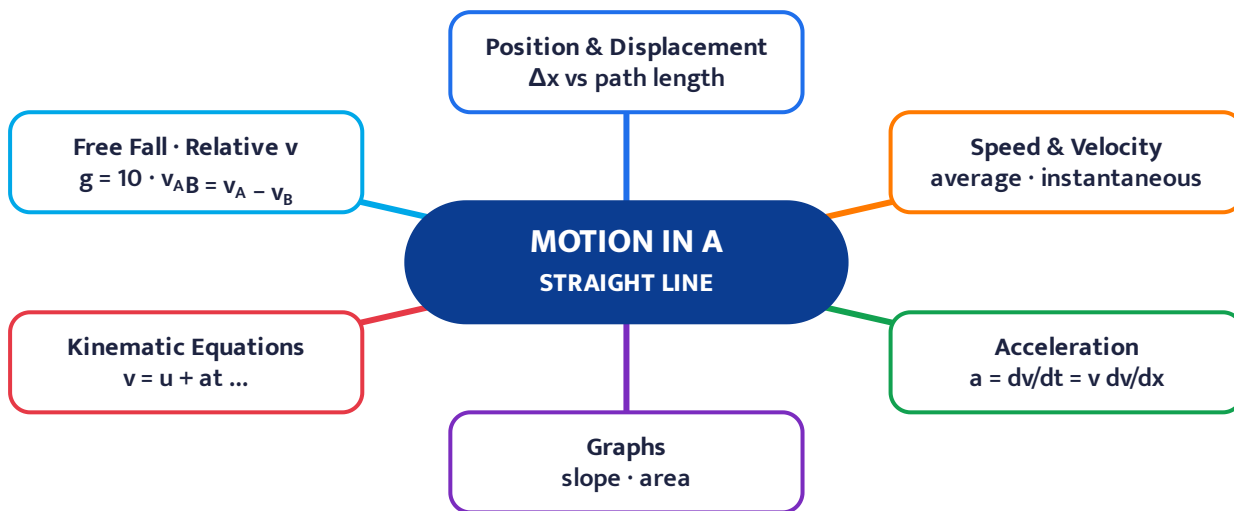
Acceleration

x-t, v-t, a-t graphs

Kinematic equations

Free fall

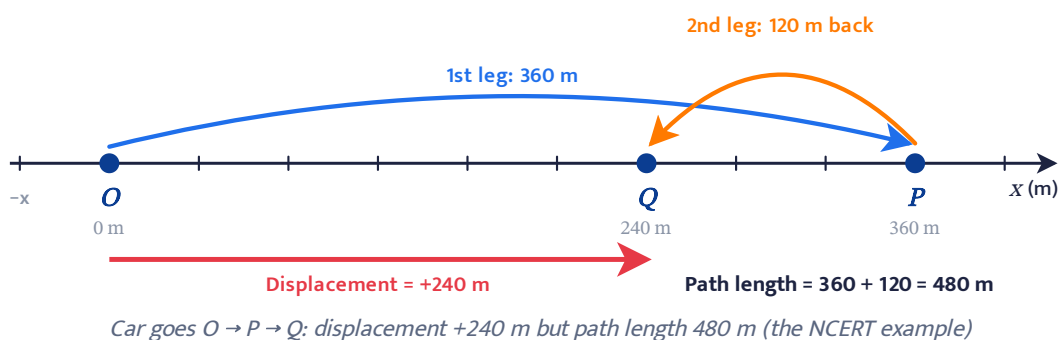
Relative velocity



Chapter at a glance

2.1 Position, Path Length and Displacement

To describe motion we choose a **reference point (origin)** and an axis – this is a **frame of reference**. **Position** x is the coordinate of the object. **Displacement** $\Delta x = x_2 - x_1$ is the change in position (a vector: sign gives direction). **Path length (distance)** is the total length actually travelled (a scalar, never negative).



Displacement

Vector – has direction (+/-)

Can be +, - or zero; depends only on end points

 $|\text{displacement}| \leq \text{distance}$

Distance (path length)

Scalar – only magnitude

Never negative; depends on the actual path

Equal only for straight-line motion without turning back

NCERT Line to Remember

An object is treated as a **point object** when its size is negligible compared with the distance it moves. Motion is always **relative** – rest and motion depend on the frame of reference chosen.

2.2 Speed and Velocity

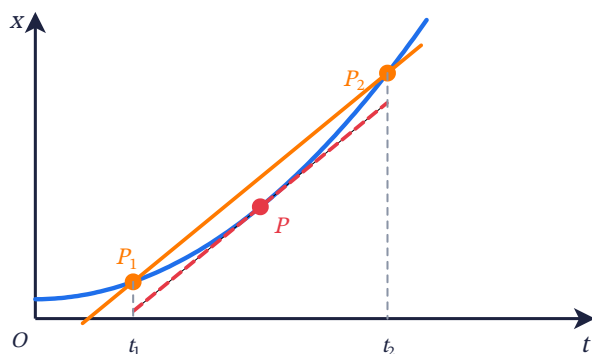
Velocity Formulas

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{x_2 - x_1}{t_2 - t_1} \quad \text{Average velocity (vector)}$$

$$\text{Average speed} = \frac{\text{total path length}}{\text{total time}} \quad \text{Average speed (scalar); always } \geq |\text{average velocity}|$$

$$v = \lim_{\Delta t \rightarrow 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt} \quad \text{Instantaneous velocity = slope of } x\text{-}t \text{ graph}$$

$$\text{speed} = |v| \quad \text{Instantaneous speed = magnitude of instantaneous velocity}$$



— slope of chord = average velocity

- - - slope of tangent = instantaneous velocity

$$v_{\text{avg}} = \Delta x / \Delta t$$

$$v = dx/dt$$

Average velocity = slope of chord P_1P_2 ; instantaneous velocity = slope of tangent at P

VS Sir's Exam Tip

- **Equal distances** at v_1, v_2 : $v_{av} = \frac{2v_1v_2}{v_1 + v_2}$ (harmonic mean).
- **Equal time intervals**: $v_{av} = \frac{v_1 + v_2}{2}$ (arithmetic mean).

Common Trap

- Round trip: average **velocity** = 0, but average **speed** $\neq$ 0.
- Average speed is NOT the magnitude of average velocity in general.

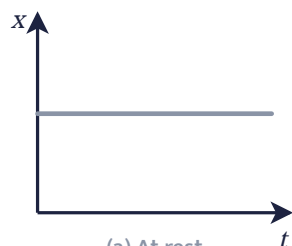
2.3 Acceleration

Acceleration (SI unit m s^{-2})

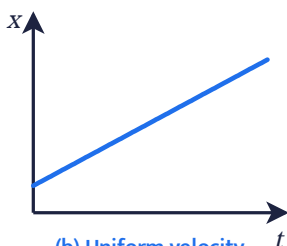
$$\bar{a} = \frac{\Delta v}{\Delta t} \quad \text{Average acceleration}$$

$$a = \frac{dv}{dt} = \frac{d^2x}{dt^2} \quad \text{Instantaneous acceleration = slope of } v\text{-}t \text{ graph}$$

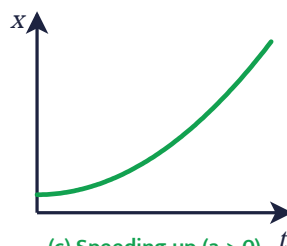
$$a = v \frac{dv}{dx} \quad \text{Use when } v \text{ is given as a function of } x$$



(a) At rest
slope = 0



(b) Uniform velocity
constant slope



(c) Speeding up ($a > 0$)
slope increasing



(d) Slowing down ($a < 0$)
slope decreasing

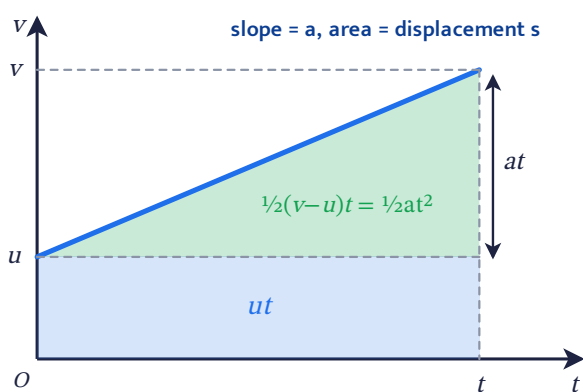
Shapes of $x\text{-}t$ graphs: the curvature tells you the sign of acceleration

NCERT Line to Remember

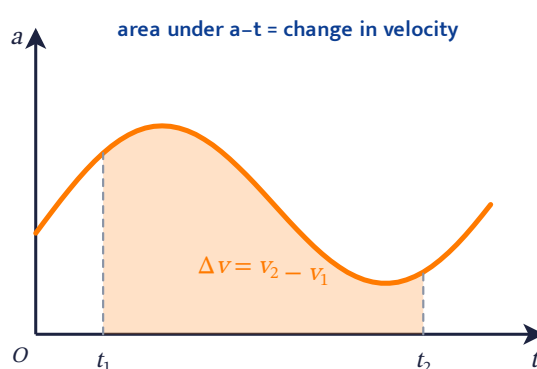
- The sign of acceleration depends on the chosen positive direction. A body **speeds up** when v and a have the **same sign** and **slows down** when they have **opposite signs**.
- A body can have **zero velocity and non-zero acceleration** (top of a vertical throw). An $x-t$ graph can never show two positions at one instant, and speed can never be negative.

2.4 Graphs – Slopes and Areas

Graph	Slope gives	Area under it gives
$x-t$	velocity v	—
$v-t$	acceleration a	displacement Δx (distance = total area)
$a-t$	jerk (not in NEET)	change in velocity Δv



$v-t$ for uniform a : area = $ut + \frac{1}{2}at^2$



$a-t$ graph: shaded area = Δv

Memory Trick

" $x \rightarrow v \rightarrow a$: slope down the ladder; $a \rightarrow v \rightarrow x$: area up the ladder." Differentiate (slope) to go $x \rightarrow v \rightarrow a$; integrate (area) to come back $a \rightarrow v \rightarrow x$.

2.5 Kinematic Equations for Uniform Acceleration

The five equations (valid ONLY for constant a)

$v = u + at$	from slope of $v-t$ graph	$s = ut + \frac{1}{2}at^2$	from area of $v-t$ graph
$v^2 = u^2 + 2as$	time-free equation	$s = \frac{(u+v)}{2}t$	acceleration-free equation
$s_n = u + \frac{a}{2}(2n-1)$	displacement in the n -th second		

Calculus method (also for variable acceleration)

From $a = \frac{dv}{dt}$: $\int_u^v dv = \int_0^t a dt \Rightarrow v = u + at$ (a constant). From $v = \frac{dx}{dt}$: $\int_{x_0}^x dx = \int_0^t (u + at) dt \Rightarrow x - x_0 = ut + \frac{1}{2}at^2$. From $a = v \frac{dv}{dx}$: $\int_u^v v dv = \int_{x_0}^x a dx \Rightarrow v^2 = u^2 + 2a(x - x_0)$. If a depends on t , simply integrate: $v = u + \int_0^t a dt$, $x = x_0 + \int_0^t v dt$.



JEE Main Extension

- **a as a function of v:** $a = f(v) \Rightarrow \int dt = \int \frac{dv}{f(v)}$. E.g. $a = -kv \Rightarrow v = u e^{-kt}$ (velocity decays exponentially).
- **a as a function of x:** $a = f(x) \Rightarrow \int v dv = \int f(x) dx$. Always separate variables before integrating.



Solved Example

The position of a particle is $x = t^3 - 6t^2 + 9t$ (x in m, t in s). Find (i) when it is momentarily at rest, (ii) its acceleration at those instants.

Solution: $v = \frac{dx}{dt} = 3t^2 - 12t + 9 = 3(t-1)(t-3) \Rightarrow v = 0$ at **t = 1 s and t = 3 s**. $a = \frac{dv}{dt} = 6t - 12 \Rightarrow a(1) = -6 \text{ m s}^{-2}, a(3) = +6 \text{ m s}^{-2}$.



Galileo's law of odd numbers

Starting from rest with constant a , distances in successive equal intervals are in the ratio **1 : 3 : 5 : 7 ...**; total distances after 1, 2, 3 ... intervals are **1 : 4 : 9 ...** ($s \propto t^2$).



Common Trap

- s_n has unit metre even though it looks like velocity – it is distance in 1 s.
- Never use $v = u + at$ when a varies with time – integrate instead.

2.6 Motion Under Gravity – Free Fall and Vertical Throw

Neglecting air resistance, every body near the Earth has the same downward acceleration g (9.8 m s^{-2} ; take **10 m s^{-2}** in NEET numericals). Take **upward as positive** $\Rightarrow a = -g$ and use the same equations.



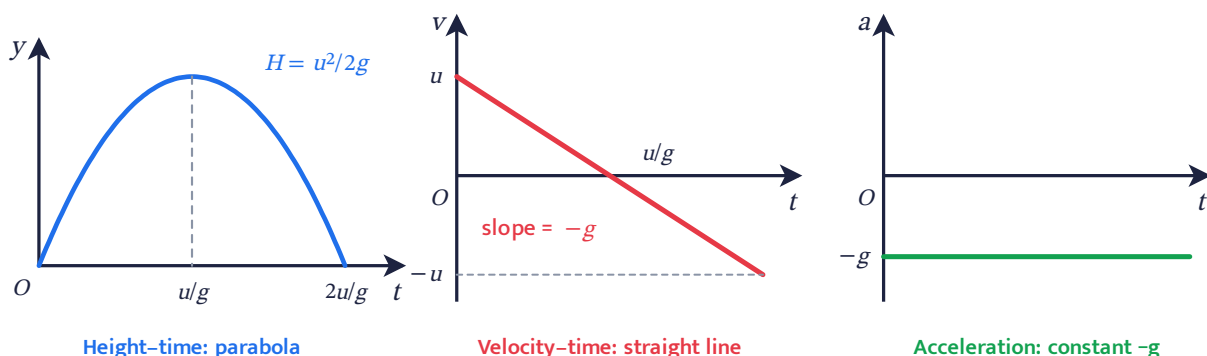
Free-fall formulas

$$t = \sqrt{\frac{2h}{g}}, \quad v = \sqrt{2gh} \quad \text{Dropped from height } h \text{ (} u = 0 \text{)}$$

$$t_{up} = \frac{u}{g}, \quad H = \frac{u^2}{2g} \quad \text{Thrown up with speed } u: \text{ time to top and maximum height}$$

$$T = \frac{2u}{g} \quad \text{Total time to return (time up = time down)}$$

$$h_n = \frac{g}{2}(2n-1) \quad \text{Fall in } n\text{-th second from rest}$$



Ball thrown vertically up (upward positive): y - t , v - t and a - t graphs

Solved Example

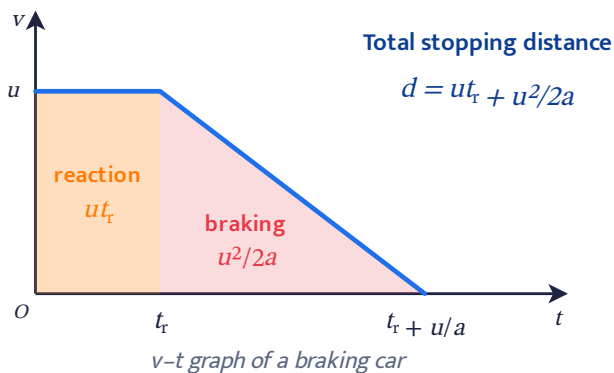
A ball is thrown vertically upward with 20 m/s from the top of a 25 m high building. When does it hit the ground and with what speed? ($g = 10 \text{ m s}^{-2}$)

Solution: Origin at roof, up positive: $-25 = 20t - 5t^2 \Rightarrow t^2 - 4t - 5 = 0 \Rightarrow t = 5 \text{ s}$. $v = 20 - 10(5) = -30 \Rightarrow$ speed **30 m/s** downward. (Check: $v^2 = 20^2 + 2(10)(25) = 900$.)

VS Sir's Exam Tip

- A body thrown up passes every height with the **same speed** going up and coming down; time up = time down.
- Ball thrown up from a tower with u hits the ground with the same speed as one thrown **down** with u :

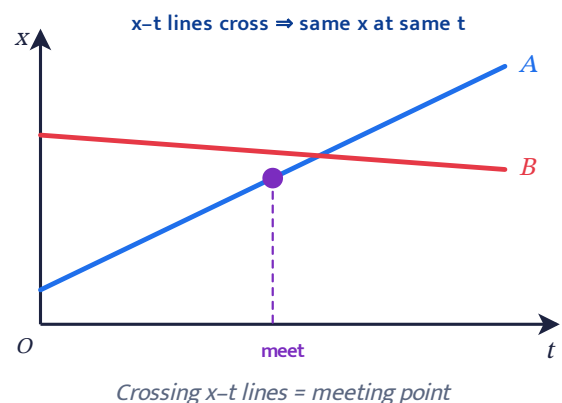
$$v = \sqrt{u^2 + 2gh}.$$

2.7 Stopping Distance and Reaction Time**Key Concept**

- **Stopping distance** (braking only): $d_s = \frac{u^2}{2a} \Rightarrow d_s \propto u^2$. Double the speed $\rightarrow$ **4 times** the stopping distance.
- **Reaction time** t_r : time between seeing and acting. The car moves $u t_r$ during it.
- **Ruler-drop test**: a ruler falls d before being caught $\Rightarrow t_r = \sqrt{2d/g}$ (21 cm $\Rightarrow \approx 0.2$ s).

2.8 Relative Velocity in One Dimension**Key Concept**

- Velocity of A **relative to** B (as seen by an observer moving with B): $v_{AB} = v_A - v_B$; and $v_{BA} = -v_{AB}$.
- x - t graph: v_{AB} = difference of slopes; **parallel** lines $\Rightarrow v_{AB} = 0$; **crossing** lines $\Rightarrow$ bodies meet.
- Time to meet = relative distance $\div$ relative speed.

**Solved Example**

Two trains, 100 m and 150 m long, run on parallel tracks at 20 m/s and 30 m/s. Find the time to cross each other if they move (i) in opposite directions, (ii) in the same direction.

Solution: Relative distance = 100 + 150 = 250 m. (i) $v_{rel} = 20 + 30 = 50 \text{ m/s} \Rightarrow t = 5 \text{ s}$. (ii) $v_{rel} = 30 - 20 = 10 \text{ m/s} \Rightarrow t = 25 \text{ s}$.

 Common Trap

- Opposite directions: relative **speed** $= v_A + v_B$; same direction: $|v_A - v_B|$. Keep signs, not just magnitudes.
- Two bodies falling freely have **zero relative acceleration** – relative to each other they move with constant velocity.

 Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** A particle moves 10 m east and then 4 m west along a straight line. Its distance and displacement are
 (a) 14 m, 14 m east (b) 14 m, 6 m west
 (c) 14 m, 6 m east (d) 6 m, 14 m east
- 2 **NCERT based** The area under a velocity–time graph between two instants gives
 (a) acceleration (b) displacement
 (c) change in velocity (d) average acceleration
- 3 **NCERT based** At the highest point of a ball thrown vertically upward,
 (a) $v = 0$ and $a = g$ downward (b) $v = 0$ and $a = 0$
 (c) $v \neq 0$ and $a = 0$ (d) $v = 0$ and $a = g$ upward
- 4 **NCERT based** A body starts from rest with uniform acceleration. The distances covered in the 1st, 2nd and 3rd seconds are in the ratio
 (a) 1 : 2 : 3 (b) 1 : 1 : 1 (c) 1 : 3 : 5 (d) 1 : 4 : 9
- 5 **NEET pattern** A car covers the first half of a distance at 30 km/h and the second half at 60 km/h. Its average speed is
 (a) 50 km/h (b) 35 km/h (c) 45 km/h (d) 40 km/h
- 6 **NEET pattern** Car A moving at 54 km/h overtakes car B moving at 36 km/h in the same direction. Taking the direction of motion as positive, the velocity of B relative to A is
 (a) -25 m/s (b) $+25$ m/s (c) $+5$ m/s (d) -5 m/s
- 7 **NEET pattern** The x – t graph of a particle is a parabola $x = kt^2$ ($k > 0$). Its v – t graph is
 (a) a line parallel to the time axis
 (b) a parabola
 (c) a straight line with negative slope
 (d) a straight line through the origin with positive slope
- 8 **NCERT based** A student catches a falling ruler after it drops 20 cm. His reaction time is ($g = 10 \text{ m s}^{-2}$)
 (a) 0.4 s (b) 0.14 s (c) 0.02 s (d) 0.2 s

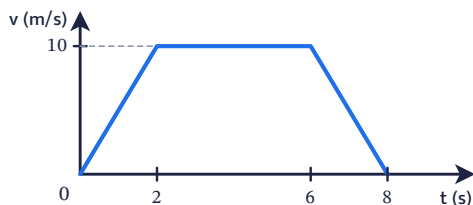
- 9 **NEET pattern** A ball thrown vertically upward with 20 m/s. Its maximum height and total time of flight are ($g = 10 \text{ m s}^{-2}$)
- (a) 10 m, 2 s (b) 40 m, 4 s (c) 20 m, 4 s (d) 20 m, 2 s

- 10 **NEET pattern** A stone dropped from a cliff reaches the ground in 4 s. The distance it falls in the last second is ($g = 10 \text{ m s}^{-2}$)
- (a) 30 m (b) 35 m (c) 80 m (d) 45 m

- 11 **NEET pattern** The position of a particle is $x = 2t^3 - 3t^2 + 4$ (SI units). Its acceleration at $t = 2 \text{ s}$ is
- (a) 18 m s^{-2} (b) 12 m s^{-2} (c) 24 m s^{-2} (d) 6 m s^{-2}

- 12 **NEET pattern** A body has initial velocity 10 m/s and uniform acceleration 4 m s^{-2} . The distance covered in the 5th second is
- (a) 100 m (b) 18 m (c) 30 m (d) 28 m

- 13 **NEET pattern** The v-t graph of a body is shown. The displacement in 8 s is



- (a) 70 m (b) 50 m (c) 60 m (d) 80 m
- 14 **NEET pattern** A police van moving at 30 km/h fires a bullet with muzzle speed 150 m/s at a thief's car moving ahead at 192 km/h in the same direction. The speed with which the bullet hits the car is
- (a) 96.7 m/s (b) 150 m/s (c) 203.3 m/s (d) 105 m/s

- 15 **JEE Main pattern** A car moving at 36 km/h can be stopped in 10 m by applying brakes. With the same braking, the same car moving at 72 km/h stops in
- (a) 10 m (b) 80 m (c) 40 m (d) 20 m

- 16 **Assertion-Reason** Assertion: In a round trip, the average velocity is zero but the average speed is not. Reason: In a round trip the displacement is zero while the path length is not.
- (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

- 17 **Assertion-Reason** Assertion: A negative acceleration always means that the body is slowing down. Reason: The sign of acceleration depends on the choice of the positive direction.
- (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

18. **JEE Main pattern** A ball is thrown vertically upward with 20 m/s from the top of a 60 m high tower. It reaches the ground after ($g = 10 \text{ m s}^{-2}$)
 (a) 4 s (b) 2 s (c) 8 s (d) 6 s
19. **JEE Main pattern** The velocity of a particle varies with position as $v = \alpha\sqrt{x}$ (α constant). Its acceleration is
 (a) $\frac{\alpha^2}{2}$ (b) α^2 (c) $\frac{\alpha^2}{2\sqrt{x}}$ (d) $2\alpha^2$
20. **JEE Main pattern** A ball is dropped from height h . At the same instant another ball is thrown upward from the ground with speed u along the same vertical line. They meet after time
 (a) $h / 2u$ (b) $2h / u$ (c) h / u (d) u / h

✓ Answer Key & Quick Solutions

- 1(c) 2(b) 3(a) 4(c) 5(d) 6(d) 7(d) 8(d) 9(c) 10(b) 11(a) 12(d) 13(c) 14(d) 15(c)
 16(a) 17(c) 18(d) 19(a) 20(c)

- (c) Distance = $10 + 4 = 14 \text{ m}$; displacement = $10 - 4 = 6 \text{ m}$ towards east.
- (b) $\int v dt = \Delta x$, the displacement. (Area under a - t gives Δv .)
- (a) Velocity is momentarily zero but gravity still acts: $a = g$ downward.
- (c) $s_n \propto (2n - 1) \Rightarrow 1 : 3 : 5$ (Galileo's odd-number law).
- (d) $\frac{2v_1 v_2}{v_1 + v_2} = \frac{2(30)(60)}{90} = 40 \text{ km/h}$.
- (d) $v_{BA} = v_B - v_A = 10 - 15 = -5 \text{ m/s}$.
- (d) $v = 2kt$ - linear in t , passes through origin, slope $2k > 0$.
- (d) $t = \sqrt{2d/g} = \sqrt{2(0.2)/10} = 0.2 \text{ s}$.
- (c) $H = u^2 / 2g = 400/20 = 20 \text{ m}$; $T = 2u / g = 4 \text{ s}$.
- (b) $h_4 = \frac{g}{2}(2 \times 4 - 1) = 5 \times 7 = 35 \text{ m}$ (total $80 \text{ m} - 45 \text{ m}$ in 3 s).
- (a) $v = 6t^2 - 6t$, $a = 12t - 6 \Rightarrow a(2) = 18 \text{ m s}^{-2}$.
- (d) $s_5 = u + \frac{a}{2}(2n - 1) = 10 + 2(9) = 28 \text{ m}$.
- (c) Area of trapezium = $\frac{1}{2}(10)(2) + 10(4) + \frac{1}{2}(10)(2) = 10 + 40 + 10 = 60 \text{ m}$.
- (d) Bullet ground speed = $150 + 8.33 = 158.33 \text{ m/s}$; car = $53.33 \text{ m/s} \Rightarrow$ relative speed = 105 m/s .
- (c) $d \propto u^2$: doubling $u \Rightarrow 4 \times 10 = 40 \text{ m}$.
- (a) Average velocity = displacement/time = 0 ; average speed = path length/time $\neq 0$. R is exactly the reason.
- (c) A body moving in $-x$ with $a < 0$ speeds up. Slowing down needs v and a of opposite signs. A false, R true.
- (d) $-60 = 20t - 5t^2 \Rightarrow t^2 - 4t - 12 = 0 \Rightarrow (t - 6)(t + 2) = 0 \Rightarrow t = 6 \text{ s}$.
- (a) $a = v \frac{dv}{dx} = \alpha\sqrt{x} \cdot \frac{\alpha}{2\sqrt{x}} = \frac{\alpha^2}{2}$ (constant).
- (c) Relative acceleration = 0 , relative speed = u , gap = $h \Rightarrow t = h/u$.

CHAPTER

3

Motion in a Plane

17

Day 3–4

NEET weightage ★★★

NCERT Class 11

Scalars & vectors

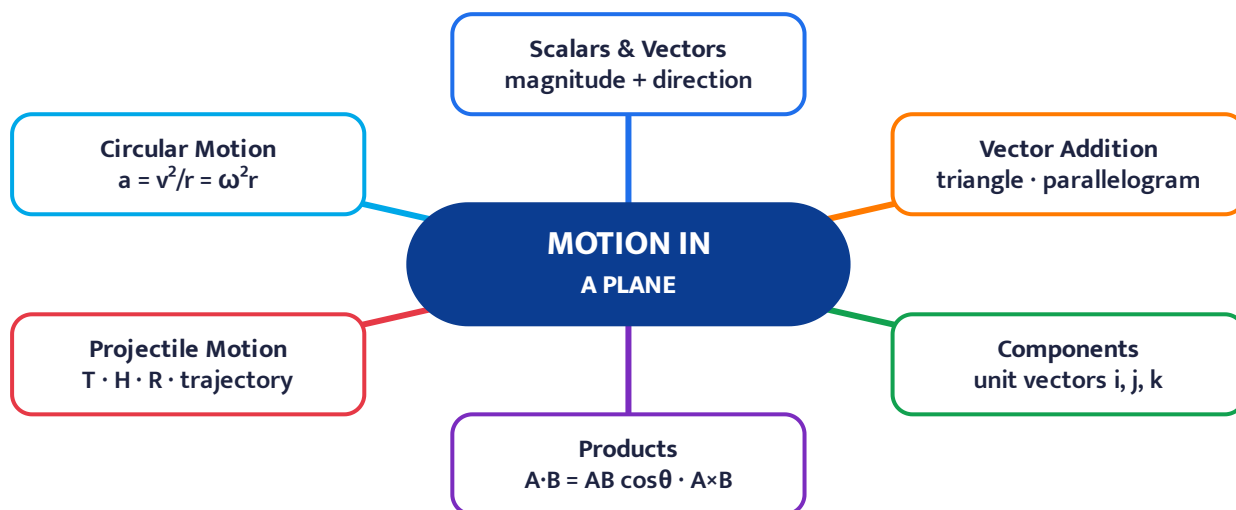
Vector addition

Dot & cross product

Projectile motion

Relative velocity

Uniform circular motion



Chapter at a glance

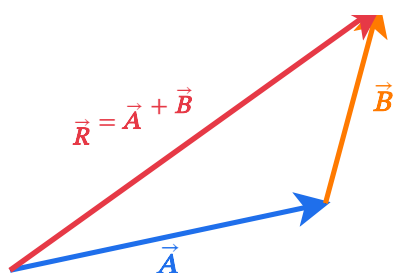
3.1 Scalars and Vectors

A **scalar** has only magnitude (mass, time, speed, work, energy, temperature). A **vector** has magnitude **and** direction and adds by the triangle law (displacement, velocity, force, momentum, torque). Current has direction but is a **scalar** because it does not obey vector addition.

NCERT Line to Remember

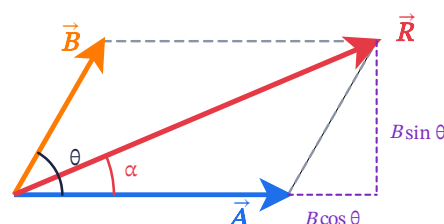
- **Equal vectors:** same magnitude and same direction (free vectors can be shifted parallel to themselves).
- **Multiplying by a real number λ :** magnitude becomes $|\lambda| A$; direction reverses if $\lambda < 0$; and $\vec{A} - \vec{A} = \vec{0}$ (**null vector**, zero magnitude).
- **Position vector $\vec{r} = x\hat{i} + y\hat{j}$; displacement $\Delta\vec{r} = \vec{r}_2 - \vec{r}_1$** is independent of the path.

3.2 Addition and Subtraction of Vectors



head of A joined to tail of B

Triangle law (head-to-tail)



Parallelogram law: R is the diagonal

Resultant of two vectors (angle θ between them)

$$R = \sqrt{A^2 + B^2 + 2AB\cos\theta} \quad \text{Magnitude of resultant}$$

$$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta} \quad \text{Direction of R from } \vec{A}$$

$$|\vec{A} - \vec{B}| = \sqrt{A^2 + B^2 - 2AB\cos\theta} \quad \text{Subtraction: } \vec{A} - \vec{B} = \vec{A} + (-\vec{B}) \quad |A - B| \leq R \leq A + B \quad \text{Range of resultant } (\theta = 180^\circ \rightarrow \min, \theta = 0^\circ \rightarrow \max)$$

Solved Example

Forces of 3 N and 5 N act at 60° . Find the resultant.

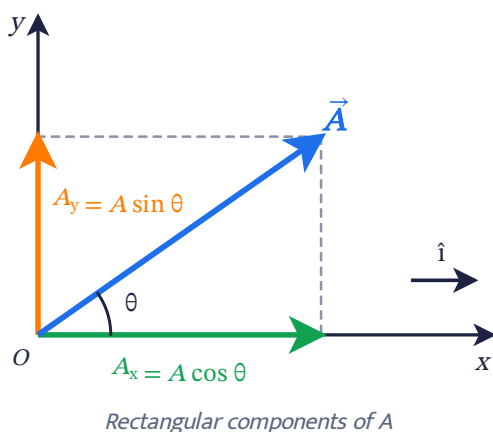
Solution: $R = \sqrt{9 + 25 + 2(3)(5)\frac{1}{2}} = \sqrt{49} = 7 \text{ N.}$

$$\tan\alpha = \frac{5(0.866)}{3 + 2.5} = 0.787 \Rightarrow \alpha \approx 38^\circ \text{ with the 3 N force.}$$

Memory Trick

- Two **equal** vectors A at angle θ : $R = 2A\cos\frac{\theta}{2}$.
- $\theta = 60^\circ \rightarrow \sqrt{3} A$; $90^\circ \rightarrow \sqrt{2} A$; $120^\circ \rightarrow A$; $180^\circ \rightarrow 0$.

3.3 Resolution of Vectors and Unit Vectors



Components

$$\vec{A} = A_x\hat{i} + A_y\hat{j} + A_z\hat{k} \quad \text{Component form}$$

$$A = \sqrt{A_x^2 + A_y^2 + A_z^2} \quad \text{Magnitude}$$

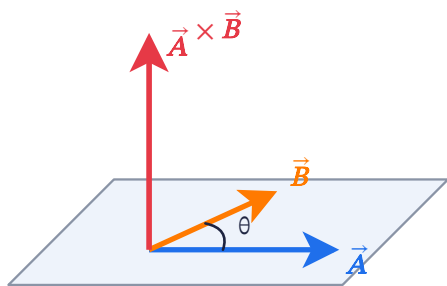
$$\tan\theta = \frac{A_y}{A_x} \quad \text{Direction in the x-y plane}$$

$$\hat{A} = \frac{\vec{A}}{|\vec{A}|} \quad \text{Unit vector: magnitude 1, no unit}$$

$$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1 \quad \text{Direction cosines } \cos\alpha = A_x / A \dots$$

3.4 Scalar (Dot) and Vector (Cross) Products

	Dot product $\vec{A} \cdot \vec{B}$	Cross product $\vec{A} \times \vec{B}$
Definition	$AB\cos\theta$ (a scalar)	$AB\sin\theta \hat{n}$ (a vector $\perp$ to both)
Components	$A_xB_x + A_yB_y + A_zB_z$	$(A_yB_z - A_zB_y)\hat{i} + (A_zB_x - A_xB_z)\hat{j} + (A_xB_y - A_yB_x)\hat{k}$
Unit vectors	$\hat{i} \cdot \hat{i} = 1, \hat{i} \cdot \hat{j} = 0$	$\hat{i} \times \hat{j} = \hat{k}, \hat{i} \times \hat{i} = \vec{0}$
Order	commutative: $\vec{A} \cdot \vec{B} = \vec{B} \cdot \vec{A}$	anti-commutative: $\vec{B} \times \vec{A} = -\vec{A} \times \vec{B}$
Zero when	$\vec{A} \perp \vec{B}$ ($\theta = 90^\circ$)	$\vec{A} \parallel \vec{B}$ ($\theta = 0^\circ$ or 180°)
Examples	work $W = \vec{F} \cdot \vec{s}$, power $\vec{F} \cdot \vec{v}$	torque $\vec{r} \times \vec{F}, \vec{L} = \vec{r} \times \vec{p}$



right-hand rule: curl $A \rightarrow B$, thumb = $A \times B$

$A \times B$ is perpendicular to the plane of A and B

VS Sir's Exam Tip

- Angle between vectors: $\cos\theta = \frac{\vec{A} \cdot \vec{B}}{AB}$.
- $|\vec{A} \times \vec{B}|$ = area of parallelogram with sides $\vec{A}, \vec{B}$; triangle = half of it.
- Projection of $\vec{A}$ on $\vec{B} = \vec{A} \cdot \hat{B}$.

Memory Trick

Cyclic order $\hat{i} \rightarrow \hat{j} \rightarrow \hat{k} \rightarrow \hat{i}$ gives +; the reverse gives -:
 $\therefore \hat{j} \times \hat{i} = -\hat{k}$.

3.5 Motion in a Plane with Constant Acceleration

Kinematics in 2D

$$\vec{v} = \frac{d\vec{r}}{dt} = v_x \hat{i} + v_y \hat{j} \quad \text{Velocity: always along the tangent to the path}$$

$$\vec{a} = \frac{d\vec{v}}{dt} = a_x \hat{i} + a_y \hat{j} \quad \text{Acceleration (need not be along the path)}$$

$$\vec{v} = \vec{v}_0 + \vec{a}t \quad \text{Constant } \vec{a} - \text{vector form}$$

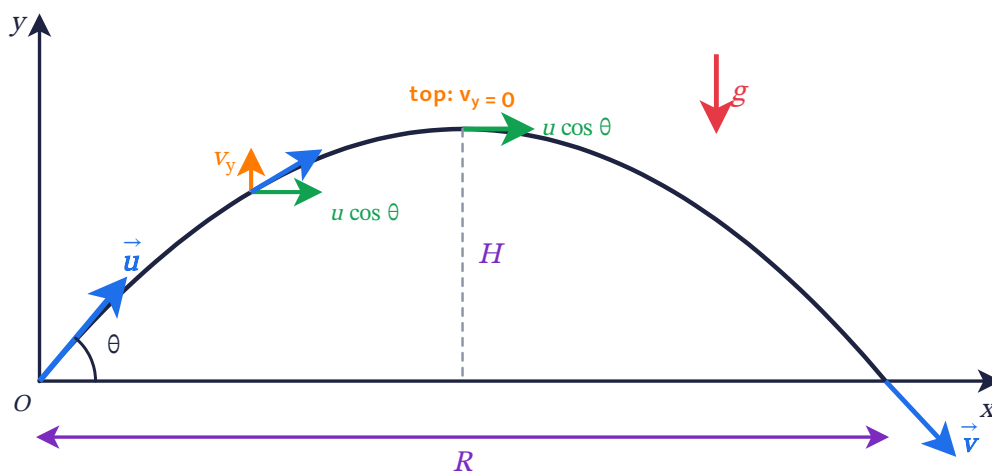
$$\vec{r} = \vec{r}_0 + \vec{v}_0 t + \frac{1}{2} \vec{a} t^2 \quad \text{Position at time } t$$

NCERT Line to Remember

Motion in a plane = **two independent 1-D motions** along x and y , occurring simultaneously. Each component obeys its own equations: $x = x_0 + v_{0x}t + \frac{1}{2}a_x t^2$, $y = y_0 + v_{0y}t + \frac{1}{2}a_y t^2$.

3.6 Projectile Motion

A body thrown into the air and moving only under gravity is a **projectile**. Horizontally: $a_x = 0 \Rightarrow v_x = u \cos\theta$ stays constant. Vertically: $a_y = -g \Rightarrow$ uniformly accelerated motion. Air resistance is neglected.



Projectile launched with speed u at angle θ : horizontal velocity constant, vertical velocity changes at rate g



Projectile on level ground ($g = 10 \text{ m s}^{-2}$)

$$T = \frac{2u \sin \theta}{g}$$

Time of flight

$$H = \frac{u^2 \sin^2 \theta}{2g}$$

Maximum height

$$R = \frac{u^2 \sin 2\theta}{g}$$

Horizontal range

$$R_{\max} = \frac{u^2}{g} (\theta = 45^\circ) \quad \text{Maximum range; then } H = R_{\max} / 4$$

$$y = x \tan \theta - \frac{gx^2}{2u^2 \cos^2 \theta} \quad \text{Trajectory - a parabola}$$

$$y = x \tan \theta \left(1 - \frac{x}{R} \right) \quad \text{Same, in terms of } R$$

$$\tan \theta = \frac{4H}{R}$$

Link between H and R

$$v = \sqrt{u^2 - 2gy}$$

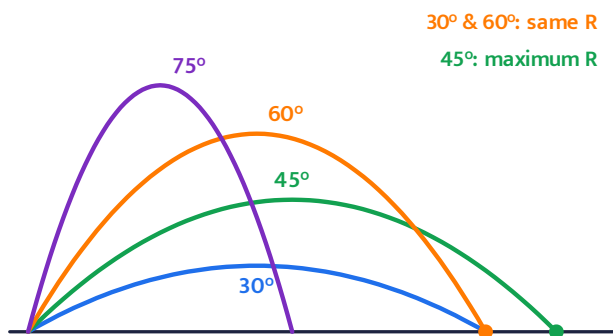
Speed at height y



Solved Example

A ball is projected at 20 m/s at 30° to the horizontal. Find T, H and R. ($g = 10 \text{ m s}^{-2}$)

Solution: $T = \frac{2(20)(0.5)}{10} = 2 \text{ s}$; $H = \frac{400(0.25)}{20} = 5 \text{ m}$; $R = \frac{400 \sin 60^\circ}{10} = 20\sqrt{3} \approx 34.6 \text{ m}$.



Same speed: θ and $(90^\circ - \theta)$ give equal ranges



Complementary angles $\theta, 90^\circ - \theta$

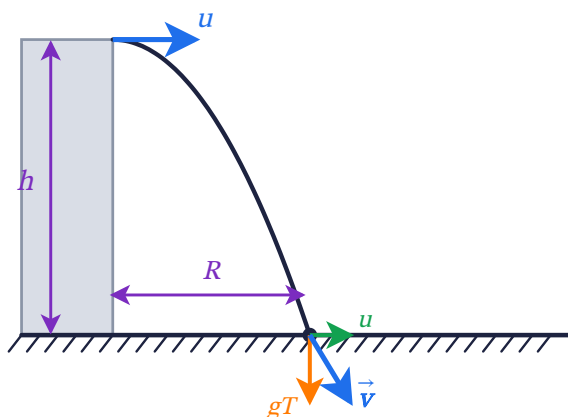
- Same range R ; different H and T .
- $H_1 + H_2 = \frac{u^2}{2g}$, $\frac{H_1}{H_2} = \tan^2 \theta$
- $R = 4\sqrt{H_1 H_2}$, $T_1 T_2 = \frac{2R}{g}$



Common Trap

- At the highest point the velocity is **not zero** – it is $u \cos \theta$ (horizontal); only $v_y = 0$. Acceleration is still g downward, so $\vec{v} \perp \vec{a}$ there.
- KE at the top = $K \cos^2 \theta$ (not zero). Momentum changes only in the vertical direction.

Projectile fired horizontally from a height h



Horizontal projection from a cliff



Horizontal projection

$$T = \sqrt{\frac{2h}{g}}$$

Same as free fall – independent of u

$$R = u \sqrt{\frac{2h}{g}}$$

Horizontal distance

$$v = \sqrt{u^2 + 2gh}$$

Speed on hitting the ground

$$\tan \beta = \frac{gT}{u}$$

Angle below horizontal

$$y = \frac{gx^2}{2u^2}$$

Path (y measured downward)

 Solved Example

A stone is thrown horizontally at 10 m/s from a 45 m high cliff. Find the time, the horizontal distance and the speed on landing.

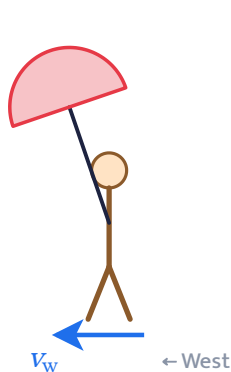
Solution: $T = \sqrt{2(45)/10} = 3 \text{ s}$; $R = 10 \times 3 = 30 \text{ m}$; $v = \sqrt{10^2 + 30^2} = \sqrt{1000} \approx 31.6 \text{ m/s}$, at $\tan^{-1}3 \approx 72^\circ$ below the horizontal.

 VS Sir's Exam Tip

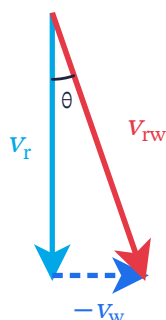
- A ball **dropped** and a ball **thrown horizontally** from the same height land **at the same time**.
- For the same u , R is maximum at 45° ; H and T are maximum at 90° . $R \propto u^2$, $H \propto u^2$, $T \propto u$.

3.7 Relative Velocity in Two Dimensions

$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$ – subtract vectorially (triangle law with $-\vec{v}_B$). **Rain-umbrella:** the umbrella must be held along the velocity of rain **relative to the person**, $\vec{v}_{rm} = \vec{v}_r - \vec{v}_m$.

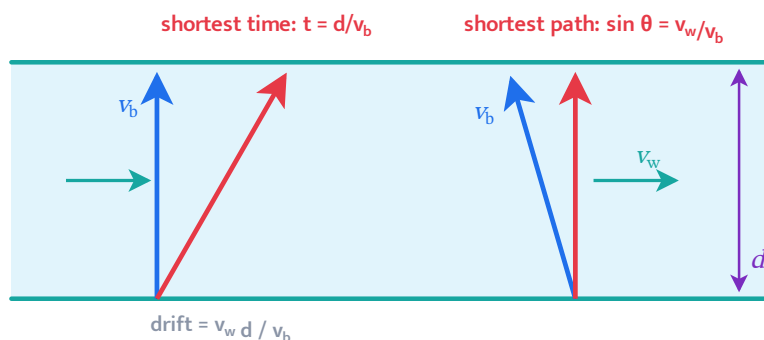


Umbrella tilted forward, along the relative velocity of rain

 **Solved Example**

Rain falls vertically at 35 m/s. A woman cycles from east to west at 12 m/s. In which direction should she hold her umbrella? (NCERT)

Solution: $\vec{v}_{rw} = \vec{v}_r - \vec{v}_w$: 35 m/s down and 12 m/s towards east. $\tan\theta = \frac{12}{35} = 0.343 \Rightarrow \theta \approx 19^\circ$ **with the vertical, towards the west** (tilted forward).

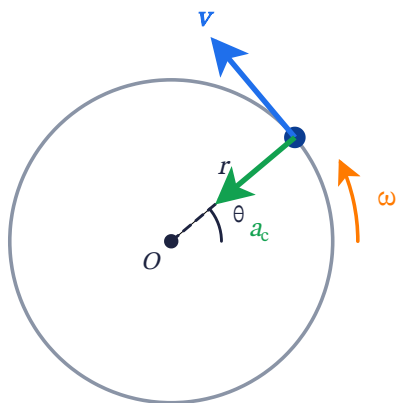


River-boat: head straight across for the least time, head upstream for the shortest path

 River-boat problems (JEE-leaning)

- River width d , water speed v_w , boat speed in still water v_b .
- **Least time:** head perpendicular to the bank: $t_{min} = \frac{d}{v_b}$, drift $x = \frac{v_w d}{v_b}$.
- **Shortest path (zero drift):** head upstream at $\sin\theta = \frac{v_w}{v_b}$ from the normal; $t = \frac{d}{\sqrt{v_b^2 - v_w^2}}$ (possible only if $v_b > v_w$).

3.8 Uniform Circular Motion



v is tangential, a points to the centre

Uniform circular motion

$$\omega = \frac{\Delta\theta}{\Delta t}, \quad v = \omega r \quad \text{Angular speed (rad/s) and linear speed}$$

$$T = \frac{2\pi}{\omega} = \frac{2\pi r}{v}, \quad f = \frac{1}{T} \quad \text{Time period and frequency}$$

$$\omega = 2\pi f \quad \text{Angular frequency}$$

$$a_c = \frac{v^2}{r} = \omega^2 r = 4\pi^2 f^2 r \quad \text{Centripetal acceleration, towards the centre}$$

NCERT Line to Remember

- In uniform circular motion the **speed is constant but velocity is not** – its direction changes continuously, so the body is accelerated.
- The centripetal acceleration has constant **magnitude** but its **direction** keeps changing (always towards the centre) – so $\vec{a}$ is **not a constant vector**.

Solved Example

An insect trapped in a circular groove of radius 12 cm moves steadily and completes 7 revolutions in 100 s. Find ω , v and the acceleration. (NCERT)

Solution: $f = 7 / 100 \text{ Hz} \Rightarrow \omega = 2\pi(0.07) = \mathbf{0.44 \text{ rad/s}}$; $v = \omega r = 0.44 \times 12 \approx \mathbf{5.3 \text{ cm/s}}$; $a = \omega^2 r \approx \mathbf{2.3 \text{ cm/s}^2}$, directed towards the centre.

Common Trap

- In UCM, work done by the centripetal force is **zero** ($\vec{F} \perp \vec{v}$), so KE is constant.
- After half a revolution, $|\Delta \vec{v}| = 2v$ (not zero); after a full revolution, $\Delta \vec{v} = 0$.

Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- NCERT based** Which of the following is a vector quantity?

(a) Displacement (b) Speed
(c) Electric current (d) Work
- NCERT based** Two vectors have magnitudes 3 and 4 units. Their resultant can NOT be

(a) 7 (b) 1 (c) 5 (d) 8
- NCERT based** $\hat{j} \times \hat{i}$ equals

(a) $\hat{k}$ (b) $-\hat{k}$ (c) $\vec{0}$ (d) $\hat{i}$

- 4 **NCERT based** In uniform circular motion, which quantity remains constant?
 (a) Acceleration (b) Speed (c) Displacement (d) Velocity
- 5 **NCERT based** At the highest point of a projectile's path ($\theta \neq 90^\circ$), the angle between velocity and acceleration is
 (a) 180° (b) 90° (c) 0° (d) 45°
- 6 **NCERT based** The angular speed of the seconds hand of a clock is
 (a) 2π rad/s (b) $\frac{\pi}{1800}$ rad/s (c) $\frac{\pi}{60}$ rad/s (d) $\frac{\pi}{30}$ rad/s
- 7 **NEET pattern** Two equal forces F act at a point with an angle of 120° between them. The resultant is
 (a) $\sqrt{3} F$ (b) $F/2$ (c) F (d) $2F$
- 8 **NEET pattern** If $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$, the angle between $\vec{A}$ and $\vec{B}$ is
 (a) 180° (b) 90° (c) 60° (d) 0°
- 9 **NEET pattern** If $|\vec{A} \times \vec{B}| = \sqrt{3} \vec{A} \cdot \vec{B}$, the angle between the vectors is
 (a) 60° (b) 45° (c) 30° (d) 90°
- 10 **NEET pattern** For what value of a are $\vec{A} = 2\hat{i} + a\hat{j} + \hat{k}$ and $\vec{B} = 4\hat{i} - 2\hat{j} - 2\hat{k}$ perpendicular?
 (a) -3 (b) 4 (c) 3 (d) 2
- 11 **NEET pattern** A ball is projected at 20 m/s at 30° with the horizontal. Its maximum height is ($g = 10 \text{ m s}^{-2}$)
 (a) 20 m (b) 5 m (c) 15 m (d) 10 m
- 12 **NEET pattern** Two projectiles are thrown with the same speed at 30° and 60° . The ratio of their maximum heights is
 (a) $1:1$ (b) $1:\sqrt{3}$ (c) $3:1$ (d) $1:3$
- 13 **NEET pattern** The range of a projectile equals four times its maximum height. The angle of projection is
 (a) 60° (b) 30° (c) 45° (d) $\tan^{-1}4$
- 14 **NEET pattern** The maximum horizontal range of a ball is 40 m. The maximum height it can reach when thrown vertically up with the same speed is
 (a) 10 m (b) 40 m (c) 20 m (d) 80 m
- 15 **NEET pattern** A ball is thrown horizontally at 20 m/s from the top of an 80 m high tower. It lands at a horizontal distance of ($g = 10 \text{ m s}^{-2}$)
 (a) 160 m (b) 20 m (c) 40 m (d) 80 m
- 16 **NEET pattern** A stone tied to a 0.5 m string is whirled in a horizontal circle at a constant speed of 10 m/s. Its centripetal acceleration is
 (a) 200 m s^{-2} (b) 50 m s^{-2} (c) 100 m s^{-2} (d) 20 m s^{-2}
- 17 **NEET pattern** A body is projected at 60° with kinetic energy K . Its kinetic energy at the highest point is
 (a) zero (b) $K/2$ (c) $K/4$ (d) $3K/4$

- 18 **NEET pattern** The position of a particle is $\vec{r} = 3t\hat{i} + 2t^2\hat{j}$ (SI units). Its speed at $t = 1$ s is
 (a) 5 m/s (b) 4 m/s (c) 7 m/s (d) 3 m/s
- 19 **Assertion-Reason** Assertion: In projectile motion, the horizontal component of velocity remains constant. Reason: No acceleration acts in the horizontal direction (air resistance neglected).
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 20 **Assertion-Reason** Assertion: A body in uniform circular motion has zero acceleration because its speed is constant. Reason: Acceleration is the rate of change of velocity, and velocity changes direction in circular motion.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 21 **JEE Main pattern** The trajectory of a projectile is $y = x - \frac{x^2}{20}$ (SI units, $g = 10 \text{ m s}^{-2}$). Its speed of projection is
 (a) $10\sqrt{2}$ m/s (b) 20 m/s (c) $20\sqrt{2}$ m/s (d) 10 m/s
- 22 **JEE Main pattern** The area of the parallelogram whose adjacent sides are $\vec{A} = 2\hat{i} + \hat{j}$ and $\vec{B} = \hat{i} + 3\hat{j}$ is
 (a) 2.5 units (b) 5 units (c) 7 units (d) $\sqrt{50}$ units
- 23 **JEE Main pattern** To a man walking at 3 km/h, rain appears to fall vertically. When he doubles his speed, the rain appears to meet him at 45° to the vertical. The actual speed of the rain is
 (a) $3\sqrt{2}$ km/h (b) $6\sqrt{2}$ km/h (c) 3 km/h (d) 6 km/h
- 24 **JEE Main pattern** A river 1 km wide flows at 3 km/h. A boat that moves at 5 km/h in still water crosses along the shortest path. The time taken is
 (a) 7.5 min (b) 15 min (c) 20 min (d) 12 min

✓ Answer Key & Quick Solutions

- 1(a) 2(d) 3(b) 4(b) 5(b) 6(d) 7(c) 8(b) 9(a) 10(c) 11(b) 12(d) 13(c) 14(c) 15(d)
 16(a) 17(c) 18(a) 19(a) 20(c) 21(a) 22(b) 23(a) 24(b)

- (a) Displacement has magnitude and direction and obeys vector addition. Current has direction but does not add as a vector.
- (d) $|A - B| \leq R \leq A + B \Rightarrow 1 \leq R \leq 7$. So 8 is impossible.
- (b) Reverse of cyclic order $\Rightarrow \hat{j} \times \hat{i} = -\hat{k}$.
- (b) Only the magnitude of velocity (speed) is constant; velocity and acceleration change direction.
- (b) Velocity is horizontal ($u \cos \theta$), acceleration g is vertical $\Rightarrow 90^\circ$.
- (d) $\omega = 2\pi / T = 2\pi / 60 = \pi / 30 \text{ rad/s}$.
- (c) $R = 2F \cos 60^\circ = F$.
- (b) Squaring: $4AB \cos \theta = 0 \Rightarrow \theta = 90^\circ$.
- (a) $AB \sin \theta = \sqrt{3} AB \cos \theta \Rightarrow \tan \theta = \sqrt{3} \Rightarrow 60^\circ$.

10. (c) $\vec{A} \cdot \vec{B} = 8 - 2a - 2 = 0 \Rightarrow a = 3$.
11. (b) $H = \frac{u^2 \sin^2 \theta}{2g} = \frac{400 \times 0.25}{20} = 5 \text{ m}$.
12. (d) $H \propto \sin^2 \theta$: $\frac{1/4}{3/4} = 1:3$ (ranges are equal).
13. (c) $\tan \theta = 4H / R = 1 \Rightarrow 45^\circ$.
14. (c) $u^2 / g = 40 \Rightarrow H = u^2 / 2g = 20 \text{ m}$.
15. (d) $T = \sqrt{2(80) / 10} = 4 \text{ s}$; $R = 20 \times 4 = 80 \text{ m}$.
16. (a) $a = v^2 / r = 100 / 0.5 = 200 \text{ m s}^{-2}$.
17. (c) $v_{\text{top}} = u \cos 60^\circ = u / 2 \Rightarrow \text{KE} = K \cos^2 60^\circ = K / 4$.
18. (a) $\vec{v} = 3\hat{i} + 4t\hat{j} \Rightarrow \text{at } t = 1: \sqrt{9 + 16} = 5 \text{ m/s}$.
19. (a) $a_x = 0 \Rightarrow v_x$ constant. R correctly explains A.
20. (c) Direction of velocity changes $\Rightarrow$ centripetal acceleration $v^2 / r \neq 0$. A false, R true.
21. (a) $\tan \theta = 1 \Rightarrow 45^\circ$; $\frac{g}{2u^2 \cos^2 \theta} = \frac{10}{u^2} = \frac{1}{20} \Rightarrow u^2 = 200 \Rightarrow u = 10\sqrt{2} \text{ m/s}$.
22. (b) $\vec{A} \times \vec{B} = (2 \cdot 3 - 1 \cdot 1)\hat{k} = 5\hat{k} \Rightarrow \text{area} = 5$.
23. (a) Rain = $(3, -v)$. At 6 km/h: relative = $(-3, -v)$; $45^\circ \Rightarrow v = 3$. Speed = $\sqrt{9 + 9} = 3\sqrt{2} \text{ km/h}$.
24. (b) $t = \frac{d}{\sqrt{v_b^2 - v_w^2}} = \frac{1}{4} \text{ h} = 15 \text{ min}$. (Least-time crossing would take 12 min.)

CHAPTER

4

Laws of Motion

17

Day 5–6

NEET weightage ★★★

NCERT Class 11

Newton's laws

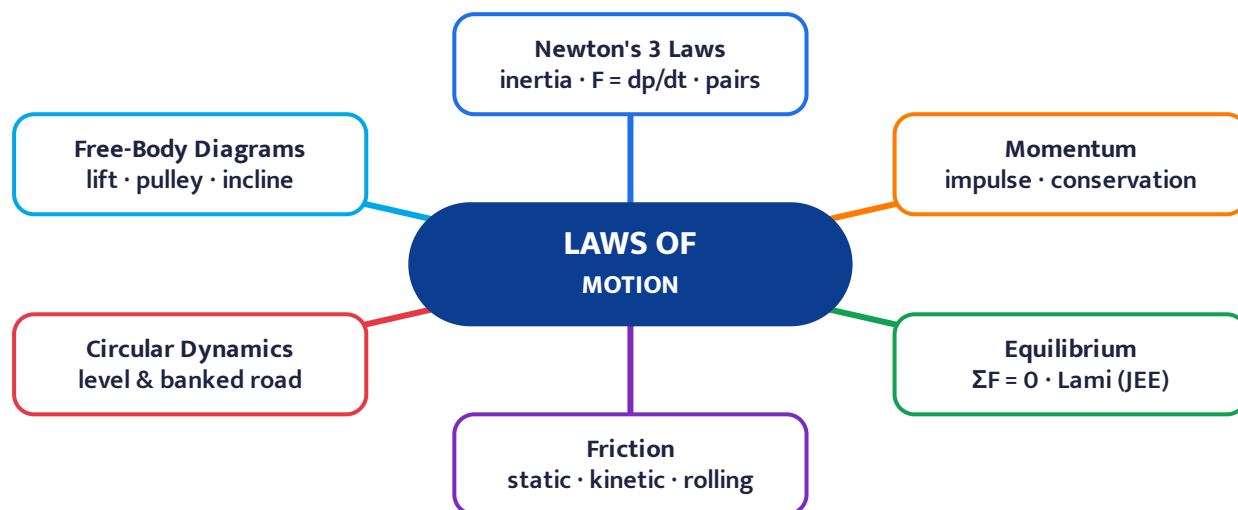
Momentum & impulse

Free-body diagrams

Apparent weight

Friction

Banking of roads



Chapter at a glance

4.1 Inertia and Newton's First Law

Aristotle believed a force is needed to **keep** a body moving. Galileo (inclined-plane experiments) showed the opposite: **a body keeps moving with uniform velocity unless an external force acts** – friction is what stops real objects. This resistance to change of state of rest or motion is **inertia**, and its measure is **mass**.

NCERT Line to Remember

First law: Every body continues in its state of rest or of uniform motion in a straight line, unless compelled by some external force to act otherwise. If $\vec{F}_{net} = 0$ then $\vec{a} = 0$.

Everyday inertia

- Passengers jerk **backward** when a bus starts suddenly, **forward** when it stops.
- Dust leaves a carpet when it is beaten (inertia of rest).
- Seat belts protect against inertia of motion.

4.2 Momentum, Newton's Second Law and Impulse

Linear momentum $\vec{p} = m\vec{v}$ (vector, unit kg m s^{-1}). The **second law**: the rate of change of momentum of a body is directly proportional to the applied force and takes place in the direction of the force. It is a vector law: $F_x = ma_x$, $F_y = ma_y$, $F_z = ma_z$.

NCERT Line to Remember

A cricketer **moves his hands back** while catching a ball: for the same Δp , a larger Δt means a smaller force. Same idea: sand pits for high jump, shock absorbers, packing with thermocol.

Second law & impulse

$$\vec{F} = \frac{d\vec{p}}{dt} = m\vec{a}$$

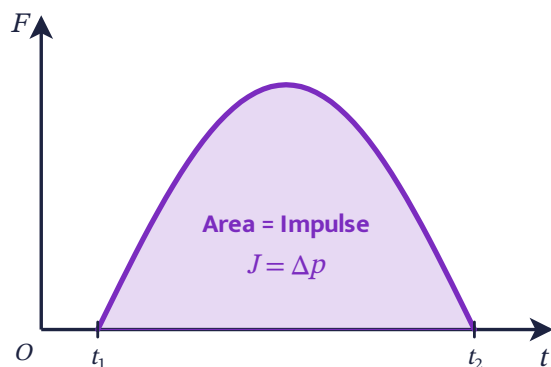
Second law (constant mass). $1 \text{ N} = 1 \text{ kg m s}^{-2}$

$$\vec{J} = \vec{F}_{av} \Delta t = \Delta \vec{p}$$

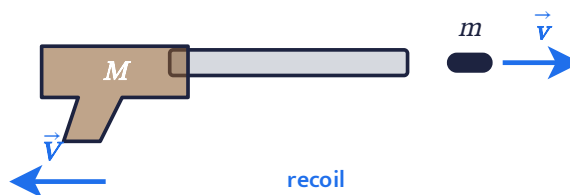
Impulse = change in momentum

$$J = \int F dt$$

Area under F-t graph



Impulse = area under the F-t curve



Before: $p = 0 \rightarrow$ After: $mv + MV = 0$

Recoil of a gun (momentum conservation)

Solved Example

A 0.15 kg ball moving at 12 m/s is hit back along its line with 20 m/s. If contact lasts 0.01 s, find the impulse and average force.

Solution: Take the return direction +ve: $J = m(v - u) = 0.15[20 - (-12)] = 4.8 \text{ N s}$.

$F_{av} = J / \Delta t = 4.8 / 0.01 = 480 \text{ N}$, opposite to the initial motion.

4.3 Third Law and Conservation of Momentum

Third law: to every action there is always an equal and opposite reaction, $\vec{F}_{AB} = -\vec{F}_{BA}$. The two forces act **simultaneously on different bodies**, so they never cancel each other.

For an isolated system ($\vec{F}_{ext} = 0$), internal forces come in third-law pairs, so the **total momentum is conserved**: $m_1\vec{u}_1 + m_2\vec{u}_2 = m_1\vec{v}_1 + m_2\vec{v}_2$. Applications: recoil of a gun, rocket propulsion, explosion of a shell, collisions.

Solved Example

A 4 kg gun fires a 20 g bullet with muzzle speed 400 m/s. Find the recoil speed of the gun.

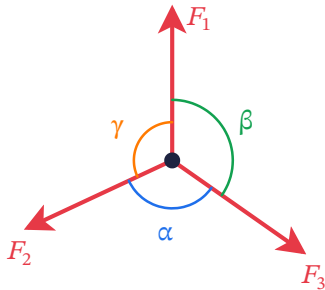
Solution: $0 = mv + MV \Rightarrow V = -\frac{mv}{M} = -\frac{0.02 \times 400}{4} = -2 \text{ m/s}$. The gun recoils at **2 m/s**. (KE of the bullet is much larger: $E \propto \frac{1}{m}$ for equal momenta.)

Common Trap

- Action-reaction pairs **never** act on the same body. Weight mg and normal force N on a book are **not** a third-law pair (both act on the book; the pair of mg is the book pulling the Earth).
- Momentum is conserved even in inelastic collisions and explosions – kinetic energy need not be.

4.4 Equilibrium of Concurrent Forces

A particle is in **equilibrium** if the net external force on it is zero: $\vec{F}_1 + \vec{F}_2 + \dots = 0$, i.e. $\sum F_x = 0$, $\sum F_y = 0$, $\sum F_z = 0$. For three forces in equilibrium, they must be **coplanar** and can be represented by the three sides of a triangle taken in order.



Three concurrent forces in equilibrium



Lami's theorem (JEE Main extra)

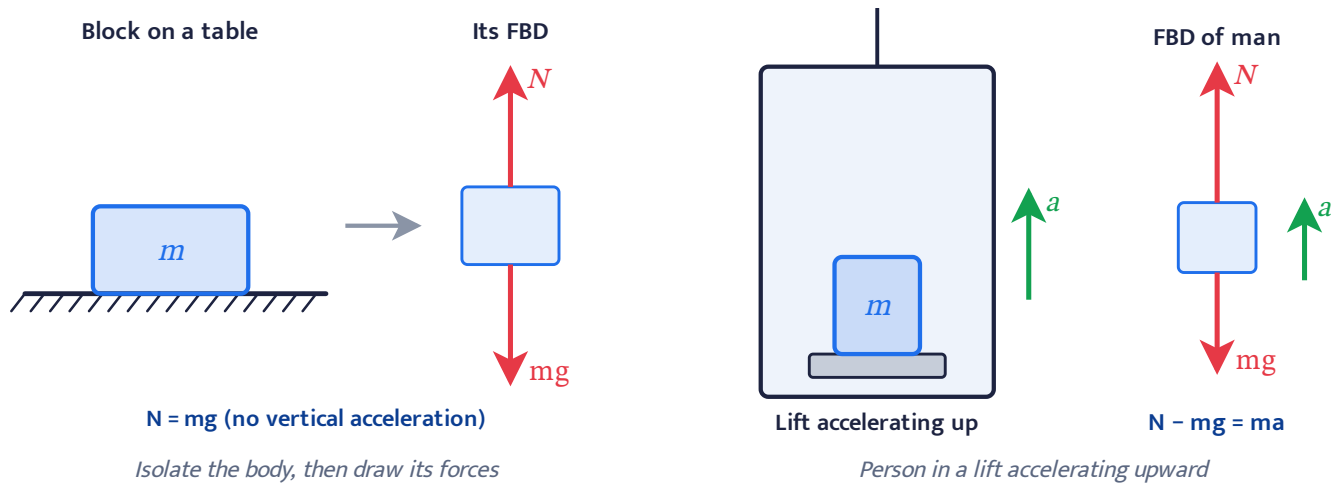
If three concurrent forces keep a body in equilibrium, each force is proportional to the sine of the angle between the other two:

$$\frac{F_1}{\sin \alpha} = \frac{F_2}{\sin \beta} = \frac{F_3}{\sin \gamma}$$

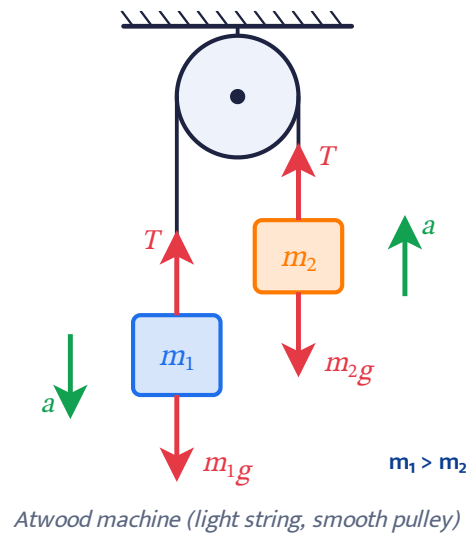
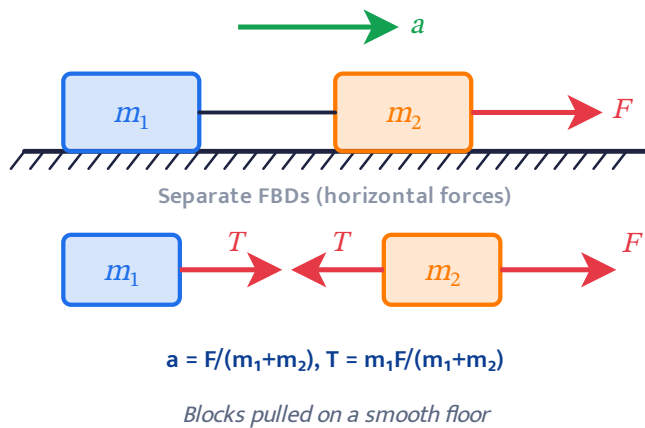
α lies between F_2 and F_3 (opposite F_1), and so on. Ideal for a weight hung by two strings.

4.5 Free-Body Diagrams, Connected Bodies and Apparent Weight

A **free-body diagram (FBD)** shows **one** isolated body and **all external forces on it** – weight, normal reaction, tension, friction, spring force, applied force. Forces the body exerts on others are **not** drawn. Contact forces: normal $N \perp$ surface, friction $\parallel$ surface; tension in a light string is the same throughout (and across a light frictionless pulley).



Lift motion	Acceleration	Apparent weight N	Scale reads
At rest / uniform velocity (up or down)	0	mg	true weight
Accelerating up or decelerating down	a upward	$m(g + a)$	more
Accelerating down or decelerating up	a downward	$m(g - a)$	less
Free fall (cable breaks)	$a = g \downarrow$	0	weightless



Connected-body results

$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$	Atwood machine acceleration	$T = \frac{2m_1m_2g}{m_1 + m_2}$	String tension (always between m_2g and m_1g)
$N_{12} = \frac{m_2F}{m_1 + m_2}$	Contact force when F pushes m_1 against m_2	$a = g\sin\theta$	Smooth incline, block sliding down

Solved Example

In an Atwood machine $m_1 = 3$ kg and $m_2 = 2$ kg. Find the acceleration and the tension ($g = 10$ m/s²).

Solution: For m_1 : $m_1g - T = m_1a$; for m_2 : $T - m_2g = m_2a$. Add: $a = \frac{(3-2)10}{5} = 2$ m/s².

$T = m_2(g + a) = 2 \times 12 = 24$ N (check: $\frac{2 \cdot 3 \cdot 2 \cdot 10}{5} = 24$).

VS Sir's Exam Tip

Standard recipe for every NEET mechanics problem: (1) choose the body/system, (2) draw its FBD, (3) pick axes along the acceleration, (4) write $\sum F = ma$ for each body, (5) add constraints (same a for a taut string, contact).

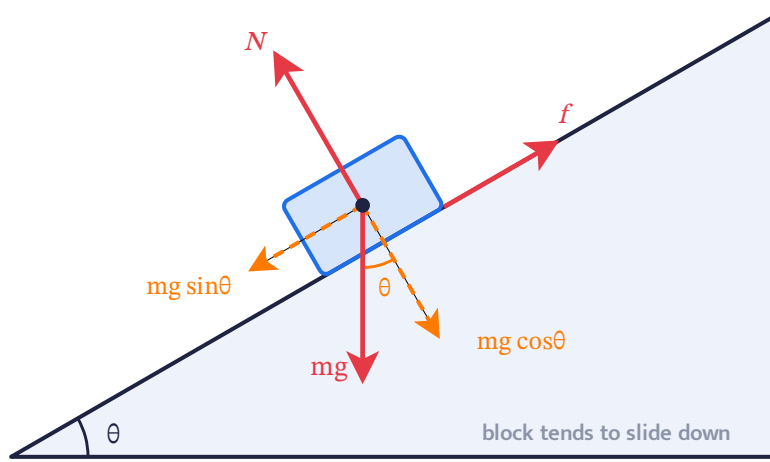
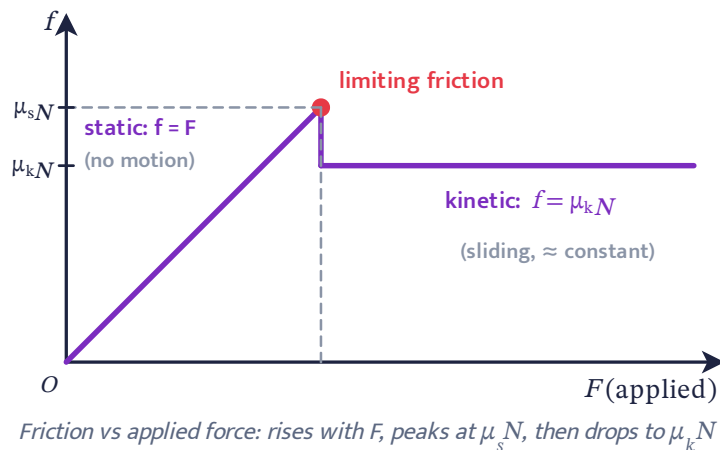
Treating the whole string-connected system as one body gives a instantly: $a = \frac{\text{net driving force}}{\text{total mass}}$.

4.6 Friction

Friction opposes **relative motion (or impending relative motion)** between surfaces in contact. **Static friction** f_s is self-adjusting: $f_s \leq \mu_s N$; it equals the applied force until the limiting value $\mu_s N$ is reached. Once sliding starts, **kinetic friction** $f_k = \mu_k N$ acts, and $\mu_k < \mu_s$. Both are (nearly) independent of the area of contact and of speed.

Friction formulas

$f_s^{\max} = \mu_s N, \quad f_k = \mu_k N$	Limiting and kinetic friction	$\tan\lambda = \mu_s$	Angle of friction λ (resultant of $f_{\max}$ and N with the normal)
$\tan\theta_r = \mu_s$	Angle of repose: block just begins to slide $\Rightarrow \theta_r = \lambda$	$a = g(\sin\theta - \mu_k \cos\theta)$	Block sliding down a rough incline
$a = g(\sin\theta + \mu_k \cos\theta)$	Retardation of a block moving up the incline		



Solved Example

A block slides down a rough incline of 37° with $\mu_k = 0.5$. Find its acceleration ($\sin 37^\circ = 0.6$, $g = 10 \text{ m/s}^2$).

Solution: $a = g(\sin \theta - \mu_k \cos \theta) = 10(0.6 - 0.5 \times 0.8) = 2 \text{ m/s}^2$ down the incline. (It slides at all because $\tan 37^\circ = 0.75 > \mu$.)

Rolling friction & lubrication

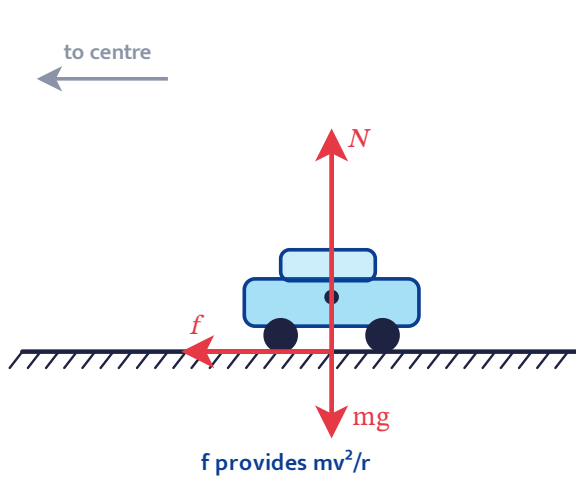
- **Rolling friction** $\ll$ sliding friction – hence wheels and ball bearings.
- Lubricants replace dry friction by much smaller fluid friction.
- Friction is necessary: walking, braking and a car turning all need it.

Common Trap

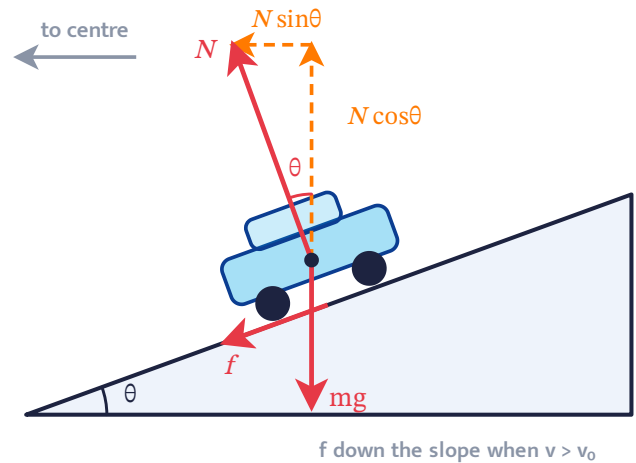
- Friction is **not** always μN : if $F < \mu_s N$, then $f = F$ (self-adjusting).
- **Pulling** a lawn roller at angle θ is easier than pushing: $N = mg - F \sin \theta$ (pull) vs $mg + F \sin \theta$ (push).
- Friction can act **along** the motion (e.g. on a box placed on an accelerating truck).

4.7 Dynamics of Uniform Circular Motion

A body moving in a circle of radius r with speed v needs a net force $\frac{mv^2}{r}$ **towards the centre** (centripetal force). It is not a new force – friction, tension, gravity or a component of the normal force supplies it.



Car on a level road: friction alone

Banked road: $N \sin \theta (+ f \cos \theta)$ gives mv^2/r f down the slope when $v > v_0$

Circular motion on roads

$$v_{\max} = \sqrt{\mu_s r g}$$

Level road (friction provides mv^2/r)

$$v_0 = \sqrt{r g \tan \theta}$$

Banked road, **no friction needed** (optimum speed)

$$v_{\max} = \sqrt{r g \frac{\mu_s + \tan \theta}{1 - \mu_s \tan \theta}}$$

Banked road with friction (maximum safe speed)

$$v_{\min} = \sqrt{r g \frac{\tan \theta - \mu_s}{1 + \mu_s \tan \theta}}$$

Minimum speed (friction up the slope)

Solved Example

A circular track of radius 300 m is banked at 15° . If $\mu_s = 0.2$, find the optimum speed and the maximum permissible speed ($\tan 15^\circ = 0.27$).

Solution: $v_0 = \sqrt{300 \times 9.8 \times 0.27} \approx 28.1 \text{ m/s}$. $v_{\max} = \sqrt{300 \times 9.8 \times \frac{0.2 + 0.27}{1 - 0.2 \times 0.27}} \approx 38.2 \text{ m/s}$ (NCERT example, $g = 9.8 \text{ m/s}^2$).

Memory Trick

"Level – μ , Banked – $\tan$ ": level road $v^2 = \mu r g$; frictionless banking $v^2 = r g \tan \theta$; with friction just replace $\tan \theta$ by $\frac{\mu + \tan \theta}{1 - \mu \tan \theta} = \tan(\theta + \lambda)$. Mass never appears!

Conical pendulum (JEE Main)

Bob on a string of length l at angle θ with the vertical: $T \cos \theta = mg$, $T \sin \theta = m \omega^2 r \Rightarrow \tan \theta = \frac{v^2}{r g}$ and period $2\pi \sqrt{\frac{l \cos \theta}{g}}$.

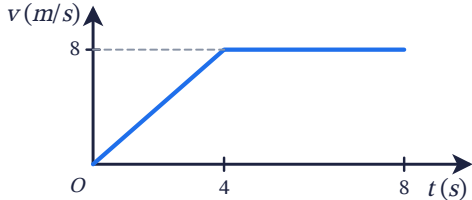
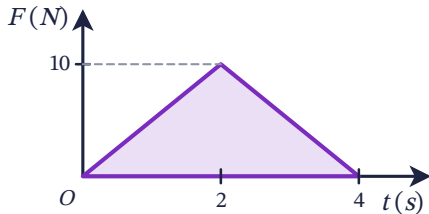
VS Sir's Exam Tip

Centrifugal force is **never** drawn in an FBD in the ground (inertial) frame. A car skids outward on a curve because friction is **insufficient**, not because an outward force appears.



Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** Inertia of a body is measured by its
 (a) weight (b) mass (c) momentum (d) velocity
- 2 **NCERT based** The v - t graph of a 2 kg body is shown. The net force on it between $t = 4$ s and $t = 8$ s is
- 
- (a) 4 N (b) 2 N (c) 0 (d) 16 N
- 3 **NCERT based** Which statement about action and reaction is correct?
 (a) They are equal only when bodies are at rest
 (b) They act on the same body and cancel
 (c) They act on different bodies, so they do not cancel
 (d) Reaction comes slightly after action
- 4 **NCERT based** A person stands on a weighing machine in a lift. The machine reads zero when the lift
 (a) falls freely (b) moves up with uniform speed
 (c) accelerates upward at g (d) moves down with uniform speed
- 5 **NEET pattern** A 60 kg man stands in a lift accelerating downward at 2 m/s^2 . His apparent weight is ($g = 10 \text{ m/s}^2$)
 (a) 480 N (b) 720 N (c) 600 N (d) 120 N
- 6 **NEET pattern** The momentum of a particle varies as $p = 3t^2 + 2t$ (SI units). The force on it at $t = 2$ s is
 (a) 14 N (b) 16 N (c) 12 N (d) 8 N
- 7 **NEET pattern** The F - t graph for a force acting on a 2 kg body initially at rest is shown. Its speed after 4 s is
- 
- (a) 20 m/s (b) 10 m/s (c) 5 m/s (d) 40 m/s
- 8 **NEET pattern** A 5 kg gun fires a 50 g bullet at 400 m/s. The recoil speed of the gun is
 (a) 40 m/s (b) 8 m/s (c) 4 m/s (d) 0.4 m/s

- 9 **NEET pattern** A 0.15 kg ball hits a bat at 12 m/s and returns along the same line at 20 m/s. The magnitude of impulse imparted is
(a) 2.4 N s (b) 1.2 N s (c) 3.0 N s (d) 4.8 N s
- 10 **NCERT based** A block of mass 2 kg lies on a floor with $\mu_s = 0.5$. A horizontal force of 6 N is applied. The friction force on the block is ($g = 10 \text{ m/s}^2$)
(a) 10 N (b) 6 N (c) 0 (d) 4 N
- 11 **NEET pattern** Two blocks of 4 kg and 6 kg are in contact on a smooth floor. A 20 N force pushes the 4 kg block. The contact force between the blocks is
(a) 10 N (b) 12 N (c) 8 N (d) 20 N
- 12 **NEET pattern** A block just begins to slide when the incline is raised to 30° . The coefficient of static friction is
(a) $\sqrt{3}$ (b) $1 / \sqrt{3}$ (c) $1/2$ (d) $\sqrt{3} / 2$
- 13 **NEET pattern** In an Atwood machine, masses 3 kg and 2 kg hang from a light string over a smooth pulley. The tension in the string is ($g = 10 \text{ m/s}^2$)
(a) 24 N (b) 20 N (c) 12 N (d) 30 N
- 14 **NEET pattern** A block slides down a rough incline of 37° with $\mu_k = 0.5$. Its acceleration is ($\sin 37^\circ = 0.6$, $g = 10 \text{ m/s}^2$)
(a) 4 m/s^2 (b) 2 m/s^2 (c) 10 m/s^2 (d) 6 m/s^2
- 15 **NEET pattern** The maximum speed with which a car can take a turn of radius 20 m on a level road with $\mu_s = 0.5$ is ($g = 10 \text{ m/s}^2$)
(a) 10 m/s (b) 20 m/s (c) 5 m/s (d) 100 m/s
- 16 **NEET pattern** A road of radius 40 m is banked at 45° . The speed at which a car needs no friction to take the turn is ($g = 10 \text{ m/s}^2$)
(a) 40 m/s (b) 14.1 m/s (c) 20 m/s (d) 10 m/s
- 17 **NEET pattern** A machine gun fires 240 bullets per minute, each of mass 10 g, at 600 m/s. The average force needed to hold the gun is
(a) 24 N (b) 2.4 N (c) 240 N (d) 1440 N
- 18 **NCERT based** Which of the following statements about friction is correct?
(a) Static friction always equals $\mu_s N$
(b) Kinetic friction is greater than limiting static friction
(c) Friction depends strongly on the area of contact
(d) Rolling friction is much smaller than sliding friction

- 19 **Assertion-Reason** Assertion: A cricketer moves his hands backward while catching a fast ball. Reason: Increasing the time of impact reduces the force for the same change in momentum.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 20 **Assertion-Reason** Assertion: The apparent weight of a person in a lift moving up with constant velocity is more than the true weight. Reason: A lift moving with constant velocity has zero acceleration.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 21 **JEE Main pattern** A 100 N weight hangs from two strings making 30° and 60° with the vertical. The tension in the string making 30° with the vertical is
 (a) $50\sqrt{3}$ N (b) 50 N (c) $100 / \sqrt{3}$ N (d) 100 N
- 22 **JEE Main pattern** A force of 12 N pulls a 4 kg block on a smooth floor through a uniform rope of mass 2 kg. The tension at the midpoint of the rope is
 (a) 12 N (b) 6 N (c) 8 N (d) 10 N
- 23 **JEE Main pattern** A road of radius 30 m is banked at 45° and $\mu_s = 0.5$. The maximum safe speed of a car on it is ($g = 10 \text{ m/s}^2$)
 (a) 30 m/s (b) 17.3 m/s (c) 21.2 m/s (d) 10 m/s
- 24 **JEE Main pattern** A 2 kg block rests on an 8 kg block lying on a smooth floor; $\mu_s = 0.3$ between the blocks. The maximum horizontal force on the lower block for which both move together is ($g = 10 \text{ m/s}^2$)
 (a) 24 N (b) 6 N (c) 30 N (d) 60 N

✓ Answer Key & Quick Solutions

- 1(b) 2(c) 3(c) 4(a) 5(a) 6(a) 7(b) 8(c) 9(d) 10(b) 11(b) 12(b) 13(a) 14(b) 15(a)
 16(c) 17(a) 18(d) 19(a) 20(c) 21(a) 22(d) 23(a) 24(c)

- (b) Mass is the quantitative measure of inertia.
- (c) Velocity is constant (8 m/s) $\Rightarrow a = 0 \Rightarrow F = 0$ (first law). (During $0-4 \text{ s}$ the force is $2 \times 2 = 4 \text{ N}$.)
- (c) Third-law forces are simultaneous, equal, opposite and on different bodies.
- (a) $N = m(g - a)$; in free fall $a = g \Rightarrow N = 0$.
- (a) $N = m(g - a) = 60 \times 8 = 480 \text{ N}$.
- (a) $F = dp/dt = 6t + 2 = 14 \text{ N}$.
- (b) Impulse = area = $\frac{1}{2} \times 4 \times 10 = 20 \text{ N s} = \Delta p \Rightarrow v = 20/2 = 10 \text{ m/s}$.
- (c) $V = mv/M = 0.05 \times 400/5 = 4 \text{ m/s}$.
- (d) $J = m(v - u) = 0.15 \times (20 + 12) = 4.8 \text{ N s}$ (velocities are opposite).
- (b) Limiting friction = $0.5 \times 20 = 10 \text{ N} > 6 \text{ N} \Rightarrow$ block stays at rest; static friction = applied force = 6 N .
- (b) $a = 20/10 = 2 \text{ m/s}^2$; contact force = $6 \times 2 = 12 \text{ N}$.

12. (b) At the angle of repose, $\mu_s = \tan 30^\circ = 1/\sqrt{3} \approx 0.58$.
13. (a) $T = 2m_1m_2g/(m_1 + m_2) = 2 \times 3 \times 2 \times 10/5 = 24 \text{ N}$.
14. (b) $a = g(\sin\theta - \mu \cos\theta) = 10(0.6 - 0.4) = 2 \text{ m/s}^2$.
15. (a) $v = \sqrt{\mu rg} = \sqrt{0.5 \times 20 \times 10} = 10 \text{ m/s}$.
16. (c) $v_0 = \sqrt{rg \tan\theta} = \sqrt{40 \times 10 \times 1} = 20 \text{ m/s}$.
17. (a) $F = (n/t) m v = (240/60) \times 0.01 \times 600 = 24 \text{ N}$.
18. (d) Rolling friction $\ll$ sliding; $\mu_k < \mu_s$; friction $\approx$ independent of area; static friction is self-adjusting up to $\mu_s N$.
19. (a) $F = \Delta p / \Delta t$: larger Δt , smaller force – exactly why he pulls his hands back.
20. (c) $a = 0 \Rightarrow N = mg$: apparent weight equals true weight. A is false, R is true.
21. (a) Angle between strings $= 90^\circ$. Lami: $T_1 / \sin(180^\circ - 60^\circ) = 100 / \sin 90^\circ \Rightarrow T_1 = 100 \sin 60^\circ = 50\sqrt{3} \text{ N}$.
22. (d) $a = 12/6 = 2 \text{ m/s}^2$. Midpoint pulls block + half rope: $T = (4 + 1) \times 2 = 10 \text{ N}$.
23. (a) $v^2 = rg \frac{\mu + \tan\theta}{1 - \mu \tan\theta} = 300 \times \frac{1.5}{0.5} = 900 \Rightarrow v = 30 \text{ m/s}$.
24. (c) Upper block can get at most $a = \mu g = 3 \text{ m/s}^2$ from friction. $F = (2 + 8) \times 3 = 30 \text{ N}$.

CHAPTER

5

Work, Energy and Power

17

Day 7-8

NEET weightage ★★★

NCERT Class 11

Work

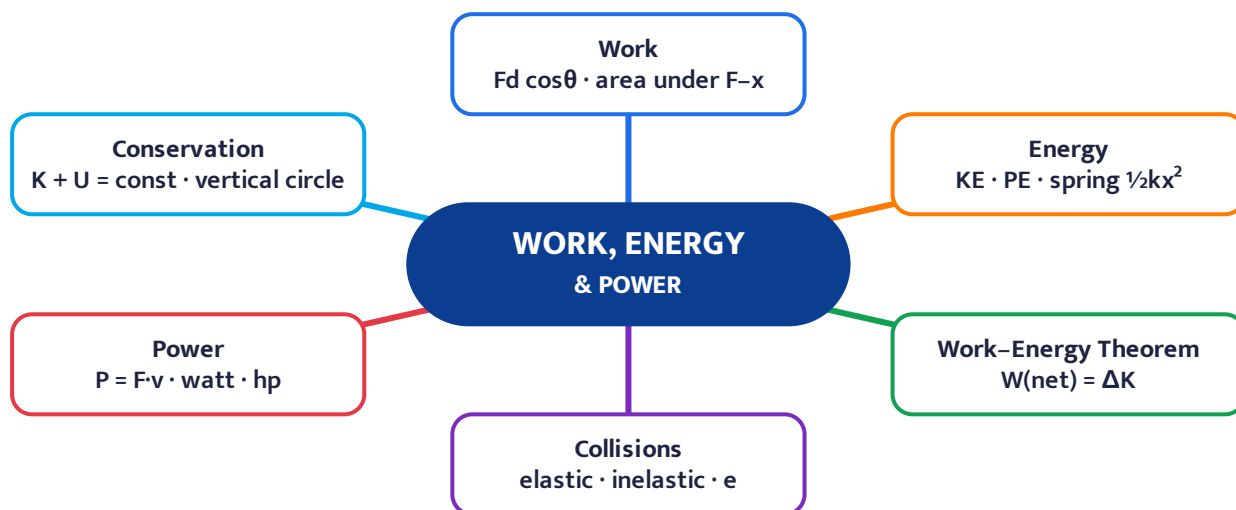
Work-energy theorem

Potential energy

Conservation of energy

Power

Collisions

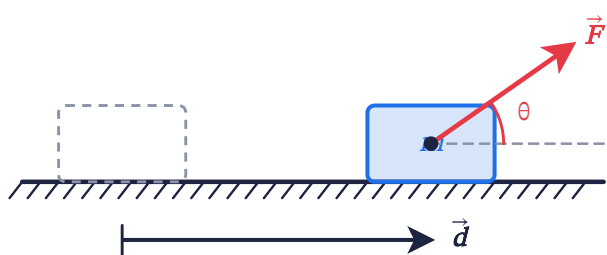


Chapter at a glance

5.1 Work Done by a Constant Force

Work is done when a force produces a displacement **along its own direction**. It is the **scalar (dot) product** of force and displacement. SI unit: joule (1 J = 1 N m); dimensions $[ML^2T^{-2}]$. Other units: 1 erg = 10^{-7} J, 1 eV = 1.6×10^{-19} J, 1 kWh = 3.6×10^6 J.

$$W = F d \cos\theta$$



Only the component $F \cos\theta$ along d does work

Work

$$W = \vec{F} \cdot \vec{d} = Fd\cos\theta \quad \text{Constant force}$$

$$W = F_x d_x + F_y d_y + F_z d_z \quad \text{Component form}$$

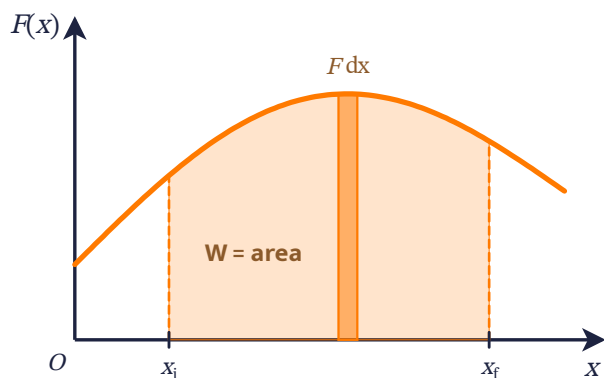


Sign of work

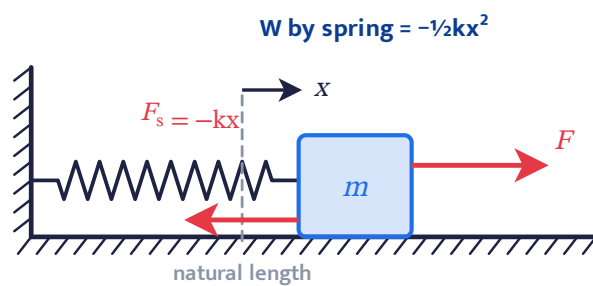
- $\theta < 90^\circ$: **positive** – gravity on a falling body.
- $\theta = 90^\circ$: **zero** – centripetal force, normal force on a horizontally moving block, a coolie walking with a load.
- $\theta > 90^\circ$: **negative** – kinetic friction on a sliding block, gravity on a rising ball.

5.2 Work Done by a Variable Force

If the force changes with position, split the path into tiny steps dx and add: $W = \int_{x_i}^{x_f} F(x) dx$ = **area under the F-x graph** (area below the x-axis counts as negative work).



Work by a variable force = shaded area



Spring force is a restoring force: $F_s = -kx$

Solved Example

A force $F = (3x^2 + 2)$ N acts on a body along the x-axis. Find the work done as it moves from $x = 1$ m to $x = 2$ m.

Solution: $W = \int_1^2 (3x^2 + 2) dx = [x^3 + 2x]_1^2 = (8 + 4) - (1 + 2) = 9 \text{ J}.$

5.3 Kinetic Energy and the Work–Energy Theorem

Kinetic energy is the energy of motion, $K = \frac{1}{2}mv^2$ (scalar, never negative). The **work–energy theorem**: the change in kinetic energy of a particle equals the work done on it by the **net force** – true for constant or variable, conservative or non-conservative forces.

Kinetic energy

$$K = \frac{1}{2}mv^2 = \frac{p^2}{2m}, \quad p = \sqrt{2mK}$$

KE–momentum relation

$$W_{\text{net}} = K_f - K_i = \Delta K$$

Work–energy theorem

$$W_g + W_{\text{friction}} + W_{\text{applied}} + \dots = \Delta K$$

Add the work of **every** force

Solved Example

A 1 kg ball is dropped from a height of 20 m and hits the ground at 18 m/s. Find the work done by air resistance ($g = 10 \text{ m/s}^2$).

Solution: $W_g + W_{\text{air}} = \Delta K \Rightarrow 200 + W_{\text{air}} = \frac{1}{2}(1)(18)^2 = 162 \Rightarrow W_{\text{air}} = -38 \text{ J}.$

Common Trap

- For the same momentum, the **lighter** body has more KE ($K = p^2 / 2m$). For the same KE, the **heavier** body has more momentum.
- KE increases by 300% $\Rightarrow K_f = 4K \Rightarrow p$ doubles (increase 100%), not 300%.

5.4 Potential Energy and Conservative Forces

Potential energy is energy stored due to position or configuration. It is defined only for **conservative forces**, for which $F(x) = -\frac{dU}{dx}$. Near the Earth's surface: $U = mgh$ (with an arbitrary zero level).

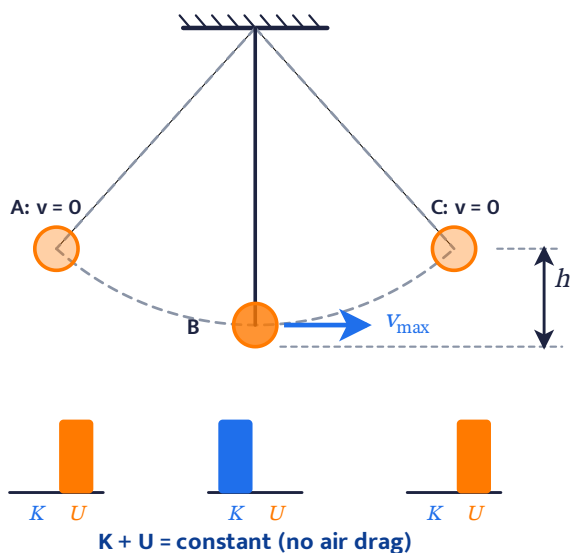
Conservative Force	Non-conservative Force
Work depends only on initial and final positions	Work depends on the path taken
Work in a closed loop = 0	Work in a closed loop $\neq 0$
Potential energy can be defined; $F = -dU/dx$	No potential energy
Gravity, spring force, electrostatic force	Friction, air drag, viscous force

VS Sir's Exam Tip

On a U - x graph: force = **negative slope**. Where $dU/dx = 0$ the force is zero (**equilibrium**): a **minimum** of U is **stable**, a **maximum** is **unstable**, a flat region is **neutral**.

5.5 Conservation of Mechanical Energy

If only conservative forces do work, the **total mechanical energy** $E = K + U$ remains constant: $K_i + U_i = K_f + U_f$. KE and PE keep converting into each other – as in a swinging pendulum or a roller-coaster. A body falling freely from height H : at height h , $v = \sqrt{2g(H-h)}$, and it reaches the ground with $v = \sqrt{2gH}$.



Pendulum: all PE at the extremes, all KE at the lowest point

NCERT Line to Remember

A bob released from angle θ_0 with string length l :
 $h = l(1 - \cos\theta_0)$ and speed at the bottom
 $v = \sqrt{2gl(1 - \cos\theta_0)}$. Mass cancels!

Common Trap

If friction/air drag acts, mechanical energy is **not** conserved – use $W_{nc} = \Delta E = \Delta K + \Delta U$. Total energy (including heat) is still conserved.

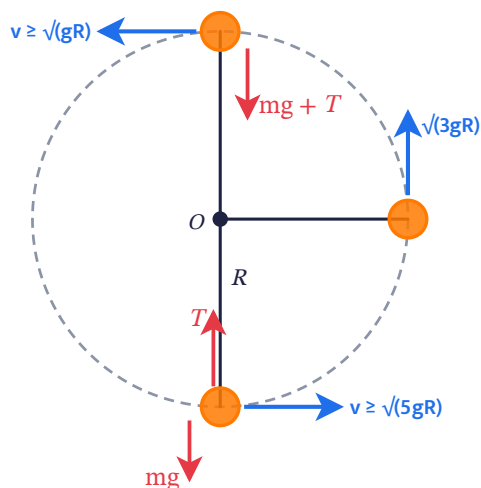
Motion in a vertical circle (NCERT-based)

A bob tied to a string of length R is whirled in a vertical circle. Energy conservation gives $v_{top}^2 = v_{bottom}^2 - 4gR$. The string stays taut at the top only if $T_{top} \geq 0$, i.e. $mg \leq \frac{mv_{top}^2}{R}$.

Solved Example

A 0.5 kg stone tied to a 1.6 m string is whirled in a vertical circle. Find the minimum speed at the bottom and the tension there in this critical case ($g = 10 \text{ m/s}^2$).

Solution: $v = \sqrt{5gR} = \sqrt{5 \times 10 \times 1.6} = \sqrt{80} \approx 8.9 \text{ m/s}$. $T = mg + \frac{mv^2}{R} = 5 + \frac{0.5 \times 80}{1.6} = 5 + 25 = 30 \text{ N}$ ($= 6mg$ ✓).



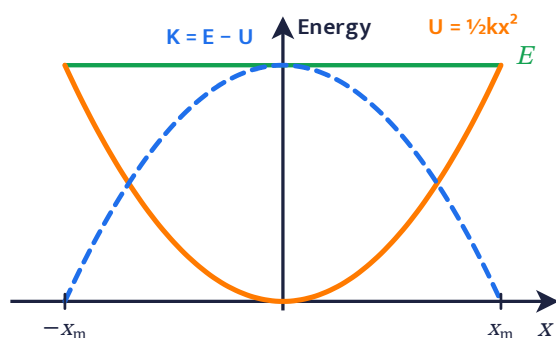
Critical speeds for a bob on a string

Vertical circle

$v_{top} \geq \sqrt{gR}$	Minimum speed at the top
$v_{bottom} \geq \sqrt{5gR}$	Minimum speed at the bottom to complete the loop
$v_{side} = \sqrt{3gR}$	At the horizontal position (critical case)
$T_{bottom} - T_{top} = 6mg$	Always (string); critical case: $T_{bottom} = 6mg$
$v_{bottom} \geq \sqrt{4gR}$	For a light rod (it can push, so $v_{top} \geq 0$)

5.6 Potential Energy of a Spring

An ideal spring obeys Hooke's law, $F_s = -kx$ (k = spring constant, N m^{-1}). The work done **by** the spring when stretched or compressed by x is $-\frac{1}{2}kx^2$, so the energy stored is $U = \frac{1}{2}kx^2$ – the same for stretching and compression. A block oscillating on a smooth floor exchanges this PE with KE.

Spring-block: U (solid), K (dashed), E constant

Spring energy

$U = \frac{1}{2}kx^2$	PE stored in a spring
$W_{1 \rightarrow 2} = \frac{1}{2}k(x_2^2 - x_1^2)$	Work to stretch from x_1 to x_2
$v_{max} = x_m \sqrt{\frac{k}{m}}$	Maximum speed (at natural length)
$k_{series}^{-1} = k_1^{-1} + k_2^{-1}$	Cutting a spring in half doubles k

Solved Example

A car of mass 1000 kg moving at 18 km/h on a smooth road hits a horizontal spring ($k = 6.25 \times 10^3 \text{ N m}^{-1}$). Find the maximum compression.

Solution: $v = 5 \text{ m/s}$. $\frac{1}{2}mv^2 = \frac{1}{2}kx_m^2 \Rightarrow x_m = v\sqrt{m/k} = 5\sqrt{1000/6250} = 5 \times 0.4 = 2 \text{ m}$ (NCERT example).

5.7 Power

Power = rate of doing work

$$P_{av} = \frac{W}{t}, \quad P = \frac{dW}{dt} \quad \text{Average and instantaneous power}$$

$$P = \vec{F} \cdot \vec{v} = Fv \cos \theta \quad \text{Power of a force on a moving body}$$

$$P = mgv \quad \text{Lifting a load at constant speed}$$

$$1 \text{ hp} = 746 \text{ W}$$

Watt = J s^{-1} ; **kWh is energy**, not power

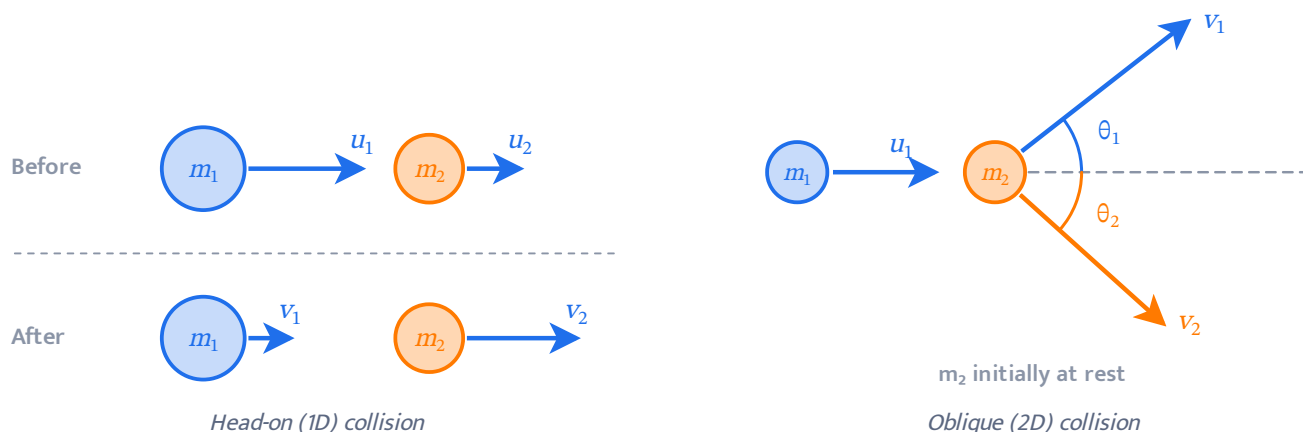
 Solved Example

An elevator of total mass 1800 kg moves up at a constant 2 m/s against a friction force of 4000 N. Find the minimum power of the motor ($g = 10 \text{ m/s}^2$).

Solution: $F = mg + f = 18000 + 4000 = 22000 \text{ N}$. $P = Fv = 44000 \text{ W} = \mathbf{44 \text{ kW} \approx 59 \text{ hp}}$ (NCERT example).

5.8 Collisions

In every collision, **linear momentum is conserved** (impulsive internal forces only). **Elastic:** total KE also conserved. **Inelastic:** some KE is lost (heat, sound, deformation). **Perfectly (completely) inelastic:** bodies stick together – maximum KE loss.

**Collision formulas**

$$v_1 = \frac{(m_1 - m_2)u_1 + 2m_2u_2}{m_1 + m_2}$$

1D elastic: velocity of m_1 after collision

$$v_2 = \frac{(m_2 - m_1)u_2 + 2m_1u_1}{m_1 + m_2}$$

1D elastic: velocity of m_2 after collision

$$v = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$$

Perfectly inelastic (common velocity)

$$\Delta K = \frac{m_1m_2}{2(m_1 + m_2)}u_1^2$$

KE lost when m_1 hits m_2 at rest and they stick

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

Coefficient of restitution: $e = 1$ elastic, $0 < e < 1$ inelastic, $e = 0$ perfectly inelastic**Elastic 1D case ($u_2 = 0$)****Result**Equal masses $m_1 = m_2$ **Velocities exchange:** $v_1 = 0, v_2 = u_1$ Heavy hits light ($m_1 \gg m_2$) $v_1 \approx u_1, v_2 \approx 2u_1$ Light hits heavy ($m_1 \ll m_2$) $v_1 \approx -u_1$ (rebounds), $v_2 \approx 0$

2D collision (m_2 at rest): $m_1u_1 = m_1v_1\cos\theta_1 + m_2v_2\cos\theta_2$ and $0 = m_1v_1\sin\theta_1 - m_2v_2\sin\theta_2$; add $\frac{1}{2}m_1u_1^2 = \frac{1}{2}m_1v_1^2 + \frac{1}{2}m_2v_2^2$ if elastic. For an **elastic oblique collision of equal masses** with one at rest, the two bodies move off **at 90° to each other** ($\theta_1 + \theta_2 = 90^\circ$).

 Solved Example

A 2 kg block moving at 3 m/s collides with a 1 kg block at rest and they stick together. Find the common velocity and the KE lost.

Solution: $v = \frac{2 \times 3}{3} = 2 \text{ m/s}$. $K_i = \frac{1}{2}(2)(9) = 9 \text{ J}$, $K_f = \frac{1}{2}(3)(4) = 6 \text{ J} \Rightarrow \text{loss} = 3 \text{ J}$ (check: $\frac{2 \times 1}{2 \times 3} \times 9 = 3 \text{ J}$).

 Memory Trick

"**Momentum** – always; **KE** – only if elastic." Equal masses, elastic, head-on $\Rightarrow$ **swap** velocities (Newton's cradle). Equal masses, elastic, oblique $\Rightarrow$ **90°** apart.

 Bouncing ball & ballistic pendulum (JEE)

- Dropped from h on a floor with restitution e : after n bounces $h_n = e^{2n}h$, $v_n = e^n \sqrt{2gh}$.
- Bullet m embeds in block M : $v = \frac{mu}{m+M}$, rise $h = \frac{v^2}{2g}$.

 VS Sir's Exam Tip

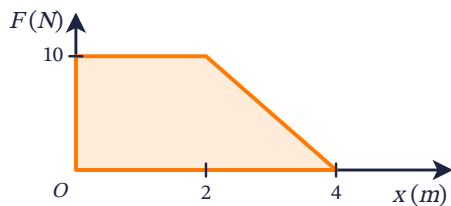
In collision MCQs first write **momentum conservation**; then use **e** (or KE conservation for elastic) as the second equation. Never conserve KE in an inelastic collision, and never forget that momentum is a **vector** – take signs for opposite directions.

 Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- NCERT based** The work done by the centripetal force on a body in uniform circular motion over one revolution is
(a) $\pi r F$ (b) negative (c) $2\pi r F$ (d) zero
- NCERT based** A force $\vec{F} = (2\hat{i} + 3\hat{j} - \hat{k})$ N displaces a body by $\vec{d} = (4\hat{i} + \hat{j} + 2\hat{k})$ m. The work done is
(a) 11 J (b) 7 J (c) 13 J (d) 9 J
- NCERT based** A body of mass 2 kg has momentum 10 kg m/s. Its kinetic energy is
(a) 100 J (b) 10 J (c) 25 J (d) 50 J
- NCERT based** Which of the following is a non-conservative force?
(a) Kinetic friction (b) Electrostatic force
(c) Gravitational force (d) Spring force
- NCERT based** For a particle moving along x , the force is zero at points where the U - x graph
(a) has zero slope (b) has negative value
(c) crosses the x -axis (d) has maximum value only
- NEET pattern** If the kinetic energy of a body increases by 300%, its momentum increases by
(a) 200% (b) 300% (c) 100% (d) 50%

- 7 **NEET pattern** The F - x graph of a force acting on a body is shown. The work done from $x = 0$ to $x = 4$ m is



- (a) 10 J (b) 30 J (c) 20 J (d) 40 J
- 8 **NEET pattern** A spring of constant 200 N/m is stretched from 5 cm to 10 cm. The work done is
(a) 0.25 J (b) 0.75 J (c) 0.5 J (d) 1.0 J
- 9 **NEET pattern** A force $F = 2x$ newton acts on a particle along the x -axis. The work done as it moves from $x = 1$ m to $x = 3$ m is
(a) 6 J (b) 4 J (c) 8 J (d) 16 J
- 10 **NEET pattern** A 1 kg ball dropped from 20 m reaches the ground with speed 18 m/s. The work done by air resistance is ($g = 10 \text{ m/s}^2$)
(a) -200 J (b) -162 J (c) 38 J (d) -38 J
- 11 **NEET pattern** A pendulum bob with string length 2 m is released from rest at 60° to the vertical. Its speed at the lowest point is ($g = 10 \text{ m/s}^2$)
(a) $2\sqrt{10}$ m/s (b) $\sqrt{10}$ m/s (c) $2\sqrt{5}$ m/s (d) 2 m/s
- 12 **NEET pattern** A motor lifts a 500 kg load vertically at a constant speed of 2 m/s. The power delivered is ($g = 10 \text{ m/s}^2$)
(a) 5 kW (b) 10 kW (c) 20 kW (d) 1 kW
- 13 **NEET pattern** A 2 kg ball moving at 4 m/s collides head-on elastically with an identical ball at rest. Their velocities after the collision are
(a) 0 and 4 m/s (b) -4 m/s and 4 m/s
(c) 2 m/s and 2 m/s (d) 4 m/s and 0
- 14 **NEET pattern** A 2 kg block moving at 3 m/s strikes a 1 kg block at rest and they stick together. The loss in kinetic energy is
(a) 6 J (b) 1.5 J (c) 3 J (d) 9 J
- 15 **NEET pattern** The minimum speed at the lowest point for a stone tied to a 0.5 m string to complete a vertical circle is ($g = 10 \text{ m/s}^2$)
(a) 10 m/s (b) $\sqrt{5}$ m/s (c) 5 m/s (d) $\sqrt{10}$ m/s
- 16 **NEET pattern** A very heavy body moving with velocity u collides elastically head-on with a very light body at rest. The light body moves off with velocity
(a) 0 (b) $u/2$ (c) u (d) $2u$

- 17 **NEET pattern** A ball is dropped from 10 m on a floor with coefficient of restitution 0.5. The height of the first rebound is
 (a) 1.25 m (b) 5 m (c) 7.5 m (d) 2.5 m
- 18 **NEET pattern** In an elastic oblique collision of two equal masses, one initially at rest, the angle between their velocities after the collision is
 (a) 60° (b) 45° (c) 180° (d) 90°
- 19 **Assertion-Reason** Assertion: Work done by friction is always negative. Reason: Friction always opposes relative motion between surfaces in contact.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 20 **Assertion-Reason** Assertion: The potential energy of a spring increases both when it is stretched and when it is compressed. Reason: The elastic potential energy $U = \frac{1}{2}kx^2$ depends on the square of the deformation.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 21 **JEE Main pattern** A body starts from rest and is driven by a machine delivering constant power. Its displacement s varies with time t as
 (a) $s \propto t$ (b) $s \propto t^{3/2}$ (c) $s \propto t^2$ (d) $s \propto t^{1/2}$
- 22 **JEE Main pattern** A 50 g bullet moving at 200 m/s embeds in a 950 g block suspended by a string. The block rises to a height of (g = 10 m/s²)
 (a) 20 m (b) 10 m (c) 5 m (d) 2.5 m
- 23 **JEE Main pattern** The potential energy of a particle is $U = (2x^2 - 4x + 5)$ J (x in m). The force on it at $x = 3$ m is
 (a) -8 N (b) 11 N (c) 0 (d) 8 N
- 24 **JEE Main pattern** A 0.5 kg stone tied to a string moves in a vertical circle. The difference between the tensions at the lowest and highest points is (g = 10 m/s²)
 (a) 60 N (b) 30 N (c) 10 N (d) 20 N



Answer Key & Quick Solutions

- 1(d) 2(d) 3(c) 4(a) 5(a) 6(c) 7(b) 8(b) 9(c) 10(d) 11(c) 12(b) 13(a) 14(c) 15(c)
 16(d) 17(d) 18(d) 19(c) 20(a) 21(b) 22(c) 23(a) 24(b)

1. (d) Force $\perp$ displacement at every instant $\Rightarrow W = 0$.
 2. (d) $W = 8 + 3 - 2 = 9$ J.
 3. (c) $K = p^2/2m = 100/4 = 25$ J.
 4. (a) Work done by friction depends on the path; it has no potential energy.

5. (a) $F = -dU/dx = 0$ wherever the slope is zero (maxima, minima or flat parts).
6. (c) $K' = 4K \Rightarrow p' = \sqrt{4} p = 2p \Rightarrow$ increase 100%.
7. (b) Area = rectangle (10×2) + triangle $(\frac{1}{2} \times 2 \times 10) = 20 + 10 = 30$ J.
8. (b) $W = \frac{1}{2}k(x_2^2 - x_1^2) = 100(0.01 - 0.0025) = 0.75$ J.
9. (c) $W = \int 2x \, dx = [x^2]$ from 1 to 3 = $9 - 1 = 8$ J.
10. (d) $\Delta K = W_g + W_{air} \Rightarrow 162 = 200 + W_{air} \Rightarrow W_{air} = -38$ J.
11. (c) $h = l(1 - \cos 60^\circ) = 1$ m; $v = \sqrt{2gh} = \sqrt{20} = 2\sqrt{5} \approx 4.47$ m/s.
12. (b) $P = mgv = 500 \times 10 \times 2 = 10\,000$ W = 10 kW.
13. (a) Equal masses, elastic, head-on $\Rightarrow$ velocities exchange: first stops, second moves at 4 m/s.
14. (c) $v = 6/3 = 2$ m/s; $\Delta K = 9 - 6 = 3$ J.
15. (c) $v = \sqrt{5gR} = \sqrt{25} = 5$ m/s.
16. (d) $v_2 = \frac{2m_1u}{m_1 + m_2} \rightarrow 2u$ when $m_1 \gg m_2$.
17. (d) Rebound speed = $e \times$ impact speed $\Rightarrow h_1 = e^2 h = 0.25 \times 10 = 2.5$ m.
18. (d) Momentum triangle + KE equation $\Rightarrow u^2 = v_1^2 + v_2^2 \Rightarrow \vec{v}_1 \perp \vec{v}_2$.
19. (c) Friction on a box resting on an accelerating truck is along its displacement $\Rightarrow$ positive work. So A is false; R is true.
20. (a) x^2 is positive for $x > 0$ and $x < 0$, so U rises either way – R explains A.
21. (b) $Pt = \frac{1}{2}mv^2 \Rightarrow v \propto t^{1/2} \Rightarrow s = \int v \, dt \propto t^{3/2}$.
22. (c) $v = (0.05 \times 200)/1 = 10$ m/s; $h = v^2/2g = 100/20 = 5$ m.
23. (a) $F = -dU/dx = -(4x - 4) = -8$ N (towards the stable equilibrium at $x = 1$ m).
24. (b) $T_{bottom} - T_{top} = 6mg = 6 \times 0.5 \times 10 = 30$ N (independent of speed).

CHAPTER

6

System of Particles and Rotational Motion

Day 9–10

NEET weightage ★★ ★

NCERT Class 11

Centre of mass

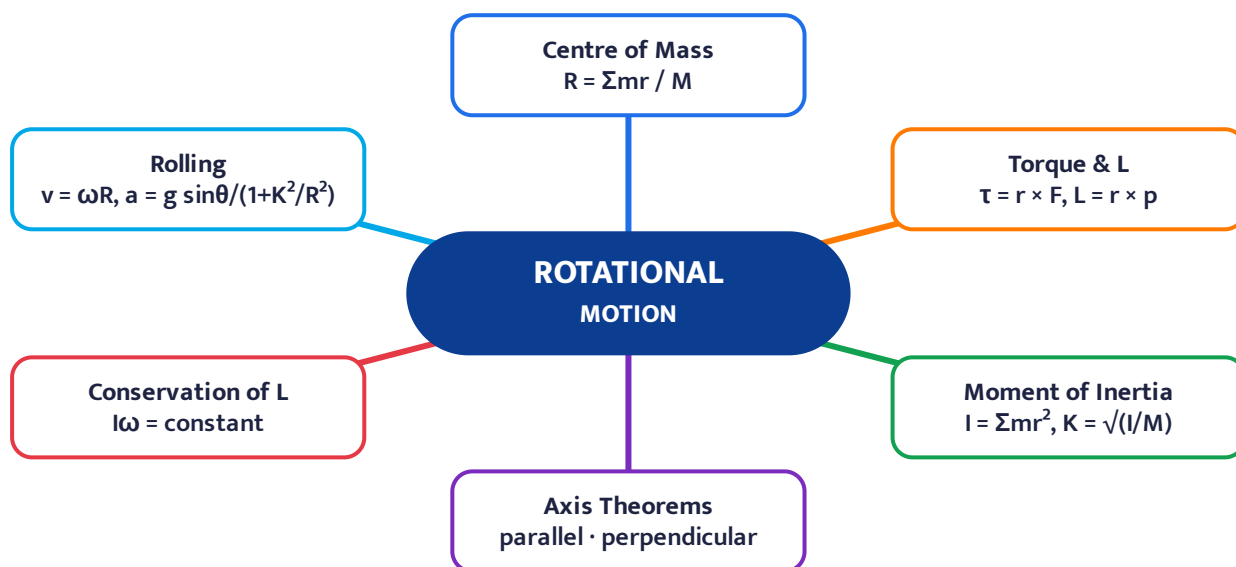
Torque & angular momentum

Moment of inertia

Axis theorems

Conservation of L

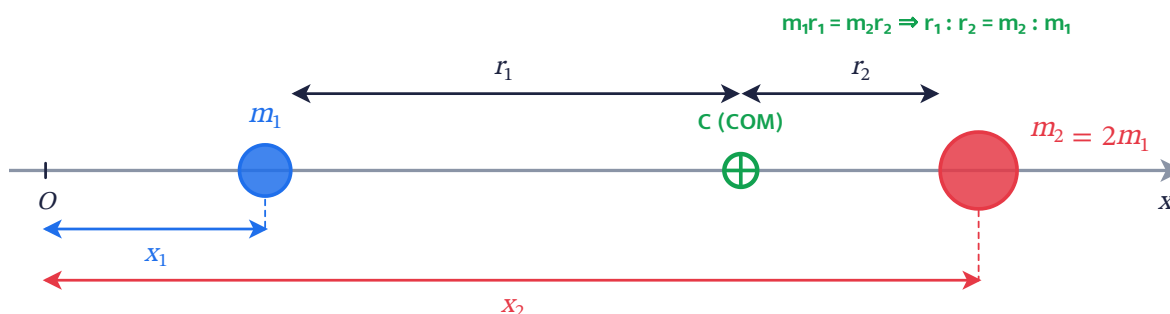
Rolling motion



Chapter at a glance

6.1 Centre of Mass

A **rigid body** keeps the distance between every pair of its particles fixed. Its general motion = **translation of the centre of mass (COM)** + **rotation about the COM**. The COM is the point where the whole mass may be taken to be concentrated for describing translation.



Two-particle system: the COM lies on the joining line, nearer the heavier mass

NCERT Line to Remember

Internal forces cannot change the motion of the COM. A shell that explodes in mid-air: its fragments fly apart, but the COM continues on the **same parabola** until a fragment hits the ground.

Common Trap

- COM need not lie inside the body (ring, L-shape, hollow sphere).
- If $F_{ext} = 0$: V_{cm} is constant, even if parts move.

Centre-of-mass formulas

$$x_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

Two particles on a line

$$\vec{R}_{cm} = \frac{\sum m_i \vec{r}_i}{M}$$

n particles (apply to x, y, z separately)

$$\vec{R}_{cm} = \frac{1}{M} \int \vec{r} dm$$

Continuous body

$$\vec{V}_{cm} = \frac{\sum m_i \vec{v}_i}{M}, \vec{P} = M\vec{V}_{cm}$$

Velocity of COM; total momentum

$$M\vec{A}_{cm} = \vec{F}_{ext}$$

COM moves as if all external forces act on total mass M

Uniform body

Position of COM

Rod of length L

Mid-point, L/2 from either end

Ring, disc, sphere, cylinder

Geometric centre (for a ring it lies outside the material)

Triangular lamina

Centroid – intersection of medians (h/3 from base)

Semicircular ring / semicircular disc

 $\frac{2R}{\pi} / \frac{4R}{3\pi}$ from centre

Solid hemisphere / hollow hemisphere

 $\frac{3R}{8} / \frac{R}{2}$ from centre

Solid cone of height h

h/4 from the base

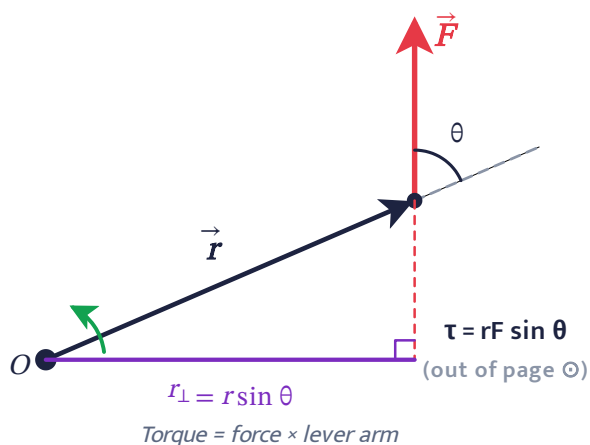
Solved Example

Masses 1 kg, 2 kg and 3 kg are placed at (0, 0), (3 m, 0) and (0, 4 m). Find the centre of mass.

Solution: $x_{cm} = \frac{1(0) + 2(3) + 3(0)}{6} = 1 \text{ m}; y_{cm} = \frac{1(0) + 2(0) + 3(4)}{6} = 2 \text{ m. COM} = (1 \text{ m}, 2 \text{ m}).$

6.2 Torque and Angular Momentum

Vector (cross) product: $\vec{A} \times \vec{B} = AB \sin \theta \hat{n}$, where $\hat{n}$ is given by the **right-hand rule**. It is not commutative: $\vec{A} \times \vec{B} = -\vec{B} \times \vec{A}; \hat{i} \times \hat{j} = \hat{k}, \hat{j} \times \hat{k} = \hat{i}, \hat{k} \times \hat{i} = \hat{j}$, and $\hat{i} \times \hat{i} = 0$.



Torque & angular momentum

$$\vec{\tau} = \vec{r} \times \vec{F}$$

Torque (moment of force), N m

$$\tau = rF \sin \theta = r_{\perp} F \quad r_{\perp} = \text{lever arm}$$

$$\vec{L} = \vec{r} \times \vec{p}$$

Angular momentum (J s)

$$L = mvr_{\perp} = mvr \sin \theta \quad \text{For circular motion: } L = mvr$$

$$\vec{\tau}_{ext} = \frac{d\vec{L}}{dt}$$

Rotational analogue of $\vec{F} = \frac{d\vec{p}}{dt}$

Solved Example

Find the torque about the origin of a force $\vec{F} = 7\hat{i} + 3\hat{j} - 5\hat{k}$ acting at $\vec{r} = \hat{i} - \hat{j} + \hat{k}$.

Solution: $\vec{\tau} = \vec{r} \times \vec{F}$; $\tau_x = (-1)(-5) - (1)(3) = 2$, $\tau_y = (1)(7) - (1)(-5) = 12$, $\tau_z = (1)(3) - (-1)(7) = 10 \Rightarrow \vec{\tau} = 2\hat{i} + 12\hat{j} + 10\hat{k} \text{ N m}$.

NCERT Line to Remember

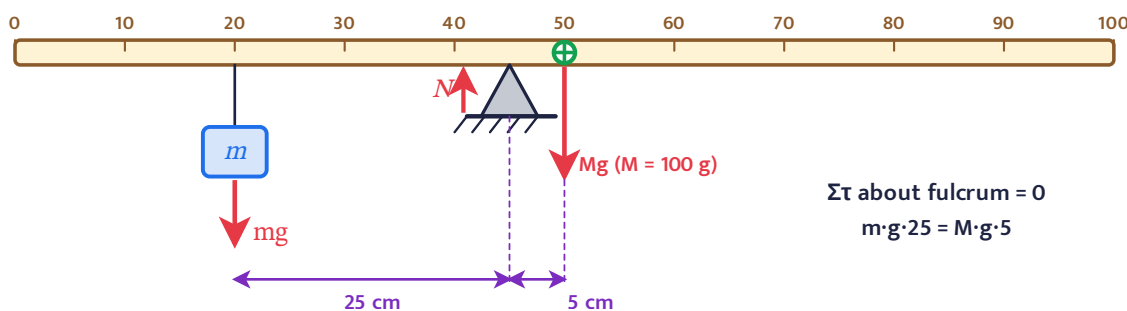
- Torque and work have the same dimensions $[ML^2T^{-2}]$, but torque is a **vector** and is never written in joule.
- A particle moving along a straight line has angular momentum mvd about a point at perpendicular distance d – constant if v is constant.
- A **central force** (along $\vec{r}$) gives zero torque, so $\vec{L}$ is conserved – the basis of Kepler's second law.

6.3 Equilibrium of a Rigid Body**Two conditions**

- Translational:** $\sum \vec{F}_{ext} = 0$ (no linear acceleration).
- Rotational:** $\sum \vec{\tau}_{ext} = 0$ about **any** point.
- Principle of moments** (lever): $F_1d_1 = F_2d_2$; mechanical advantage = load/effort = d_1 / d_2 .

Couple & centre of gravity

- Couple:** two equal, opposite, non-collinear forces. Net force = 0, torque = $F \times d$, same about every point $\rightarrow$ pure rotation.
- Centre of gravity:** point where total gravitational torque is zero; coincides with COM when g is uniform.



A uniform metre rod (100 g) balanced on a knife-edge at 45 cm with a mass m at 20 cm

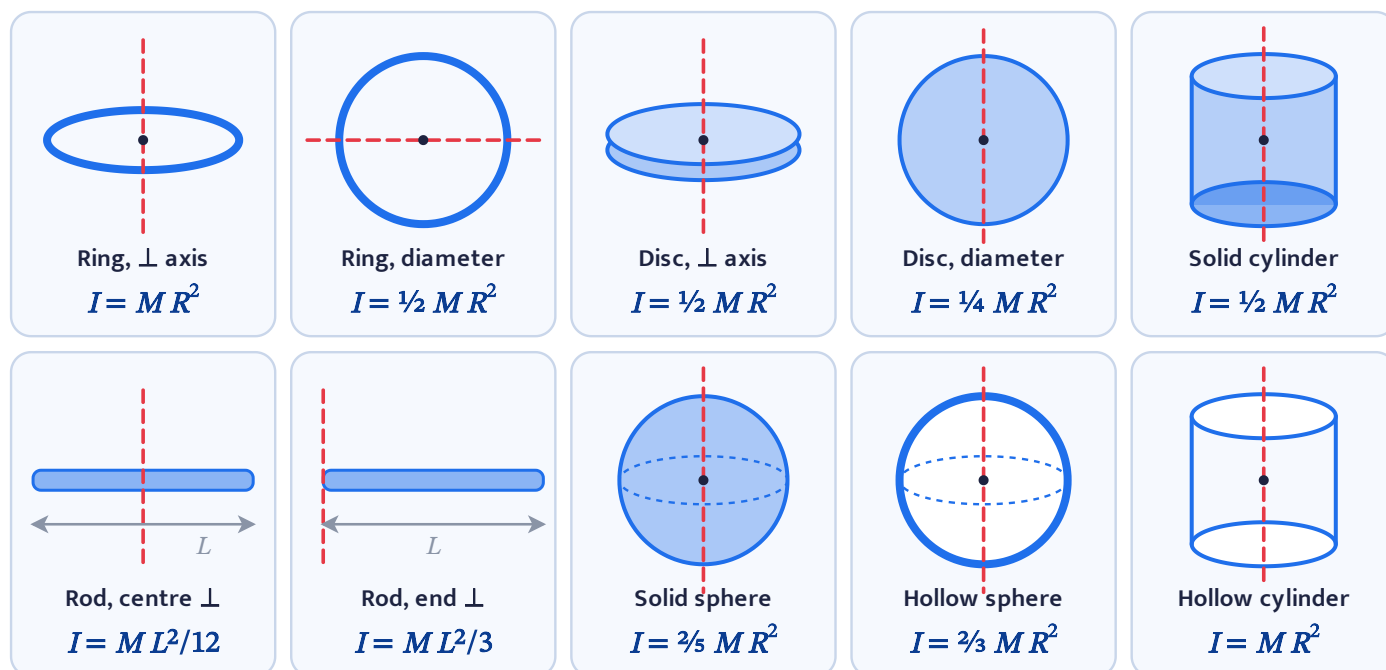
Solved Example

In the figure, find the mass m that keeps the rod horizontal.

Solution: Torques about the knife-edge: $mg(45 - 20) = Mg(50 - 45) \Rightarrow m \times 25 = 100 \times 5 \Rightarrow m = 20 \text{ g}$. (N passes through the pivot – zero torque.)

6.4 Moment of Inertia & Radius of Gyration

Moment of inertia $I = \sum m_i r_i^2$ (r_i = perpendicular distance from the axis) is the **rotational inertia** – it plays the role of mass in rotation. It depends on the **mass, its distribution about the axis, and the position/orientation of the axis** – not a fixed property like mass. SI unit kg m^2 , dimension $[ML^2]$. **Radius of gyration** K : $I = MK^2 \Rightarrow K = \sqrt{I/M}$.



Moments of inertia of standard uniform bodies (red dashed line = axis)

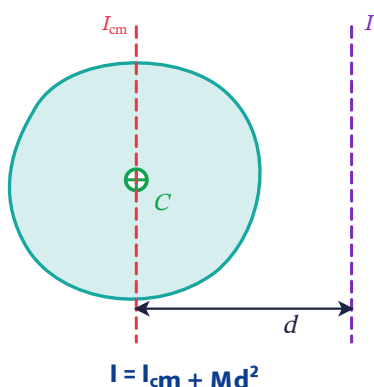
Memory Trick

"Ring 1, Disc $\frac{1}{2}$, Hollow $\frac{2}{3}$, Solid $\frac{2}{5}$ " – coefficient of MR^2 about the symmetry axis. Mass farther from the axis $\Rightarrow$ larger I : ring > hollow sphere > disc > solid sphere (same M, R).

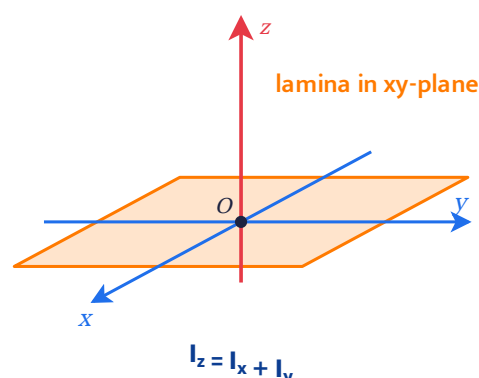
Radius of gyration

$K_{\text{ring}} = R$	ring ($\perp$ axis)
$K_{\text{disc}} = R / \sqrt{2}$	disc ($\perp$ axis)
$K_{\text{sphere}} = \sqrt{2/5} R$	solid sphere
$K_{\text{rod}} = L / \sqrt{12}$	rod about centre

6.5 Theorems of Parallel and Perpendicular Axes



Parallel axes: valid for any body; one axis must pass through the COM



Perpendicular axes: only for a plane lamina; I_x, I_y in its plane

Common Trap

- Parallel-axis theorem: d is measured **from the COM axis**. Between two non-COM axes you cannot write $I_2 = I_1 + Md^2$ directly.
- Perpendicular-axis theorem does **not** apply to a sphere or cylinder (3-D bodies).
- I about the COM axis is the **minimum** among all parallel axes.

Body and axis	Theorem used	I
Ring – tangent $\perp$ to plane	parallel	$MR^2 + MR^2 = 2MR^2$
Ring – tangent in its plane	parallel	$\frac{1}{2}MR^2 + MR^2 = \frac{3}{2}MR^2$
Disc – tangent $\perp$ to plane	parallel	$\frac{3}{2}MR^2$
Disc – tangent in its plane	parallel	$\frac{1}{4}MR^2 + MR^2 = \frac{5}{4}MR^2$
Rod – $\perp$ axis through one end	parallel	$\frac{ML^2}{12} + M\left(\frac{L}{2}\right)^2 = \frac{ML^2}{3}$
Solid sphere – tangent	parallel	$\frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$
Hollow sphere – tangent	parallel	$\frac{5}{3}MR^2$
Ring/disc – diameter	perpendicular	$I_d = \frac{I_z}{2} = \frac{1}{2}MR^2 / \frac{1}{4}MR^2$
Rectangular plate $a \times b$ – $\perp$ axis through centre	perpendicular	$\frac{M(a^2 + b^2)}{12}$

6.6 Rotation about a Fixed Axis – Kinematics & Dynamics

Every particle of the body moves in a circle about the axis with the **same** ω and α ; its linear speed $v = \omega r$ and tangential acceleration $a_t = \alpha r$ grow with distance r . Every linear result has a rotational twin:

Linear motion	Rotation about a fixed axis
Displacement x , velocity v , acceleration a	Angle θ , angular velocity ω , angular acceleration α
$v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$	$\omega = \omega_0 + \alpha t$, $\theta = \omega_0 t + \frac{1}{2}\alpha t^2$, $\omega^2 = \omega_0^2 + 2\alpha\theta$
Mass m	Moment of inertia I
Force $F = ma$	Torque $\tau = I\alpha$
Momentum $p = mv$; $F = \frac{dp}{dt}$	Angular momentum $L = I\omega$; $\tau = \frac{dL}{dt}$
Kinetic energy $\frac{1}{2}mv^2 = \frac{p^2}{2m}$	Rotational KE $\frac{1}{2}I\omega^2 = \frac{L^2}{2I}$
Work $W = Fs$; power $P = Fv$; $W = \Delta KE$	Work $W = \tau\theta$; power $P = \tau\omega$; $W = \Delta KE_{rot}$

Solved Example

The angular speed of a motor wheel rises uniformly from 1200 rpm to 3120 rpm in 16 s. Find (i) its angular acceleration, (ii) the number of revolutions made.

Solution: $\omega_0 = 2\pi \frac{1200}{60} = 40\pi$ rad/s, $\omega = 104\pi$ rad/s. (i) $\alpha = \frac{64\pi}{16} = 4\pi$ rad/s². (ii)

$\theta = 40\pi(16) + \frac{1}{2}(4\pi)(16)^2 = 1152\pi$ rad = **576 revolutions**.

VS Sir's Exam Tip

For a rod pivoted at one end and released from horizontal, use energy: $mg\frac{L}{2} = \frac{1}{2}\left(\frac{mL^2}{3}\right)\omega^2 \Rightarrow \omega = \sqrt{3g/L}$, and the free end moves with $v = \omega L = \sqrt{3gL}$. Initial angular acceleration: $\alpha = \frac{mg(L/2)}{mL^2/3} = \frac{3g}{2L}$.

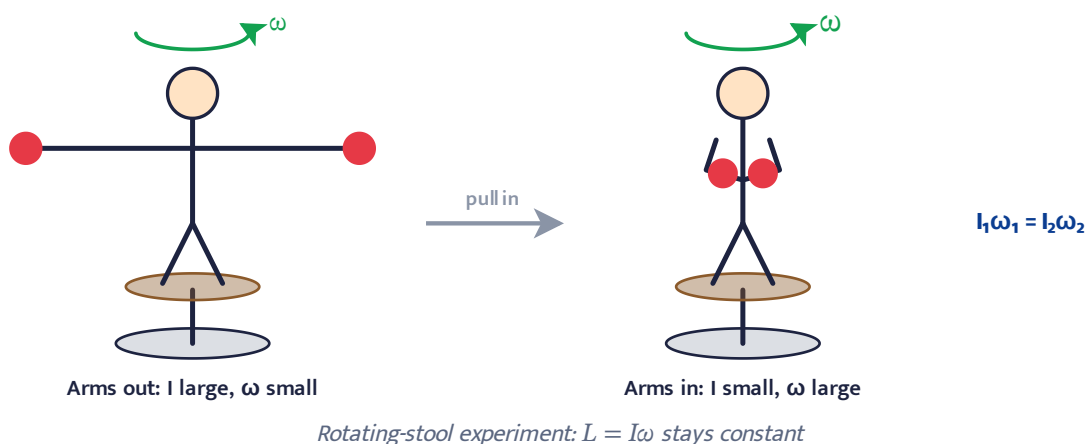
Solved Example

A cord wound round a flywheel (uniform disc) of mass 20 kg and radius 20 cm is pulled with a steady force of 25 N. Find the angular acceleration and the kinetic energy gained when 2 m of cord unwinds.

Solution: $I = \frac{1}{2}MR^2 = \frac{1}{2}(20)(0.2)^2 = 0.4 \text{ kg m}^2$. $\tau = FR = 5 \text{ N m} \Rightarrow \alpha = \tau / I = 12.5 \text{ rad/s}^2$. Work by the pull = $F \times s = 25 \times 2 = 50 \text{ J}$ = gain in rotational KE (no friction).

6.7 Conservation of Angular Momentum

If the net external torque is zero, $\frac{d\vec{L}}{dt} = 0 \Rightarrow \vec{L} = \text{constant}$. For rotation about a fixed axis: $I_1\omega_1 = I_2\omega_2$. When I decreases, ω increases – an **ice skater** or **ballet dancer** pulls the arms in to spin faster; a **diver** tucks the body to somersault faster; a person on a **rotating stool** holding dumbbells speeds up on pulling the arms in.

**Solved Example**

A skater spins at 2 rev/s with $I = 4 \text{ kg m}^2$. Pulling in the arms makes $I = 2.5 \text{ kg m}^2$. Find the new spin rate and the ratio of final to initial KE.

Solution: $4 \times 2 = 2.5 \omega_2 \Rightarrow \omega_2 = 3.2 \text{ rev/s}$. $KE = \frac{L^2}{2I}$ with L fixed $\Rightarrow \frac{KE_2}{KE_1} = \frac{I_1}{I_2} = 1.6$. The extra KE comes from the work done by the skater's muscles.

Common Trap

- L is conserved but **KE is not** – pulling arms in increases KE.
- Two discs coupled coaxially: $\omega = \frac{I_1\omega_1 + I_2\omega_2}{I_1 + I_2}$; KE is **lost** (like a perfectly inelastic collision).

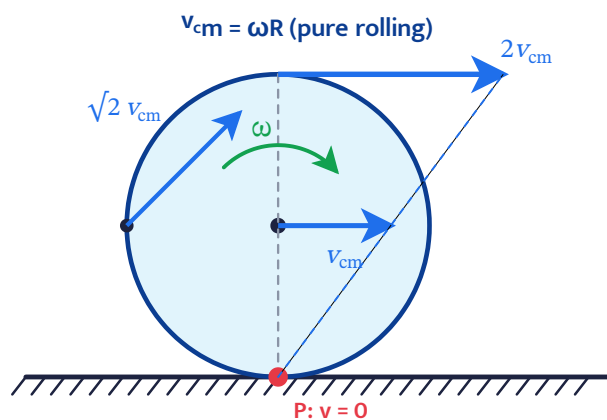
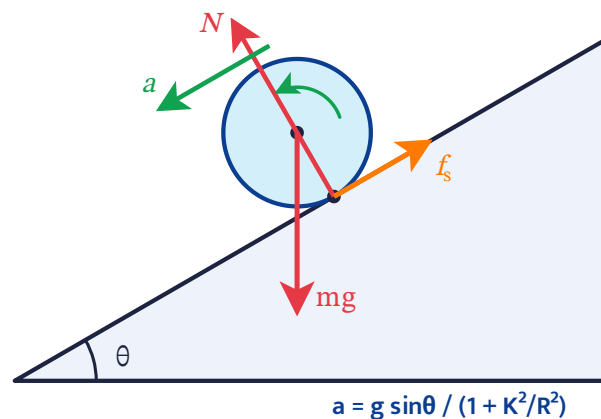
VS Sir's Exam Tip

If Earth shrank with mass fixed, $I \propto R^2 \Rightarrow T \propto R^2$. Radius halved $\Rightarrow$ day = $24/4 = 6 \text{ h}$. Melting polar ice spreads mass towards the equator $\Rightarrow I \uparrow \Rightarrow$ day slightly longer.

6.8 Rolling Motion**Note**

Rolling motion has been **removed from the rationalised NCERT text** and is not listed explicitly in the NMC NEET syllabus. It is still asked in **JEE Main** and in NEET questions on rigid-body rotation – revise the key results below.

Rolling = translation of the COM + rotation about the COM. In **pure rolling (no slipping)** the contact point is momentarily at rest: $v_{cm} = \omega R$. The body turns instantaneously about the contact point P, so the speed of any point $\propto$ its distance from P.

Velocities in pure rolling: top $2v_{cm}$, contact 0Rolling down an incline: static friction f_s acts up the slope

Rolling formulas

$$KE = \frac{1}{2}mv_{cm}^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}mv_{cm}^2 \left(1 + \frac{K^2}{R^2}\right) \quad \text{Total KE of a rolling body}$$

$$a = \frac{g \sin \theta}{1 + K^2/R^2}, \quad v = \sqrt{\frac{2gh}{1 + K^2/R^2}} \quad \text{Down an incline of height } h \text{ (from rest)}$$

$$f_s = \frac{mg \sin \theta}{1 + R^2/K^2}, \quad \mu_{min} = \frac{\tan \theta}{1 + R^2/K^2} \quad \text{Friction needed for pure rolling (does no work)}$$

Solved Example

A solid sphere rolls without slipping from rest down an incline of vertical height 7 m. Find its speed at the bottom ($g = 10 \text{ m/s}^2$).

Solution: $v = \sqrt{\frac{2gh}{1 + 2/5}} = \sqrt{\frac{10gh}{7}} = \sqrt{\frac{10 \times 10 \times 7}{7}} = 10 \text{ m/s}$ (sliding without friction would give $\sqrt{2gh} = 11.8 \text{ m/s}$).

Body	K^2 / R^2	a on incline	KE_{rot} / KE_{total}	v at bottom
Solid sphere	$2/5$	$\frac{5}{7}g \sin \theta$	$2/7$	$\sqrt{10gh/7}$
Disc / solid cylinder	$1/2$	$\frac{2}{3}g \sin \theta$	$1/3$	$\sqrt{4gh/3}$
Hollow sphere	$2/3$	$\frac{3}{5}g \sin \theta$	$2/5$	$\sqrt{6gh/5}$
Ring / hollow cylinder	1	$\frac{1}{2}g \sin \theta$	$1/2$	$\sqrt{gh}$

JEE Main Extension

Rolling KE can be written as rotation about the contact point: $KE = \frac{1}{2}I_P\omega^2$ with $I_P = I_{cm} + MR^2$. For a ring, $KE = mv^2$; for a disc, $\frac{3}{4}mv^2$; for a solid sphere, $\frac{7}{10}mv^2$.

🎯 Race down the incline

- Smaller $K^2 / R^2 \Rightarrow$ larger $a \Rightarrow$ reaches first: **solid sphere $\rightarrow$ disc $\rightarrow$ hollow sphere $\rightarrow$ ring**.
- Result is independent of mass and radius – only the **shape** matters.
- All reach with the **same KE** (mgh) if masses are equal.

⚠ Common Trap

- On a **frictionless** incline no body can roll – all slide with $a = g\sin\theta$ and reach together.
- Static friction in pure rolling does **no work** – mechanical energy is conserved.

📝 Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1** **NCERT based** The centre of mass of a uniform circular ring lies

(a) it is not defined
(b) anywhere on the ring
(c) on the rim, nearest the point of support
(d) at its geometric centre, where there is no mass
- 2** **NEET pattern** Masses 2 kg and 3 kg are at $x = 0$ and $x = 5$ m. The centre of mass is at $x =$

(a) 3 m (b) 2.5 m (c) 2 m (d) 3.5 m
- 3** **NCERT based** Moment of inertia of a thin uniform rod of mass M and length L about an axis perpendicular to it through one end is

(a) $\frac{ML^2}{3}$ (b) $\frac{ML^2}{12}$ (c) $\frac{ML^2}{2}$ (d) ML^2
- 4** **NCERT based** A force $\vec{F} = (3\hat{i} + 4\hat{j})$ N acts at the point $\vec{r} = (2\hat{i} + \hat{j})$ m. The torque about the origin is

(a) $-5\hat{k}$ N m (b) $10\hat{k}$ N m (c) $11\hat{k}$ N m (d) $5\hat{k}$ N m
- 5** **NEET pattern** A solid sphere, a hollow sphere, a disc and a ring (same mass and radius) roll down the same incline from rest without slipping. Which reaches the bottom first?

(a) Disc (b) Hollow sphere
(c) Ring (d) Solid sphere
- 6** **NEET pattern** For a solid sphere rolling without slipping, the ratio of rotational KE to total KE is

(a) $5/7$ (b) $2/7$ (c) $2/5$ (d) $1/2$
- 7** **NEET pattern** A disc of moment of inertia I rotates freely with angular speed ω . An identical disc, initially at rest, is gently placed on it coaxially. The loss in kinetic energy is

(a) $\frac{1}{8}I\omega^2$ (b) zero (c) $\frac{1}{4}I\omega^2$ (d) $\frac{1}{2}I\omega^2$

- 8 **NCERT based** The angular momentum of a system of particles is conserved when
 (a) the linear momentum is zero
 (b) the net external force is zero
 (c) the net external torque is zero
 (d) the total kinetic energy is constant
- 9 **JEE Main pattern** The moment of inertia of a uniform disc of mass M , radius R about a tangent in its own plane is
 (a) $\frac{5}{4}MR^2$ (b) $\frac{3}{4}MR^2$ (c) $\frac{1}{4}MR^2$ (d) $\frac{3}{2}MR^2$
- 10 **NCERT based** The radius of gyration of a uniform disc about an axis through its centre and perpendicular to its plane is
 (a) R (b) $R / \sqrt{2}$ (c) $R / 2$ (d) $\sqrt{2/5} R$
- 11 **NEET pattern** A dancer spinning with angular speed ω folds her arms, reducing her moment of inertia by 40%. Her new angular speed is
 (a) $\frac{3}{5}\omega$ (b) 0.6ω (c) 1.4ω (d) $\frac{5}{3}\omega$
- 12 **NEET pattern** A wheel starts from rest with a constant angular acceleration of $\pi \text{ rad/s}^2$. The number of revolutions it makes in the first 10 s is
 (a) 100 (b) 12.5 (c) 50 (d) 25
- 13 **NEET pattern** A constant torque of 10 N m acts on a wheel ($I = 2 \text{ kg m}^2$) initially at rest. Its kinetic energy after 4 s is
 (a) 100 J (b) 200 J (c) 800 J (d) 400 J
- 14 **JEE Main pattern** A uniform rod of length L , pivoted at one end, is released from the horizontal position. Its angular speed when it becomes vertical is
 (a) $\sqrt{3g/2L}$ (b) $\sqrt{g/L}$ (c) $\sqrt{6g/L}$ (d) $\sqrt{3g/L}$
- 15 **JEE Main pattern** A solid cylinder rolls without slipping down an incline of 30° . Its acceleration is ($g = 10 \text{ m/s}^2$)
 (a) 2.5 m/s^2 (b) 3.33 m/s^2 (c) 3.57 m/s^2 (d) 5 m/s^2
- 16 **Assertion-Reason** Assertion (A): In pure rolling, the point of the body in contact with the ground is momentarily at rest. Reason (R): In pure rolling, $v_{cm} = \omega R$, so the translational and rotational velocities of the contact point cancel.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 17 **NCERT based** The perpendicular-axis theorem can be applied to
 (a) plane laminae only (b) all three-dimensional bodies
 (c) solid spheres only (d) any body, about any axis

- 18 **NEET pattern** A particle of mass m moves with constant velocity v along the line $y = a$ (parallel to the x -axis). Its angular momentum about the origin
 (a) is zero (b) is constant, of magnitude mva
 (c) decreases as x increases (d) increases as x increases
- 19 **JEE Main pattern** A projectile would land 100 m away. At the highest point it explodes into two equal fragments; one falls vertically down (starting from rest). The other lands at a distance of
 (a) 100 m (b) 150 m (c) 200 m (d) 125 m
- 20 **JEE Main pattern** A disc of mass 2 kg rolls without slipping with $v_{cm} = 4$ m/s. Its total kinetic energy is
 (a) 32 J (b) 16 J (c) 24 J (d) 8 J
- 21 **Assertion-Reason** Assertion (A): The moment of inertia of a body is a fixed quantity, like its mass. Reason (R): Moment of inertia depends on the distribution of mass about the axis of rotation.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 22 **NEET pattern** A uniform beam AB, 4 m long and of mass 20 kg, rests on supports at its ends. A 10 kg load is placed 1 m from A. The reaction at A is ($g = 10$ m/s²)
 (a) 125 N (b) 150 N (c) 175 N (d) 200 N
- 23 **JEE Main pattern** The moment of inertia of a solid sphere of mass M and radius R about a tangent is
 (a) $\frac{5}{3}MR^2$ (b) $\frac{2}{5}MR^2$ (c) $\frac{7}{5}MR^2$ (d) $\frac{3}{2}MR^2$
- 24 **NEET pattern** If the Earth were to contract to half its present radius with no change in mass, the length of the day would become
 (a) 3 h (b) 12 h (c) 48 h (d) 6 h

✓ Answer Key & Quick Solutions

- 1(d) 2(a) 3(a) 4(d) 5(d) 6(b) 7(c) 8(c) 9(a) 10(b) 11(d) 12(d) 13(d) 14(d) 15(b)
 16(a) 17(a) 18(b) 19(b) 20(c) 21(c) 22(c) 23(c) 24(d)

1. (d) By symmetry the COM is at the centre – it need not lie on the material.
2. (a) $x_{cm} = \frac{2(0) + 3(5)}{5} = 3$ m (nearer the heavier mass).
3. (a) $\frac{ML^2}{12} + M\left(\frac{L}{2}\right)^2 = \frac{ML^2}{3}$.
4. (d) $\tau_z = xF_y - yF_x = 2(4) - 1(3) = 5 \Rightarrow 5\hat{k}$ N m.
5. (d) $a = \frac{g \sin \theta}{1 + K^2/R^2}$; smallest K^2/R^2 ($2/5$) $\Rightarrow$ solid sphere.
6. (b) $\frac{K^2/R^2}{1 + K^2/R^2} = \frac{2/5}{7/5} = \frac{2}{7}$.
7. (c) $\omega' = \omega/2$. $KE_f = \frac{1}{2}(2I)\frac{\omega^2}{4} = \frac{1}{4}I\omega^2$; loss $= \frac{1}{2}I\omega^2 - \frac{1}{4}I\omega^2 = \frac{1}{4}I\omega^2$.
8. (c) $\vec{\tau}_{ext} = \frac{d\vec{L}}{dt} = 0 \Rightarrow \vec{L}$ constant. Zero net force does not imply zero torque (couple).

9. (a) Diameter: $\frac{1}{4}MR^2$ ($\perp$ axis theorem); tangent parallel to it: $\frac{1}{4}MR^2 + MR^2 = \frac{5}{4}MR^2$.
10. (b) $MK^2 = \frac{1}{2}MR^2 \Rightarrow K = R / \sqrt{2}$.
11. (d) $I_2 = 0.6I_1$; $I_1\omega = 0.6I_1\omega_2 \Rightarrow \omega_2 = \omega / 0.6 = \frac{5}{3}\omega$.
12. (d) $\theta = \frac{1}{2}\pi(10)^2 = 50\pi \text{ rad} = 50\pi / 2\pi = 25 \text{ rev}$.
13. (d) $\alpha = 5 \text{ rad/s}^2$, $\omega = 20 \text{ rad/s} \Rightarrow KE = \frac{1}{2}(2)(20)^2 = 400 \text{ J}$ (check: $W = \tau\theta = 10 \times 40$).
14. (d) $mg\frac{L}{2} = \frac{1}{2} \cdot \frac{mL^2}{3}\omega^2 \Rightarrow \omega = \sqrt{3g/L}$.
15. (b) $a = \frac{g\sin 30^\circ}{1 + \frac{1}{2}} = \frac{5}{1.5} = 3.33 \text{ m/s}^2$.
16. (a) Contact point: v_{cm} forward + ωR backward = 0. R correctly explains A.
17. (a) $I_z = I_x + I_y$ needs all mass in the xy-plane.
18. (b) $L = mvr_{\perp} = mva$; the lever arm a never changes.
19. (b) COM still lands at 100 m. First piece lands at 50 m $\Rightarrow \frac{50+x}{2} = 100 \Rightarrow x = 150 \text{ m}$.
20. (c) $KE = \frac{1}{2}mv^2(1 + \frac{1}{2}) = \frac{3}{4}(2)(16) = 24 \text{ J}$.
21. (c) I changes with the axis (rod: $ML^2 / 12$ vs $ML^2 / 3$), so A is false; R is true.
22. (c) Torques about B: $R_A(4) = 200(2) + 100(3) = 700 \Rightarrow R_A = 175 \text{ N}$ (and $R_B = 125 \text{ N}$).
23. (c) $\frac{2}{5}MR^2 + MR^2 = \frac{7}{5}MR^2$.
24. (d) $I\omega$ constant, $I \propto R^2 \Rightarrow T \propto R^2 \Rightarrow T = 24 / 4 = 6 \text{ h}$.

CHAPTER

7

Gravitation

Day 11-12

Day 11-12

NEET weightage ★★☆☆

NCERT Class 11

Kepler's laws

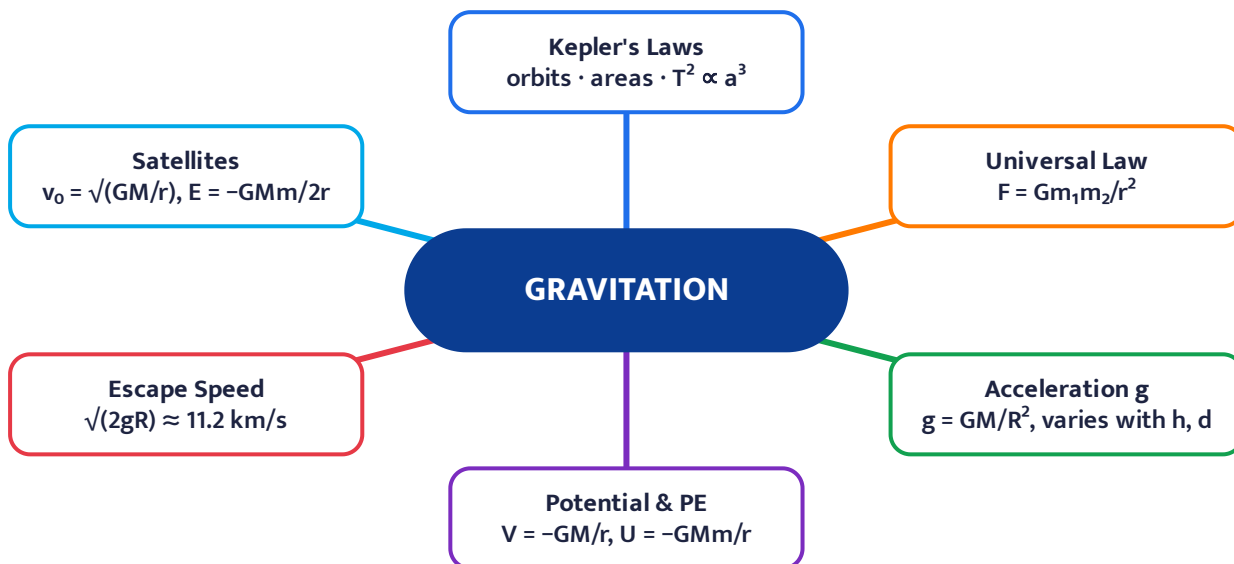
Law of gravitation

Variation of g

Potential & PE

Escape speed

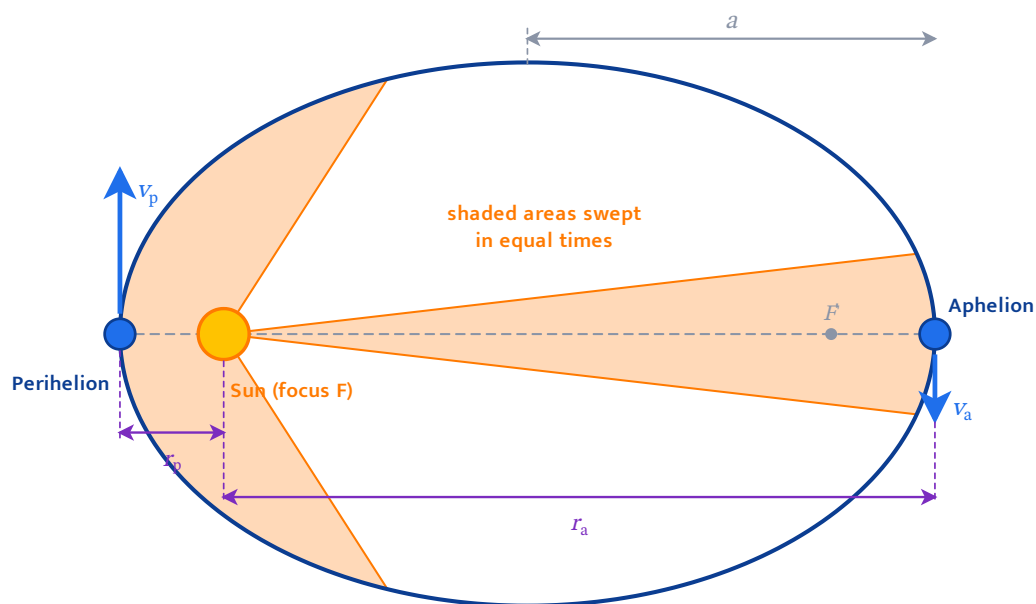
Satellites



Chapter at a glance

7.1 Kepler's Laws of Planetary Motion

- **Law of orbits:** every planet moves in an **ellipse** with the Sun at one **focus**.
- **Law of areas:** the line joining the Sun and a planet sweeps **equal areas in equal times** – areal velocity $\frac{dA}{dt} = \frac{L}{2m}$ is constant. It follows from **conservation of angular momentum** (gravity is a central force).
- **Law of periods:** $T^2 \propto a^3$, where a is the **semi-major axis**.



Law of areas: the two shaded sectors are swept in equal times, so the planet is fastest at perihelion

**Kepler's formulas**

$$v_p r_p = v_a r_a \quad \text{from } L = mvr = \text{constant}$$

$$a = \frac{r_p + r_a}{2} \quad \text{semi-major axis}$$

$$\frac{T_1^2}{T_2^2} = \frac{a_1^3}{a_2^3} \quad \text{compare two planets}$$

**Solved Example**

A planet is 4 times as far from the Sun as the Earth. Find its year.

Solution: $T \propto a^{3/2} \Rightarrow T = 1 \times 4^{3/2} = 8 \text{ Earth-years.}$

7.2 Universal Law of Gravitation

Every particle attracts every other particle with a force proportional to the product of their masses and inversely proportional to the square of the distance between them. The force is **central, always attractive, independent of the medium**, and the pair obeys Newton's third law.

**Newton's law of gravitation**

$$F = \frac{Gm_1m_2}{r^2}$$

$$G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}; [G] = [M^{-1}L^3T^{-2}]$$

$$\vec{F}_{12} = -\frac{Gm_1m_2}{r^2}\hat{r}_{12}$$

Vector form (force on 2 due to 1 points towards 1)

$$\vec{F} = \vec{F}_1 + \vec{F}_2 + \dots$$

Principle of superposition

**NCERT Line to Remember**

- A uniform **spherical shell** attracts an outside point mass as if all its mass were at its centre.
- Force on a point mass **inside** a uniform shell is **zero**.
- G was first measured by **Cavendish** (torsion balance).

 G **g**

Universal constant

Varies with place

Scalar

Vector (towards centre)

 $\text{N m}^2 \text{ kg}^{-2}$ m s^{-2} 6.67×10^{-11} $\approx 9.8 \text{ on Earth}$ **7.3 Acceleration due to Gravity and its Variation**

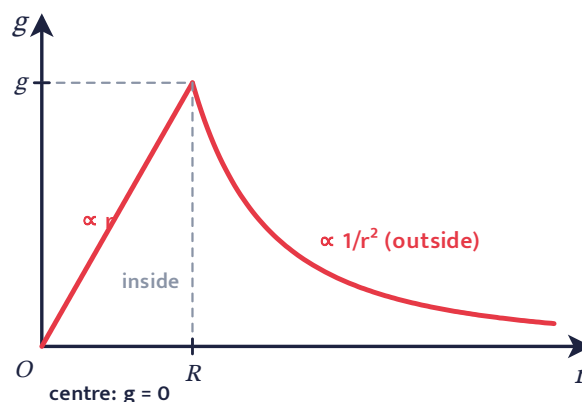
Treating Earth as a uniform sphere of mass M and radius R : $mg = \frac{GMm}{R^2} \Rightarrow g = \frac{GM}{R^2} = \frac{4}{3}\pi G\rho R$. g is independent of the falling body's mass.

**Variation of g**

$$g_h = \frac{g}{(1 + h/R)^2} = g \frac{R^2}{(R + h)^2} \quad \text{Height } h \text{ (exact)}$$

$$g_h \approx g \left(1 - \frac{2h}{R}\right) \quad \text{Height, } h \ll R$$

$$g_d = g \left(1 - \frac{d}{R}\right) \quad \text{Depth } d \text{ (exact); } g = 0 \text{ at centre}$$



g vs distance r from Earth's centre: maximum at the surface

 Solved Example

At what height does g become 64% of its surface value? ($R = 6400$ km)

Solution: $\left(\frac{R}{R+h}\right)^2 = 0.64 \Rightarrow \frac{R}{R+h} = 0.8 \Rightarrow h = \frac{R}{4} = 1600$ km. (The approximation $1 - 2h/R$ fails here since h is not small.)

 Common Trap

- For small h : g at height $h = g$ at depth $2h$ (both fall by the same amount).
- g **decreases** both above and below the surface; it is maximum at the surface.
- At height $h = R$: $g/4$ (not $g/2$).

 JEE Main Extension

Earth's rotation reduces effective g :

$g' = g - \omega^2 R \cos^2 \lambda$ (λ = latitude). Least at the equator, unaffected at the poles. If Earth spun with $\omega = \sqrt{g/R}$ (day ≈ 1.4 h), bodies at the equator would feel weightless.

7.4 Gravitational Field, Potential and Potential Energy

Gravitational field (intensity) $E = F/m$ = force per unit mass (= g). **Gravitational potential** V = work done per unit mass in bringing a mass from infinity to the point (zero at infinity, negative elsewhere). **Potential energy** $U = mV$ of the pair; $\vec{E} = -\frac{dV}{dr}$.

 Field, potential, PE

$$E = \frac{GM}{r^2} \quad (r \geq R), \quad E = \frac{GM}{R^2} \quad (r < R) \quad \text{Field of a uniform solid sphere}$$

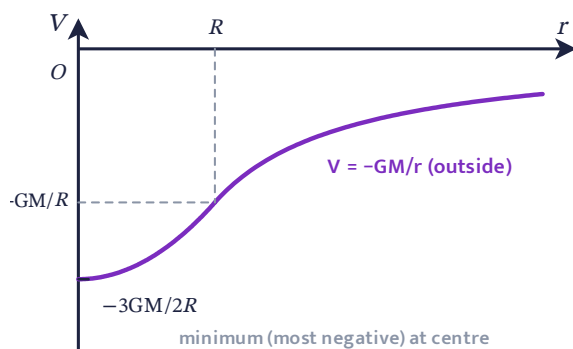
$$V = -\frac{GM}{r} \quad (r \geq R), \quad V_{\text{centre}} = -\frac{3GM}{2R} \quad \text{Potential (outside; at centre)}$$

$$U = -\frac{GMm}{r} \quad \text{PE of mass } m \text{ at distance } r \text{ (zero at } \infty)$$

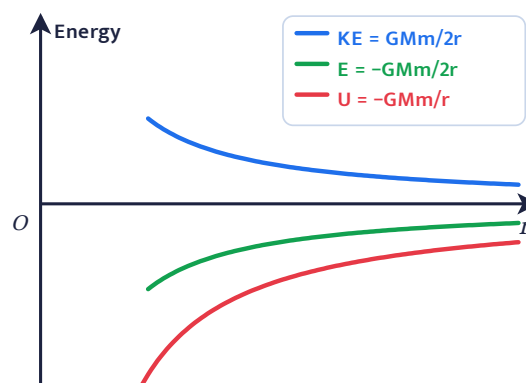
$$\Delta U = \frac{mgh}{1 + h/R} \approx mgh \quad \text{Raising } m \text{ from surface to height } h$$

 VS Sir's Exam Tip

Raising a body to $h = R$ needs $\Delta U = \frac{mgR}{2}$ - **not** mgR , because g weakens with height. Use mgh only when $h \ll R$.



Potential V vs r for a uniform solid sphere



Energies of a satellite vs orbit radius r

7.5 Escape Speed

Minimum speed with which a body must be projected from the surface to escape to infinity (total energy ≥ 0):

$$\frac{1}{2}mv_e^2 - \frac{GMm}{R} = 0.$$

Escape speed

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \approx 11.2 \text{ km/s for Earth}$$

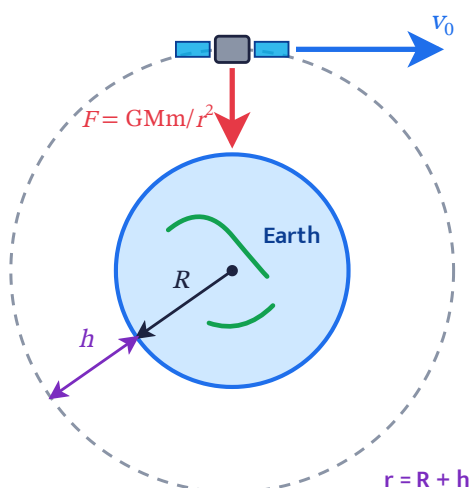
$$v_e = R\sqrt{\frac{8\pi G\rho}{3}} \propto R\sqrt{\rho} \quad \text{in terms of density}$$

$$v_e = \sqrt{2} v_0 \quad v_0 = \text{orbital speed near surface}$$

NCERT Line to Remember

- v_e is **independent of the mass** of the body and of the **direction** of projection (ignoring air).
- Moon: $v_e \approx 2.3 \text{ km/s}$ – about five times smaller, so gas molecules escape and the **Moon has no atmosphere**.

7.6 Earth Satellites – Orbital Speed, Period and Energy



Gravity supplies the centripetal force

Satellite formulas

$$\frac{mv_0^2}{r} = \frac{GMm}{r^2}$$

$$r = R + h$$

$$v_0 = \sqrt{\frac{GM}{r}} = \sqrt{\frac{gR^2}{R+h}} \quad \text{Orbital speed (independent of m)}$$

$$T = 2\pi\sqrt{\frac{r^3}{GM}} \quad \text{Period } (T^2 \propto r^3)$$

$$v_0 = \sqrt{gR} \approx 7.9 \text{ km/s} \quad \text{Near surface; } T \approx 84 \text{ min}$$

Quantity	Formula	Relation to E
Kinetic energy	$K = \frac{GMm}{2r}$	$K = -E$
Potential energy	$U = -\frac{GMm}{r}$	$U = 2E$
Total energy	$E = K + U = -\frac{GMm}{2r}$	negative $\Rightarrow$ bound
Binding energy	$BE = -E = \frac{GMm}{2r}$	energy to free it

Solved Example

Find the orbital speed and period of a satellite at height $h = R$ above the Earth ($R = 6400 \text{ km}$, $g = 10 \text{ m/s}^2$).

Solution: $v_0 = \sqrt{\frac{gR^2}{2R}} = \sqrt{\frac{gR}{2}} = \sqrt{3.2 \times 10^7} \approx 5.66 \text{ km/s}$. $T = \frac{2\pi(2R)}{v_0} = \frac{2\pi \times 1.28 \times 10^7}{5657} \approx 1.42 \times 10^4 \text{ s} \approx 3.95 \text{ h}$.

Solved Example

A 400 kg satellite orbits at radius $2R$. How much energy is needed to shift it to radius $4R$? ($R = 6400$ km, $g = 10$ m/s²)

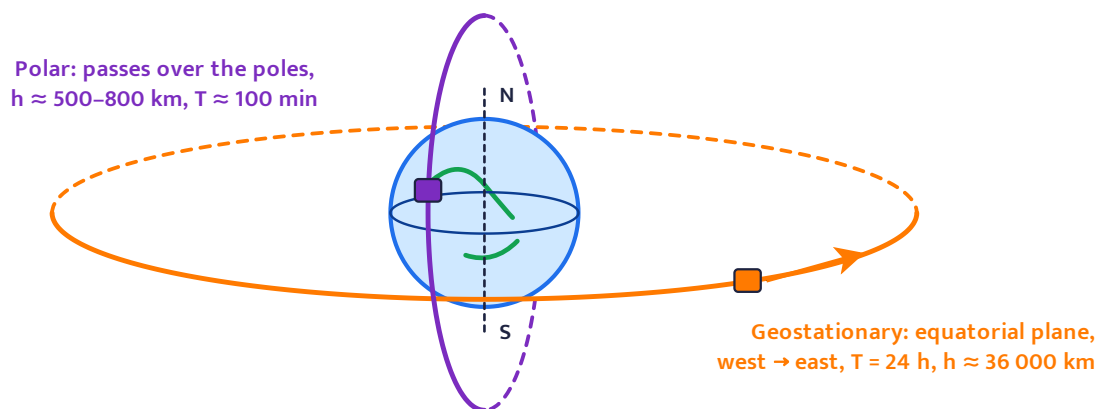
Solution: $\Delta E = E_f - E_i = -\frac{GMm}{8R} + \frac{GMm}{4R} = \frac{GMm}{8R} = \frac{mgR}{8} = \frac{400 \times 10 \times 6.4 \times 10^6}{8} = 3.2 \times 10^9$ J. KE falls by this amount; PE rises by twice it.

Memory Trick

"**K : U : E = 1 : -2 : -1**". Higher orbit $\Rightarrow$ slower ($v \propto r^{-1/2}$), longer period ($T \propto r^{3/2}$), **less KE but more total energy** (E less negative).

7.7 Geostationary & Polar Satellites, Weightlessness**Note**

These topics are **not in the rationalised NCERT text**; they are kept here as quick facts because they still appear in concept-based questions.



Geostationary orbit (equatorial) and polar orbit

Satellite types

- **Geostationary:** $T = 24$ h, circular, in the **equatorial plane**, moving **west to east**; appears fixed over one place – used for communication/TV (radius $\approx 42\,000$ km).
- **Polar:** low orbit passing over the poles; Earth rotates beneath, so it scans the whole globe – remote sensing, weather.

Weightlessness

- In an orbiting satellite, the astronaut and the cabin fall freely with the **same acceleration** g_h , so the floor exerts no normal force $\Rightarrow$ apparent weight zero.
- Gravity is **not** zero there – it provides the centripetal force.

Numbers to remember

Earth: $R \approx 6400$ km, $M \approx 6 \times 10^{24}$ kg, $g = 9.8$ m/s² · $v_e = 11.2$ km/s · $v_0 = 7.9$ km/s (near surface, $T \approx 84$ min) · geostationary: $T = 24$ h, $r \approx 6.6R$ ($\approx 42\,000$ km) · Moon: $g_{\text{moon}} \approx g/6$, $v_e \approx 2.3$ km/s.



Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** Kepler's second law (law of areas) is a consequence of the conservation of
 - (a) angular momentum
 - (b) linear momentum
 - (c) kinetic energy
 - (d) mass
- 2 **NEET pattern** The gravitational force between two masses is F . If the distance between them is halved, the force becomes
 - (a) $4F$
 - (b) $F/2$
 - (c) $2F$
 - (d) $F/4$
- 3 **NEET pattern** A planet revolves round the Sun in an orbit whose semi-major axis is 4 times that of the Earth. Its period of revolution is
 - (a) 16 years
 - (b) 2 years
 - (c) 8 years
 - (d) 4 years
- 4 **NCERT based** The acceleration due to gravity at the centre of the Earth is
 - (a) $g/2$
 - (b) infinite
 - (c) zero
 - (d) g
- 5 **NEET pattern** At what height above the Earth's surface does the acceleration due to gravity become one-fourth of its surface value? (R = Earth's radius)
 - (a) $R/2$
 - (b) $2R$
 - (c) $R/4$
 - (d) R
- 6 **NEET pattern** The value of g at a small height h above the surface equals its value at a depth d below the surface. Then
 - (a) $d = h/2$
 - (b) $d = 2h$
 - (c) $d = h$
 - (d) $d = 4h$
- 7 **NEET pattern** A planet has 8 times the mass and twice the radius of the Earth. If escape speed from Earth is 11.2 km/s, that from the planet is
 - (a) 11.2 km/s
 - (b) 44.8 km/s
 - (c) 22.4 km/s
 - (d) 5.6 km/s
- 8 **NEET pattern** The total energy of a satellite in a circular orbit is E . Its kinetic energy is
 - (a) E
 - (b) $-2E$
 - (c) $-E$
 - (d) $2E$
- 9 **JEE Main pattern** The energy needed to move a satellite of mass m from a circular orbit of radius $2R$ to one of radius $3R$ around the Earth (mass M) is
 - (a) $\frac{GMm}{6R}$
 - (b) $\frac{GMm}{12R}$
 - (c) $\frac{GMm}{3R}$
 - (d) $\frac{GMm}{8R}$
- 10 **NCERT based** The escape speed of a body from the Earth depends on
 - (a) the mass and radius of the Earth only
 - (b) the angle of projection
 - (c) the mass of the body
 - (d) the mass of the body and the angle of projection

- 11 **NEET pattern** The orbital speed of a satellite very close to the Earth's surface is 7.9 km/s. The escape speed from the Earth is about
 (a) 7.9 km/s (b) 5.6 km/s (c) 11.2 km/s (d) 15.8 km/s
- 12 **Assertion-Reason** Assertion (A): The Moon has no atmosphere. Reason (R): The escape speed on the Moon (≈ 2.3 km/s) is small, so gas molecules there easily escape.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 13 **NEET pattern** A body weighs 72 N on the Earth's surface. The gravitational force on it at a height equal to half the Earth's radius is
 (a) 36 N (b) 48 N (c) 32 N (d) 24 N
- 14 **JEE Main pattern** The increase in potential energy of a body of mass m raised from the Earth's surface to a height equal to the Earth's radius R is
 (a) mgR (b) $\frac{mgR}{4}$ (c) $2mgR$ (d) $\frac{mgR}{2}$
- 15 **NCERT based** How does g vary with distance r from the centre of the Earth (uniform density)?
 (a) Increases linearly up to R , then decreases as $1/r^2$
 (b) Decreases as $1/r^2$ everywhere
 (c) Increases as $1/r$ up to R , then constant
 (d) Constant inside, then decreases as $1/r^2$
- 16 **NEET pattern** Two satellites A and B revolve in circular orbits of radii r and $4r$. The ratio of their orbital speeds $v_A:v_B$ is
 (a) 2 : 1 (b) 1 : 4 (c) 4 : 1 (d) 1 : 2
- 17 **Assertion-Reason** Assertion (A): The value of g increases as we go down into the Earth. Reason (R): At depth d , $g_d = g(1 - d/R)$.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 18 **JEE Main pattern** A geostationary satellite
 (a) moves over the poles with a period of 24 h
 (b) moves east to west in the equatorial plane with a period of 12 h
 (c) can be placed at any height if its speed is suitable
 (d) moves west to east in the equatorial plane with a period of 24 h
- 19 **JEE Main pattern** If the radius of the Earth shrinks by 1% with its mass unchanged, the value of g at its surface
 (a) increases by about 1% (b) remains unchanged
 (c) increases by about 2% (d) decreases by about 2%

20. **NEET pattern** A comet has speed v_1 at its closest distance r_1 from the Sun. At its farthest distance r_2 , its speed is
 (a) $\frac{v_1 r_2}{r_1}$ (b) $v_1 \left(\frac{r_1}{r_2}\right)^2$ (c) $v_1 \sqrt{\frac{r_1}{r_2}}$ (d) $\frac{v_1 r_1}{r_2}$
21. **JEE Main pattern** A body is projected vertically upward from the Earth's surface with half the escape speed. The maximum height reached is (R = Earth's radius)
 (a) R (b) $R/3$ (c) $R/4$ (d) $R/2$
22. **NEET pattern** The binding energy of a 1000 kg satellite orbiting at a height equal to the Earth's radius is ($R = 6400$ km, $g = 10$ m/s²)
 (a) 1.6×10^{10} J (b) 6.4×10^{10} J (c) 3.2×10^{10} J (d) 0.8×10^{10} J
23. **NCERT based** According to Kepler's third law, the square of the period of a planet is proportional to the cube of
 (a) its own radius
 (b) its perihelion distance
 (c) the semi-major axis of its orbit
 (d) the semi-minor axis of its orbit
24. **JEE Main pattern** A satellite just above the Earth's surface has a period of about 84 min. The period of a satellite at a height $3R$ above the surface is about
 (a) 252 min (b) 336 min (c) 168 min (d) 672 min

✓ Answer Key & Quick Solutions

- 1(a) 2(a) 3(c) 4(c) 5(d) 6(b) 7(c) 8(c) 9(b) 10(a) 11(c) 12(a) 13(c) 14(d) 15(a) 16(a)
 17(c) 18(d) 19(c) 20(d) 21(b) 22(a) 23(c) 24(d)

1. (a) Gravity is central $\Rightarrow$ zero torque $\Rightarrow L$ constant $\Rightarrow dA/dt = L/2m$ constant.
2. (a) $F \propto 1/r^2 \Rightarrow F' = F/(1/2)^2 = 4F$.
3. (c) $T \propto a^{3/2} = 4^{3/2} = 8$.
4. (c) $g_d = g(1 - d/R)$; at $d = R$, $g_d = 0$.
5. (d) $\left(\frac{R}{R+h}\right)^2 = \frac{1}{4} \Rightarrow R+h = 2R \Rightarrow h = R$.
6. (b) $g(1 - 2h/R) = g(1 - d/R) \Rightarrow d = 2h$.
7. (c) $v_e \propto \sqrt{M/R} = \sqrt{8/2} = 2 \Rightarrow 22.4$ km/s.
8. (c) $K = \frac{GMm}{2r}$, $E = -\frac{GMm}{2r} \Rightarrow K = -E$ (positive, since $E < 0$).
9. (b) $\Delta E = -\frac{GMm}{6R} + \frac{GMm}{4R} = \frac{GMm}{12R}$.
10. (a) $v_e = \sqrt{2GM/R}$ – no dependence on the body's mass or direction.
11. (c) $v_e = \sqrt{2}$ $v_0 = 1.414 \times 7.9 \approx 11.2$ km/s.
12. (a) A sizeable fraction of gas molecules (lighter ones especially) have thermal speeds comparable to 2.3 km/s, so over time the gas escapes – R explains A.
13. (c) $W' = W\left(\frac{R}{1.5R}\right)^2 = 72 \times \frac{4}{9} = 32$ N.
14. (d) $\Delta U = \frac{mgh}{1+h/R} = \frac{mgR}{2}$.
15. (a) Inside: $g = \frac{GM}{R^3} r \propto r$; outside: $g = \frac{GM}{r^2}$. Maximum at the surface.

16. (a) $v \propto 1 / \sqrt{r} \Rightarrow v_A / v_B = \sqrt{4r / r} = 2.$

17. (c) R is the correct formula and shows g **decreases** with depth, so A is false.

18. (d) It must co-rotate with Earth: equatorial plane, west $\rightarrow$ east, $T = 24$ h, fixed height $\approx 36\,000$ km.

19. (c) $g \propto R^{-2} \Rightarrow \frac{\Delta g}{g} = -2 \frac{\Delta R}{R} = +2\%.$

20. (d) At both extreme points $\vec{v} \perp \vec{r}$: $mv_1 r_1 = mv_2 r_2 \Rightarrow v_2 = v_1 r_1 / r_2.$

21. (b) $\frac{1}{2} m \frac{v_e^2}{4} = \frac{GMm}{4R} = \frac{GMm}{R} - \frac{GMm}{R+h} \Rightarrow R+h = \frac{4R}{3} \Rightarrow h = R/3.$

22. (a) $BE = \frac{GMm}{2(2R)} = \frac{mgR}{4} = \frac{1000 \times 10 \times 6.4 \times 10^6}{4} = 1.6 \times 10^{10} \text{ J}.$

23. (c) $T^2 \propto a^3$, a = semi-major axis = mean of r_p and r_a .

24. (d) $r = 4R$; $T \propto r^{3/2} \Rightarrow T = 84 \times 4^{3/2} = 84 \times 8 = 672 \text{ min} \approx 11.2 \text{ h}.$

CHAPTER

8

Mechanical Properties of Solids

Day 13

NEET weightage ★★☆☆

NCERT Class 11

Stress & strain

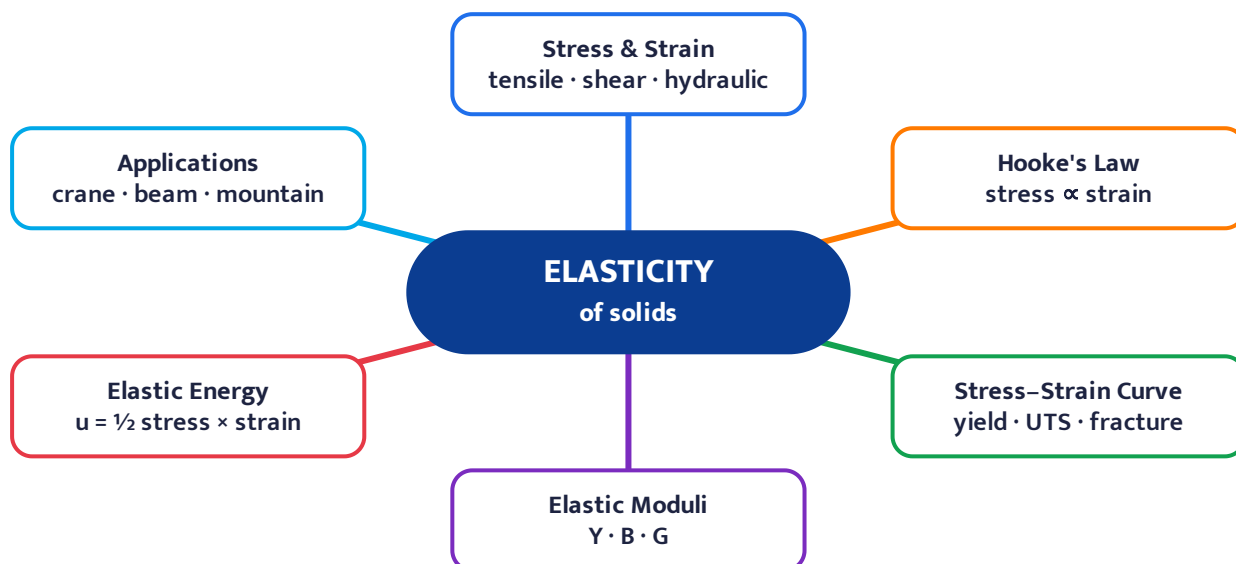
Hooke's law

Stress-strain curve

Y, B, G moduli

Elastic energy

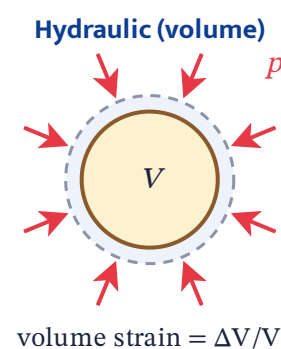
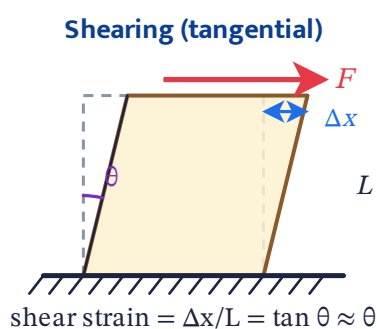
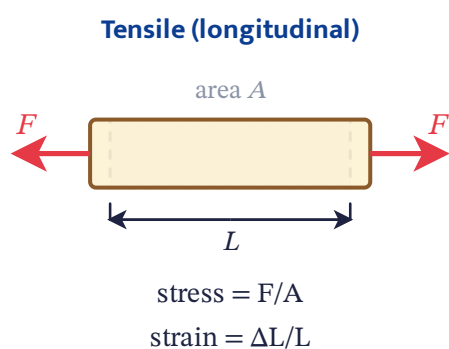
Applications



Chapter at a glance

8.1 Elasticity, Plasticity & Stress

Elasticity is the property of a body to regain its original size and shape when the deforming force is removed. A body that does **not** regain its shape (putty, mud) is **plastic**. When a force deforms a body, an internal **restoring force** develops; the restoring force per unit area is called **stress**.



Three types of stress and the strain each produces

Stress	Force acts	Strain produced	Modulus
Tensile / compressive	normal to area (pull / push)	Longitudinal $\Delta L / L$	Young's Y
Shearing (tangential)	parallel to area	Shear $\Delta x / L = \theta$	Shear G (rigidity)
Hydraulic	normal, from all sides (pressure p)	Volume $\Delta V / V$	Bulk B

NCERT Line to Remember

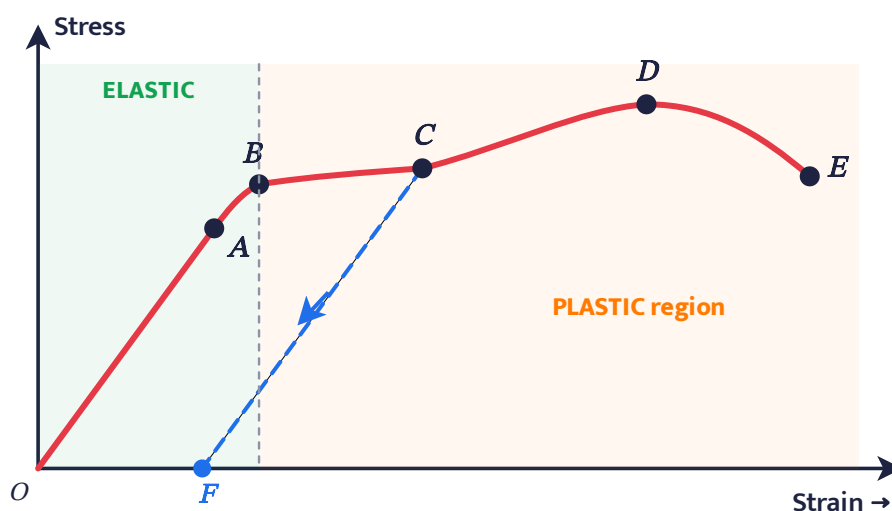
Stress has units $\text{N m}^{-2} = \text{Pa}$ and dimensions $[ML^{-1}T^{-2}]$ (same as pressure). **Strain is a ratio – no units, no dimensions.** In shear, the body changes **shape**, not volume; in hydraulic stress it changes **volume**, not shape.

Common Trap

Stress is **not** simply "applied force/area" – it is the internal **restoring** force per unit area (equal and opposite to the applied force in equilibrium). Pressure acts only normally; stress can be normal **or** tangential.

8.2 Hooke's Law & the Stress–Strain Curve

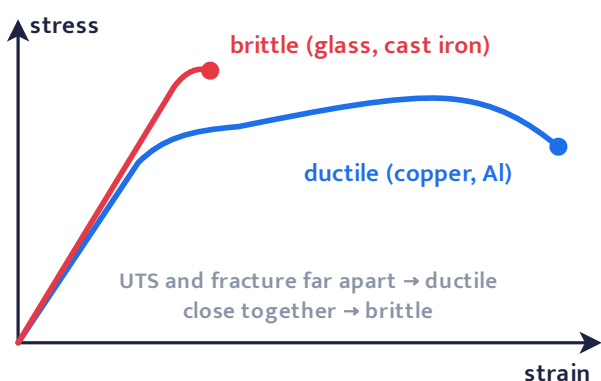
Hooke's law: for small deformations, stress is directly proportional to strain: $\text{stress} = k \times \text{strain}$, where k is the **modulus of elasticity**. It holds only up to the **proportional limit**.



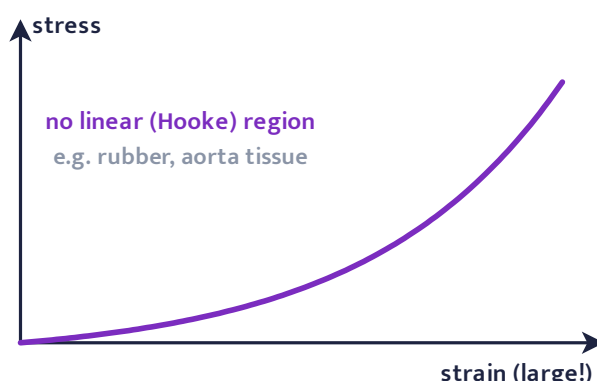
- A** proportional limit
- B** yield point = elastic limit
- C** a point in the plastic region
- D** ultimate tensile strength
- E** fracture point
- OF** permanent set ($CF \parallel OA$)

Stress–strain curve of a metal (tensile test). Yield strength σ_y at B, ultimate tensile strength σ_u at D.

- O → A:** straight line, Hooke's law obeyed; A = **proportional limit**.
- A → B:** not linear, but the body still returns fully on unloading. B = **yield point (elastic limit)**; stress there = **yield strength** σ_y .
- B → D:** plastic region – strain rises quickly. Unloading from C retraces a line parallel to OA and leaves a **permanent set** OF.
- D:** maximum stress = **ultimate tensile strength** σ_u . Beyond D the wire necks and breaks at **E (fracture point)** even if the load is reduced.



Ductile vs brittle



Elastomer (NCERT: aorta)

**Classify by the curve**

- **Ductile:** D and E far apart – large plastic range (Cu, Al, mild steel) → wires, sheets.
- **Brittle:** D and E close – breaks soon after elastic limit (glass, cast iron).
- **Elastomers:** huge elastic strain, no Hooke region (rubber, aorta).

**VS Sir's Exam Tip**

"More elastic" means **larger modulus** – it resists deformation more. So **steel is more elastic than rubber** (for the same stress, steel shows far smaller strain). A **perfectly rigid body** has $Y = \infty$.

8.3 Elastic Moduli – Y, B and G**Moduli of elasticity (unit: Pa = N m⁻²)**

$$Y = \frac{F/A}{\Delta L/L} = \frac{FL}{A\Delta L}$$

Young's modulus (tensile / compressive)

$$\Delta L = \frac{FL}{AY} = \frac{FL}{\pi r^2 Y}$$

Extension of a wire – same material: $\Delta L \propto L / r^2$

$$G = \frac{F/A}{\Delta x/L} = \frac{F}{A\theta}$$

Shear modulus (modulus of rigidity); usually $G \approx Y/3$

$$B = -\frac{p}{\Delta V/V}$$

Bulk modulus (minus: volume decreases as p rises)

$$k = \frac{1}{B}$$

Compressibility

Material	Y (10 ⁹ Pa)	G (10 ⁹ Pa)	B (10 ⁹ Pa)
Steel	200	84	160
Copper	120	42	140
Aluminium	70	25	72
Glass	65	23	37
Water / Air (STP)	—	—	2.2 / 1.0 × 10 ⁻⁴

**NCERT Line to Remember**

- Solids are the least compressible, gases the most: $B_{\text{solid}} > B_{\text{liquid}} \gg B_{\text{gas}}$.
- Liquids and gases have **only bulk modulus** – they cannot sustain shear or tensile stress (Y and G not defined).
- Y and G depend only on the **material** (and temperature), not on length or area.

**Poisson's ratio (NCERT note)**

$\sigma = -\frac{\Delta d/d}{\Delta L/L}$ = lateral strain / longitudinal strain (dimensionless). For steel ≈ 0.28 – 0.30 ; practical range 0 to 0.5 (rubber ≈ 0.5 → volume nearly unchanged). $\frac{\Delta V}{V} = \frac{\Delta L}{L}(1 - 2\sigma)$.

Solved Example

A structural steel rod of radius 10 mm and length 1.0 m is pulled by a force of 100 kN. Find the stress, elongation and strain ($Y = 2.0 \times 10^{11}$ Pa).

Solution: $A = \pi r^2 = 3.14 \times 10^{-4} \text{ m}^2$. Stress $= F/A = \frac{10^5}{3.14 \times 10^{-4}} = 3.18 \times 10^8 \text{ Pa}$.

$\Delta L = \frac{\text{stress} \times L}{Y} = \frac{3.18 \times 10^8 \times 1}{2 \times 10^{11}} = 1.59 \text{ mm}$. Strain $= 1.59 \times 10^{-3} = \mathbf{0.16\%}$.

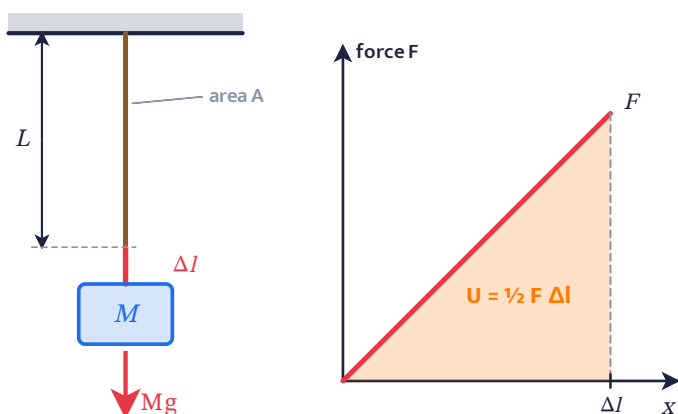
Solved Example

Find the fractional compression of water at the bottom of an ocean 1 km deep. ($B_{\text{water}} = 2.2 \times 10^9 \text{ Pa}$, $\rho = 10^3 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$)

Solution: Extra pressure $p = h\rho g = 10^3 \times 10^3 \times 10 = 10^7 \text{ Pa}$. $\frac{\Delta V}{V} = \frac{p}{B} = \frac{10^7}{2.2 \times 10^9} = 4.5 \times 10^{-3} \approx \mathbf{0.45\%}$.

Common Trap

- Doubling the length of a wire doubles the **extension**, not Y . Y is a material constant.
- Same material, same load: $\Delta L \propto \frac{L}{r^2}$. Same volume: $\Delta L \propto L^2$ (since $A = V/L \propto \frac{1}{r^4}$).
- A wire cut into halves: each half has the **same breaking load** (depends on area) but half the extension.

8.4 Elastic Potential Energy in a Stretched Wire

Work done = area under F - x line

**Elastic energy**

$$U = \frac{1}{2} F \Delta L \quad \text{Energy stored in the wire}$$

$$u = \frac{1}{2} \text{ stress} \times \text{strain} \quad \text{Energy per unit volume}$$

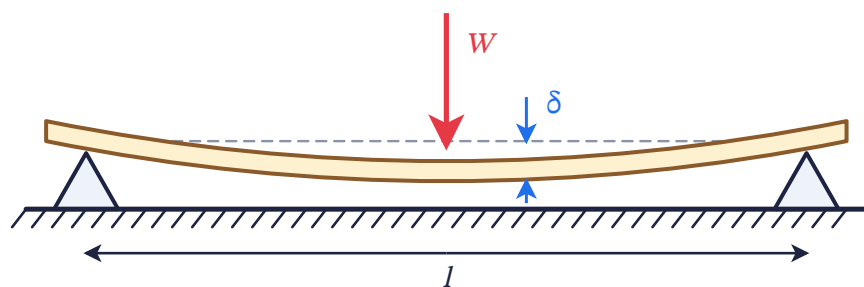
$$u = \frac{\sigma^2}{2Y} = \frac{1}{2} Y \epsilon^2 \quad \text{Using } \sigma = Y\epsilon$$

Memory Trick

Elastic energy looks like a spring: $\frac{1}{2} kx^2 \leftrightarrow \frac{1}{2} Y\epsilon^2$ per volume. "**Half** $\times$ stress $\times$ strain" – the half comes from the triangle (force grows from 0 to F). If a load is simply released, gravity does work $Mg\Delta L$: only half is stored, the other half is lost as heat.

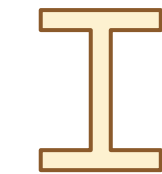
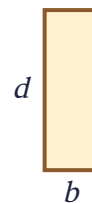
8.5 Applications of Elastic Behaviour

- **Crane ropes:** to lift load W safely, $\pi r^2 \sigma_y \geq W$. For 10 t with $\sigma_y = 300 \text{ MPa}$, $r \approx 1 \text{ cm}$; with a **safety factor of 10**, $r \approx 3 \text{ cm}$. A rope is made of **many thin wires braided** – flexible and strong.
- **Bridges & buildings:** beams must not bend too much or break under traffic, wind and their own weight.
- **Mountains:** stress at the base must stay below the rock's elastic limit $\Rightarrow$ maximum height on Earth $\approx \mathbf{10 \text{ km}}$.



Beam loaded at the centre: depression δ . Deep, I-shaped sections resist bending best.

cross-sections



I-section girder

$$\delta = \frac{Wl^3}{4bd^3Y} \Rightarrow \delta \propto \frac{1}{bd^3}$$

VS Sir's Exam Tip

Depth d enters as d^3 , breadth only as b – so a beam is placed **on its edge** (large d). Doubling d cuts the sag 8 times; doubling b only halves it. Too-deep beams can **buckle**, hence the **I-section**: large depth, low weight.

JEE Main: thermal stress & own weight

- Rod clamped between rigid walls, heated by ΔT :
stress = $Y\alpha\Delta T$, force = $YA\alpha\Delta T$ (independent of length).
- Rod hanging under its own weight:

$$\Delta L = \frac{MgL}{2AY} = \frac{\rho gL^2}{2Y} \text{ (half of a point load } Mg\text{).}$$

Solved Example

A copper wire of length 2.2 m and a steel wire of length 1.6 m, both of diameter 3.0 mm, are joined end to end. Under a load the net elongation is 0.70 mm. Find the load. ($Y_{\text{Cu}} = 1.1 \times 10^{11}$, $Y_{\text{steel}} = 2.0 \times 10^{11}$ Pa)

Solution: Same F in both (series): $\Delta L = \frac{F}{A} \left(\frac{L_1}{Y_1} + \frac{L_2}{Y_2} \right)$. $A = 7.07 \times 10^{-6} \text{ m}^2$; bracket
 $= 2.0 \times 10^{-11} + 0.8 \times 10^{-11} = 2.8 \times 10^{-11}$. $F = \frac{7 \times 10^{-4} \times 7.07 \times 10^{-6}}{2.8 \times 10^{-11}} \approx \mathbf{177 \text{ N}}$ ($\approx 18 \text{ kg}$).

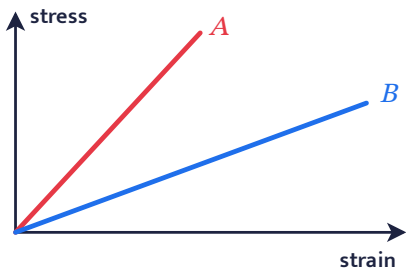
Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** The ratio of stress to strain within the elastic limit is called
- (a) Poisson's ratio (b) modulus of elasticity
 (c) compressibility (d) yield strength

- 2 **NCERT based** Steel is said to be more elastic than rubber because
- (a) for the same stress, steel shows a larger strain
 (b) rubber has a larger Young's modulus
 (c) for the same stress, steel shows a smaller strain
 (d) steel is denser than rubber

- 3 **NCERT based** For which of the following does the stress–strain curve show no linear (Hooke's law) region, though the body regains its shape?
- (a) Copper (b) Mild steel
 (c) Elastic tissue of the aorta (d) Glass

- 4 **NCERT based** Which of the following has the largest compressibility?
 (a) Glass (b) Steel (c) Air (d) Water
- 5 **NEET pattern** A steel wire 2 m long with cross-section 1 mm^2 carries a load of 10 kg. The elongation is ($Y = 2 \times 10^{11} \text{ Pa}, g = 10 \text{ m s}^{-2}$)
 (a) 1 mm (b) 10 mm (c) 0.5 mm (d) 0.1 mm
- 6 **NEET pattern** A wire of length L and radius r stretches by Δl under a force F . Another wire of the same material with length $2L$ and radius $2r$, under the same force, stretches by
 (a) Δl (b) $\Delta l/2$ (c) $2\Delta l$ (d) $\Delta l/4$
- 7 **NEET pattern** Stress-strain graphs of wires A and B are straight lines through the origin; A makes a larger angle with the strain axis. Then
- 
- (a) $Y_A > Y_B$ (b) $Y_A < Y_B$
 (c) $Y_A = Y_B$ (d) cannot be compared
- 8 **NEET pattern** Two wires of the same material have radii in the ratio 1 : 2 and lengths in the ratio 2 : 1. Under the same load, the ratio of their extensions is
 (a) 1 : 8 (b) 4 : 1 (c) 8 : 1 (d) 2 : 1
- 9 **NEET pattern** The pressure on a volume of water is increased by $1.1 \times 10^7 \text{ Pa}$. If $B = 2.2 \times 10^9 \text{ Pa}$, the percentage decrease in volume is
 (a) 0.5% (b) 0.2% (c) 0.05% (d) 5%
- 10 **NEET pattern** The elastic energy stored per unit volume in a steel wire under a strain of 10^{-3} is ($Y = 2 \times 10^{11} \text{ Pa}$)
 (a) 10^8 J m^{-3} (b) 10^5 J m^{-3} (c) $2 \times 10^8 \text{ J m}^{-3}$ (d) $2 \times 10^5 \text{ J m}^{-3}$
- 11 **NEET pattern** A force gradually increased to 20 N stretches a wire by 1 mm. The elastic energy stored in the wire is
 (a) 0.01 J (b) 0.02 J (c) 0.04 J (d) 0.1 J
- 12 **NEET pattern** A rectangular beam supported at its ends sags by δ under a central load. If its breadth is doubled and depth halved, the sag becomes
 (a) 8δ (b) 4δ (c) $\delta/4$ (d) 2δ
- 13 **NEET pattern** A cube of aluminium of side 10 cm has its lower face fixed. A tangential force of 10^4 N acts on the top face. The shear strain is ($G = 2.5 \times 10^{10} \text{ Pa}$)
 (a) 4×10^{-6} (b) 4×10^{-5} (c) 4×10^{-4} (d) 2.5×10^{-5}

- 14 **NEET pattern** Which statement about the stress–strain curve of a metal is correct?
 (a) Brittle materials have a large plastic region
 (b) Fracture occurs at the maximum stress point for ductile metals
 (c) Hooke's law holds right up to the fracture point
 (d) Beyond the yield point the wire keeps a permanent set on unloading
- 15 **NCERT based** The minimum radius of a steel rope of a crane to lift 10^4 kg (yield strength 3×10^8 Pa, $g = 10$ m s^{-2} , no safety factor) is about
 (a) 3 cm (b) 0.3 cm (c) 10 cm (d) 1 cm
- 16 **Assertion–Reason** Assertion: Steel is more elastic than rubber. Reason: For a given stress, the strain produced in steel is much smaller than in rubber.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 17 **Assertion–Reason** Assertion: Young's modulus of a wire doubles when its length is doubled. Reason: The extension of a wire is $\Delta L = FL / (AY)$.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 18 **JEE Main pattern** A uniform rod of length L , density ρ and Young's modulus Y hangs vertically from a rigid support. Its extension due to its own weight is
 (a) $\frac{\rho g L}{2Y}$ (b) $\frac{\rho g L^2}{2Y}$ (c) $\frac{\rho g L^2}{Y}$ (d) $\frac{2\rho g L^2}{Y}$
- 19 **JEE Main pattern** A steel rod is clamped between two rigid walls and heated through 50°C . The thermal stress developed is ($Y = 2 \times 10^{11}$ Pa, $\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$)
 (a) 2.4×10^8 Pa (b) 6×10^7 Pa (c) 1.2×10^8 Pa (d) 1.2×10^6 Pa
- 20 **JEE Main pattern** A wire has Poisson's ratio 0.3. When stretched to a longitudinal strain of 2×10^{-3} , the percentage change in its volume is
 (a) 0.08% increase (b) 0.2% increase
 (c) 0.08% decrease (d) zero

✓ Answer Key & Quick Solutions

- 1(b) 2(c) 3(c) 4(c) 5(a) 6(b) 7(a) 8(c) 9(a) 10(b) 11(a) 12(b) 13(b) 14(d) 15(d)
 16(a) 17(c) 18(b) 19(c) 20(a)

1. (b) By Hooke's law stress/strain = constant = modulus of elasticity.
 2. (c) Larger Y = stress/strain $\Rightarrow$ smaller strain for the same stress $\Rightarrow$ more elastic.
 3. (c) Elastomers such as aorta tissue and rubber are elastic over large strains but do not obey Hooke's law.

4. (c) Compressibility = $1/B$; B of air ($\sim 10^5$ Pa) is the smallest.

5. (a) $\Delta L = \frac{FL}{AY} = \frac{100 \times 2}{10^{-6} \times 2 \times 10^{11}} = 10^{-3} \text{ m} = 1 \text{ mm}.$

6. (b) $\Delta l \propto L / r^2 \Rightarrow 2 / 4 = 1 / 2 \Rightarrow \Delta l / 2.$

7. (a) Slope of stress-strain line = Y ; A is steeper $\Rightarrow Y_A > Y_B.$

8. (c) $\frac{\Delta l_1}{\Delta l_2} = \frac{L_1}{L_2} \cdot \frac{r_2^2}{r_1^2} = 2 \times 4 = 8.$

9. (a) $\Delta V / V = p / B = 5 \times 10^{-3} = 0.5\%.$

10. (b) $u = \frac{1}{2} Y \epsilon^2 = \frac{1}{2} (2 \times 10^{11}) (10^{-6})^2 = 10^5 \text{ J m}^{-3}.$

11. (a) $U = \frac{1}{2} F \Delta l = \frac{1}{2} \times 20 \times 10^{-3} = 0.01 \text{ J}.$

12. (b) $\delta \propto \frac{1}{bd^3} \Rightarrow \frac{1}{2 \times (1/8)} = 4 \Rightarrow 4\delta.$

13. (b) $\theta = \frac{F}{AG} = \frac{10^4}{10^{-2} \times 2.5 \times 10^{10}} = 4 \times 10^{-5}.$

14. (d) Past the elastic limit the deformation is plastic, leaving a permanent set; the other statements are false.

15. (d) $r = \sqrt{\frac{Mg}{\pi \sigma_y}} = \sqrt{\frac{10^5}{\pi \times 3 \times 10^8}} \approx 1.03 \times 10^{-2} \text{ m} \approx 1 \text{ cm}.$

16. (a) Smaller strain for the same stress means a larger modulus, i.e. more elastic. R is the correct explanation.

17. (c) Y is a material property; doubling L doubles ΔL , not Y . A false, R true.

18. (b) Tension varies linearly from 0 to Mg , so the average is $Mg/2$: $\Delta L = \frac{MgL}{2AY} = \frac{\rho g L^2}{2Y}.$

19. (c) $\sigma = Y \alpha \Delta T = 2 \times 10^{11} \times 1.2 \times 10^{-5} \times 50 = 1.2 \times 10^8 \text{ Pa}.$

20. (a) $\frac{\Delta V}{V} = \epsilon(1 - 2\sigma) = 2 \times 10^{-3} \times 0.4 = 8 \times 10^{-4} = 0.08\% \text{ increase}.$

CHAPTER

9

Mechanical Properties of Fluids

17

Day 14–15

NEET weightage ★★☆☆

NCERT Class 11

Pressure & Pascal

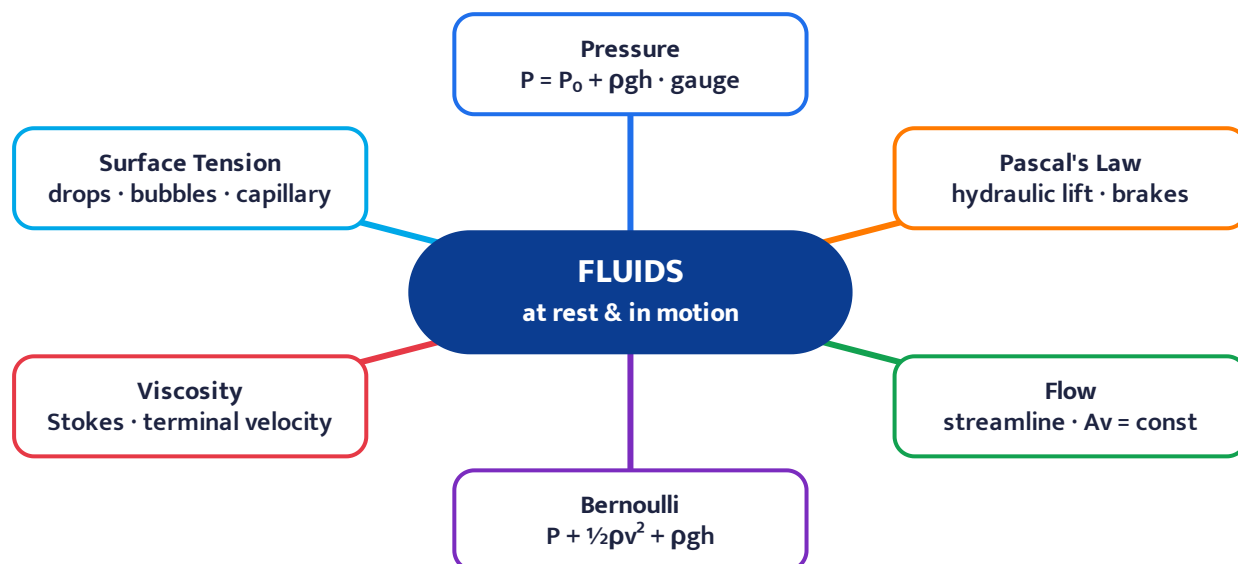
Continuity

Bernoulli's principle

Viscosity & Stokes

Surface tension

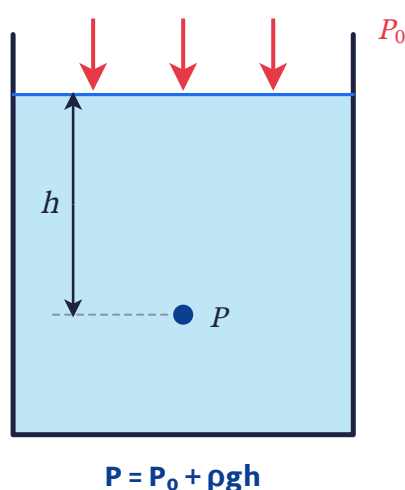
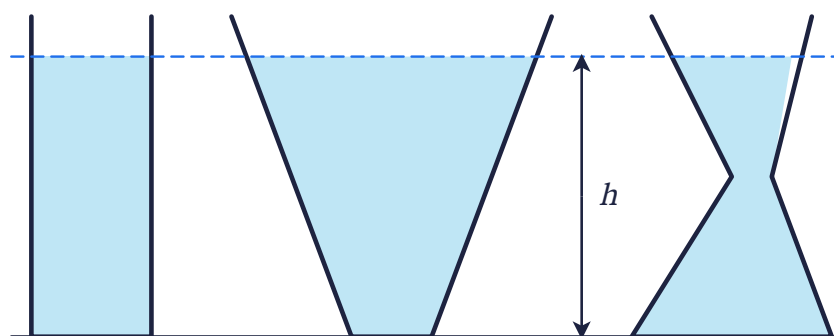
Capillarity



Chapter at a glance

9.1 Pressure and its Variation with Depth

Liquids and gases are **fluids** – they flow and take the shape of the container. A fluid at rest exerts a force **normal** to any surface in contact. **Pressure** $P = F / A$ is a **scalar**; SI unit pascal ($1 \text{ Pa} = 1 \text{ N m}^{-2}$). $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$; $1 \text{ bar} = 10^5 \text{ Pa}$; $1 \text{ torr} = 1 \text{ mm of Hg}$.

Hydrostatic paradox: same $h \Rightarrow$ same P at the base

Pressure depends on depth, not on shape or amount of liquid

Pressure at depth h – and why the vessel shape does not matter

Pressure formulas

$$P = P_0 + \rho gh$$

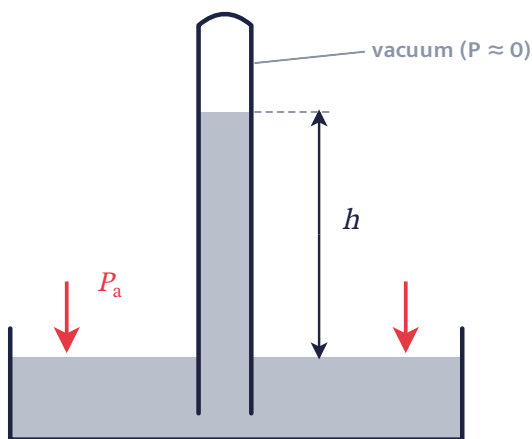
Absolute pressure at depth h (P_0 = atmospheric)

$$P_g = P - P_0 = \rho gh$$

Gauge pressure (what a tyre gauge reads)

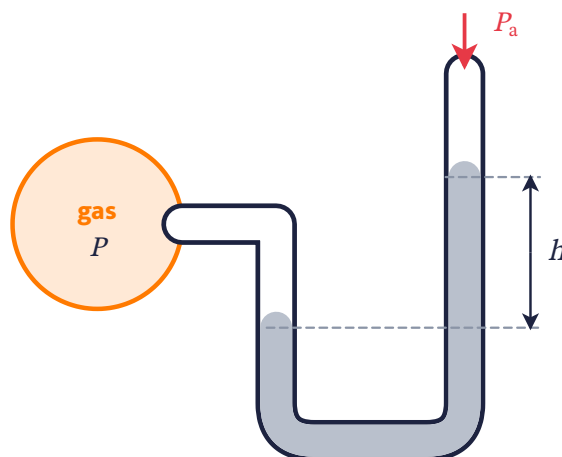
$$P_a = \rho_{\text{Hg}} g h = 13.6 \times 10^3 \times 9.8 \times 0.76 \approx 1.013 \times 10^5 \text{ Pa}$$

Mercury barometer: 76 cm of Hg at sea level



$$P_a = \rho gh \quad (h = 76 \text{ cm of Hg})$$

Mercury barometer (Torricelli)



$$\text{gauge pressure } P - P_a = \rho gh$$

Open-tube manometer

NCERT Line to Remember

- Pressure is the same at all points **at the same horizontal level** in a liquid at rest.
- Atmospheric pressure falls with altitude (air gets thinner).
- A **manometer** measures pressure differences; a **barometer** measures atmospheric pressure.

Archimedes (brief)

Buoyant force = weight of fluid displaced:

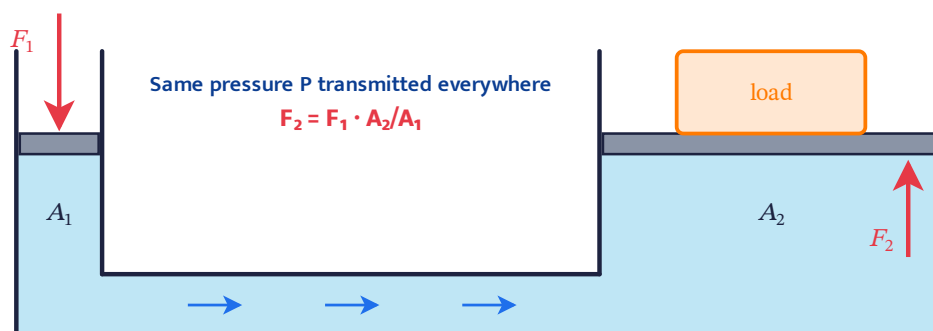
$$F_B = \rho_{\ell} V_{\text{sub}} g. \text{ Apparent weight} = W - F_B.$$

A body floats when $F_B = W$: fraction submerged

$$= \rho_{\text{body}} / \rho_{\ell}.$$

9.2 Pascal's Law & Hydraulic Machines

Pascal's law: pressure applied to an enclosed fluid is transmitted **undiminished** to every portion of the fluid and to the walls of the container. This multiplies force in hydraulic lifts, presses and **hydraulic brakes**.



Hydraulic lift: a small force on the small piston lifts a heavy load on the large piston

Solved Example

Two syringes of piston areas 1.0 cm^2 and 10 cm^2 are connected by a water-filled tube. A force of 10 N acts on the small piston. Find the force on the large piston. If the small piston moves in by 6 cm , how far does the large one move out?

Solution: $F_2 = F_1 \frac{A_2}{A_1} = 10 \times 10 = 100 \text{ N}$. Volume is conserved:
 $A_1 d_1 = A_2 d_2 \Rightarrow d_2 = \frac{6}{10} = 0.6 \text{ cm}$.

Common Trap

A hydraulic lift multiplies **force, not energy**: $F_1 d_1 = F_2 d_2$. The big piston moves a much smaller distance.

9.3 Streamline Flow & Equation of Continuity

In **steady (streamline) flow**, the velocity at each point does not change with time; the path of a fluid particle is a **streamline**, and **two streamlines never cross**. At high speeds the flow becomes **turbulent** (eddies, churning). For an incompressible fluid, mass entering = mass leaving:

$$A_1 v_1 = A_2 v_2 = \text{constant (volume flux)} \Rightarrow v \propto \frac{1}{A}$$

VS Sir's Exam Tip

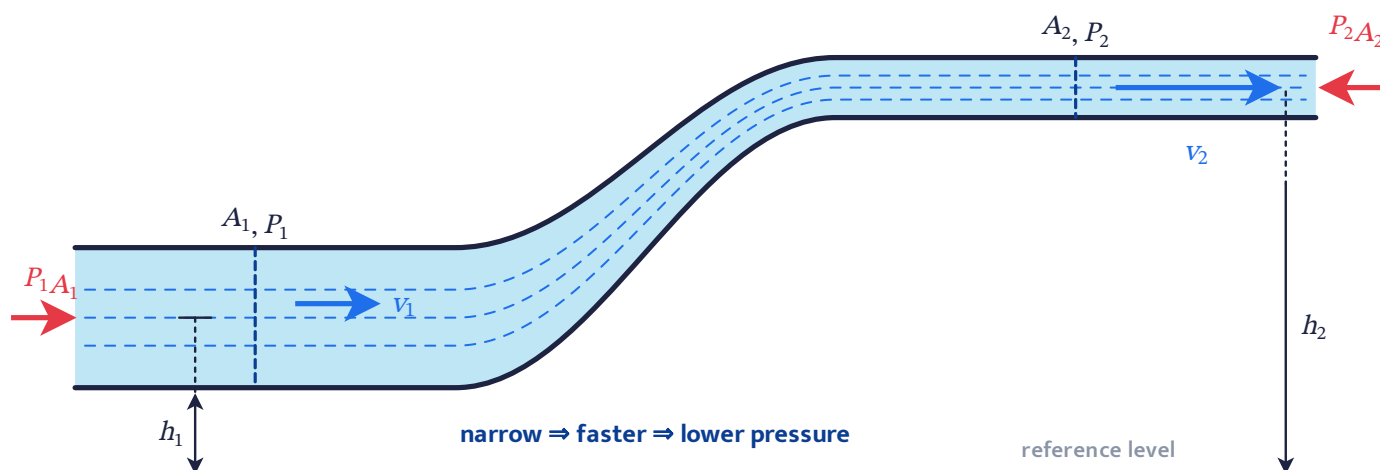
Narrow part $\Rightarrow$ streamlines crowd together $\Rightarrow$ **higher speed**. Why a finger on a garden-hose mouth makes the jet go farther. For circular pipes $v \propto 1/d^2$: halve the diameter $\rightarrow 4\times$ speed.

JEE Main: Reynolds number

$R_e = \frac{\rho v d}{\eta}$ (dimensionless). $R_e < 1000$: streamline; $1000-2000$: unsteady; > 2000 : turbulent. Critical speed $v_c = R_e \eta / (\rho d)$.

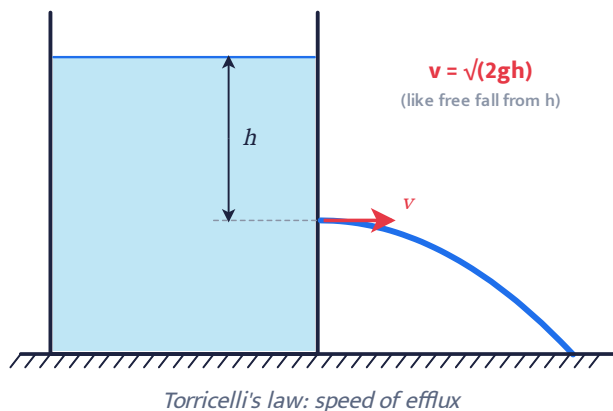
9.4 Bernoulli's Principle

For the steady flow of an ideal (incompressible, non-viscous) fluid along a streamline, the sum of pressure, kinetic energy per unit volume and potential energy per unit volume is constant. It is the **work-energy theorem** applied to a fluid.



Flow through a pipe of varying cross-section and height

$$P_1 + \frac{1}{2}\rho v_1^2 + \rho g h_1 = P_2 + \frac{1}{2}\rho v_2^2 + \rho g h_2 = \text{constant}$$



Applications

- **Speed of efflux:** $v = \sqrt{2gh}$ (tank open to air). Range from a hole at depth h in a tank of height H :
 $R = 2\sqrt{h(H-h)}$, maximum at $h = H/2$.
- **Dynamic lift:** aerofoil wing – faster air above $\Rightarrow$ lower pressure $\Rightarrow$ upward lift. A **spinning ball** swerves (Magnus effect).
- **Roofs blown off** in storms: fast wind above $\Rightarrow$ low pressure outside, higher pressure inside pushes the roof up.
- **Blood flow:** a narrowed artery speeds up blood, lowers pressure and the artery may collapse (heart attack).

Solved Example

Water flows through a horizontal pipe. At a point where the cross-section is 10 cm^2 the speed is 1 m s^{-1} and pressure is $2 \times 10^5 \text{ Pa}$. Find the pressure where the cross-section is 5 cm^2 .

Solution: Continuity: $v_2 = v_1 \frac{A_1}{A_2} = 2 \text{ m s}^{-1}$. Bernoulli (same height):

$$P_2 = P_1 - \frac{1}{2}\rho(v_2^2 - v_1^2) = 2 \times 10^5 - \frac{1}{2}(1000)(4 - 1) = 1.985 \times 10^5 \text{ Pa}.$$

JEE Main: Venturi meter

Measures flow speed in a pipe of areas A and a :

$$v_1 = \sqrt{\frac{2\Delta P}{\rho\left(\frac{A^2}{a^2} - 1\right)}}, \text{ with } \Delta P = \rho_m g h \text{ read on a}$$

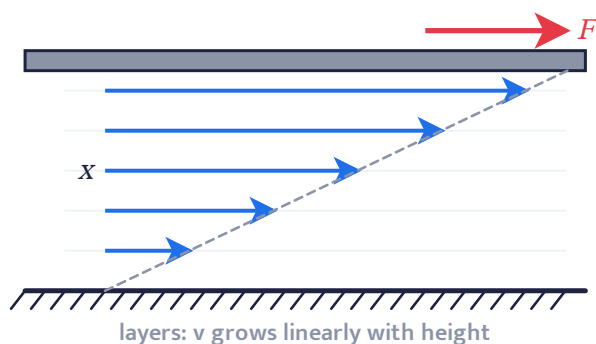
manometer. Same principle: **atomiser/sprayer**, carburettor.

Common Trap

- Bernoulli needs **steady, streamline, incompressible, non-viscous** flow – it fails for turbulent flow.
- Faster fluid $\Rightarrow$ **lower** pressure (not higher!).

9.5 Viscosity

Viscosity is the internal friction between adjacent layers of a fluid in relative motion. Near a fixed surface the fluid is at rest; the velocity increases layer by layer.



Viscosity

$$F = \eta A \frac{dv}{dx}$$

Newton's law of viscosity

$$\eta = \frac{F/A}{v/l}$$

Coefficient of viscosity = shear stress / strain rate

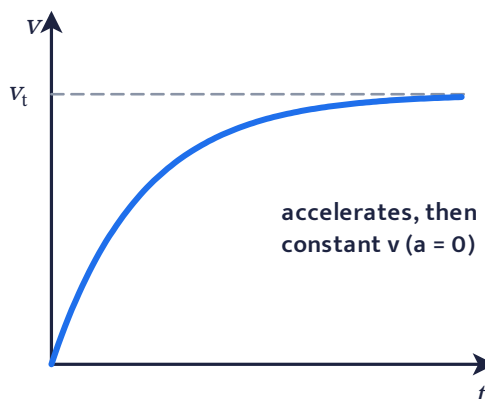
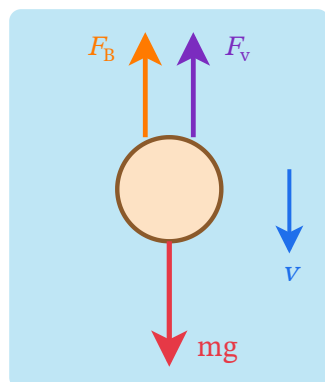
$$[\eta] = [ML^{-1}T^{-1}]$$

SI: Pa s ($= \text{N s m}^{-2}$); 1 poise = 0.1 Pa s

NCERT Line to Remember

Viscosity of **liquids decreases** with temperature, while that of **gases increases**. Blood is more viscous than water; honey/glycerine are highly viscous.

Stokes' law and terminal velocity



$$mg = F_B + 6\pi\eta rv_t$$

Forces on a sphere falling in a viscous fluid, and its v - t graph

Falling through a fluid

$$F_v = 6\pi\eta rv$$

Stokes' law (small sphere, slow streamline flow)

$$v_t = \frac{2r^2(\rho - \sigma)g}{9\eta}$$

Terminal velocity (ρ = sphere, σ = fluid density); $v_t \propto r^2$

Solved Example

A steel ball of radius 1 mm (density $8 \times 10^3 \text{ kg m}^{-3}$) falls in a liquid of density $2 \times 10^3 \text{ kg m}^{-3}$ and viscosity 1.0 Pa s. Find its terminal velocity.

Solution: $v_t = \frac{2(10^{-3})^2(6000)(10)}{9(1.0)} = 1.33 \times 10^{-2} \text{ m s}^{-1} \approx 1.3 \text{ cm s}^{-1}$.

Memory Trick

8 drops → 1 drop: volume $\times 8 \Rightarrow$ radius $\times 2 \Rightarrow v_t \times 4$. In general, n drops merge $\Rightarrow v_t$ becomes $n^{2/3}$ times. If $\sigma > \rho$ (air bubble in water) v_t is upward.

9.6 Surface Tension

A molecule inside a liquid is pulled equally from all sides, but one at the surface is pulled **inwards**. So the surface has extra energy and behaves like a **stretched membrane** – it tends to have the minimum area (drops are spherical). **Surface tension** S = force per unit length = surface energy per unit area.

Surface tension formulas

$$S = \frac{F}{l} = \frac{W}{\Delta A}$$

Unit $\text{N m}^{-1} = \text{J m}^{-2}$; $[MT^{-2}]$

$$W = S \times 2\Delta A$$

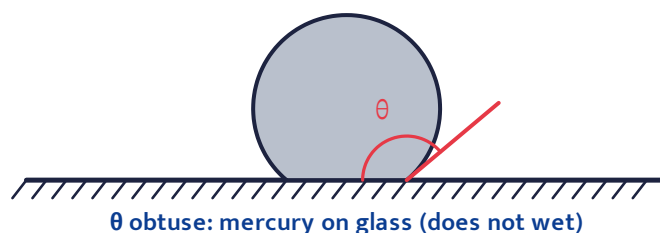
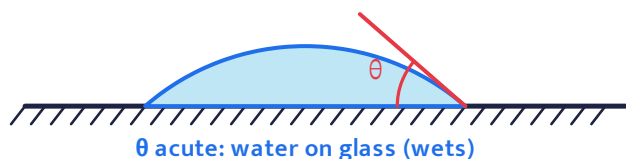
Soap film/bubble has **two** surfaces

$$P_i - P_o = \frac{2S}{r}$$

Liquid drop; air bubble inside a liquid

$$P_i - P_o = \frac{4S}{r}$$

Soap bubble in air (two surfaces)



Angle of contact θ – measured inside the liquid, between the solid surface and the tangent to the liquid surface

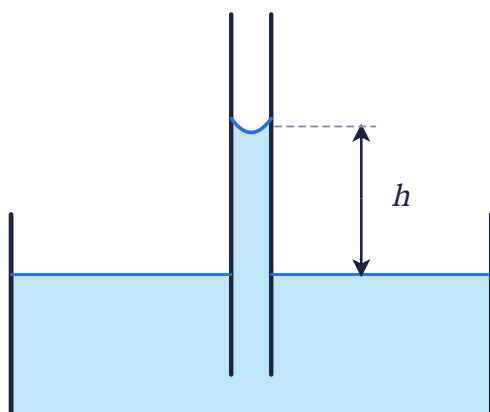
NCERT Line to Remember

- $\theta < 90^\circ$ if adhesion (liquid–solid) > cohesion: liquid **wets** the solid (water–glass $\approx 0^\circ$).
- $\theta > 90^\circ$ if cohesion > adhesion: mercury–glass $\approx 140^\circ$, water on a waxy/oily leaf.
- **Detergents/soaps** lower S and θ so water penetrates cloth; **waterproofing** agents increase θ .
- S **decreases with temperature** (zero at the critical temperature).

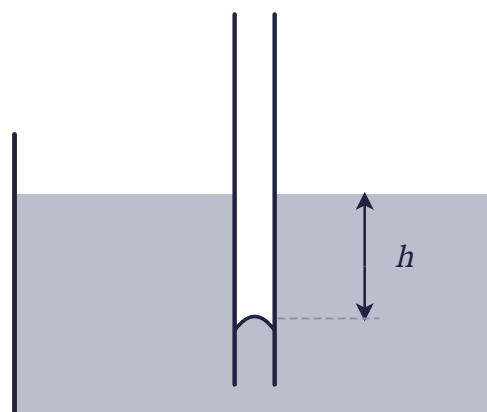
Common Trap

- Soap bubble: $4S / r$ (two surfaces). Air bubble **inside** water: only $2S / r$ (one surface).
- Smaller bubble $\Rightarrow$ **higher** inside pressure: when two bubbles are joined by a tube, air flows from the **small to the large** one.

Capillary rise



Water–glass: $\theta < 90^\circ \rightarrow$ rises
concave meniscus



Mercury–glass: $\theta \approx 140^\circ \rightarrow$ falls
convex meniscus

Liquid in a narrow glass tube: water rises, mercury is depressed

$$h = \frac{2S \cos \theta}{r \rho g} \quad \Rightarrow \quad h \propto \frac{1}{r} \quad (\text{Jurin's law: } hr = \text{const})$$

Solved Example

Find the height to which water rises in a glass capillary of radius 0.1 mm. ($S = 0.07 \text{ N m}^{-1}$, $\theta = 0^\circ$, $\rho = 10^3 \text{ kg m}^{-3}$, $g = 10 \text{ m s}^{-2}$)

Solution: $h = \frac{2(0.07)(1)}{(10^{-4})(10^3)(10)} = 0.14 \text{ m} = \mathbf{14 \text{ cm}}$.

VS Sir's Exam Tip

If the tube is shorter than h , water does **not** overflow – it rises to the top and the meniscus flattens (radius of curvature increases). In a **freely falling lift** ($g = 0$), water rises to the full length of the tube.



Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** The working of a hydraulic lift is based on
 (a) Pascal's law (b) Bernoulli's principle
 (c) Archimedes' principle (d) Stokes' law
- 2 **NCERT based** The SI unit of the coefficient of viscosity is
 (a) Pa m (b) N m s⁻¹ (c) N m⁻¹ (d) Pa s
- 3 **NCERT based** When temperature rises, the viscosity of
 (a) liquids increases and gases decreases
 (b) both liquids and gases decreases
 (c) liquids decreases and gases increases
 (d) both liquids and gases increases
- 4 **NCERT based** Mercury is depressed in a glass capillary because
 (a) its surface tension is low
 (b) its density is high
 (c) its angle of contact with glass is zero
 (d) its angle of contact with glass is obtuse
- 5 **NCERT based** The excess pressure inside a soap bubble of radius r and surface tension S is
 (a) $8S / r$ (b) $4S / r$ (c) S / r (d) $2S / r$
- 6 **NCERT based** Adding detergent to water
 (a) increases the angle of contact with cloth
 (b) increases its surface tension
 (c) decreases its surface tension
 (d) does not change surface tension
- 7 **NEET pattern** The absolute pressure at a depth of 10 m in a lake is ($P_0 = 10^5$ Pa, $g = 10$ m s⁻²)
 (a) 2×10^5 Pa (b) 1.1×10^5 Pa (c) 3×10^5 Pa (d) 1×10^5 Pa
- 8 **NEET pattern** In a hydraulic lift the pistons have areas 5 cm² and 500 cm². The force needed on the small piston to lift a car of mass 1000 kg is
 (a) 100 N (b) 10^4 N (c) 1000 N (d) 10 N
- 9 **NEET pattern** Water flows at 2 m s⁻¹ in a pipe of diameter 4 cm which narrows to 2 cm. The speed in the narrow part is
 (a) 16 m s⁻¹ (b) 8 m s⁻¹ (c) 4 m s⁻¹ (d) 1 m s⁻¹
- 10 **NEET pattern** Water leaks from a small hole 5 m below the free surface of a large open tank. The speed of efflux is ($g = 10$ m s⁻²)
 (a) 7.1 m s⁻¹ (b) 5 m s⁻¹ (c) 50 m s⁻¹ (d) 10 m s⁻¹

- 11 **NEET pattern** A small steel ball is dropped from rest into a long column of glycerine. Its velocity–time graph
- (a) rises linearly for ever
 - (b) rises and then becomes constant (terminal velocity)
 - (c) rises, reaches a peak and falls to zero
 - (d) is constant from the start
- 12 **NEET pattern** Eight identical raindrops, each falling with terminal velocity 5 cm s^{-1} , coalesce to form a single drop. Its terminal velocity is
- (a) 40 cm s^{-1}
 - (b) 5 cm s^{-1}
 - (c) 10 cm s^{-1}
 - (d) 20 cm s^{-1}
- 13 **NEET pattern** Water rises to a height h in a capillary of radius r . In a capillary of radius $r/2$ of the same glass, it rises to
- (a) $4h$
 - (b) $2h$
 - (c) $h/2$
 - (d) h
- 14 **NEET pattern** Water rises in a glass capillary of radius 0.2 mm . ($S = 0.07 \text{ N m}^{-1}$, $\theta = 0^\circ$, $g = 10 \text{ m s}^{-2}$) The height of rise is
- (a) 7 cm
 - (b) 3.5 cm
 - (c) 14 cm
 - (d) 0.7 cm
- 15 **NEET pattern** The work done to blow a soap bubble of radius 1 cm ($S = 0.03 \text{ N m}^{-1}$) is
- (a) $3.8 \times 10^{-5} \text{ J}$
 - (b) $7.5 \times 10^{-5} \text{ J}$
 - (c) $7.5 \times 10^{-4} \text{ J}$
 - (d) $1.5 \times 10^{-4} \text{ J}$
- 16 **NEET pattern** Air flows over the top of an aeroplane wing at 60 m s^{-1} and under it at 50 m s^{-1} . The pressure difference across the wing is (air density 1.2 kg m^{-3})
- (a) 1320 Pa
 - (b) 660 Pa
 - (c) 6.6 Pa
 - (d) 60 Pa
- 17 **NEET pattern** A steel ball of radius 1 mm and density $8 \times 10^3 \text{ kg m}^{-3}$ falls in oil of density $2 \times 10^3 \text{ kg m}^{-3}$ and viscosity 2.0 Pa s . Its terminal velocity is ($g = 10 \text{ m s}^{-2}$)
- (a) 13.3 mm s^{-1}
 - (b) 8.9 mm s^{-1}
 - (c) 6.7 mm s^{-1}
 - (d) 3.3 mm s^{-1}
- 18 **NEET pattern** Which of the following is NOT explained by Bernoulli's principle?
- (a) Lift on an aeroplane wing
 - (b) Floating of a ship
 - (c) Roofs blown off in a storm
 - (d) Swing of a spinning ball
- 19 **JEE Main pattern** In a horizontal Venturi tube the area ratio is $A_1 / A_2 = 2$. Water enters the wide part at 1 m s^{-1} . The pressure difference between the wide and narrow parts is
- (a) 3000 Pa
 - (b) 500 Pa
 - (c) 1500 Pa
 - (d) 2000 Pa
- 20 **JEE Main pattern** Water ($\eta = 10^{-3} \text{ Pa s}$) flows at 5 cm s^{-1} through a pipe of diameter 1 cm . The Reynolds number and nature of flow are
- (a) 50, streamline
 - (b) 500, streamline
 - (c) 5000, turbulent
 - (d) 500, turbulent
- 21 **JEE Main pattern** Two soap bubbles of radii 3 cm and 4 cm coalesce in vacuum under isothermal conditions. The radius of the new bubble is
- (a) 7 cm
 - (b) 5 cm
 - (c) 3.5 cm
 - (d) 4.5 cm

22. **JEE Main pattern** A tank is filled with water to a height of 2 m. A small hole is made 0.5 m below the water surface. The water jet strikes the ground at a horizontal distance of

(a) 1 m (b) $\sqrt{3}$ m (c) 1.5 m (d) 2 m

23. **Assertion–Reason** Assertion: Small raindrops fall more slowly than large ones. Reason: Terminal velocity $v_t \propto r^2$.

(a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

24. **Assertion–Reason** Assertion: Mercury does not wet glass. Reason: Mercury is a liquid metal of very high density.

(a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

✓ Answer Key & Quick Solutions

1(a) 2(d) 3(c) 4(d) 5(b) 6(c) 7(a) 8(a) 9(b) 10(d) 11(b) 12(d) 13(b) 14(a) 15(b)
 16(b) 17(c) 18(b) 19(c) 20(b) 21(b) 22(b) 23(a) 24(b)

- (a) Pressure applied to an enclosed liquid is transmitted undiminished – Pascal's law.
- (d) $\eta = (F/A)/(dv/dx) \Rightarrow \text{Pa} \div \text{s}^{-1} = \text{Pa s}$ (1 poise = 0.1 Pa s).
- (c) Liquids: weaker cohesion when hot $\Rightarrow \eta$ falls. Gases: more molecular momentum exchange $\Rightarrow \eta$ rises.
- (d) $\theta \approx 140^\circ \Rightarrow \cos \theta < 0 \Rightarrow h$ negative (depression).
- (b) A soap bubble has two free surfaces $\Rightarrow 2 \times 2S/r = 4S/r$.
- (c) Detergent molecules lower S (and θ), so water wets and penetrates the fabric.
- (a) $P = P_0 + \rho gh = 10^5 + 10^3(10)(10) = 2 \times 10^5 \text{ Pa}$.
- (a) $F_1 = Mg A_1 / A_2 = 10^4 \times 5 / 500 = 100 \text{ N}$.
- (b) $v_2 = v_1(d_1/d_2)^2 = 2 \times 4 = 8 \text{ m s}^{-1}$.
- (d) $v = \sqrt{2gh} = \sqrt{100} = 10 \text{ m s}^{-1}$.
- (b) Viscous drag grows with v until $mg = F_B + F_v$; then $a = 0$ and $v = v_t$ (exponential approach).
- (d) Volume $\times 8 \Rightarrow R = 2r$; $v_t \propto r^2 \Rightarrow 4 \times 5 = 20 \text{ cm s}^{-1}$.
- (b) $h \propto 1/r \Rightarrow$ halving r doubles h .
- (a) $h = 2S / (r\rho g) = 0.14 / (2 \times 10^{-4} \times 10^4) = 0.07 \text{ m} = 7 \text{ cm}$.
- (b) $W = S \times 2(4\pi r^2) = 0.03 \times 8\pi \times 10^{-4} = 7.5 \times 10^{-5} \text{ J}$.
- (b) $\Delta P = \frac{1}{2}\rho(v_1^2 - v_2^2) = 0.6(3600 - 2500) = 660 \text{ Pa}$.
- (c) $v_t = 2(10^{-6})(6000)(10) / (9 \times 2) = 6.7 \times 10^{-3} \text{ m s}^{-1}$.
- (b) Floating is due to buoyancy (Archimedes); the others are pressure differences due to speed differences.
- (c) $v_2 = 2 \text{ m s}^{-1}$; $\Delta P = \frac{1}{2}(1000)(4 - 1) = 1500 \text{ Pa}$.
- (b) $R_e = \rho v d / \eta = (10^3)(0.05)(0.01) / 10^{-3} = 500 < 1000 \Rightarrow$ streamline.
- (b) PV conserved with $P = 4S/r \Rightarrow r^2 = r_1^2 + r_2^2 \Rightarrow r = 5 \text{ cm}$.
- (b) $R = 2\sqrt{h(H-h)} = 2\sqrt{0.5 \times 1.5} = \sqrt{3} \approx 1.73 \text{ m}$.
- (a) $v_t \propto r^2$ – smaller drops reach a much smaller terminal velocity. R explains A.
- (b) Both are true, but non-wetting is due to cohesion (Hg–Hg) exceeding adhesion (Hg–glass), not density.

CHAPTER

10

Thermal Properties of Matter

Day 16–17

Day 16–17

NEET weightage ★★☆☆

NCERT Class 11

Temperature scales

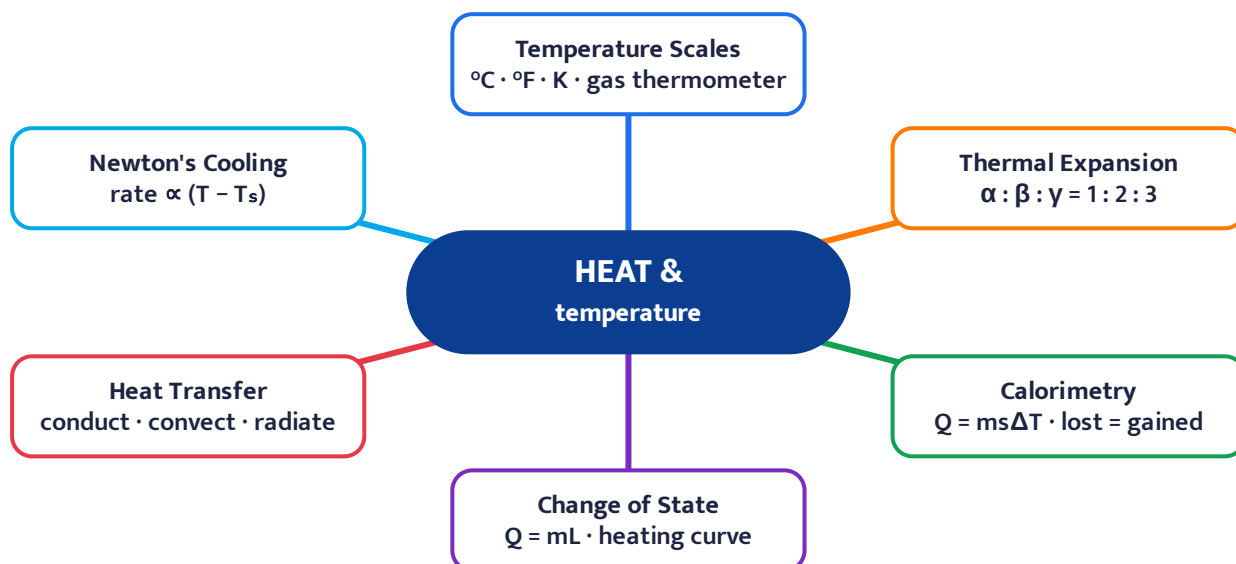
Thermal expansion

Calorimetry

Latent heat

Conduction • radiation

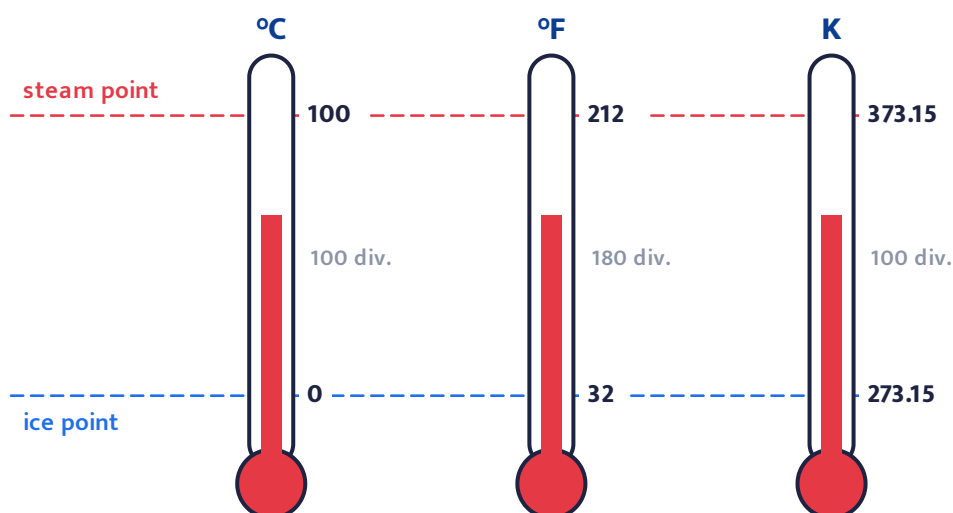
Newton's cooling



Chapter at a glance

10.1 Temperature, Heat and Their Measurement

Temperature is a measure of the "hotness" of a body; it decides the direction of heat flow. **Heat** is energy transferred between a system and its surroundings **because of a temperature difference**. Heat flows from higher to lower temperature until **thermal equilibrium** is reached.



Linear relation

$$F = \left(\frac{9}{5}\right) C + 32$$

$$K = C + 273.15$$

$$-40^{\circ}\text{C} = -40^{\circ}\text{F}$$

Celsius, Fahrenheit and Kelvin scales between the two fixed points

Temperature conversion

$$\frac{T_C - 0}{100} = \frac{T_F - 32}{180} = \frac{T_K - 273.15}{100}$$

General conversion (reading – lower fixed point)/(interval)

$$\Delta T_C = \Delta T_K = \frac{5}{9} \Delta T_F$$

A change of 1 °C = 1 K = 1.8 °F

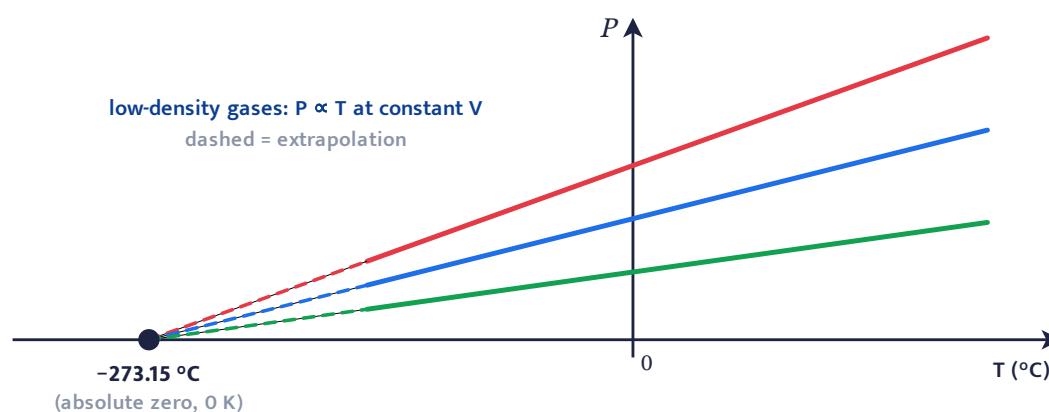
Solved Example

A faulty thermometer reads 20 at the ice point and 220 at the steam point. What does it read when the true temperature is 50 °C?

Solution: $\frac{X - 20}{220 - 20} = \frac{50 - 0}{100} \Rightarrow X = 20 + 100 = 120.$

Ideal-gas thermometer and absolute temperature

For a low-density gas, $PV = \mu RT$. At constant volume, $P \propto T$: the P–T lines of all gases extrapolate to zero pressure at -273.15 °C – the **absolute zero**, the basis of the Kelvin (absolute) scale.



Constant-volume gas thermometer: pressure vs temperature

NCERT Line to Remember

The **triple point of water** (ice, water and vapour coexist) is **273.16 K** at 0.006 atm – the modern fixed point of the Kelvin scale. $T_K = T_C + 273.15$.

10.2 Thermal Expansion

Most substances expand on heating. The fractional change in length, area or volume is proportional to the temperature change:

Coefficients of expansion (NCERT symbols: $\alpha_l, \alpha_A, \alpha_V$)

$$\frac{\Delta l}{l} = \alpha \Delta T$$

Linear expansion (α : K^{-1}); metals
 $\sim 10^{-5} \text{ K}^{-1}$

$$\frac{\Delta A}{A} = \beta \Delta T, \beta = 2\alpha$$

Area (superficial) expansion

$$\frac{\Delta V}{V} = \gamma \Delta T, \gamma = 3\alpha$$

Volume (cubical) expansion

$$\gamma_{\text{gas}} = \frac{1}{T}$$

Ideal gas at constant P ($= 3.7 \times 10^{-3} \text{ K}^{-1}$ at 0 °C)

Memory Trick

$\alpha : \beta : \gamma = 1 : 2 : 3$ – one dimension, two dimensions, three dimensions. A hole in a plate **expands** on heating exactly like the metal it was cut from.

Common Trap

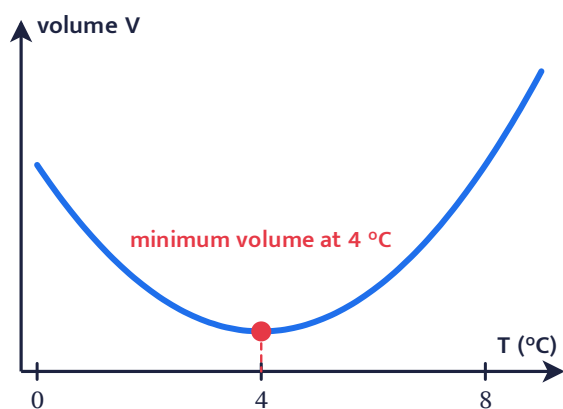
Liquid in a vessel: **apparent** expansion

$\gamma_{\text{app}} = \gamma_{\text{liquid}} - \gamma_{\text{vessel}}$. Expansion of the vessel is not zero! Thermal stress (rod between walls) = $Y\alpha\Delta T$ – see Ch 8.

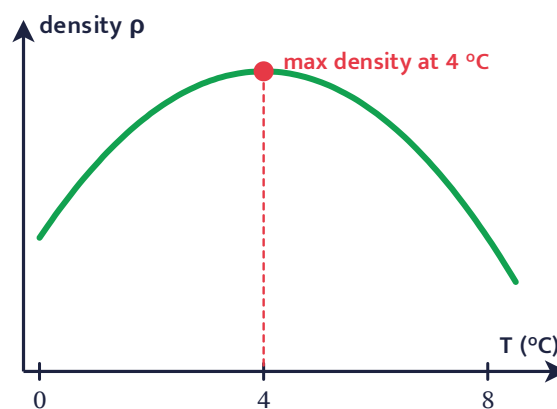
Solved Example

A hole of diameter 4.24 cm is drilled in a copper sheet at 27 °C. Find its diameter at 227 °C. ($\alpha_{\text{Cu}} = 1.70 \times 10^{-5} \text{ K}^{-1}$)

Solution: The hole expands like copper: $\Delta d = d \alpha \Delta T = 4.24 \times 1.7 \times 10^{-5} \times 200 = 0.0144 \text{ cm} \Rightarrow d \approx \mathbf{4.254 \text{ cm}}$.

Anomalous expansion of water

Volume of water vs temperature



Density of water vs temperature

NCERT Line to Remember

Water **contracts** when heated from 0 °C to 4 °C and expands above 4 °C, so its **density is maximum at 4 °C**. In winter the top layer of a lake freezes first while water at 4 °C stays at the bottom – aquatic life survives.

10.3 Specific Heat Capacity & Calorimetry**Heat and temperature change**

$$Q = m s \Delta T$$

Heat to change temperature; s = specific heat ($\text{J kg}^{-1} \text{K}^{-1}$)

$$S = \frac{\Delta Q}{\Delta T} = ms$$

Heat capacity of the body (J K^{-1})

$$C = \frac{1}{\mu} \frac{\Delta Q}{\Delta T}$$

Molar specific heat (C_p, C_v for gases)

heat lost = heat gained

Principle of calorimetry (isolated system)

NCERT Line to Remember

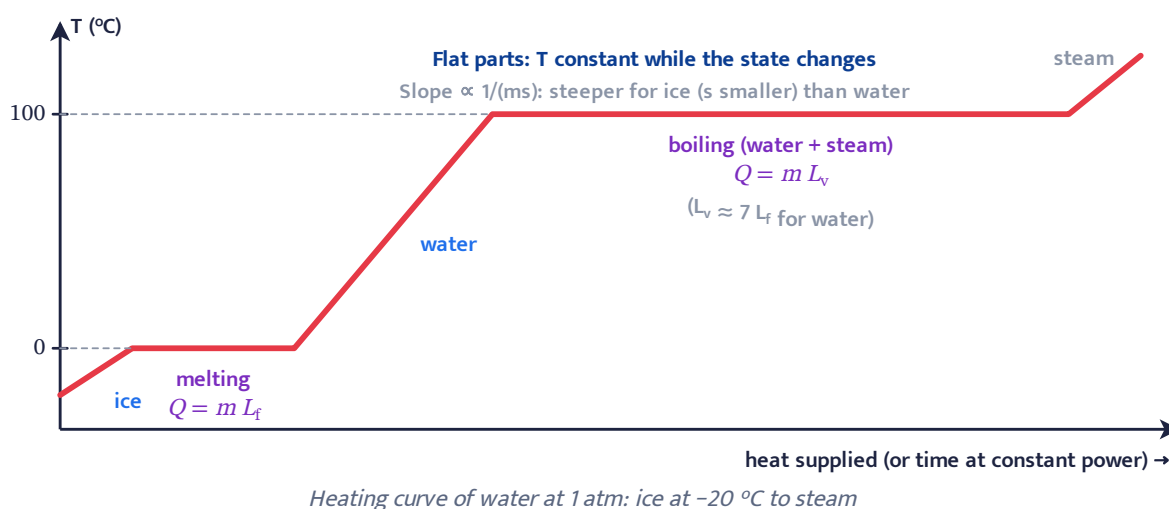
- Water has the **highest specific heat** of common substances, $4186 \text{ J kg}^{-1} \text{K}^{-1}$ ($1 \text{ cal g}^{-1} \text{°C}^{-1}$) – used as a coolant in radiators and hot-water bags.
- Land heats and cools faster than the sea (lower s) → sea and land breezes; coastal climates are moderate.

VS Sir's Exam Tip

In mixing problems, first check whether the ice melts **completely**: compare heat available (cooling the hot part to 0 °C) with mL_f needed. If not enough, final temperature is **0 °C** with some ice left.

10.4 Change of State & Latent Heat

During a change of state (melting, boiling, sublimation) heat is absorbed **without any change of temperature**. The heat per unit mass is the **latent heat** L : $Q = mL$. For water $L_f = 3.33 \times 10^5 \text{ J kg}^{-1}$ and $L_v = 22.6 \times 10^5 \text{ J kg}^{-1}$.



Solved Example

How much heat is needed to convert 3 kg of ice at -12°C into steam at 100°C ? ($s_{\text{ice}} = 2100$, $s_{\text{water}} = 4186 \text{ J kg}^{-1} \text{ K}^{-1}$, $L_f = 3.35 \times 10^5$, $L_v = 2.256 \times 10^6 \text{ J kg}^{-1}$)

Solution: $Q = 3[2100(12) + 3.35 \times 10^5 + 4186(100) + 2.256 \times 10^6] = 3(3.0348 \times 10^6) \approx 9.1 \times 10^6 \text{ J}$. Note: boiling alone takes about 74% of it.

NCERT Line to Remember

- Melting point **decreases** with pressure for ice → **regelation** (wire passes through an ice block; skating).
- Boiling point **increases** with pressure → a **pressure cooker** cooks faster; at hills water boils below 100°C .
- Sublimation:** solid → vapour directly (dry ice, camphor, iodine).

Common Trap

Steam at 100°C burns more than water at 100°C : it gives up L_v (2256 J g^{-1}) **before** cooling. In the heating curve, temperature stays flat during a phase change even though heat is being supplied.

10.5 Heat Transfer – Conduction

Conduction transfers heat between adjacent parts of a body by molecular collisions, **without bulk motion** of matter. In the steady state the heat current through a bar is

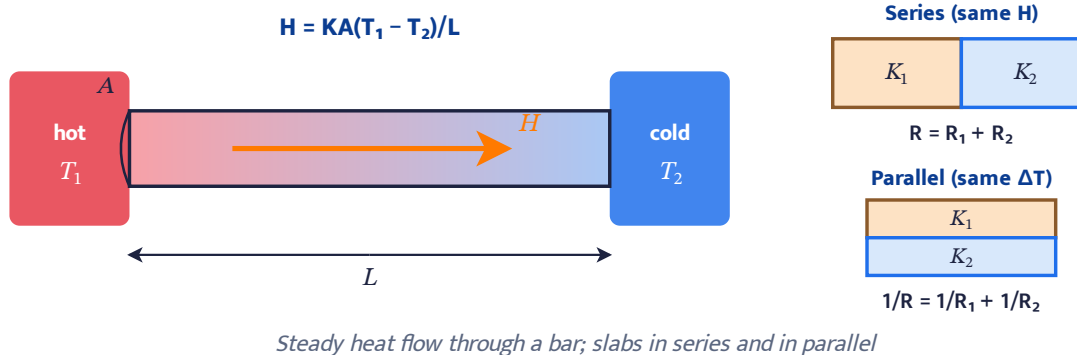
Conduction Formulas

$$H = \frac{KA(T_1 - T_2)}{L} \quad K = \text{thermal conductivity (W m}^{-1} \text{ K}^{-1}\text{)}$$

$$R = \frac{L}{KA}, \quad H = \frac{\Delta T}{R} \quad \text{Thermal resistance – like Ohm's law}$$

$$K_s = \frac{L_1 + L_2}{\frac{L_1}{K_1} + \frac{L_2}{K_2}} \left(= \frac{2K_1K_2}{K_1 + K_2} \right) \quad \text{Series; bracket for equal lengths}$$

$$K_p = \frac{K_1A_1 + K_2A_2}{A_1 + A_2} \left(= \frac{K_1 + K_2}{2} \right) \quad \text{Parallel; bracket for equal areas}$$



Solved Example

An iron bar ($K_1 = 79 \text{ W m}^{-1} \text{ K}^{-1}$) and a brass bar ($K_2 = 109 \text{ W m}^{-1} \text{ K}^{-1}$), each 0.1 m long with area 0.02 m^2 , are joined end to end. The free ends are at 373 K and 273 K. Find the junction temperature and the heat current.

Solution: Same H in series, equal L, A : $K_1(373 - T_0) = K_2(T_0 - 273) \Rightarrow T_0 = \frac{79(373) + 109(273)}{188} \approx 315 \text{ K}$.
 $H = \frac{79(0.02)(58)}{0.1} \approx 916 \text{ W}$.

NCERT Line to Remember

- Good conductors: silver (406), copper (385), aluminium (205) $\text{W m}^{-1} \text{ K}^{-1}$. Poor: glass, wood, air (0.024) – air trapped in wool/feathers insulates.
- Copper-bottom cooking pots spread heat evenly; houses with thick walls of low K stay cool.

Convection

Heat carried by **actual motion of the fluid**: natural (density differences, gravity) or forced (fan, pump, blood circulation). Examples: **sea breeze** by day, **land breeze** at night, **trade winds**. Not possible in solids or in vacuum.

10.6 Radiation

Radiation transfers heat by electromagnetic waves (mostly infrared) at the speed of light – it needs **no medium**. Every body above 0 K radiates. A **blackbody** absorbs all radiation falling on it and is the best emitter ($e = 1$). Dark, rough surfaces are good absorbers **and** good emitters.

Radiation laws (T in kelvin!)

$$H = e\sigma AT^4 \quad \text{Stefan-Boltzmann law; } \sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}, 0 \leq e \leq 1$$

$$H_{\text{net}} = e\sigma A(T^4 - T_s^4) \quad \text{Net loss to surroundings at } T_s$$

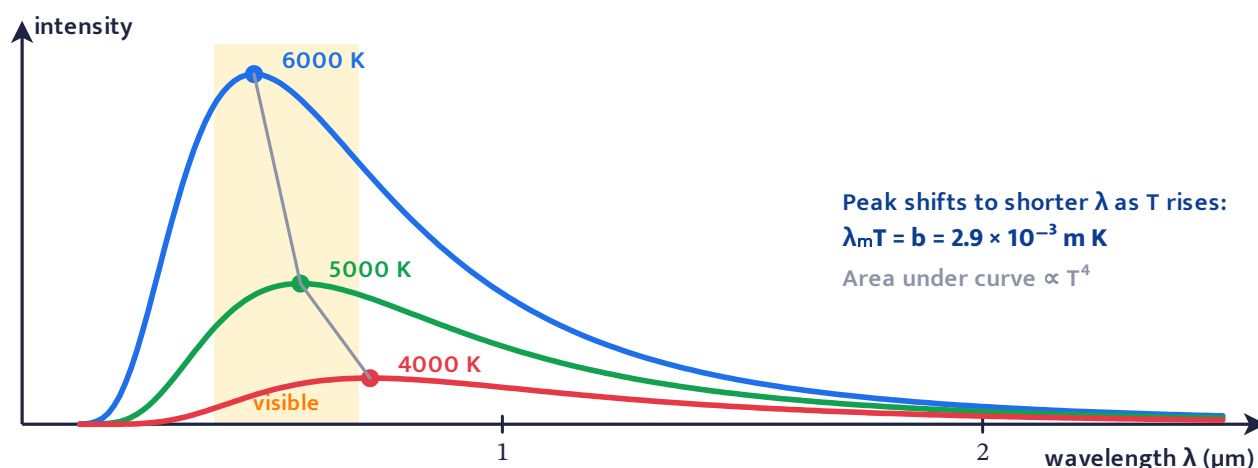
$$\lambda_m T = b = 2.9 \times 10^{-3} \text{ m K} \quad \text{Wien's displacement law}$$

NCERT Line to Remember

- Sun: $\lambda_m \approx 0.5 \mu\text{m} \Rightarrow$ surface $T \approx 5800 \text{ K}$. Moon: $\lambda_m \approx 14 \mu\text{m} \Rightarrow T \approx 200 \text{ K}$.
- A heated iron glows dull red $\rightarrow$ orange $\rightarrow$ white as T rises (λ_m decreases).
- **Greenhouse effect:** CO_2, CH_4 , water vapour trap the infrared emitted by Earth, warming it.

Common Trap

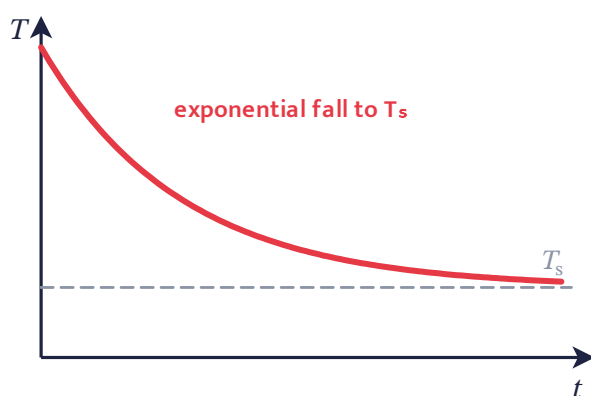
Always use **kelvin** in T^4 and in Wien's law. Doubling the Celsius temperature does NOT double T . Also, power radiated $\propto AT^4$ – a bigger sphere radiates more even at the same T ($A \propto r^2$).



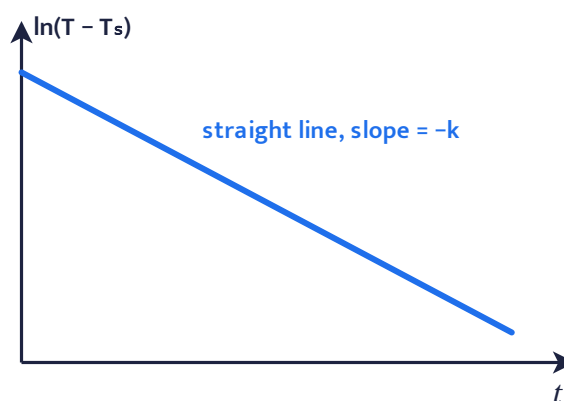
Blackbody spectrum at three temperatures – Wien's displacement law

10.7 Newton's Law of Cooling

For a **small** temperature difference, the rate of loss of heat of a body is proportional to the difference between its temperature T and that of the surroundings T_s .



Temperature vs time

 $\ln(T - T_s)$ vs time is linear

Newton's law of cooling

$$-\frac{dT}{dt} = k(T - T_s)$$

Differential form; k depends on area, nature of surface, ms

$$T - T_s = (T_0 - T_s)e^{-kt}$$

Exponential cooling

$$\frac{T_1 - T_2}{t} = k \left[\frac{T_1 + T_2}{2} - T_s \right]$$

Average-temperature form – use in NEET numericals



Solved Example

A body cools from 60°C to 50°C in 10 min in a room at 25°C . How long will it take to cool from 50°C to 40°C ?

Solution: $\frac{10}{10} = k(55 - 25) \Rightarrow k = \frac{1}{30}$. Then
 $\frac{10}{t} = \frac{1}{30}(45 - 25) \Rightarrow t = 15 \text{ min.}$



VS Sir's Exam Tip

Equal temperature drops take **longer and longer** as the body approaches room temperature. Hot tea cools faster in the first minutes than later.



Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- 1 **NCERT based** At what temperature do the Celsius and Fahrenheit scales give the same reading?
 (a) 40° (b) 0° (c) -32° (d) -40°
- 2 **NCERT based** For an isotropic solid, the coefficients of linear (α), area (β) and volume (γ) expansion are in the ratio
 (a) 1 : 1 : 1 (b) 3 : 2 : 1 (c) 1 : 2 : 3 (d) 1 : 4 : 9
- 3 **NCERT based** The density of water is maximum at
 (a) 4°C (b) -4°C (c) 0°C (d) 100°C
- 4 **NCERT based** As the temperature of a blackbody increases, the wavelength at which it emits maximum energy
 (a) decreases (b) remains the same
 (c) first increases then decreases (d) increases
- 5 **NCERT based** Food cooks faster in a pressure cooker because
 (a) specific heat of water increases
 (b) latent heat of vaporisation becomes zero
 (c) the boiling point of water falls with pressure
 (d) the boiling point of water rises with pressure
- 6 **NCERT based** The triple point of water is
 (a) 273.15 K (b) 0 K (c) 373.15 K (d) 273.16 K
- 7 **NEET pattern** Normal body temperature 98.6°F on the Celsius scale is
 (a) 37°C (b) 38°C (c) 36.6°C (d) 35.6°C
- 8 **NEET pattern** A 10 m steel rail ($\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$) is heated through 50°C . Its increase in length is
 (a) 0.6 mm (b) 3 mm (c) 6 mm (d) 12 mm
- 9 **NEET pattern** A metal plate with a circular hole is heated. The diameter of the hole
 (a) decreases
 (b) increases
 (c) remains unchanged
 (d) increases or decreases depending on α
- 10 **NEET pattern** 200 g of water at 80°C is mixed with 100 g of water at 20°C . Neglecting heat losses, the final temperature is
 (a) 40°C (b) 60°C (c) 50°C (d) 70°C
- 11 **NEET pattern** 10 g of ice at 0°C is added to 50 g of water at 40°C . The final temperature is ($L_f = 80 \text{ cal g}^{-1}$, $s_w = 1 \text{ cal g}^{-1} \text{ }^\circ\text{C}^{-1}$)
 (a) 10°C (b) 0°C (c) 20°C (d) 26.7°C

- 12 **NEET pattern** Heat required to convert 1 kg of ice at -10°C into water at 0°C is ($s_{ice} = 2100 \text{ J kg}^{-1} \text{ K}^{-1}$, $L_f = 3.34 \times 10^5 \text{ J kg}^{-1}$)
 (a) 334 kJ (b) 355 kJ (c) 376 kJ (d) 21 kJ
- 13 **NEET pattern** Two rods of equal length and area with conductivities K and $2K$ are joined in series. The equivalent conductivity is
 (a) $4K/3$ (b) $2K/3$ (c) $3K/2$ (d) $3K$
- 14 **NEET pattern** Two slabs of equal thickness and area, conductivities $2K$ and K , are in contact. The outer face of the $2K$ slab is at 100°C and that of the K slab at 0°C . The interface temperature is
 (a) 33.3°C (b) 66.7°C (c) 75°C (d) 50°C
- 15 **NEET pattern** If the absolute temperature of a blackbody is doubled, the power radiated by it becomes
 (a) 8 times (b) 16 times (c) 2 times (d) 4 times
- 16 **NEET pattern** The spectrum of a star peaks at 500 nm . Its surface temperature is about ($b = 2.9 \times 10^{-3} \text{ m K}$)
 (a) 1450 K (b) 5800 K (c) 2900 K (d) 10000 K
- 17 **NEET pattern** A body cools from 60°C to 50°C in 10 min in surroundings at 25°C . The time to cool from 50°C to 40°C is
 (a) 10 min (b) 15 min (c) 20 min (d) 12 min
- 18 **NEET pattern** Ice at -20°C is heated at a constant rate until it becomes water at 50°C . On the T - t graph, the slope for ice is greater than that for water because
 (a) specific heat of ice is less than that of water
 (b) ice is denser than water
 (c) ice has latent heat
 (d) specific heat of ice is more than that of water
- 19 **JEE Main pattern** The coefficient of volume expansion of an ideal gas at constant pressure at 27°C is
 (a) 27^{-1} K^{-1} (b) $3.7 \times 10^{-3} \text{ K}^{-1}$ (c) $3.7 \times 10^{-2} \text{ K}^{-1}$ (d) $3.3 \times 10^{-3} \text{ K}^{-1}$
- 20 **JEE Main pattern** A pendulum clock with a steel rod ($\alpha = 1.2 \times 10^{-5} \text{ K}^{-1}$) keeps correct time at 20°C . At 30°C , the time it loses per day is about
 (a) 5.2 s (b) 10.4 s (c) 52 s (d) 2.6 s
- 21 **JEE Main pattern** Two spheres of the same material have radii 1 m and 4 m and temperatures 4000 K and 2000 K . The ratio of the energy radiated per second by them is
 (a) 1 : 16 (b) 1 : 4 (c) 1 : 1 (d) 4 : 1
- 22 **JEE Main pattern** A glass flask of volume 1000 cm^3 is completely filled with mercury at 0°C . When heated to 100°C , the mercury that overflows is ($\gamma_{Hg} = 1.82 \times 10^{-4} \text{ K}^{-1}$, $\alpha_{glass} = 9 \times 10^{-6} \text{ K}^{-1}$)
 (a) 17.3 cm^3 (b) 18.2 cm^3 (c) 15.5 cm^3 (d) 2.7 cm^3

- 23** **Assertion-Reason** Assertion: In winter, lakes freeze from the top while the water below remains liquid. Reason: Water has its maximum density at 4 °C.
- (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

- 24** **Assertion-Reason** Assertion: Newton's law of cooling holds for any temperature difference between a body and its surroundings. Reason: For small temperature differences, the rate of cooling is proportional to the temperature difference.
- (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

✓ Answer Key & Quick Solutions

1(d) 2(c) 3(a) 4(a) 5(d) 6(d) 7(a) 8(c) 9(b) 10(b) 11(c) 12(b) 13(a) 14(b) 15(b)
 16(b) 17(b) 18(a) 19(d) 20(a) 21(c) 22(c) 23(a) 24(c)

1. (d) $F = \frac{9}{5}C + 32$ with $F = C \Rightarrow -\frac{4}{5}C = 32 \Rightarrow C = -40$.
2. (c) $\beta = 2\alpha$ and $\gamma = 3\alpha$.
3. (a) Anomalous expansion: water contracts from 0 to 4 °C, so density peaks at 4 °C.
4. (a) Wien's law: $\lambda_m \propto 1/T$.
5. (d) Higher pressure $\Rightarrow$ water boils above 100 °C $\Rightarrow$ food is at a higher temperature.
6. (d) Triple point = 273.16 K (0.01 °C) at 0.006 atm.
7. (a) $C = (5/9)(98.6 - 32) = (5/9)(66.6) = 37$ °C.
8. (c) $\Delta l = l\alpha\Delta T = 10 \times 1.2 \times 10^{-5} \times 50 = 6 \times 10^{-3}$ m.
9. (b) Heating is a uniform enlargement – the hole expands as if it were made of the same metal.
10. (b) $T = (200 \times 80 + 100 \times 20)/300 = 60$ °C.
11. (c) Heat to cool water to 0 °C = 2000 cal > 800 cal to melt ice. $50(40 - T) = 800 + 10T \Rightarrow T = 20$ °C.
12. (b) $Q = 1(2100)(10) + 3.34 \times 10^5 = 21000 + 334000 = 355$ kJ.
13. (a) $K_s = 2K_1K_2 / (K_1 + K_2) = 4K^2 / 3K = 4K / 3$.
14. (b) $2K(100 - T) = K(T - 0) \Rightarrow T = \frac{200}{3} = 66.7$ °C.
15. (b) $H \propto T^4 \Rightarrow 2^4 = 16$.
16. (b) $T = b / \lambda_m = 2.9 \times 10^{-3} / 5 \times 10^{-7} = 5800$ K.
17. (b) $10/10 = k(55 - 25) \Rightarrow k = 1/30$; $10/t = (45 - 25)/30 \Rightarrow t = 15$ min.
18. (a) Slope $dT/dt = P/(ms)$; $s_{ice} \approx 2100 < s_{water} = 4186$ J kg⁻¹ K⁻¹.
19. (d) $\gamma = 1/T = 1/300$ K⁻¹ = 3.3×10^{-3} K⁻¹.
20. (a) Fractional change in period = $\frac{1}{2}\alpha\Delta\theta = 6 \times 10^{-5}$; loss = $86400 \times 6 \times 10^{-5} \approx 5.2$ s.
21. (c) $H \propto r^2T^4$; $(1/4)^2 \times (2)^4 = 1 \Rightarrow 1:1$.
22. (c) $\gamma_{glass} = 3\alpha = 2.7 \times 10^{-5}$; overflow = $V(\gamma_{Hg} - \gamma_g)\Delta T = 1000(1.55 \times 10^{-4})(100) = 15.5$ cm³.
23. (a) The densest water (4 °C) sinks; colder, lighter water stays on top and freezes first. R explains A.
24. (c) The law is an approximation of Stefan's law valid only for small differences. A false, R true.

CHAPTER

11

Thermodynamics

Day 18–19

Day 18–19

NEET weightage ★★★

NCERT Class 11

Zeroth & first law

Cp, Cv, γ

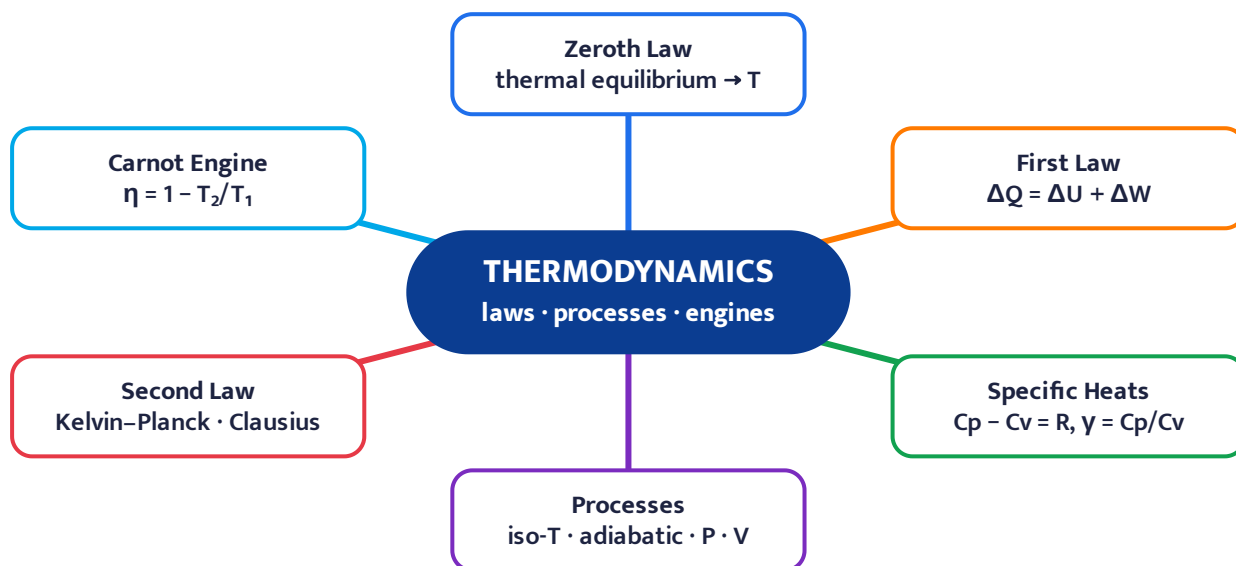
Isothermal & adiabatic

Work = area on PV

Second law

Engine & refrigerator

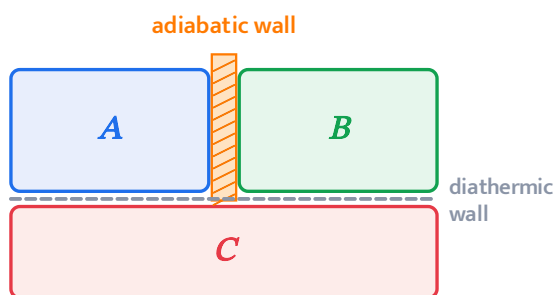
Carnot engine



Chapter at a glance

11.1 Thermal Equilibrium & the Zeroth Law

Two systems are in **thermal equilibrium** when, in thermal contact through a conducting (diathermic) wall, their macroscopic variables (P, V, T) stop changing. An **adiabatic wall** is an insulating wall that allows no heat flow.



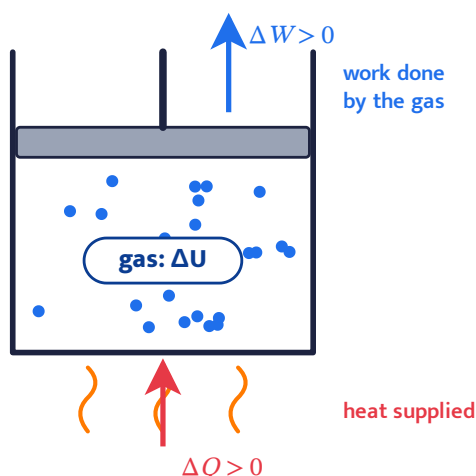
A and B are each in equilibrium with C

Zeroth law

- If two systems are each in thermal equilibrium with a **third** system, they are in thermal equilibrium with **each other**.
- It gives the concept of **temperature**: systems in equilibrium have the same T . This is why a thermometer works.
- Named "zeroth" because it was recognised after the first and second laws but is logically prior.

11.2 Heat, Internal Energy and Work

Internal energy U = sum of kinetic and potential energies of the molecules, measured in the frame of the centre of mass. It is a **state variable** – it depends only on the state (P, V, T), not on how the state was reached. For an **ideal gas**, U depends on T only.



NCERT sign convention

Heat and work – energy in transit

- Heat flows because of a **temperature difference**; **work** is transfer by mechanical means (e.g. moving a piston).
- Heat and work are **not** state variables – they depend on the **path**. A body "has" internal energy, never "heat".
- $\Delta Q > 0$: heat **given to** the system.
- $\Delta W > 0$: work done **by** the system (expansion).

State variables & equation of state

- An equation connecting state variables is an **equation of state**, e.g. $PV = \mu RT$ for an ideal gas.
- Extensive** variables scale with the size of the system (U, V, mass). **Intensive** variables do not (P, T, density).

11.3 First Law of Thermodynamics

Heat supplied to a system is partly used to raise its internal energy and partly to do work on the surroundings. It is the **law of conservation of energy** applied to thermal systems.

First-law toolkit

$$\Delta Q = \Delta U + \Delta W \quad \text{First law (NCERT form)}$$

$$\Delta W = P \Delta V \quad \text{Work by a gas against pressure } P$$

$$W = \int_{V_1}^{V_2} P dV \quad \text{Work for any quasi-static process}$$

$$\Delta U = \mu C_v \Delta T \quad \text{Ideal gas – true for **every** process}$$

Solved Example

A gas absorbs 500 J of heat and at the same time 200 J of work is done **on** the gas. Find the change in internal energy.

Solution: $\Delta Q = +500 \text{ J}$, $\Delta W = -200 \text{ J}$ (work done on the gas). $\Delta U = \Delta Q - \Delta W = 500 - (-200) = 700 \text{ J}$ (increase).

Common Trap

- Chemistry books write $\Delta U = q + w$ (w = work done **on** the system). NCERT Physics uses $\Delta Q = \Delta U + \Delta W$ (ΔW = work **by** the system). Stick to one convention in a problem.
- $\Delta U = \mu C_v \Delta T$ holds even at constant pressure – do **not** write $\mu C_p \Delta T$ for ΔU .

11.4 Specific Heat Capacity

Specific heat capacity $s = \frac{1}{m} \frac{\Delta Q}{\Delta T}$ ($\text{J kg}^{-1} \text{K}^{-1}$). **Molar specific heat** $C = \frac{1}{\mu} \frac{\Delta Q}{\Delta T}$ ($\text{J mol}^{-1} \text{K}^{-1}$). For gases, the heat needed depends on the process, so we define C_v (constant volume) and C_p (constant pressure). Water has $s = 4186 \text{ J kg}^{-1} \text{K}^{-1}$ ($1 \text{ cal g}^{-1} \text{°C}^{-1}$).

Specific heats of an ideal gas

$C_p - C_v = R$	Mayer's relation (ideal gas, per mole)	$\gamma = \frac{C_p}{C_v} > 1$	Ratio of specific heats
$C_v = \frac{R}{\gamma - 1}, C_p = \frac{\gamma R}{\gamma - 1}$	Useful rearrangement	$c_p - c_v = \frac{R}{M}$	Per unit mass (M = molar mass)

Why $C_p > C_v$: at constant volume all heat raises U ; at constant pressure the gas also expands and does work $P\Delta V = \mu R\Delta T$. So $\mu C_p \Delta T = \mu C_v \Delta T + \mu R\Delta T \Rightarrow C_p - C_v = R$.

Gas (rigid molecules)	C_v	C_p	γ	Examples
Monatomic	$\frac{3}{2}R$	$\frac{5}{2}R$	$\frac{5}{3} \approx 1.67$	He, Ne, Ar
Diatomic	$\frac{5}{2}R$	$\frac{7}{2}R$	$\frac{7}{5} = 1.4$	$\text{H}_2, \text{O}_2, \text{N}_2$
Polyatomic (non-linear)	$3R$	$4R$	$\frac{4}{3} \approx 1.33$	NH_3, CH_4

Solved Example

2 mol of a diatomic ideal gas is heated at constant pressure. What fraction of the heat supplied goes into internal energy?

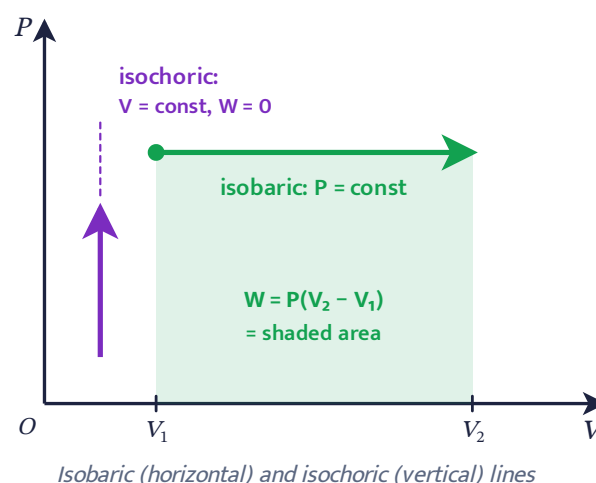
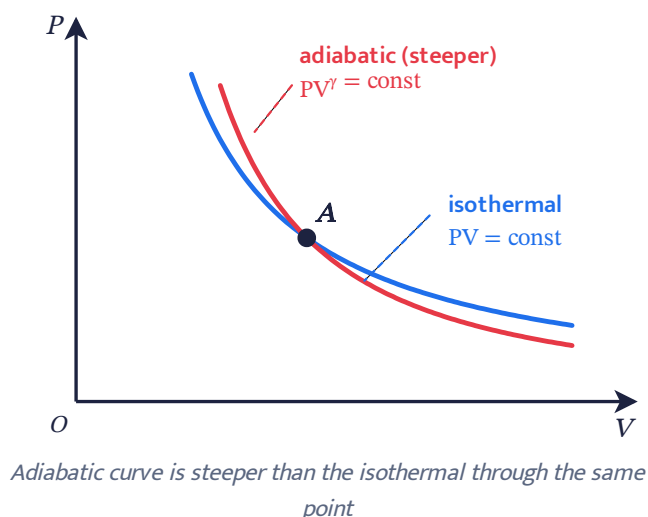
Solution: $\frac{\Delta U}{\Delta Q} = \frac{C_v}{C_p} = \frac{1}{\gamma} = \frac{5}{7}$; the remaining $\frac{2}{7}$ becomes work (independent of the number of moles).

Memory Trick

3-5-7 rule: monatomic $C_v:C_p = 3:5$, diatomic $5:7$ (in units of $R/2$). $\gamma = 5/3, 7/5$ – top number = bottom + 2.

11.5 Thermodynamic Processes

A **quasi-static** process is infinitely slow, so the system passes through a sequence of equilibrium states and can be drawn as a curve on a P - V diagram. The four standard processes:



Process	Condition	Work done by gas W	ΔU	ΔQ
Isothermal	$T = \text{const}, PV = \text{const}$	$\mu RT \ln \frac{V_2}{V_1}$	0	$= W$
Adiabatic	$\Delta Q = 0, PV^\gamma = \text{const}$	$\frac{P_1 V_1 - P_2 V_2}{\gamma - 1} = \frac{\mu R(T_1 - T_2)}{\gamma - 1}$	$-W$	0
Isobaric	$P = \text{const}$	$P(V_2 - V_1) = \mu R \Delta T$	$\mu C_v \Delta T$	$\mu C_p \Delta T$
Isochoric	$V = \text{const}$	0	$\mu C_v \Delta T$	$= \Delta U$
Cyclic	returns to start	enclosed area	0	$= W$

Adiabatic relations

$$PV^\gamma = \text{const} \quad \text{Adiabatic: } P-V$$

$$TV^{\gamma-1} = \text{const} \quad \text{Adiabatic: } T-V$$

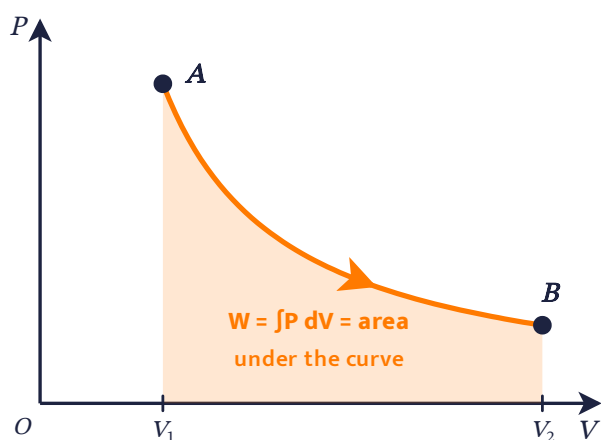
$$P^{1-\gamma} T^\gamma = \text{const} \quad \text{Adiabatic: } P-T$$

$$\left(\frac{dP}{dV}\right)_{ad} = \gamma \left(\frac{dP}{dV}\right)_{iso} = -\gamma \frac{P}{V} \quad \text{Adiabatic slope is } \gamma \text{ times isothermal slope}$$

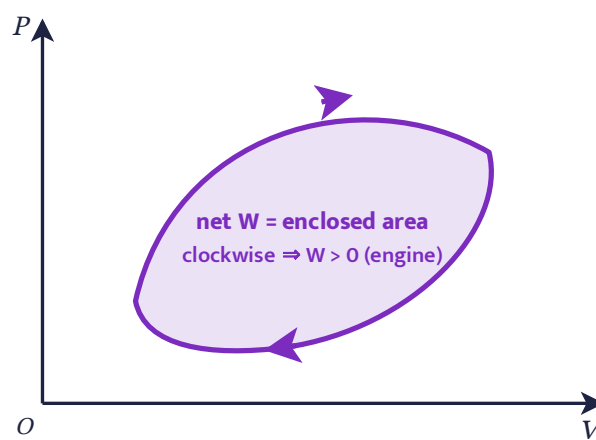
Solved Example

2 mol of an ideal gas expands isothermally at 300 K from 10 L to 20 L. Find the work done by the gas and the heat absorbed. ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$, $\ln 2 = 0.693$)

Solution: $W = \mu RT \ln \frac{V_2}{V_1} = 2 \times 8.314 \times 300 \times 0.693 \approx 3.46 \text{ kJ}$. Since $\Delta U = 0$, heat absorbed $\Delta Q = W \approx 3.46 \text{ kJ}$.



Work done by the gas = area under the P-V curve



Cyclic process: $\Delta U = 0$, work = area of the loop

Common Trap

- Expansion to the same final volume:
 $W_{\text{isobaric}} > W_{\text{isothermal}} > W_{\text{adiabatic}}$ (compare areas).
- Anticlockwise cycle on P-V $\Rightarrow$ net work **negative** (refrigerator).
- Molar heat capacity: isothermal $C = \infty$, adiabatic $C = 0$.

VS Sir's Exam Tip

- Adiabatic expansion $\Rightarrow$ gas **cools** (does work at the cost of U). Adiabatic compression $\Rightarrow$ gas **heats** (diesel engine, bicycle pump).
- Isothermal needs a thin, conducting, slow setup; adiabatic needs an insulated or very fast one.

11.6 Second Law, Heat Engines & Refrigerators

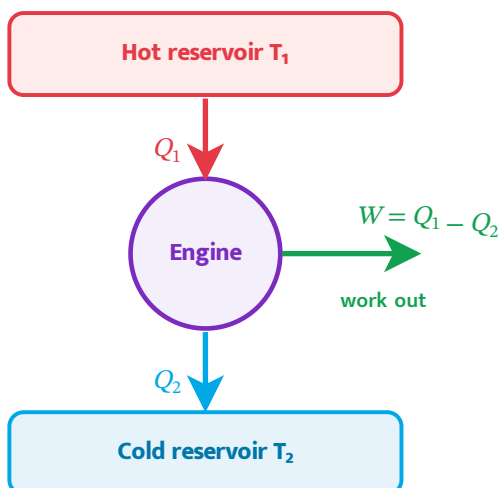
The first law allows processes that never happen (heat flowing by itself from cold to hot); the **second law** rules them out.

Kelvin–Planck statement

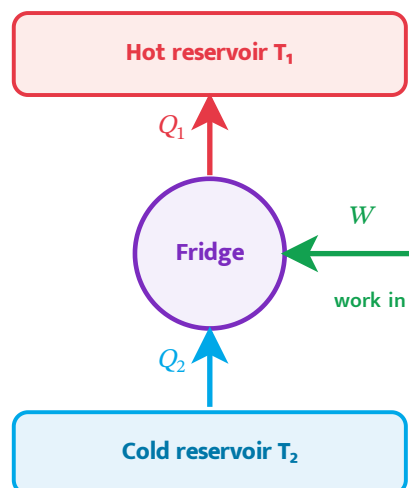
No process is possible whose **sole result** is the absorption of heat from a reservoir and its **complete conversion into work**. $\Rightarrow$ No engine has $\eta = 1$.

Clausius statement

No process is possible whose **sole result** is the transfer of heat from a **colder** object to a **hotter** object. $\Rightarrow$ A refrigerator needs work input.



Heat engine: takes Q_1 , rejects Q_2 , delivers W



Refrigerator: work W pumps Q_2 from cold to hot

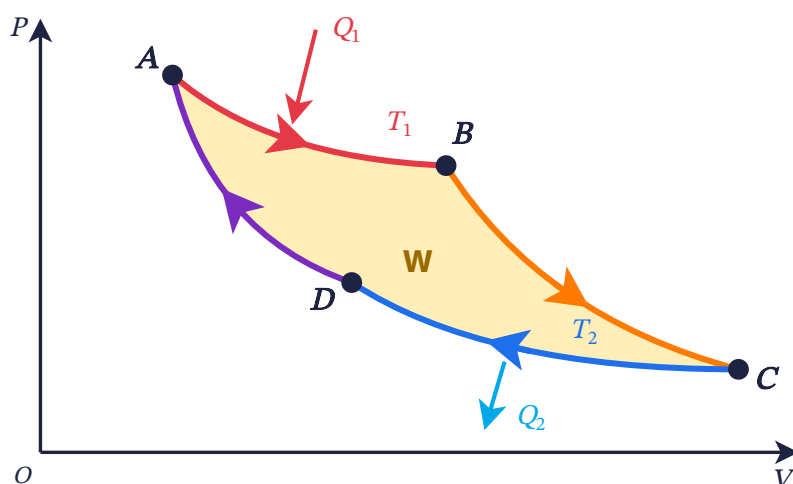
Engine and refrigerator

$$\eta = \frac{W}{Q_1} = 1 - \frac{Q_2}{Q_1} \quad \text{Efficiency of a heat engine}$$

$$\alpha = \frac{Q_2}{W} = \frac{Q_2}{Q_1 - Q_2} \quad \text{Coefficient of performance of a refrigerator}$$

11.7 Reversible Processes & the Carnot Engine

A **Carnot engine** is an ideal **reversible** engine working between a source at T_1 and a sink at T_2 . Its cycle has two isothermal and two adiabatic steps.



Carnot cycle on a P - V diagram (clockwise $\Rightarrow$ positive work)

- **A $\rightarrow$ B isothermal expansion**
at T_1 , absorbs Q_1 from source
 - **B $\rightarrow$ C adiabatic expansion**
 T falls from T_1 to T_2
 - **C $\rightarrow$ D isothermal compression**
at T_2 , rejects Q_2 to sink
 - **D $\rightarrow$ A adiabatic compression**
 T rises from T_2 to T_1
- $W = Q_1 - Q_2 = \text{enclosed area}$**

Reversible and irreversible processes

A process is **reversible** if it can be retraced so that **both the system and the surroundings** return to their original states. It must be **quasi-static** and free of **dissipative effects** (friction, viscosity). Real processes – free expansion, heat flow across a finite ΔT , mixing, burning, friction – are **irreversible**.

Carnot results

$\frac{Q_2}{Q_1} = \frac{T_2}{T_1}$	Carnot cycle (reversible)	$\eta = 1 - \frac{T_2}{T_1}$	Carnot efficiency (T in kelvin)
$\alpha = \frac{T_2}{T_1 - T_2}$	Carnot refrigerator COP	$\alpha = \frac{1 - \eta}{\eta}$	Same reservoirs, reversible machine

NCERT Line to Remember

Carnot's theorem: (a) no engine working between two given temperatures can have efficiency greater than a Carnot engine; (b) the efficiency of a Carnot engine is **independent of the working substance**.

Solved Example

A Carnot engine works between 500 K and 300 K and absorbs 1000 J per cycle from the source. Find its efficiency, the work per cycle and the heat rejected.

Solution: $\eta = 1 - \frac{300}{500} = 0.4 = 40\%$. $W = \eta Q_1 = 400 \text{ J}$. $Q_2 = Q_1 - W = 600 \text{ J}$.

Solved Example

A Carnot refrigerator keeps its freezer at -13°C in a room at 27°C . Find its coefficient of performance.

Solution: $T_2 = 260 \text{ K}$, $T_1 = 300 \text{ K}$. $\alpha = \frac{T_2}{T_1 - T_2} = \frac{260}{40} = 6.5$.

Common Trap

- Always convert to **kelvin** before using $\eta = 1 - T_2 / T_1$.
- $\eta = 1$ would need $T_2 = 0 \text{ K}$ – impossible. COP can be greater than 1; η can never be.

Memory Trick

Carnot cycle order: "**I-A-I-A**" – Isothermal expansion, Adiabatic expansion, Isothermal compression, Adiabatic compression. Heat is exchanged only in the "I" steps.

JEE Main Extension

A **heat pump** uses the refrigerator cycle to heat a room: $\text{COP}_{hp} = \frac{Q_1}{W} = \frac{T_1}{T_1 - T_2} = \alpha + 1$. Changing T_2 by ΔT affects η more than changing T_1 by the same ΔT .

Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

1

NCERT based The zeroth law of thermodynamics leads to the concept of

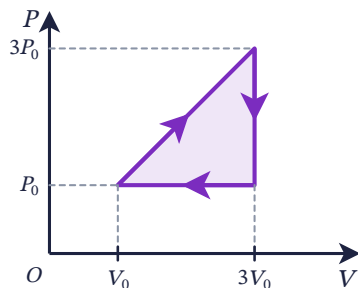
- | | |
|----------|---------------------|
| (a) work | (b) internal energy |
| (c) heat | (d) temperature |

- 2 **NCERT based** In the NCERT form $\Delta Q = \Delta U + \Delta W$, the quantity ΔW is positive when
 (a) work is done on the system (b) heat is removed from system
 (c) work is done by the system (d) the gas is compressed
- 3 **NEET pattern** 100 J of heat is given to a system and 40 J of work is done on the system. The change in its internal energy is
 (a) -60 J (b) 40 J (c) 60 J (d) 140 J
- 4 **NEET pattern** When an ideal gas expands adiabatically, its temperature
 (a) increases (b) remains constant
 (c) first rises then falls (d) decreases
- 5 **NEET pattern** For a rigid diatomic ideal gas, the ratio C_p / C_v is
 (a) 1.33 (b) 1.67 (c) 1.29 (d) 1.40
- 6 **NEET pattern** The molar specific heat of a gas in an isothermal process is
 (a) C_p (b) infinite (c) zero (d) C_v
- 7 **NCERT based** At the same point on a P-V diagram, the slope of an adiabatic curve is
 (a) $(\gamma - 1) \times$ isothermal slope (b) $(1/\gamma) \times$ isothermal slope
 (c) $\gamma \times$ isothermal slope (d) equal to isothermal slope
- 8 **NCERT based** Which statement is the Kelvin-Planck form of the second law?
 (a) Heat cannot flow by itself from a colder to a hotter body
 (b) No engine can convert all the heat absorbed from a reservoir completely into work
 (c) Energy can be neither created nor destroyed
 (d) Two systems in equilibrium with a third are in equilibrium with each other
- 9 **NCERT based** In which process is the entire heat supplied to an ideal gas used to increase its internal energy?
 (a) Isochoric (b) Isothermal (c) Adiabatic (d) Isobaric
- 10 **NEET pattern** A rigid diatomic gas is heated at constant pressure. The fraction of heat converted into work is
 (a) 2/7 (b) 3/5 (c) 5/7 (d) 2/5
- 11 **NEET pattern** 2 mol of a monatomic ideal gas is heated at constant pressure through 10 K. The heat supplied is ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)
 (a) 249 J (b) 416 J (c) 166 J (d) 582 J
- 12 **NEET pattern** One mole of an ideal gas expands isothermally at 300 K to double its volume. The work done by the gas is ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)
 (a) 1.73 kJ (b) 3.46 kJ (c) 0.75 kJ (d) 2.49 kJ
- 13 **NEET pattern** A Carnot engine works between 227 °C and 27 °C. Its efficiency is
 (a) 60% (b) 40% (c) 12% (d) 88%

- 14 **NEET pattern** An engine of efficiency 25% rejects 300 J of heat per cycle to the sink. The work done per cycle is
 (a) 400 J (b) 75 J (c) 225 J (d) 100 J

- 15 **NEET pattern** A Carnot refrigerator works between -3°C and 27°C . Its coefficient of performance is
 (a) 10 (b) 8 (c) 9 (d) 0.1

- 16 **NEET pattern** An ideal gas is taken through the cycle $A \rightarrow B \rightarrow C \rightarrow A$ shown. The net work done by the gas in one cycle is



- (a) $2P_0V_0$ (b) $4P_0V_0$ (c) $-2P_0V_0$ (d) P_0V_0
- 17 **JEE Main pattern** 1 mol of an ideal gas ($\gamma = 1.4$) expands adiabatically and its temperature falls from 400 K to 300 K. The work done by the gas is ($R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$)
 (a) 2.91 kJ (b) 1.25 kJ (c) 0.83 kJ (d) 2.08 kJ

- 18 **JEE Main pattern** A monatomic ideal gas at 300 K is compressed adiabatically to $1/8$ of its volume. Its final temperature is
 (a) 600 K (b) 9600 K (c) 1200 K (d) 2400 K

- 19 **JEE Main pattern** An ideal gas expands from the same initial state to the same final volume by isobaric (1), isothermal (2) and adiabatic (3) processes. The correct order of work done is
 (a) $W_2 > W_1 > W_3$ (b) $W_3 > W_2 > W_1$ (c) $W_1 > W_2 > W_3$ (d) $W_1 = W_2 = W_3$

- 20 **JEE Main pattern** The efficiency of a Carnot engine is $1/6$. On lowering the sink temperature by 62 K, the efficiency becomes $1/3$. The source and sink temperatures are
 (a) 372 K, 310 K (b) 330 K, 268 K (c) 372 K, 330 K (d) 310 K, 248 K

- 21 **JEE Main pattern** For hydrogen $c_p - c_v = a$ and for oxygen $c_p - c_v = b$, where c denotes specific heat per unit mass. Then
 (a) $a = 16b$ (b) $a = b$ (c) $b = 16a$ (d) $a = 4b$

- 22 **Assertion-Reason** Assertion: In an isothermal expansion of an ideal gas, heat is absorbed even though the temperature does not change. Reason: In an isothermal process of an ideal gas $\Delta U = 0$, so $\Delta Q = \Delta W$.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A true, R false
 (d) A false, R true

23 **Assertion-Reason** Assertion: Free expansion of a gas into vacuum is an irreversible process. Reason: In free expansion of an ideal gas $\Delta Q = 0$ and $\Delta W = 0$, so its temperature stays the same.

- (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A true, R false
 (d) A false, R true

24 **Assertion-Reason** Assertion: A Carnot engine can have 100% efficiency if the sink is kept at 0°C . Reason: The efficiency of a Carnot engine is $\eta = 1 - T_2 / T_1$.

- (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A true, R false
 (d) A false, R true

✓ Answer Key & Quick Solutions

1(d) 2(c) 3(d) 4(d) 5(d) 6(b) 7(c) 8(b) 9(a) 10(a) 11(b) 12(a) 13(b) 14(d) 15(c)
 16(a) 17(d) 18(c) 19(c) 20(a) 21(a) 22(a) 23(b) 24(d)

- (d) Systems in mutual thermal equilibrium share a common property – temperature.
- (c) ΔW is work done BY the system, e.g. during expansion.
- (d) $\Delta W = -40\text{ J}$; $\Delta U = \Delta Q - \Delta W = 100 + 40 = 140\text{ J}$.
- (d) $\Delta Q = 0 \Rightarrow \Delta U = -\Delta W < 0$ for expansion, so T falls.
- (d) $C_v = 5R/2$, $C_p = 7R/2 \Rightarrow \gamma = 7/5 = 1.4$.
- (b) Heat is exchanged but $\Delta T = 0$, so $C = \Delta Q/(\mu\Delta T) \rightarrow \infty$.
- (c) Isothermal: $dP/dV = -P/V$; adiabatic: $dP/dV = -\gamma P/V$.
- (b) Kelvin–Planck: no 100% conversion of heat into work. The cold→hot statement is Clausius's.
- (a) V constant $\Rightarrow \Delta W = 0 \Rightarrow \Delta Q = \Delta U$.
- (a) $W/Q = R/C_p = R/(7R/2) = 2/7$.
- (b) $Q = \mu C_p \Delta T = 2 \times (5/2)(8.314) \times 10 \approx 416\text{ J}$.
- (a) $W = RT \ln 2 = 8.314 \times 300 \times 0.693 \approx 1729\text{ J} \approx 1.73\text{ kJ}$.
- (b) $T_1 = 500\text{ K}$, $T_2 = 300\text{ K}$; $\eta = 1 - 300/500 = 40\%$. (Using $^\circ\text{C}$ gives the wrong 88%.)
- (d) $Q_2 = (1 - \eta)Q_1 \Rightarrow Q_1 = 300/0.75 = 400\text{ J}$; $W = 400 - 300 = 100\text{ J}$.
- (c) $\alpha = T_2/(T_1 - T_2) = 270/30 = 9$.
- (a) Triangle area $= \frac{1}{2} \times 2V_0 \times 2P_0 = 2P_0V_0$; the cycle is clockwise, so work is positive.
- (d) $W = \mu R(T_1 - T_2)/(\gamma - 1) = 8.314 \times 100/0.4 \approx 2078\text{ J}$.
- (c) $TV^{\gamma-1} = \text{const}$: $T_2 = 300 \times 8^{2/3} = 300 \times 4 = 1200\text{ K}$.
- (c) On a P–V diagram the isobar lies highest and the adiabatic (steepest fall) lowest, so area is largest for isobaric.
- (a) $T_2/T_1 = 5/6$ and $(T_2 - 62)/T_1 = 2/3 \Rightarrow 62/T_1 = 1/6 \Rightarrow T_1 = 372\text{ K}$, $T_2 = 310\text{ K}$.
- (a) $c_p - c_v = R/M \Rightarrow a/b = M(\text{O}_2)/M(\text{H}_2) = 32/2 = 16$.
- (a) Both true; since $\Delta U = 0$, the heat absorbed equals the work done by the gas – R explains A.
- (b) Both true, but irreversibility is due to passing through non-equilibrium states, not due to $\Delta U = 0$.
- (d) $0^\circ\text{C} = 273\text{ K} \neq 0\text{ K}$, so $\eta < 1$. A is false, R is true.

CHAPTER

12

Kinetic Theory

Day 20

NEET weightage ★★☆☆

NCERT Class 11

Ideal gas equation

Pressure of a gas

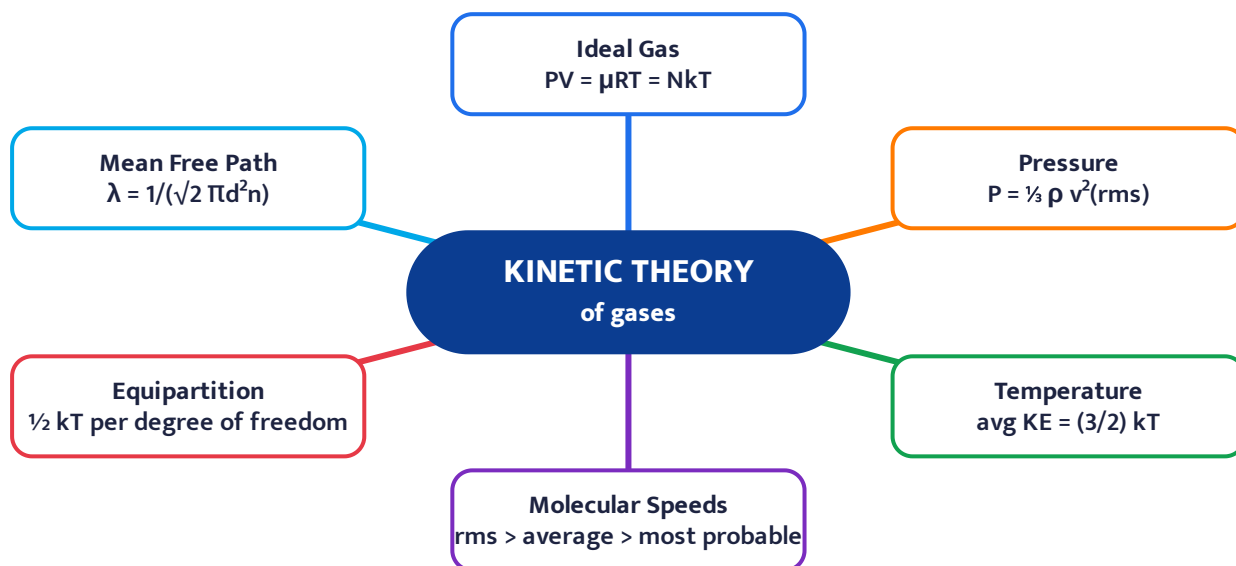
KE & temperature

rms speed

Equipartition

Specific heats

Mean free path



Chapter at a glance

12.1 Molecular Nature of Matter

Matter is made of atoms and molecules (Dalton's atomic theory, Gay-Lussac's law, **Avogadro's law**: equal volumes of all gases at the same T and P contain equal numbers of molecules). Atomic size $\approx 1 \text{ \AA} = 10^{-10} \text{ m}$. In a gas the molecules are far apart (about ten times their size) and interact only during collisions – so gases are the easiest phase to model. **Kinetic theory** explains the macroscopic behaviour of a gas (P, T, specific heat) from the random motion of its molecules.

12.2 Behaviour of Gases – Ideal Gas Equation

Ideal gas

$$PV = \mu RT$$

Ideal gas equation, μ = number of moles

$$PV = Nk_B T$$

N = number of molecules

$$\mu = \frac{M}{M_0} = \frac{N}{N_A}$$

M = mass, M_0 = molar mass

$$P = \frac{\rho RT}{M_0}$$

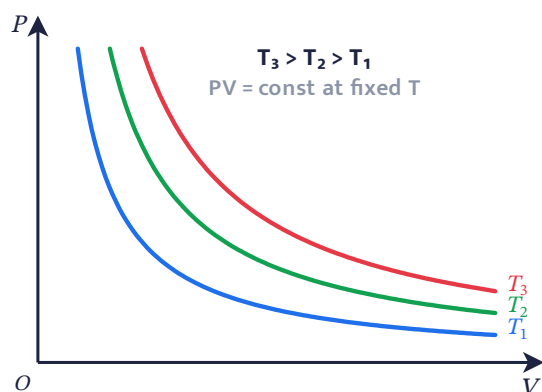
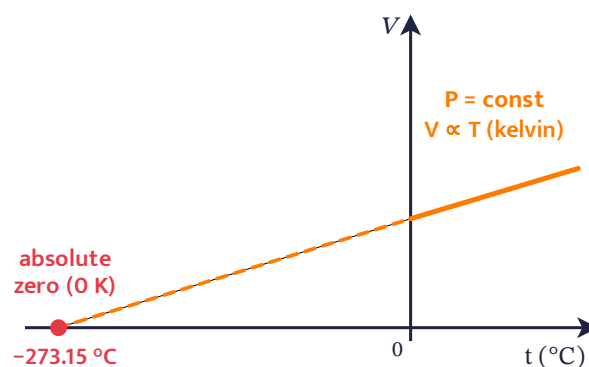
In terms of density ρ

$$R = N_A k_B = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$$

Universal gas constant

$$k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

Boltzmann constant

Boyle's law: P - V isotherms are hyperbolasCharles' law: V - t line meets t -axis at -273.15°C

Law	Condition	Statement
Boyle's law	T, μ fixed	$PV = \text{const}$
Charles' law	P, μ fixed	$V \propto T$
Gay-Lussac's (pressure) law	V, μ fixed	$P \propto T$
Avogadro's law	P, T fixed	$V \propto N$; 1 mol at STP occupies 22.4 L
Dalton's law of partial pressures	mixture of ideal gases	$P = P_1 + P_2 + \dots = (\mu_1 + \mu_2 + \dots) \frac{RT}{V}$

Solved Example

2 mol of an ideal gas at 300 K is kept in a vessel of volume 8.3 L. Find its pressure.

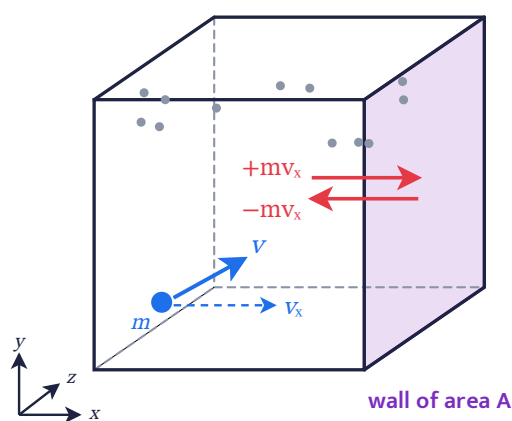
Solution: $P = \frac{\mu RT}{V} = \frac{2 \times 8.314 \times 300}{8.3 \times 10^{-3}} \approx 6.0 \times 10^5 \text{ Pa} (\approx 6 \text{ atm}).$

NCERT Line to Remember

Real gases approach ideal behaviour at **low pressure and high temperature** (molecules far apart, interactions negligible).

12.3 Kinetic Theory of an Ideal Gas – Pressure**Assumptions of kinetic theory**

- A gas consists of a very large number of identical molecules in **random motion**.
- Molecular size is negligible (point masses); **no intermolecular forces** except during collisions, so they move in straight lines between collisions.
- Collisions with each other and with the walls are **perfectly elastic**; collision time is negligible.

Molecule of velocity v hitting the shaded wall**Derivation in 3 steps**

- Momentum given to the wall per collision: $2mv_x$.
- In time Δt , molecules within $v_x \Delta t$ moving towards the wall (half of them) hit it: number $= \frac{1}{2} n A v_x \Delta t$.
- Force/area: $P = nm \overline{v_x^2}$; isotropy gives $\overline{v_x^2} = \frac{1}{3} \overline{v^2}$, so $P = \frac{1}{3} nm \overline{v^2}$.



Pressure of an ideal gas

$$P = \frac{1}{3}nm\overline{v^2}$$

n = number density, m = mass of a molecule

$$P = \frac{1}{3}\rho v_{rms}^2$$

$\rho = nm$ = density of gas

$$PV = \frac{2}{3}E$$

E = total translational KE

$$P = \frac{2}{3} \frac{E}{V}$$

Pressure = $\frac{2}{3} \times$ KE per unit volume

12.4 Kinetic Interpretation of Temperature

Comparing $PV = \frac{2}{3}E$ with $PV = Nk_B T$ gives $E = \frac{3}{2}Nk_B T$. **The average kinetic energy of a molecule is proportional to the absolute temperature** and is independent of P, V and the nature of the gas.



Temperature and molecular speeds

$$\overline{KE} = \frac{1}{2}m\overline{v^2} = \frac{3}{2}k_B T$$

Average translational KE per molecule

$$v_{rms} = \sqrt{\frac{3k_B T}{m}} = \sqrt{\frac{3RT}{M_0}} = \sqrt{\frac{3P}{\rho}}$$

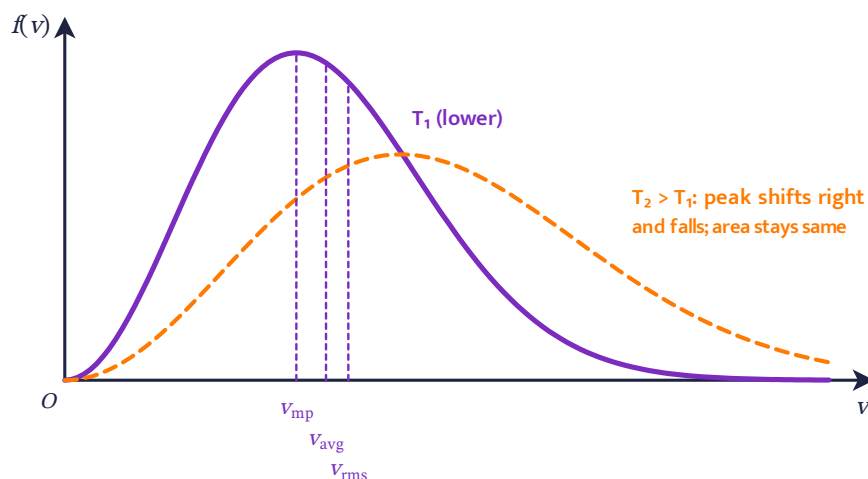
Root-mean-square speed

$$v_{avg} = \sqrt{\frac{8RT}{\pi M_0}}$$

Mean (average) speed

$$v_{mp} = \sqrt{\frac{2RT}{M_0}}$$

Most probable speed



Distribution of molecular speeds: $v_{mp} < v_{avg} < v_{rms}$



Memory Trick

- $v_{rms} : v_{avg} : v_{mp} = \sqrt{3} : \sqrt{8/\pi} : \sqrt{2} \approx 1.73 : 1.60 : 1.41$ – "Rich Always More" (rms > avg > mp).
- All speeds $\propto \sqrt{T/M_0}$.



Solved Example

Find the rms speed of nitrogen molecules at 27°C ($M_0 = 28 \text{ g/mol}$).

Solution: $v_{rms} = \sqrt{\frac{3 \times 8.314 \times 300}{0.028}} \approx 517 \text{ m/s}$. (Use M_0 in kg/mol!)

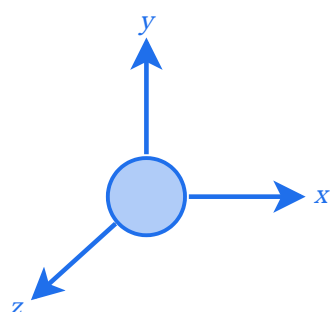


Common Trap

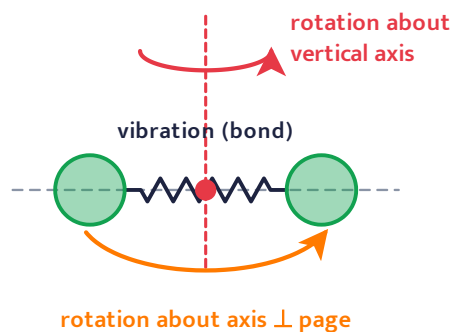
- At the same T , **all gases have the same average KE per molecule**, but lighter molecules move faster ($v_{rms} \propto 1/\sqrt{M_0}$).
- Doubling T (kelvin) makes v_{rms} only $\sqrt{2}$ times; to double v_{rms} , T must become 4 times.
- KE of 1 mole = $\frac{3}{2}RT$ (translational); KE per molecule = $\frac{3}{2}k_B T$ – do not mix R and k_B .

12.5 Law of Equipartition of Energy

Degrees of freedom (f) = number of independent ways a molecule can possess energy. **Law of equipartition:** in thermal equilibrium at temperature T , the energy is shared equally among all modes – each quadratic (translational or rotational) term contributes $\frac{1}{2}k_B T$ per molecule; each **vibrational** mode contributes $k_B T$ (kinetic + potential).



Monatomic: $f = 3$
(3 translational)



Rigid diatomic: $f = 5$
(3 trans + 2 rot; vibration adds 2 at high T)

Degrees of freedom of monatomic and diatomic molecules

12.6 Specific Heat Capacity of Gases and Solids

For one mole with f degrees of freedom (each $\frac{1}{2}k_B T$): $U = \frac{f}{2}RT$, so $C_v = \frac{dU}{dT} = \frac{f}{2}R$, $C_p = C_v + R$ and $\gamma = 1 + \frac{2}{f}$.

Molecule	f	U per mole	C_v	C_p	γ
Monatomic (He, Ar)	3	$\frac{3}{2}RT$	$\frac{3}{2}R$	$\frac{5}{2}R$	$5/3 \approx 1.67$
Diatomic, rigid (O_2, N_2)	5	$\frac{5}{2}RT$	$\frac{5}{2}R$	$\frac{7}{2}R$	$7/5 = 1.40$
Diatomic + vibration	$5 + 1 \text{ vib}$	$\frac{7}{2}RT$	$\frac{7}{2}R$	$\frac{9}{2}R$	$9/7 \approx 1.29$
Polyatomic, rigid non-linear	6	$3RT$	$3R$	$4R$	$4/3 \approx 1.33$

Solids and water

- Each atom of a solid oscillates in 3 dimensions: energy $3k_B T$ per atom $\Rightarrow U = 3RT$ per mole $\Rightarrow C = 3R \approx 25 \text{ J mol}^{-1} \text{ K}^{-1}$ (**Dulong–Petit law**). ($C_p \approx C_v$ for solids since ΔV is negligible.)
- Water (3 atoms, treated like a solid): $C \approx 9R \approx 75 \text{ J mol}^{-1} \text{ K}^{-1}$.
- Classical values fail at low T (vibrational modes "freeze") – explained by quantum theory.

VS Sir's Exam Tip

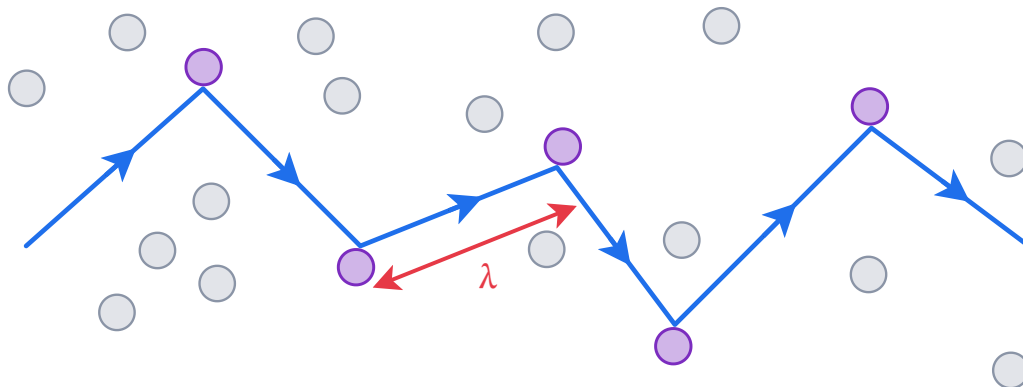
- Quick γ : $\gamma = 1 + 2/f$. $f = 3 \rightarrow 1.67$, $f = 5 \rightarrow 1.4$, $f = 6 \rightarrow 1.33$.
- Larger $f \Rightarrow$ larger C_v but **smaller γ** .
- Total KE of 1 mol of rigid diatomic gas $= \frac{5}{2}RT$, of which translational $= \frac{3}{2}RT$.

JEE Main Extension

Mixture of gases: $C_{v, \text{mix}} = \frac{\mu_1 C_{v1} + \mu_2 C_{v2}}{\mu_1 + \mu_2}$, $\gamma_{\text{mix}} = \frac{C_{p, \text{mix}}}{C_{v, \text{mix}}}$. Also $\frac{\mu_1 + \mu_2}{\gamma_{\text{mix}} - 1} = \frac{\mu_1}{\gamma_1 - 1} + \frac{\mu_2}{\gamma_2 - 1}$.

12.7 Mean Free Path

A molecule of diameter d collides with every molecule whose centre lies within a cylinder of radius d along its path. The **mean free path** λ is the average distance travelled between two successive collisions.



Free path = distance between two successive collisions; λ is its average

Zig-zag path of a gas molecule

Mean free path

$$\lambda = \frac{1}{\sqrt{2} \pi d^2 n}$$

n = number density (molecules per m^3)

$$\lambda = \frac{k_B T}{\sqrt{2} \pi d^2 P}$$

Using $n = P / k_B T$: $\lambda \propto T/P$

Solved Example

Estimate the mean free path of air molecules at STP ($d \approx 2 \text{ \AA}$, $n \approx 2.7 \times 10^{25} \text{ m}^{-3}$).

Solution: $\lambda = \frac{1}{1.414 \times 3.14 \times (2 \times 10^{-10})^2 \times 2.7 \times 10^{25}} \approx 2 \times 10^{-7} \text{ m}$ – about 1000 times the molecular size.

Common Trap

- λ depends on d as $1/d^2$ – doubling the diameter makes λ one-fourth.
- In a **closed** vessel (n fixed), heating does **not** change λ ; at fixed P , $\lambda \propto T$.

Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

1 **NCERT based** The average translational kinetic energy of a gas molecule at absolute temperature T is

(a) $\frac{3}{2} k_B T$

(b) $k_B T$

(c) $\frac{1}{2} k_B T$

(d) $\frac{3}{2} RT$

2 **NCERT based** Which of the following is **not** an assumption of the kinetic theory of an ideal gas?

(a) Molecular size is negligible

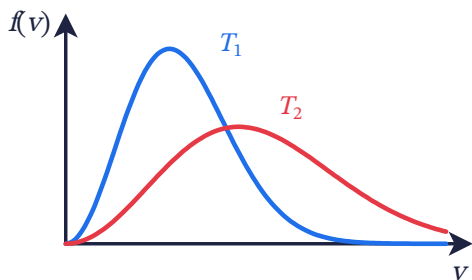
(b) Collisions are perfectly elastic

(c) Molecules attract each other between collisions

(d) Molecules are in random motion

- 3 **NCERT based** The number of degrees of freedom of a rigid diatomic molecule at room temperature is
 (a) 7 (b) 3 (c) 6 (d) 5
- 4 **NCERT based** According to the Dulong–Petit law, the molar specific heat of a solid is about
 (a) $\frac{5}{2}R$ (b) $3R$ (c) $\frac{3}{2}R$ (d) R
- 5 **NCERT based** At constant temperature, if the pressure of a gas is doubled, its mean free path
 (a) becomes half (b) becomes double
 (c) becomes four times (d) remains the same
- 6 **NEET pattern** The pressure P of an ideal gas and its translational kinetic energy per unit volume E are related by
 (a) $P = \frac{3}{2}E$ (b) $P = \frac{1}{3}E$ (c) $P = E$ (d) $P = \frac{2}{3}E$
- 7 **NEET pattern** For a gas, the ratio $v_{rms}:v_{avg}:v_{mp}$ is
 (a) 1:1:1 (b) $\sqrt{3}:\sqrt{8/\pi}:\sqrt{2}$ (c) $\sqrt{3}:\sqrt{2}:\sqrt{8/\pi}$ (d) $\sqrt{2}:\sqrt{8/\pi}:\sqrt{3}$
- 8 **NEET pattern** The temperature of a gas is raised from 27 °C to 927 °C. Its rms speed becomes
 (a) $\sqrt{(927/27)}$ times (b) 2 times
 (c) $\sqrt{2}$ times (d) 4 times
- 9 **NEET pattern** The rms speed of oxygen molecules ($M_0 = 32$ g/mol) at 27 °C is about ($R = 8.314$ J mol⁻¹ K⁻¹)
 (a) 15.3 m/s (b) 446 m/s (c) 484 m/s (d) 395 m/s
- 10 **NEET pattern** At what temperature will the rms speed of hydrogen molecules equal that of oxygen molecules at 47 °C?
 (a) 5 K (b) 20 K (c) 80 K (d) 160 K
- 11 **NEET pattern** A gas in a closed rigid vessel is at 27 °C and 2 atm. It is heated to 127 °C. The new pressure is
 (a) 1.5 atm (b) 9.4 atm (c) 2.67 atm (d) 3.0 atm
- 12 **NEET pattern** The number of molecules in 1 m³ of an ideal gas at STP (1.013×10^5 Pa, 273 K) is ($k_B = 1.38 \times 10^{-23}$ J/K)
 (a) 6.02×10^{23} (b) 2.69×10^{25} (c) 4.46×10^1 (d) 2.69×10^{22}
- 13 **NEET pattern** If the vibrational mode of a diatomic molecule is also excited, the molar specific heat C_v of the gas becomes
 (a) $\frac{3}{2}R$ (b) $\frac{7}{2}R$ (c) $\frac{5}{2}R$ (d) $\frac{9}{2}R$
- 14 **JEE Main pattern** The average translational KE of a gas molecule at 27 °C is ($k_B = 1.38 \times 10^{-23}$ J/K)
 (a) 4.14×10^{-21} J (b) 2.07×10^{-21} J (c) 6.21×10^{-21} J (d) 5.59×10^{-22} J
- 15 **JEE Main pattern** The total internal energy of 2 mol of oxygen (rigid diatomic) at 300 K is ($R = 8.314$ J mol⁻¹ K⁻¹)
 (a) 17.5 kJ (b) 6.23 kJ (c) 7.48 kJ (d) 12.5 kJ

- 16 **NEET pattern** The speed-distribution curves of a gas at two temperatures T_1 and T_2 are shown. Then



- (a) $T_2 > T_1$ (b) $T_1 > T_2$
 (c) $T_1 = T_2$ (d) cannot be decided
- 17 **JEE Main pattern** One mole of helium is mixed with one mole of oxygen (rigid). The value of γ for the mixture is
 (a) 1.53 (b) 1.40 (c) 1.50 (d) 1.67
- 18 **JEE Main pattern** The rms speed of the molecules of a diatomic gas is v . If the temperature (kelvin) is doubled and the molecules dissociate into atoms, the rms speed becomes
 (a) $2v$ (b) $v / \sqrt{2}$ (c) $\sqrt{2} v$ (d) $4v$
- 19 **Assertion-Reason** Assertion: At a given temperature, the rms speed of hydrogen molecules is greater than that of oxygen molecules. Reason: At the same temperature, molecules of all gases have the same average kinetic energy.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A true, R false
 (d) A false, R true
- 20 **Assertion-Reason** Assertion: The molar specific heat of a gas at constant pressure is greater than that at constant volume. Reason: At constant pressure, part of the heat supplied is used by the gas to do work in expanding.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A true, R false
 (d) A false, R true

✓ Answer Key & Quick Solutions

- 1(a) 2(c) 3(d) 4(b) 5(a) 6(d) 7(b) 8(b) 9(c) 10(b) 11(c) 12(b) 13(b) 14(c) 15(d)
 16(a) 17(c) 18(a) 19(a) 20(a)

- Each of 3 translational degrees contributes $\frac{1}{2}k_B T$.
- Kinetic theory assumes no intermolecular forces except during collisions.
- 3 translational + 2 rotational = 5.
- Each atom oscillates along 3 directions, $k_B T$ each $\Rightarrow 3k_B T \Rightarrow C = 3R \approx 25 \text{ J mol}^{-1} \text{ K}^{-1}$.
- $\lambda = k_B T / (\sqrt{2} \pi d^2 P) \propto 1/P$ at fixed T.
- $P = \frac{1}{3} \rho \overline{v^2} = \frac{2}{3} \left(\frac{1}{2} \rho \overline{v^2} \right) = \frac{2}{3} E$.

7. (b) $v \propto \sqrt{3}, \sqrt{8/\pi}, \sqrt{2}$ times $\sqrt{RT/M_0} \Rightarrow 1.73 : 1.60 : 1.41$.
8. (b) T: 300 K $\rightarrow$ 1200 K (4 times); $v_{rms} \propto \sqrt{T} \Rightarrow 2$ times.
9. (c) $v_{rms} = \sqrt{3 \times 8.314 \times 300 / 0.032} \approx 484$ m/s. (395 m/s is v_{mp} , 446 m/s is v_{avg} .)
10. (b) $T_H / M_H = T_O / M_O \Rightarrow T_H = 320 \times 2 / 32 = 20$ K.
11. (c) V const $\Rightarrow P \propto T$: $P = 2 \times 400/300 \approx 2.67$ atm. (Always use kelvin.)
12. (b) $n = P / k_B T = 1.013 \times 10^5 / (1.38 \times 10^{-23} \times 273) \approx 2.69 \times 10^{25}$ (Loschmidt number).
13. (b) 3 trans (3/2) + 2 rot (1) + 1 vib ($k_B T$) $\Rightarrow U = \frac{7}{2} RT \Rightarrow C_v = \frac{7}{2} R$.
14. (c) $\frac{3}{2} k_B T = 1.5 \times 1.38 \times 10^{-23} \times 300 = 6.21 \times 10^{-21}$ J.
15. (d) $U = \mu \frac{5}{2} RT = 2 \times 2.5 \times 8.314 \times 300 \approx 12.5$ kJ.
16. (a) Higher T: peak shifts to higher speed and becomes lower (area constant). The flatter curve is T_2 .
17. (c) $C_v = \frac{1.5R + 2.5R}{2} = 2R, C_p = 3R \Rightarrow \gamma = 1.5$.
18. (a) $v_{rms} \propto \sqrt{T/M_0}$: $T \times 2$ and $M_0 \div 2 \Rightarrow \sqrt{4} = 2$ times.
19. (a) Equal $\frac{1}{2}mv^2 \Rightarrow$ smaller m gives larger v_{rms} . R explains A.
20. (a) $C_p = C_v + R$; the extra R accounts for the work $P\Delta V$. R explains A.

CHAPTER

13

Oscillations

Day 21–22

NEET weightage ★★★

NCERT Class 11

Periodic motion

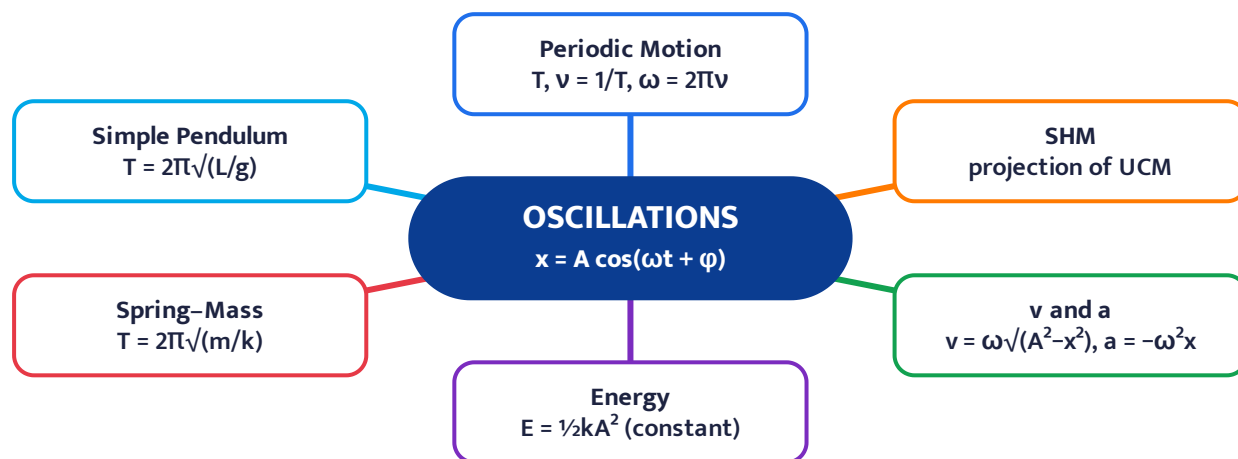
SHM equation

Velocity & acceleration

Energy in SHM

Spring-mass system

Simple pendulum



Chapter at a glance

13.1 Periodic and Oscillatory Motion

A motion that **repeats itself at regular intervals of time** is **periodic**. If the body moves **to and fro about a mean (equilibrium) position**, the periodic motion is **oscillatory** (vibration if the frequency is high). At the mean position the net force on the body is zero.

T, ν and ω

T	Period (s) – time for one oscillation
$\nu = \frac{1}{T}$	Frequency (Hz = s ⁻¹)
$\omega = 2\pi\nu = \frac{2\pi}{T}$	Angular frequency (rad/s)

NCERT Line to Remember

- Every oscillatory motion is periodic, but every periodic motion need not be oscillatory – **uniform circular motion** and the revolution of planets are periodic but **not** oscillatory.
- Any periodic function can be expressed as a superposition of sine and cosine functions of different periods (Fourier).

Displacement here means the change of any physical quantity with time – position of a block, angle of a pendulum, pressure in a sound wave, voltage in an AC circuit. $\sin\omega t$ and $\cos\omega t$ are the simplest periodic functions, each with period $T = 2\pi / \omega$.

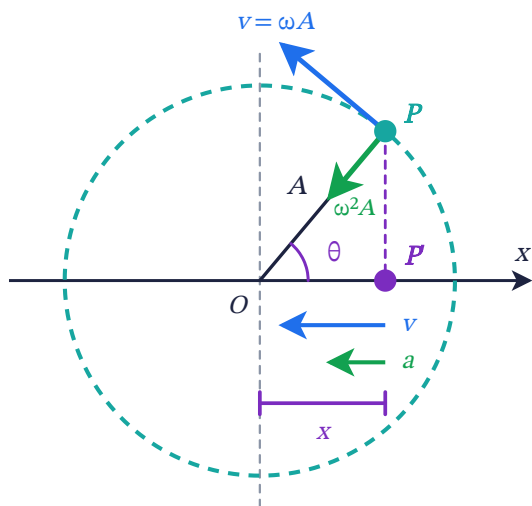
Function of time	Type of motion	Period
$\sin\omega t - \cos\omega t = \sqrt{2} \sin(\omega t - \pi/4)$	SHM	$2\pi / \omega$
$3\cos(\pi/4 - 2\omega t)$	SHM	π / ω
$\sin^3\omega t, \cos\omega t + \cos 3\omega t + \cos^2\omega t$	periodic, not SHM	$2\pi / \omega$
$e^{-\omega^2 t^2}, 1 + \omega t + \omega^2 t^2$	non-periodic	—

13.2 Simple Harmonic Motion (SHM)

SHM is the simplest oscillation: the displacement is a **sinusoidal function of time**,

$$x(t) = A \cos(\omega t + \phi)$$

Here A = **amplitude** (maximum displacement), $(\omega t + \phi)$ = **phase** (fixes position and direction of motion at time t), ϕ = **phase constant** (phase at $t = 0$, set by initial x and v), $\omega = 2\pi / T$ = **angular frequency** (set by the system: m, k or L, g).



P moves on the circle with constant ω ;
its shadow P' on the diameter executes SHM

$$\theta = \omega t + \phi$$

$$x = A \cos(\omega t + \phi)$$

$$v = -\omega A \sin(\omega t + \phi)$$

$$a = -\omega^2 A \cos(\omega t + \phi) \\ = -\omega^2 x$$

SHM as the projection of uniform circular motion on a diameter (reference circle)

NCERT Line to Remember

Simple harmonic motion is the projection of uniform circular motion on a diameter of the circle in which the latter motion takes place. The radius of the circle = amplitude A ; angular speed of the reference particle = ω of the SHM.

Solved Example

A particle moves as $x = 0.05 \cos(4\pi t + \pi/4)$ m. Find the amplitude, frequency, period, initial phase, maximum speed and maximum acceleration.

Solution: $A = 0.05$ m; $\omega = 4\pi \Rightarrow \nu = \omega / 2\pi = 2$ Hz, $T = 0.5$ s; $\phi = \pi/4$. $v_{\max} = \omega A = 0.2\pi \approx 0.63$ m/s; $a_{\max} = \omega^2 A = 0.8\pi^2 \approx 7.9$ m/s².

13.3 Velocity and Acceleration in SHM

Velocity and acceleration

$$v = \frac{dx}{dt} = -\omega A \sin(\omega t + \phi)$$

Velocity – leads x by $\pi/2$

$$|v| = \omega \sqrt{A^2 - x^2}$$

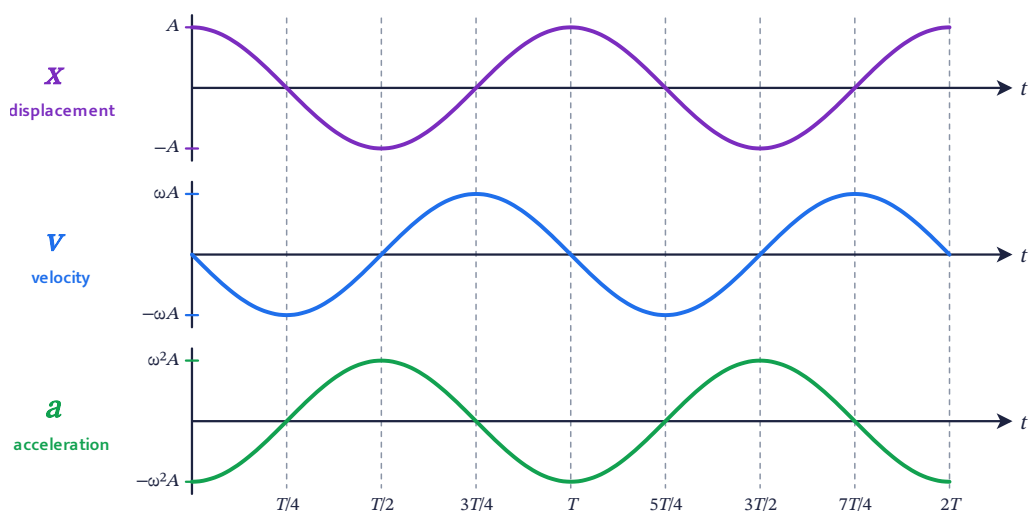
Speed at displacement x ; $v_{\max} = \omega A$ at $x = 0$

$$a = \frac{dv}{dt} = -\omega^2 A \cos(\omega t + \phi) = -\omega^2 x$$

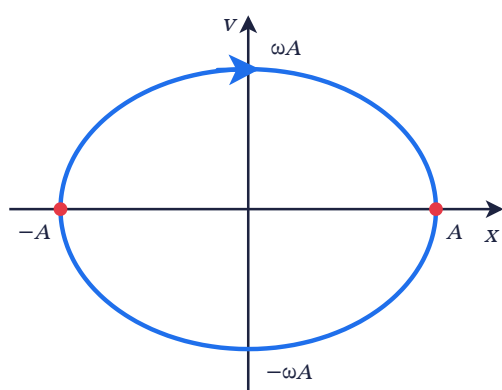
Acceleration – always towards mean, phase difference π with x

$$|a|_{\max} = \omega^2 A$$

At the extreme positions $x = \pm A$

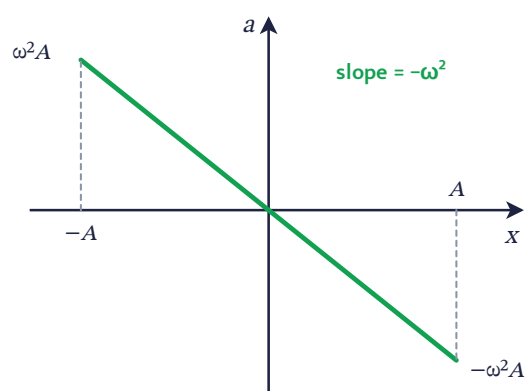


x , v and a versus t for $x = A\cos\omega t$: v leads x by $\pi/2$, a leads v by $\pi/2$ (a is opposite to x)



ellipse (circle if units scaled)

$$v-x \text{ graph: } \frac{x^2}{A^2} + \frac{v^2}{\omega^2 A^2} = 1 \text{ (ellipse)}$$



slope = $-\omega^2$

straight line through origin

$$a-x \text{ graph: } a = -\omega^2 x \text{ (straight line, negative slope)}$$

Quantity	Mean position ($x = 0$)	Extreme position ($x = \pm A$)
Speed	maximum, ωA	zero
Acceleration / restoring force	zero	maximum, $\omega^2 A / kA$
Kinetic energy	maximum = E	zero
Potential energy	zero (minimum)	maximum = E

⚠ Common Trap

- Velocity and acceleration are **not** in phase: at the mean position v is maximum but $a = 0$; at the extremes $v = 0$ but a is maximum.
- $a = -\omega^2 x$: the minus sign means a is **opposite to x** , i.e. always towards the mean.

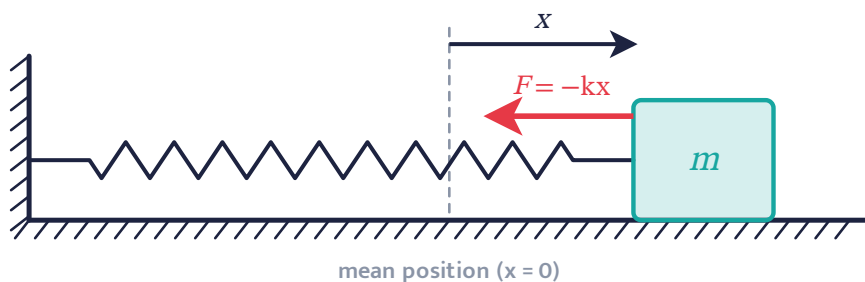
🎯 VS Sir's Exam Tip

- $x = a\sin\omega t + b\cos\omega t$ is SHM with $A = \sqrt{a^2 + b^2}$.
- Time from mean to $A/2 = T/12$; from $A/2$ to $A = T/6$; mean to $A/\sqrt{2} = T/8$.

13.4 Force Law for SHM

From Newton's second law, $F = ma = -m\omega^2 x$. So a body executes SHM when the force on it is a **restoring force proportional to displacement**:

$$F = -kx, \quad k = m\omega^2 \Rightarrow \omega = \sqrt{\frac{k}{m}}, \quad T = 2\pi\sqrt{\frac{m}{k}}$$



$$F = -kx$$

$$a = -(k/m)x$$

$$\omega = \sqrt{(k/m)}, \quad T = 2\pi\sqrt{(m/k)}$$

Block on a smooth floor: the spring pulls it back towards the mean position – a linear restoring force

One-line test for SHM

Reduce the equation to $a = -(\text{positive constant})x$. Then $\omega^2 = \text{that constant}$. Example: $a = -16x \Rightarrow \omega = 4$ rad/s, $T = 2\pi / 4 = \pi / 2$ s. Forces like $F = -kx^3$ or $F = -k|x|$ give oscillation but **not** SHM.

13.5 Energy in SHM

Energy formulas

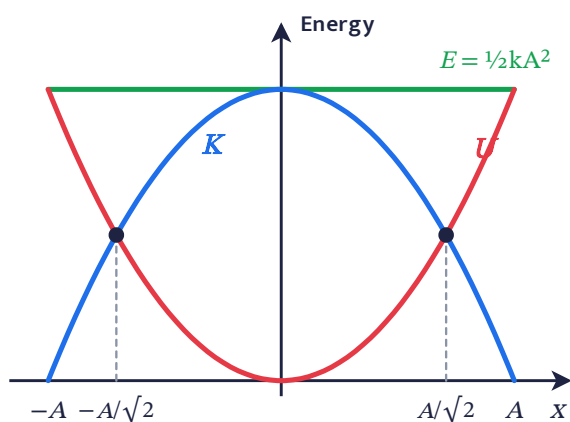
$$U = \frac{1}{2}kx^2 = \frac{1}{2}kA^2\cos^2(\omega t + \phi) \quad \text{Potential energy (zero at mean)}$$

$$K = \frac{1}{2}mv^2 = \frac{1}{2}k(A^2 - x^2) \quad \text{Kinetic energy}$$

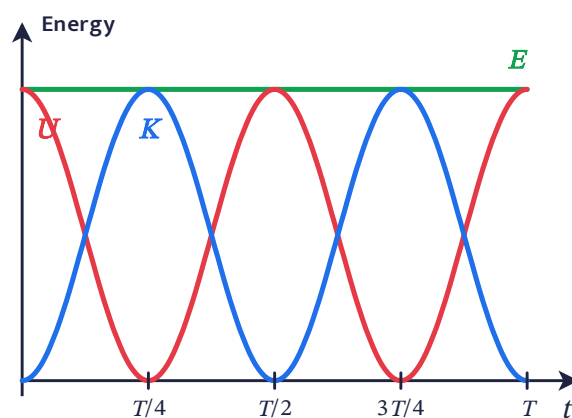
$$E = K + U = \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2A^2 \quad \text{Total energy – constant, independent of } x \text{ and } t$$

$$K = U \text{ at } x = \pm \frac{A}{\sqrt{2}} \quad \text{Equal sharing of energy}$$

$$\langle K \rangle = \langle U \rangle = \frac{E}{2} \quad \text{Time-averaged over one period}$$



Energy vs displacement: U parabola, K inverted parabola, E flat



Energy vs time: K and U oscillate at 2ν , E stays constant

Solved Example

A 2 kg block on a spring ($k = 200 \text{ N/m}$) oscillates with amplitude 0.1 m. Find ω , $v_{\max}$, total energy, and K and U at $x = 0.05 \text{ m}$.

Solution: $\omega = \sqrt{200/2} = 10 \text{ rad/s}$; $v_{\max} = \omega A = 1 \text{ m/s}$; $E = \frac{1}{2}(200)(0.1)^2 = 1 \text{ J}$. At $x = A/2$: $U = \frac{1}{2}(200)(0.05)^2 = 0.25 \text{ J}$, $K = E - U = 0.75 \text{ J}$ (ratio 3 : 1).

Common Trap

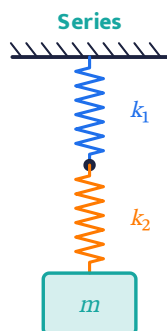
- K and U oscillate with period $T/2$ (frequency 2ν); the total energy E does **not** oscillate.
- $E \propto A^2$ and $E \propto \omega^2$: doubling A makes E four times.

Memory Trick

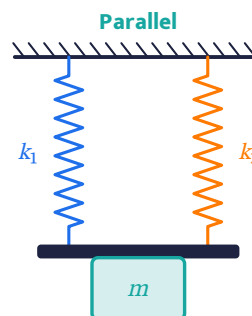
- $x = A/2 \rightarrow K : U = 3 : 1$
- $x = A/\sqrt{2} \rightarrow K : U = 1 : 1$
- $x = \sqrt{3}A/2 \rightarrow K : U = 1 : 3$
- General: $K/U = (A^2 - x^2)/x^2$

13.6 Spring–Mass System

A mass m attached to a light spring of **spring constant (force constant)** k executes SHM with $T = 2\pi\sqrt{m/k}$ – whether horizontal or vertical. For a vertical spring the equilibrium shifts down by $\Delta l = mg/k$, but the period is unchanged.



$$1/k = 1/k_1 + 1/k_2$$



$$k = k_1 + k_2$$

Springs in series (same force, extensions add) and in parallel (same extension, forces add)

Spring formulas

$$T = 2\pi\sqrt{\frac{m}{k}} = 2\pi\sqrt{\frac{\Delta l}{g}} \quad \text{Vertical spring: } \Delta l = \text{static extension}$$

$$\frac{1}{k_s} = \frac{1}{k_1} + \frac{1}{k_2}$$

Series combination (softer)

$$k_p = k_1 + k_2$$

Parallel combination (stiffer); also a block **between** two springs

$$k' = nk$$

Spring cut into n equal parts: each part ($k \propto 1/l$)

Solved Example

Two springs of constants 100 N/m and 300 N/m support a 1 kg mass. Find the period when they are (i) in parallel, (ii) in series.

Solution: (i) $k_p = 400 \text{ N/m} \Rightarrow T = 2\pi\sqrt{1/400} = \pi/10 \approx 0.31 \text{ s}$. (ii) $k_s = \frac{100 \times 300}{400} = 75 \text{ N/m} \Rightarrow T = 2\pi\sqrt{1/75} \approx 0.73 \text{ s}$.

! Common Trap

- Spring–block period does **not** depend on g : same on the Moon, in a lift or on an incline.
- Cutting a spring makes each piece **stiffer** (k increases), so T decreases.

**JEE Main Extension**

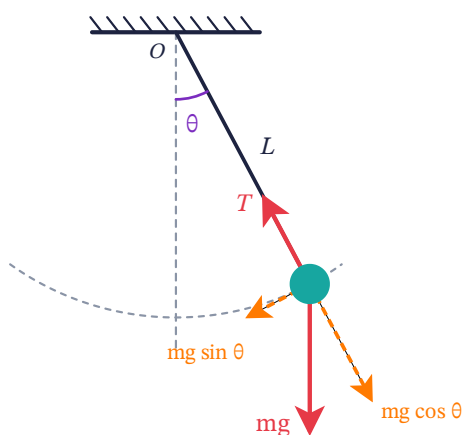
If the spring has mass m_s (uniform), use

$T = 2\pi\sqrt{\frac{m + m_s/3}{k}}$. Two masses joined by a spring on a smooth floor: use reduced mass $\mu = \frac{m_1 m_2}{m_1 + m_2}$ in place of m .

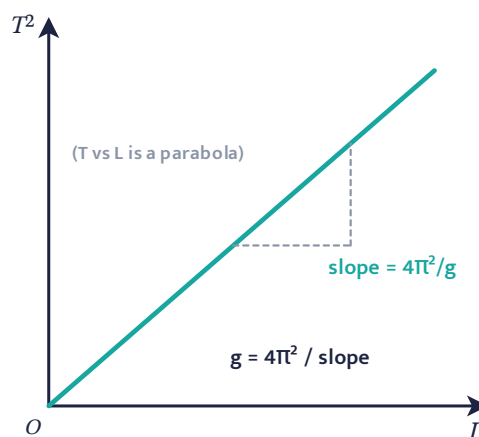
13.7 The Simple Pendulum

A point bob of mass m on a light, inextensible string of length L . When displaced by angle θ , the tangential component $mg\sin\theta$ provides a restoring torque $\tau = -mgL\sin\theta$ about the support. With $I = mL^2$:

$$\alpha = -\frac{g}{L}\sin\theta \approx -\frac{g}{L}\theta \Rightarrow \omega = \sqrt{\frac{g}{L}}, \quad T = 2\pi\sqrt{\frac{L}{g}}$$



Forces on the bob: T and mg ; $mg\sin\theta$ restores



T^2 vs L : straight line through origin

NCERT Line to Remember

The approximation $\sin\theta \approx \theta$ (θ in radian) makes the motion SHM; it holds for **small angles** (error $\approx 1\%$ at 15°). T is **independent of the mass** of the bob and (for small oscillations) of the **amplitude**.

**Pendulum in different situations**

$$T = 2\pi\sqrt{\frac{L}{g_{eff}}}$$

General form – replace g by effective g

$$g_{eff} = g + a$$

Lift accelerating **upward** (T decreases)

$$g_{eff} = g - a$$

Lift accelerating **downward** (T increases)

$$g_{eff} = 0 \Rightarrow T \rightarrow \infty$$

Free fall / orbiting satellite: no oscillation

$$g_{eff} = \sqrt{g^2 + a^2}$$

Vehicle with horizontal acceleration a

$$T = 2s \Rightarrow L = \frac{g}{\pi^2} \approx 0.99 \text{ m}$$

Seconds pendulum (≈ 1 m on Earth)

 Solved Example

A pendulum has period 2 s on the ground ($g = 10 \text{ m/s}^2$). Find its period (i) in a lift accelerating upward at 2.5 m/s^2 , (ii) on the Moon ($g_M = g/6$).

Solution: (i) $T' = T\sqrt{g/(g+a)} = 2\sqrt{10/12.5} = 2\sqrt{0.8} \approx 1.79 \text{ s}$. (ii) $T' = T\sqrt{6} = 2\sqrt{6} \approx 4.9 \text{ s}$.

 VS Sir's Exam Tip

- Pendulum clock **loses** time when T increases: in summer (L expands), on a mountain or at the equator (g smaller).
- L is measured from the support to the **centre** of the bob.

**Damped & forced oscillations – outside NEET scope**

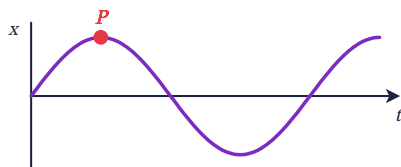
Deleted from the rationalised NCERT and not in the NEET-UG syllabus. Awareness only: damping makes the amplitude decay ($Ae^{-bt/2m}$); a periodic driving force at the natural frequency gives **resonance** (large amplitude).

**Practice Questions**

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

- NCERT based** Which of the following functions represents SHM?
 (a) $1 + \omega t + \omega^2 t^2$ (b) $e^{-\omega^2 t^2}$ (c) $\sin \omega t - \cos \omega t$ (d) $\sin^3 \omega t$
- NCERT based** The period of the motion $x = 3\cos(\pi/4 - 2\omega t)$ is
 (a) $4\pi/\omega$ (b) $2\pi/\omega$ (c) $\pi/2\omega$ (d) π/ω
- NCERT based** The phase difference between displacement and acceleration of a particle in SHM is
 (a) 0 (b) $\pi/4$ (c) π (d) $\pi/2$
- NCERT based** For a particle in SHM, the graph of velocity against displacement is
 (a) an ellipse (b) a hyperbola
 (c) a straight line (d) a parabola
- NEET pattern** A particle in SHM has amplitude 4 cm and maximum speed 8 cm/s. Its speed at $x = 2$ cm is
 (a) $2\sqrt{3}$ cm/s (b) 6 cm/s (c) $4\sqrt{3}$ cm/s (d) 4 cm/s
- NEET pattern** A particle executes SHM with amplitude 5 cm and period $\pi/5$ s. Its maximum acceleration is
 (a) 10 m/s^2 (b) 0.5 m/s^2 (c) 5 m/s^2 (d) 50 m/s^2
- NEET pattern** At what displacement are the kinetic and potential energies of a particle in SHM equal?
 (a) $A/2$ (b) $\sqrt{3} A/2$ (c) $A/4$ (d) $A/\sqrt{2}$
- NEET pattern** A particle of total energy E is in SHM. When its displacement is half the amplitude, its kinetic energy is
 (a) $3E/4$ (b) $E/2$ (c) $E/\sqrt{2}$ (d) $E/4$
- NEET pattern** If the amplitude of an oscillator is doubled (frequency unchanged), its total energy becomes
 (a) 2 times (b) unchanged (c) half (d) 4 times

- 10 **NEET pattern** The x - t graph of a particle in SHM is shown. At point P,



- (a) v is maximum, $a = 0$
 (b) $v = 0$, $a = 0$
 (c) v and a both maximum
 (d) $v = 0$, a is maximum (towards mean)
- 11 **NEET pattern** A mass m on a spring has period 2 s. When 3 kg is added to it, the period becomes 4 s. The value of m is
 (a) 0.5 kg (b) 2 kg (c) 3 kg (d) 1 kg
- 12 **NEET pattern** A spring of constant k is cut into two equal halves, which are then connected in parallel to a mass m . If the original period was T , the new period is
 (a) $T / 4$ (b) $2T$ (c) $T / \sqrt{2}$ (d) $T / 2$
- 13 **NEET pattern** A simple pendulum has period T in a stationary lift. If the lift accelerates upward with $g/3$, the period becomes
 (a) $\frac{3}{4}T$ (b) $\sqrt{\frac{2}{3}}T$ (c) $\frac{2}{\sqrt{3}}T$ (d) $\frac{\sqrt{3}}{2}T$
- 14 **NEET pattern** A simple pendulum is suspended in a freely falling lift. Its period of oscillation is
 (a) infinite (it does not oscillate)
 (b) same as on ground
 (c) half of that on ground
 (d) zero
- 15 **NCERT based** Taking $g = \pi^2 \text{ m/s}^2$, the length of a seconds pendulum is
 (a) 0.25 m (b) 2 m (c) 1 m (d) 0.5 m
- 16 **NEET pattern** If the length of a simple pendulum is increased by 21%, its period increases by
 (a) 10.5% (b) 21% (c) 42% (d) 10%
- 17 **JEE Main pattern** A particle moves as $x = 3\sin\omega t + 4\cos\omega t$ (in cm). Its amplitude is
 (a) 3.5 cm (b) 7 cm (c) 1 cm (d) 5 cm
- 18 **JEE Main pattern** The minimum time taken by a particle in SHM (period T) to go from the mean position to half the amplitude is
 (a) $T / 8$ (b) $T / 12$ (c) $T / 6$ (d) $T / 4$
- 19 **JEE Main pattern** A 1 kg block on a spring of constant 100 N/m oscillates with amplitude 10 cm. Its speed when it is 6 cm from the mean position is
 (a) 1.0 m/s (b) 0.6 m/s (c) 0.8 m/s (d) 0.4 m/s

- 20 **JEE Main pattern** A simple pendulum of period T hangs in a car moving with horizontal acceleration g . Its new period is
 (a) $T / 2^{1/4}$ (b) $T / \sqrt{2}$ (c) $\sqrt{2} T$ (d) T
- 21 **JEE Main pattern** A mass hung from a spring stretches it by 2.5 cm at equilibrium ($g = 10 \text{ m/s}^2$). The period of its vertical oscillations is
 (a) $\pi / 5 \text{ s}$ (b) $\pi / 10 \text{ s}$ (c) $\pi / 20 \text{ s}$ (d) $\pi \text{ s}$
- 22 **NEET pattern** The acceleration of a particle is $a = -16x$ (SI units). Its time period is
 (a) $\pi / 2 \text{ s}$ (b) $\pi / 4 \text{ s}$ (c) $\pi / 8 \text{ s}$ (d) $2\pi \text{ s}$
- 23 **Assertion-Reason** Assertion: Uniform circular motion is periodic but not oscillatory. Reason: In oscillatory motion a body moves to and fro about a mean position.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 24 **Assertion-Reason** Assertion: In SHM, kinetic and potential energies each oscillate with frequency 2ν . Reason: The total mechanical energy of an undamped oscillator is conserved.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

✓ Answer Key & Quick Solutions

- 1(c) 2(d) 3(c) 4(a) 5(c) 6(c) 7(d) 8(a) 9(d) 10(d) 11(d) 12(d) 13(d) 14(a) 15(c)
 16(d) 17(d) 18(b) 19(c) 20(a) 21(b) 22(a) 23(a) 24(b)

- (c) $\sin \omega t - \cos \omega t = \sqrt{2} \sin(\omega t - \pi/4)$ – a single sinusoid.
- (d) Angular frequency $= 2\omega \Rightarrow T = 2\pi/2\omega = \pi/\omega$.
- (c) $a = -\omega^2 x$: a is always opposite to $x \Rightarrow \pi$.
- (a) $x^2/A^2 + v^2/(\omega^2 A^2) = 1$ – an ellipse.
- (c) $\omega = 8/4 = 2 \text{ rad/s}$; $v = 2\sqrt{16-4} = 2\sqrt{12} = 4\sqrt{3} \text{ cm/s}$.
- (c) $\omega = 2\pi/T = 10 \text{ rad/s}$; $a_{\max} = \omega^2 A = 100 \times 0.05 = 5 \text{ m/s}^2$.
- (d) $\frac{1}{2}k(A^2 - x^2) = \frac{1}{2}kx^2 \Rightarrow x = A/\sqrt{2}$.
- (a) $U = E(x/A)^2 = E/4 \Rightarrow K = 3E/4$.
- (d) $E = \frac{1}{2}m\omega^2 A^2 \propto A^2 \Rightarrow 4E$.
- (d) P is a crest ($x = +A$, extreme): slope of $x-t = 0$, and $|a| = \omega^2 A$ directed towards the mean (negative).
- (d) $T \propto \sqrt{m} \Rightarrow (m+3)/m = 4 \Rightarrow m = 1 \text{ kg}$.
- (d) Each half: $2k$; parallel: $4k \Rightarrow T' = T/\sqrt{4} = T/2$.
- (d) $g_{\text{eff}} = 4g/3 \Rightarrow T' = T\sqrt{3/4} = (\sqrt{3}/2)T$.
- (a) $g_{\text{eff}} = 0 \Rightarrow T = 2\pi\sqrt{L/0} \rightarrow \infty$.
- (c) $T = 2 \text{ s} \Rightarrow L = gT^2/4\pi^2 = \pi^2 \cdot 4/4\pi^2 = 1 \text{ m}$.

16. (d) $T \propto \sqrt{L}$: $\sqrt{1.21} = 1.10 \Rightarrow 10\%$ (exact, not the $\frac{1}{2} \times 21\%$ approximation).

17. (d) $A = \sqrt{3^2 + 4^2} = 5 \text{ cm}$.

18. (b) $A/2 = A \sin \omega t \Rightarrow \omega t = \pi/6 \Rightarrow t = (\pi/6)(T/2\pi) = T/12$.

19. (c) $\omega = 10 \text{ rad/s}$; $v = 10\sqrt{(0.1^2 - 0.06^2)} = 10 \times 0.08 = 0.8 \text{ m/s}$.

20. (a) $g_{\text{eff}} = \sqrt{g^2 + g^2} = \sqrt{2} g \Rightarrow T' = T / (\sqrt{2})^{1/2} = T / 2^{1/4}$.

21. (b) $T = 2\pi\sqrt{\Delta l / g} = 2\pi\sqrt{0.0025} = 2\pi(0.05) = \pi / 10 \text{ s}$.

22. (a) $\omega^2 = 16 \Rightarrow \omega = 4 \text{ rad/s} \Rightarrow T = 2\pi/4 = \pi/2 \text{ s}$.

23. (a) Both true; circular motion has no to-and-fro motion about a mean position, which is exactly why it is not oscillatory.

24. (b) Both true, but the frequency 2ν comes from $\sin^2$ and $\cos^2$ terms, not from energy conservation.

CHAPTER

14

Waves

Day 17

Day 23–24

NEET weightage ★★☆☆

NCERT Class 11

Transverse & longitudinal

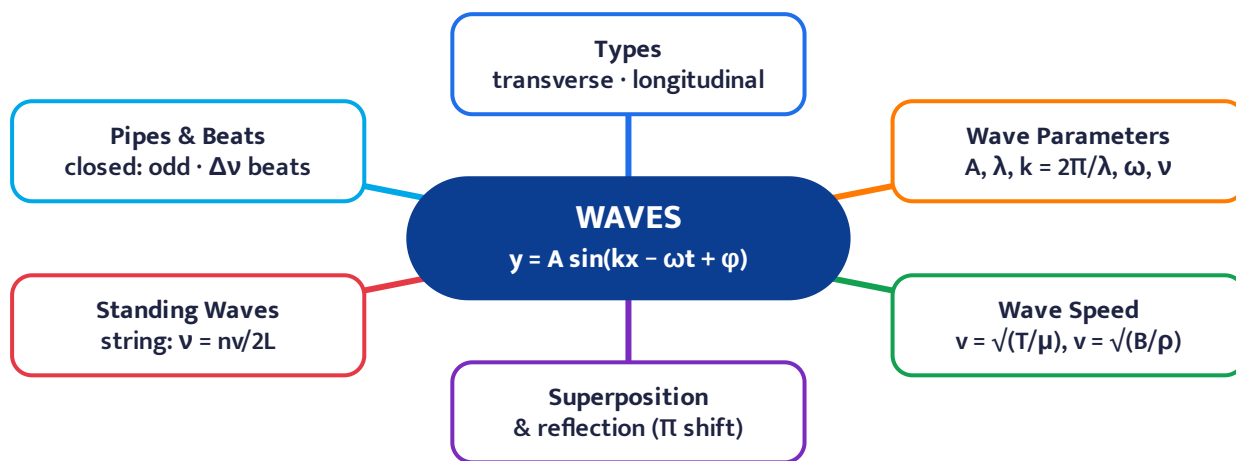
Wave equation

Speed of waves

Superposition & reflection

Standing waves

Organ pipes & beats



Chapter at a glance

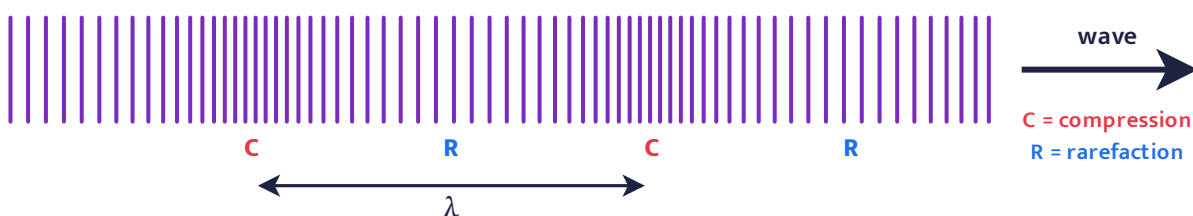
14.1 Waves and Their Types

A **wave** is a disturbance that travels through a medium, carrying **energy and momentum without any net transport of matter**. Each particle of the medium only oscillates about its mean position and passes the disturbance on to the next. **Mechanical waves** need a material medium (sound, waves on strings); electromagnetic waves (light) do not.

Transverse wave (string): particles move $\perp$ to the wave



Longitudinal wave (air / spring): particles move $\parallel$ to the wave



Transverse wave on a string and longitudinal (sound) wave in air

NCERT Line to Remember

Fluids (liquids and gases) cannot sustain shearing stress, so **transverse waves cannot travel inside a fluid**; sound in air is purely **longitudinal**. In a solid both kinds can travel (earthquakes send S-waves and P-waves).

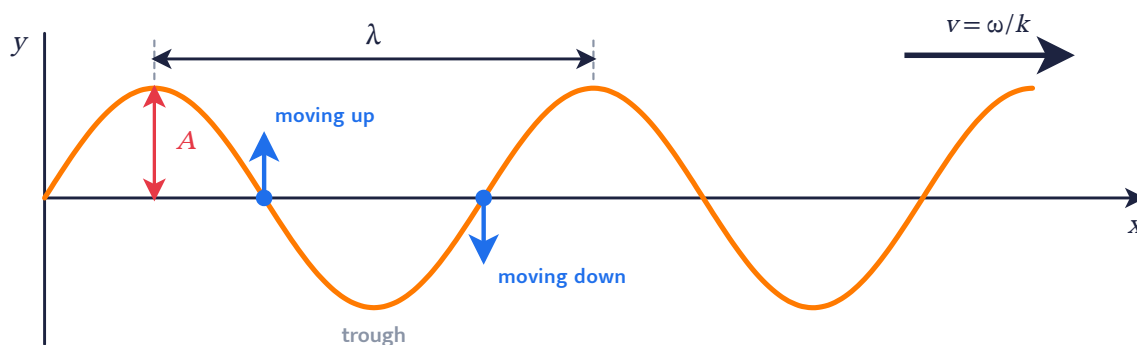
Feature	Transverse wave	Longitudinal wave
Particle motion	perpendicular to propagation	along the propagation
Pattern	crests and troughs	compressions and rarefactions
Needs	shear (rigidity) modulus	bulk modulus (compressibility)
Travels in	solids, strings, liquid surface – not inside gases	solids, liquids and gases
Examples	wave on a string, ripples	sound, spring (slinky) pulses

14.2 Displacement Relation in a Progressive Wave

A sinusoidal wave travelling in the **+x direction** is described by

$$y(x, t) = A \sin(kx - \omega t + \phi)$$

For a wave going in the **-x direction**, write $(kx + \omega t + \phi)$. Rule: **opposite signs** of kx and $\omega t \rightarrow +x$; **same signs** $\rightarrow -x$.



Snapshot of a wave moving to the right: A , λ and the particle velocities

Wave parameters

A	Amplitude – maximum displacement of a particle
$(kx - \omega t + \phi)$	Phase ; ϕ = initial phase (phase constant)
$k = \frac{2\pi}{\lambda}$	Angular wave number (rad/m); λ = wavelength
$\omega = \frac{2\pi}{T} = 2\pi\nu$	Angular frequency ; ν = frequency, T = period
$v = \frac{\omega}{k} = \nu\lambda = \frac{\lambda}{T}$	Wave speed – the wave moves one λ in one T
$\Delta\phi = \frac{2\pi}{\lambda} \Delta x$	Phase difference between two points Δx apart
$v_p = \frac{\partial y}{\partial t}$, $(v_p)_{\max} = A\omega$	Particle velocity ($\neq$ wave speed v)

Solved Example

A wave on a string is $y = 0.005 \sin(80x - 3t)$ (SI units). Find the amplitude, wavelength, period, frequency and wave speed.

Solution: $A = 5 \text{ mm}$; $\lambda = 2\pi / 80 \approx 7.85 \text{ cm}$; $T = 2\pi / 3 \approx 2.09 \text{ s}$; $\nu = 1 / T \approx 0.48 \text{ Hz}$; $v = \omega / k = 3 / 80 \approx 0.0375 \text{ m/s}$ (3.75 cm/s), wave moving along $+x$.

! Common Trap

- **Wave speed** $v = \omega / k$ is constant for a medium; **particle speed** $\omega \sqrt{A^2 - y^2}$ keeps changing. Don't mix them.
- Frequency is set by the **source** and does not change when a wave enters another medium; v and λ change.

🎯 VS Sir's Exam Tip

- Read speed directly: $v = (\text{coefficient of } t) / (\text{coefficient of } x)$.
- Wave moving right: points just **ahead of a crest** move **up**, points behind it move down.

14.3 Speed of a Travelling Wave

Wave speed depends only on the **properties of the medium** – an elastic (restoring) property and an inertial (mass) property: $v = \sqrt{\text{elastic property} / \text{inertial property}}$.

🔧 Speed of waves

$$v = \sqrt{\frac{T}{\mu}}$$

Transverse wave on a stretched string; T = tension, μ = mass per unit length

$$v = \sqrt{\frac{B}{\rho}}$$

Longitudinal (sound) wave in a fluid; B = bulk modulus

$$v = \sqrt{\frac{Y}{\rho}}$$

Longitudinal wave in a thin solid rod

$$v = \sqrt{\frac{P}{\rho}} \approx 280 \text{ m/s}$$

Newton's formula (isothermal, $B = P$) – too low

$$v = \sqrt{\frac{\gamma P}{\rho}} \approx 331 \text{ m/s}$$

Laplace correction (adiabatic, $B = \gamma P$) – matches experiment

$$v \propto \sqrt{T_{abs}} \Rightarrow \frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}}$$

Gas: $v = \sqrt{\gamma RT / M}$ – depends on absolute temperature

📘 Why Laplace was right

Compressions and rarefactions in sound are so fast that **no heat flows** – the process is **adiabatic**, not isothermal. For air $\gamma = 1.4$, raising Newton's 280 m/s to about 331 m/s at STP.

💡 What changes v of sound in a gas?

- **Pressure** (at constant T): **no effect** – P/ρ stays constant.
- **Temperature**: $v \propto \sqrt{T}$ ($\approx +0.6$ m/s per $^\circ\text{C}$ near room temperature).
- **Humidity**: moist air is less dense $\rightarrow v$ **increases**.
- $v_{solid} > v_{liquid} > v_{gas}$ (steel ≈ 5900 , water ≈ 1500 , air ≈ 340 m/s).

🔧 Solved Example

A steel wire 0.72 m long has mass 5.0×10^{-3} kg. Under a tension of 60 N, find the speed of transverse waves on it.

Solution: $\mu = 5 \times 10^{-3} / 0.72 = 6.9 \times 10^{-3}$ kg/m. $v = \sqrt{T/\mu} = \sqrt{60 / 6.9 \times 10^{-3}} \approx \mathbf{93 \text{ m/s}}$.

🔧 Solved Example

Estimate the speed of sound in air at STP by Newton's formula and by Laplace's formula ($P = 1.01 \times 10^5$ Pa, $\rho = 1.29$ kg/m³, $\gamma = 1.4$).

Solution: Newton: $\sqrt{1.01 \times 10^5 / 1.29} \approx \mathbf{280 \text{ m/s}}$ (about 15% low). Laplace: $280 \times \sqrt{1.4} \approx \mathbf{331 \text{ m/s}}$ ✓.

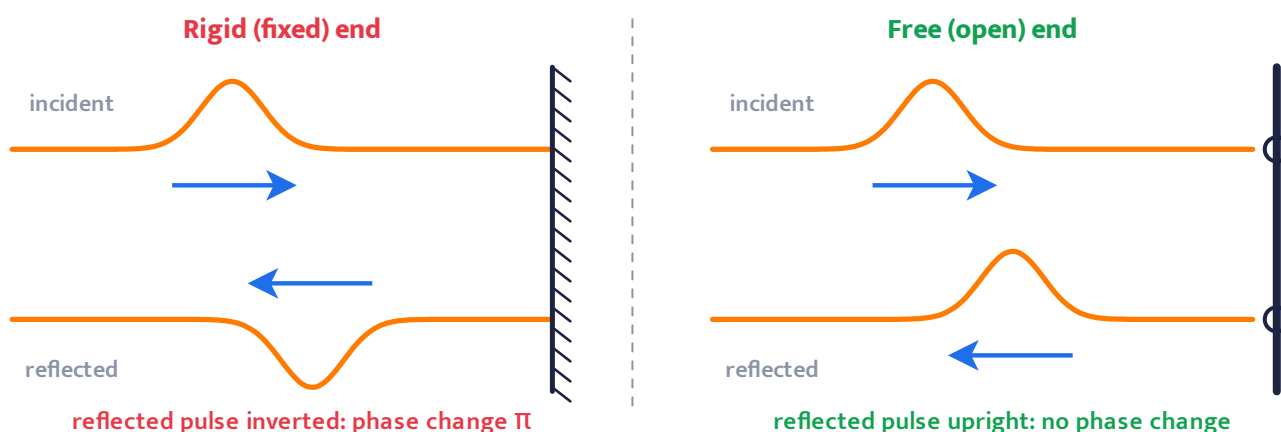
14.4 Superposition and Reflection of Waves

Principle of superposition: when two or more waves overlap, the net displacement is the **algebraic sum** of the individual displacements, $y = y_1 + y_2 + \dots$; after crossing, each wave continues unchanged. For two waves of equal A and ω differing in phase by ϕ :

$$y = \left[2A \cos \frac{\phi}{2} \right] \sin \left(kx - \omega t + \frac{\phi}{2} \right)$$

$\phi = 0 \rightarrow$ amplitude **2A** (constructive interference); $\phi = \pi \rightarrow$ amplitude **0** (destructive interference). In general

$$A_R = \sqrt{A_1^2 + A_2^2 + 2A_1A_2\cos\phi}.$$



Reflection of a pulse at a rigid end (inverted) and at a free end (upright)

NCERT Line to Remember

- At a **rigid boundary** (fixed end, denser medium) the reflected wave has a **phase change of π** (180°): incident $A\sin(kx - \omega t) \rightarrow$ reflected $-A\sin(kx + \omega t)$.
- At an **open boundary** (free end, rarer medium) reflection takes place **without phase change**.

14.5 Standing Waves and Normal Modes

Two identical waves travelling in **opposite directions** superpose to give a **standing (stationary) wave** – no energy flows along it and the waveform does not move:

$$y = 2A \sin kx \cos \omega t$$

Every point oscillates with frequency ω but with amplitude $2A \sin kx$ that depends on position. **Nodes** (zero amplitude) are at $x = 0, \frac{\lambda}{2}, \lambda, \dots$; **antinodes** (amplitude $2A$) at $x = \frac{\lambda}{4}, \frac{3\lambda}{4}, \dots$. Adjacent nodes are $\lambda/2$ apart; a node and the next antinode are $\lambda/4$ apart. All points between two nodes vibrate **in phase**; points in adjacent loops are opposite in phase.

Vibrating string (sonometer)

$$v_n = \frac{nv}{2L} = \frac{n}{2L} \sqrt{\frac{T}{\mu}}, \quad n = 1, 2, 3, \dots$$

String fixed at both ends – **all harmonics** present

$$\lambda_n = \frac{2L}{n}$$

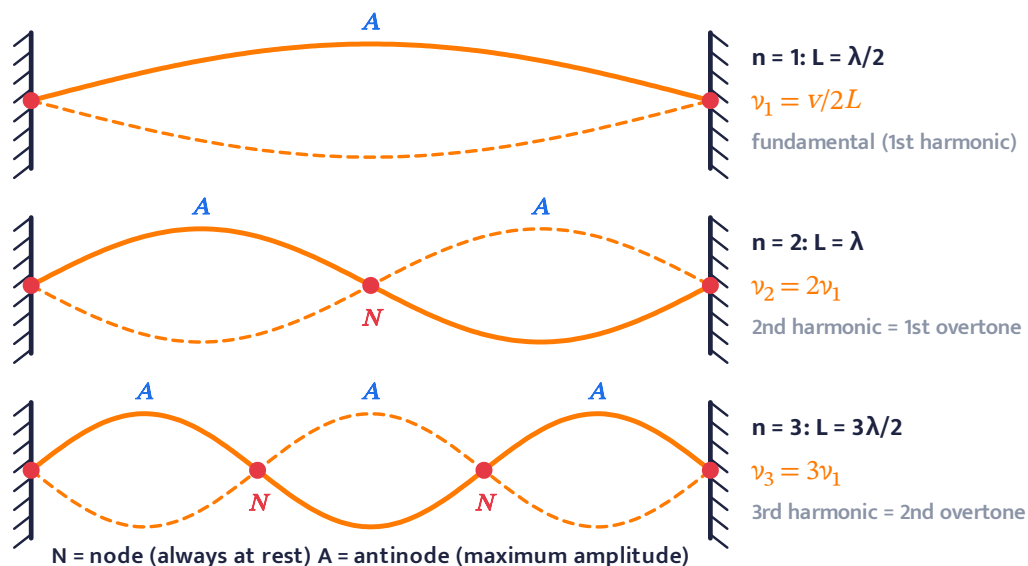
Allowed wavelengths; n = number of loops

! Common Trap

- **Overtone $\neq$ harmonic.** For a string/open pipe, p -th overtone = $(p + 1)$ -th harmonic. For a closed pipe, p -th overtone = $(2p + 1)$ -th harmonic.
- Distance between two consecutive nodes is $\lambda/2$, **not** λ .

🧠 Memory Trick

- Sonometer laws: $v \propto \frac{1}{L}$, $v \propto \sqrt{T}$, $v \propto \frac{1}{\sqrt{\mu}}$.
- **Loops = harmonic number** on a string.



Normal modes of a string fixed at both ends – both ends are nodes, so $L = n\lambda / 2$

🔧 Solved Example

A 1 m long string fixed at both ends has a mass of 10 g and tension 400 N. Find its fundamental frequency and the frequency of the 2nd overtone.

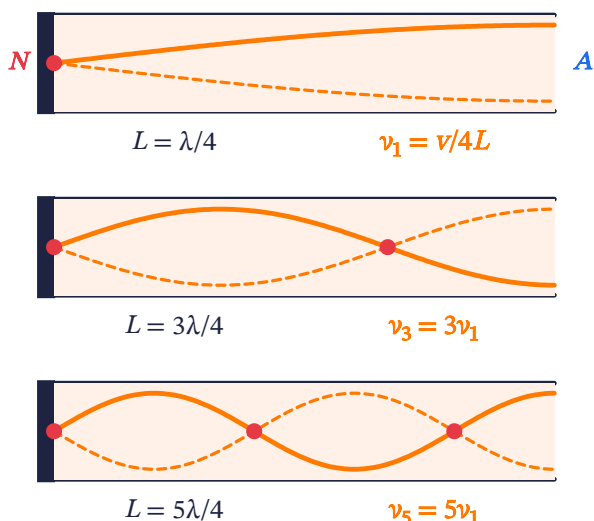
Solution: $\mu = 0.01 \text{ kg/m}$; $v = \sqrt{400/0.01} = 200 \text{ m/s}$; $v_1 = v / 2L = 100 \text{ Hz}$. 2nd overtone = 3rd harmonic = **300 Hz**.

14.6 Organ Pipes – Closed and Open

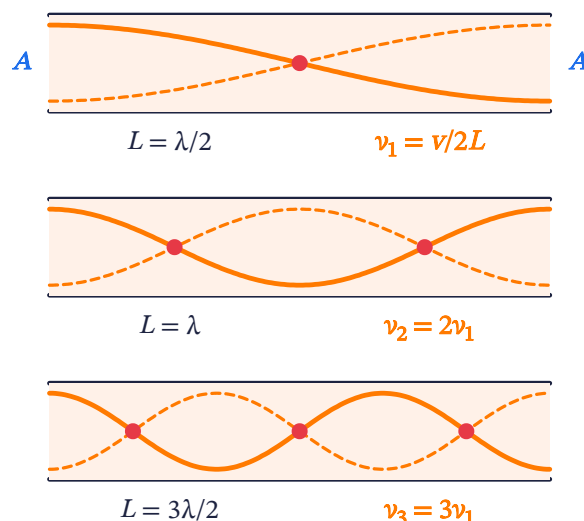
An air column also supports standing (longitudinal) waves. A **closed end** is a **displacement node** (air cannot move); an **open end** is a **displacement antinode**.

System	Allowed frequencies	Harmonics present	Ratio
String, both ends fixed	$v = \frac{nv}{2L}$	all	1 : 2 : 3 : ...
Open pipe (both ends open)	$v = \frac{nv}{2L}$	all	1 : 2 : 3 : ...
Closed pipe (one end closed)	$v = \frac{(2n-1)v}{4L}$	odd only	1 : 3 : 5 : ...

Closed pipe (one end closed)



Open pipe (both ends open)



First three modes: closed pipe (only odd harmonics) and open pipe (all harmonics). Red dots = displacement nodes

Solved Example

A pipe 30 cm long is open at both ends. Which harmonic resonates with a 1.1 kHz source? Will it resonate if one end is closed? ($v = 330$ m/s)

Solution: Open: $\nu_1 = 330 / (2 \times 0.3) = 550$ Hz, so 1100 Hz = **2nd harmonic**. Closed: $\nu_1 = 330 / (4 \times 0.3) = 275$ Hz; allowed 275, 825, 1375 Hz... 1100 Hz is an **even** multiple (4×275), so **no resonance**.

VS Sir's Exam Tip

- Same length: $\nu_{open} = 2 \nu_{closed}$ (fundamentals).
- Closing one end of an open pipe **halves** its fundamental.
- An open pipe of length L has the same fundamental as a closed pipe of length $L/2$.

JEE Main Extension

End correction: the antinode forms slightly outside an open end, $e \approx 0.6r$. Closed pipe: $\nu_1 = \frac{v}{4(L + 0.6r)}$; open pipe: $\nu_1 = \frac{v}{2(L + 1.2r)}$. In a resonance tube, $\lambda = 2(l_2 - l_1)$ removes e .

14.7 Beats

When two sound waves of **slightly different frequencies** ν_1 and ν_2 (and nearly equal amplitudes) superpose, the loudness waxes and wanes periodically – this is **beats**. The resultant is $s = [2a \cos \pi(\nu_1 - \nu_2)t] \cos 2\pi \bar{\nu} t$, a wave of average frequency $\bar{\nu} = \frac{\nu_1 + \nu_2}{2}$ whose amplitude varies slowly.

Beats

$$\nu_{beat} = |\nu_1 - \nu_2| \quad \text{Beat frequency (beats per second)}$$

$$T_{beat} = \frac{1}{|\nu_1 - \nu_2|}$$

Time between two successive maxima

Common Trap

- **Loading (wax)** a fork lowers ν ; **filig** its prongs raises ν . Increasing tension in a string raises ν .
- The ear detects beats only if $|\nu_1 - \nu_2| \lesssim 10$ Hz.

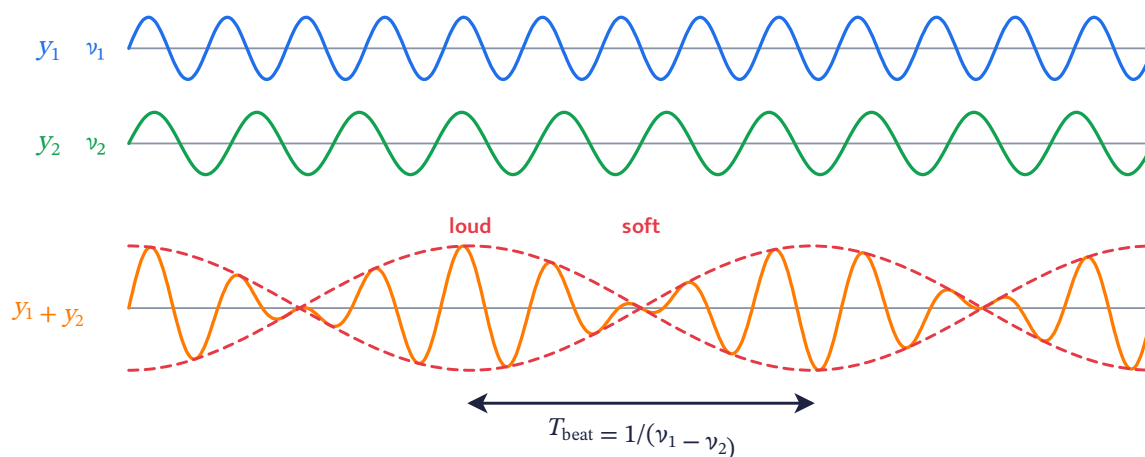
Doppler effect – JEE Main only

Not in the NEET-UG syllabus. Apparent frequency $\nu' = \nu \frac{v \pm v_o}{v \mp v_s}$ (upper signs: source and observer approaching each other).

Solved Example

A tuning fork A of 256 Hz gives 4 beats/s with fork B. On loading B with a little wax, the beat frequency increases to 6. Find the original frequency of B.

Solution: $B = 256 \pm 4 = 260$ or 252 Hz. Wax **lowers** B's frequency. If $B = 260$, lowering it would first **reduce** the beats; since beats increased, $B = 252$ Hz.



Beats: superposition of two close frequencies gives a slowly varying envelope (red, dashed)

Practice Questions

Pattern-based questions modelled on NEET and JEE Main previous-year trends. Answers on the next page.

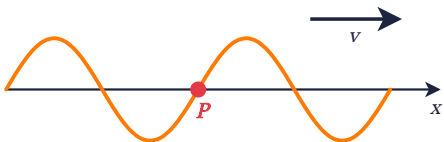
- 1 **NCERT based** Sound waves in air are
 - (a) electromagnetic
 - (b) transverse
 - (c) partly transverse, partly longitudinal
 - (d) longitudinal
- 2 **NCERT based** Transverse mechanical waves can NOT travel through

(a) a steel rod	(b) a stretched string
(c) the surface of water	(d) a gas
- 3 **NEET pattern** The equation of a wave is $y = 0.05\sin(20\pi t - 0.5\pi x)$ (SI units). The speed of the wave is

(a) 0.025 m/s	(b) 40 m/s	(c) 10 m/s	(d) 20 m/s
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- 4 **NEET pattern** For the wave $y = 0.02\sin(\frac{\pi}{2}x - 100\pi t)$ (SI units), the wavelength and frequency are

(a) 0.25 m, 50 Hz	(b) 4 m, 50 Hz
(c) 2 m, 50 Hz	(d) 4 m, 100 Hz
- 5 **NEET pattern** A transverse wave travels at 100 m/s on a stretched string. If the tension is made four times, the new speed is

(a) 400 m/s	(b) 141 m/s	(c) 200 m/s	(d) 50 m/s
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- 6 **NCERT based** A string of mass 2.5 kg and length 20 m is under a tension of 200 N. A pulse produced at one end reaches the other end in
 (a) 1 s (b) 0.25 s (c) 2 s (d) 0.5 s
- 7 **NCERT based** Laplace corrected Newton's formula for the speed of sound by assuming that sound propagation in air is
 (a) adiabatic (b) isobaric (c) isochoric (d) isothermal
- 8 **NEET pattern** The speed of sound in a gas is 360 m/s at 27 °C. At what temperature will it be 720 m/s?
 (a) 54 °C (b) 927 °C (c) 1200 °C (d) 108 °C
- 9 **NCERT based** If the pressure of air is doubled at constant temperature, the speed of sound in it
 (a) remains unchanged (b) becomes half
 (c) doubles (d) becomes $\sqrt{2}$ times
- 10 **NEET pattern** A sound wave of frequency 660 Hz travels in air at 330 m/s. The phase difference between two points 12.5 cm apart along the wave is
 (a) $\pi / 4$ (b) $\pi / 2$ (c) 2π (d) π
- 11 **NCERT based** When a wave on a string is reflected from a rigid (fixed) end, the phase change on reflection is
 (a) 2π (b) 0 (c) π (d) $\pi / 2$
- 12 **NEET pattern** The figure shows a snapshot of a transverse wave travelling in the +x direction. At this instant the particle at P is

 (a) moving along +x
 (b) at rest
 (c) moving downward with maximum speed
 (d) moving upward with maximum speed
- 13 **NEET pattern** A string 50 cm long is fixed at both ends. Waves travel on it at 100 m/s. The frequency of its second harmonic is
 (a) 50 Hz (b) 100 Hz (c) 200 Hz (d) 400 Hz
- 14 **NEET pattern** The fundamental frequency of a closed organ pipe is 100 Hz. Which of the following frequencies can it NOT produce?
 (a) 200 Hz (b) 700 Hz (c) 300 Hz (d) 500 Hz
- 15 **NEET pattern** An open pipe and a closed pipe have the same length. The ratio of their fundamental frequencies (open : closed) is
 (a) 2 : 1 (b) 1 : 2 (c) 1 : 1 (d) 4 : 1

- 16 **NEET pattern** The fundamental frequency of an open organ pipe is 300 Hz. If one end is closed, the fundamental frequency becomes
 (a) 900 Hz (b) 600 Hz (c) 300 Hz (d) 150 Hz
- 17 **NEET pattern** A tuning fork of 256 Hz produces 4 beats/s with another fork. When the second fork is loaded with wax, the beat frequency increases. The original frequency of the second fork is
 (a) 248 Hz (b) 260 Hz (c) 252 Hz (d) 256 Hz
- 18 **JEE Main pattern** A string of length 1.5 m fixed at both ends vibrates in 3 loops at 450 Hz. The speed of the transverse wave on it is
 (a) 450 m/s (b) 225 m/s (c) 900 m/s (d) 300 m/s
- 19 **JEE Main pattern** Two waves of the same frequency and amplitude A superpose with a phase difference of $2\pi/3$. The amplitude of the resultant wave is
 (a) $\sqrt{3}A$ (b) 0 (c) 2A (d) A
- 20 **JEE Main pattern** In a sonometer, the length of the wire is halved and the tension is made four times. The fundamental frequency becomes
 (a) 2 times (b) unchanged (c) 8 times (d) 4 times
- 21 **JEE Main pattern** A pipe 25 cm long is closed at one end ($v = 340$ m/s). The frequency of its second overtone is
 (a) 680 Hz (b) 1700 Hz (c) 1020 Hz (d) 1360 Hz
- 22 **JEE Main pattern** Two sound waves of wavelengths 1.00 m and 1.01 m in a gas produce 3 beats per second. The speed of sound in the gas is
 (a) 330 m/s (b) 336 m/s (c) 303 m/s (d) 300 m/s
- 23 **Assertion–Reason** Assertion: The speed of sound in humid air is greater than in dry air (same temperature and pressure). Reason: The density of moist air is less than that of dry air.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 24 **Assertion–Reason** Assertion: Transverse waves can travel through air. Reason: Air cannot sustain a shearing stress.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false

✓ Answer Key & Quick Solutions

- 1(d) 2(d) 3(b) 4(b) 5(c) 6(d) 7(a) 8(b) 9(a) 10(b) 11(c) 12(c) 13(c) 14(a) 15(a)
 16(d) 17(c) 18(a) 19(d) 20(d) 21(b) 22(c) 23(a) 24(c)

1. (d) Air cannot sustain shear, so sound in air consists only of compressions and rarefactions.

2. (d) Gases have no shear modulus, so they cannot carry transverse waves.
3. (b) $v = \omega / k = 20\pi / 0.5\pi = 40 \text{ m/s}$.
4. (b) $\lambda = 2\pi/k = 2\pi/(\pi/2) = 4 \text{ m}$; $v = \omega/2\pi = 100\pi/2\pi = 50 \text{ Hz}$.
5. (c) $v \propto \sqrt{T} \Rightarrow v' = 100 \times \sqrt{4} = 200 \text{ m/s}$.
6. (d) $\mu = 2.5/20 = 0.125 \text{ kg/m}$; $v = \sqrt{(200/0.125)} = 40 \text{ m/s}$; $t = 20/40 = 0.5 \text{ s}$.
7. (a) Compressions/rarefactions are too fast for heat exchange $\Rightarrow$ adiabatic, $B = \gamma P$.
8. (b) $v \propto \sqrt{T} : T = 300 \times 4 = 1200 \text{ K} = 927^\circ\text{C}$.
9. (a) At constant T , $\rho \propto P$, so $\sqrt{\gamma P / \rho}$ is unchanged.
10. (b) $\lambda = 330/660 = 0.5 \text{ m}$; $\Delta\phi = 2\pi(0.125)/0.5 = \pi/2$.
11. (c) Rigid boundary $\Rightarrow$ reflected wave is inverted $\Rightarrow$ phase change π .
12. (c) P is at the mean position just ahead of a trough (behind a crest), so it moves down with maximum speed $A\omega$;
 $v_p = -v \partial y / \partial x < 0$ as the slope at P is positive.
13. (c) $\nu_2 = 2v / 2L = 2 \times 100 / (2 \times 0.5) = 200 \text{ Hz}$.
14. (a) Closed pipe gives only odd harmonics: 100, 300, 500, 700 ... 200 Hz is an even harmonic.
15. (a) $v/2L : v/4L = 2 : 1$.
16. (d) $v/4L = \frac{1}{2}(v/2L) = 150 \text{ Hz}$.
17. (c) Wax lowers v ; beats rise only if the fork was already below 256: $256 - 4 = 252 \text{ Hz}$.
18. (a) Each loop $= \lambda/2 = 0.5 \text{ m} \Rightarrow \lambda = 1 \text{ m}$; $v = v\lambda = 450 \text{ m/s}$.
19. (d) $A_R = 2A\cos(\phi/2) = 2A\cos 60^\circ = A$.
20. (d) $\nu \propto \sqrt{T} / L = \sqrt{4/(1/2)} = 4$ times.
21. (b) $\nu_1 = 340 / (4 \times 0.25) = 340 \text{ Hz}$; 2nd overtone = 5th harmonic = 1700 Hz.
22. (c) $v(\frac{1}{1.00} - \frac{1}{1.01}) = 3 \Rightarrow v = 3 \times 101 = 303 \text{ m/s}$.
23. (a) Water vapour ($M = 18$) is lighter than air ($M \approx 29$), so ρ falls and $v = \sqrt{\gamma P / \rho}$ rises; R explains A.
24. (c) Because air has no shear modulus, transverse waves cannot travel in it: A is false, R is true.

DAY

27

NEET + JEE PHYSICS • CLASS 11 • FULL SYLLABUS

Full Mock Test

45 questions

60 minutes

+4 correct • -1 wrong • 0 unattempted

One sitting – no breaks

Fill the score table after checking

Sit at a table with a timer, a pencil and rough sheets only – no book, no phone. Mark answers on a separate sheet (like an OMR). After 60 minutes, stop, check with the key, and fill the score table.

Before you start

Instructions

- 45 questions from all 14 chapters of Class 11, in chapter order – weightage like NEET (mechanics heavy).
- Time: **60 minutes** in one sitting. Start a timer now.
- Marking: **+4** correct, **-1** wrong, **0** unattempted.
- Do not look at the answer key until the timer stops.

VS Sir's Exam Tip

- Round 1: attempt every question you can do in < 1 min.
- Round 2: return to the numericals you skipped.
- Round 3: tough ones – guess only if you can eliminate 2 options.

Score table (fill after checking)

Chapter group	Qs	Correct	Wrong	Score = 4C – W
Ch 1–3: Units, Kinematics	9 (Q1–9)			
Ch 4–5: Laws of motion, Work–energy	8 (Q10–17)			
Ch 6–7: Rotation, Gravitation	8 (Q18–25)			
Ch 8–10: Solids, Fluids, Thermal properties	8 (Q26–33)			
Ch 11–12: Thermodynamics, Kinetic theory	6 (Q34–39)			
Ch 13–14: Oscillations, Waves	6 (Q40–45)			
TOTAL (max 180)	45			

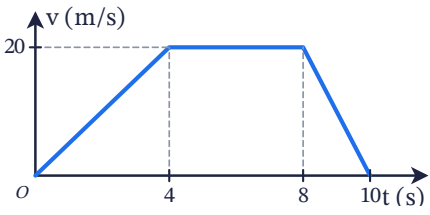
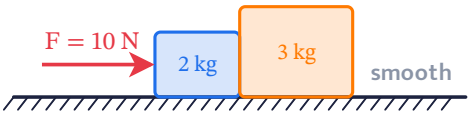
Time taken: _____ min • Attempted: _____ / 45 • Accuracy = Correct ÷ Attempted = _____ %

Mock Test – 45 Questions

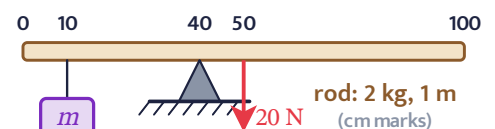
Original NEET-pattern questions modelled on previous-year trends. Timer ON: 60 minutes. Answers after Q45.

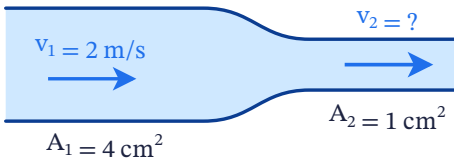
- 1 **NCERT based** If h is Planck's constant, m a mass and c the speed of light, the quantity $\frac{h}{mc}$ has the dimensions of
 (a) length (b) frequency (c) velocity (d) time

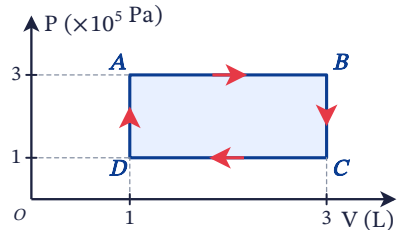
- 2 **NEET pattern** The length and breadth of a plate are (5.0 ± 0.1) cm and (2.0 ± 0.1) cm. Its area is
 (a) (10.0 ± 0.7) cm² (b) (10.0 ± 0.1) cm² (c) (10.0 ± 0.5) cm² (d) (10.0 ± 0.2) cm²

- 3 **NEET pattern** A ball is dropped from a height of 80 m ($g = 10 \text{ m/s}^2$). The distance it covers in the **last second** of its fall is
 (a) 40 m (b) 35 m (c) 45 m (d) 25 m
- 4 **NEET pattern** The velocity–time graph of a car is shown. Its average speed over the 10 s is
 (a) 12 m/s (b) 20 m/s (c) 14 m/s (d) 10 m/s
- 
- 5 **JEE Main pattern** The position of a particle is $x = t^3 - 6t^2 + 9t$ (x in m, t in s). The **distance** travelled in the first 4 s is
 (a) 8 m (b) 16 m (c) 4 m (d) 12 m
- 6 **NCERT based** For a projectile, the maximum height equals the horizontal range when the angle of projection is
 (a) $\tan^{-1}(4)$ (b) 45° (c) $\tan^{-1}(1/4)$ (d) $\tan^{-1}(2)$
- 7 **NEET pattern** A ball is projected from level ground at 20 m/s, 30° above the horizontal. The change in its velocity between projection and landing is
 (a) $10\sqrt{3}$ m/s, horizontal (b) 10 m/s, vertically downward
 (c) zero (d) 20 m/s, vertically downward
- 8 **NEET pattern** Rain falls vertically at 30 m/s. A cyclist rides east at 10 m/s. To protect himself he should hold his umbrella tilted
 (a) vertically
 (b) at $\tan^{-1}(1/3)$ from vertical, towards east
 (c) at $\tan^{-1}(1/3)$ from vertical, towards west
 (d) at $\tan^{-1}(3)$ from vertical, towards east
- 9 **NEET pattern** A particle moves on a circle of radius 2 m with its speed increasing at 3 m/s^2 . When its speed is 4 m/s, the magnitude of its acceleration is
 (a) 11 m/s^2 (b) 8 m/s^2 (c) $\sqrt{73} \text{ m/s}^2$ (d) 5 m/s^2
- 10 **NCERT based** Two blocks on a smooth floor are pushed by a 10 N force as shown. The force exerted by the 2 kg block on the 3 kg block is
 (a) 6 N (b) 4 N (c) 5 N (d) 10 N
- 
- 11 **NEET pattern** Masses of 3 kg and 5 kg hang from the ends of a light string over a smooth, light pulley. The tension in the string is ($g = 10 \text{ m/s}^2$)
 (a) 40 N (b) 37.5 N (c) 30 N (d) 50 N
- 12 **NEET pattern** A 5 kg block rests on a floor with $\mu_s = 0.4$ and $\mu_k = 0.25$. A horizontal force of 15 N acts on it. The friction force on the block is ($g = 10 \text{ m/s}^2$)
 (a) 12.5 N (b) 20 N (c) zero (d) 15 N

- 13 **NEET pattern** A 0.5 kg ball hits a wall normally at 10 m/s and rebounds at 8 m/s. If contact lasts 0.1 s, the average force on the wall is
(a) 9 N (b) 10 N (c) 80 N (d) 90 N
- 14 **NCERT based** A force $F = (2x + 3)$ N acts along the x-axis. The work done as the body moves from $x = 1$ m to $x = 3$ m is
(a) 12 J (b) 18 J (c) 10 J (d) 14 J
- 15 **NEET pattern** If the linear momentum of a body increases by 20%, its kinetic energy increases by
(a) 40% (b) 21% (c) 44% (d) 20%
- 16 **JEE Main pattern** A 1 kg ball moving at 6 m/s collides head-on **elastically** with a 2 kg ball at rest. The fraction of kinetic energy transferred to the 2 kg ball is
(a) $2/3$ (b) $4/9$ (c) $8/9$ (d) $1/9$
- 17 **NEET pattern** A 0.2 kg stone tied to a string is whirled in a vertical circle. The difference between the maximum and minimum tension in the string is ($g = 10 \text{ m/s}^2$)
(a) 6 N (b) 4 N (c) 12 N (d) 2 N
- 18 **NCERT based** Masses 1 kg, 2 kg and 3 kg are placed at (0, 0), (2 m, 0) and (0, 3 m). The centre of mass is at
(a) $(2/3 \text{ m}, 1 \text{ m})$ (b) $(1 \text{ m}, 1 \text{ m})$
(c) $(2/3 \text{ m}, 3/2 \text{ m})$ (d) $(3/2 \text{ m}, 2/3 \text{ m})$
- 19 **NEET pattern** The moment of inertia of a uniform disc (mass M , radius R) about a tangent **in its plane** is
(a) $\frac{5}{4}MR^2$ (b) $\frac{3}{4}MR^2$ (c) $\frac{3}{2}MR^2$ (d) $\frac{1}{4}MR^2$
- 20 **NEET pattern** A solid sphere rolls without slipping. The ratio of its rotational kinetic energy to its total kinetic energy is
(a) $2/5$ (b) $2/7$ (c) $5/7$ (d) $1/2$
- 21 **NCERT based** A uniform 2 kg metre rod rests on a knife-edge at the 40 cm mark. What mass hung at the 10 cm mark keeps it horizontal?
(a) 0.5 kg (b) 0.67 kg (c) 1.5 kg (d) 1 kg
- 22 **JEE Main pattern** A skater spinning freely pulls her arms in, reducing her moment of inertia by 40%. Her rotational kinetic energy becomes
(a) $5/3$ times (b) unchanged (c) $3/5$ times (d) $25/9$ times
- 23 **NEET pattern** The ratio of g at a depth $R/2$ below Earth's surface to g at a height $R/2$ above it (R = radius of Earth) is
(a) $9/8$ (b) 1 (c) $8/9$ (d) $3/2$
- 24 **NCERT based** A satellite in a circular orbit of radius r has a period of 2 h. Another satellite of the same planet in an orbit of radius $4r$ has a period of
(a) 32 h (b) 4 h (c) 16 h (d) 8 h



- 25** **Assertion–Reason** **Assertion (A):** The escape speed of a body from Earth depends on the angle at which it is projected.
Reason (R): The escape speed of a body depends on its mass.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 26** **NCERT based** A steel wire 2 m long with cross-section 1 mm^2 carries a load of 10 kg. If $Y = 2 \times 10^{11} \text{ N/m}^2$, its extension is ($g = 10 \text{ m/s}^2$)
 (a) 1 mm (b) 10 mm (c) 2 mm (d) 0.1 mm
- 27** **NEET pattern** Two wires of the same material have lengths L and $2L$ and radii $2r$ and r . Under the same force, the ratio of their extensions is
 (a) 1 : 2 (b) 8 : 1 (c) 1 : 4 (d) 1 : 8
- 28** **NEET pattern** Two identical raindrops each fall with terminal velocity 5 cm/s. If they coalesce into one drop, the new terminal velocity is
 (a) $\approx 6.3 \text{ cm/s}$ (b) $\approx 7.9 \text{ cm/s}$ (c) 10 cm/s (d) $\approx 7.1 \text{ cm/s}$
- 29** **NEET pattern** Water flows through the horizontal pipe shown. The pressure difference between the wide and narrow sections is ($\rho = 1000 \text{ kg/m}^3$)
 (a) 30 kPa (b) 6 kPa (c) 15 kPa (d) 32 kPa
- 
- 30** **Assertion–Reason** **Assertion (A):** Water rises in a clean glass capillary, but mercury is depressed in it.
Reason (R): The angle of contact of water with glass is acute, while that of mercury with glass is obtuse.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 31** **NCERT based** A metal has linear expansivity $2 \times 10^{-5} \text{ K}^{-1}$. When heated through 50 K, its volume increases by
 (a) 0.2% (b) 0.3% (c) 0.03% (d) 0.1%
- 32** **NEET pattern** 100 g of ice at 0°C is mixed with 100 g of water at 60°C in an insulated vessel. The mass of ice left unmelted is ($L_f = 80 \text{ cal/g}$, $c_w = 1 \text{ cal/g } ^\circ \text{C}$)
 (a) 50 g (b) zero (c) 25 g (d) 75 g
- 33** **NEET pattern** The temperature of a black body is raised from 27°C to 327°C . The power it radiates becomes
 (a) 16 times (b) 4 times (c) $(327 / 27)^4$ times (d) 2 times
- 34** **NCERT based** 400 J of heat is supplied to a gas and 150 J of work is done on the gas. The change in its internal energy is
 (a) -250 J (b) 400 J (c) 250 J (d) 550 J

- 35** **Assertion-Reason** **Assertion (A):** When an ideal gas expands adiabatically, its temperature falls.
Reason (R): In an adiabatic process no heat enters, so the work done by the gas comes from its internal energy.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 36** **NEET pattern** A Carnot engine works between 527°C and 127°C and absorbs 1000 J from the source per cycle. The work done per cycle is
 (a) $\approx 759\text{ J}$ (b) 800 J (c) 250 J (d) 500 J
- 37** **NEET pattern** A gas is taken through the cycle ABCDA shown. The net work done by the gas per cycle is
 (a) $+800\text{ J}$ (b) $+400\text{ J}$ (c) -400 J (d) $+200\text{ J}$
- 
- 38** **NEET pattern** At what temperature will hydrogen molecules have the same rms speed as oxygen molecules at 127°C ?
 (a) 1600 K (b) 100 K (c) 200 K (d) 25 K
- 39** **JEE Main pattern** One mole of a monatomic ideal gas is mixed with one mole of a (rigid) diatomic ideal gas. The ratio $\gamma = C_p / C_v$ of the mixture is
 (a) 1.40 (b) 1.53 (c) 1.50 (d) 1.67
- 40** **NCERT based** A particle executes SHM with amplitude 5 cm and period 2 s . Its speed when it is 3 cm from the mean position is
 (a) $3\pi\text{ cm/s}$ (b) $4\pi\text{ cm/s}$ (c) $5\pi\text{ cm/s}$ (d) $2\pi\text{ cm/s}$
- 41** **NEET pattern** A mass on a spring oscillates with period T . The spring is cut into two equal halves and the same mass is hung from one half. The new period is
 (a) $T / \sqrt{2}$ (b) $2T$ (c) $T / 2$ (d) $\sqrt{2} T$
- 42** **Assertion-Reason** **Assertion (A):** In SHM, the acceleration of the particle is maximum at the mean position.
Reason (R): In SHM, the speed of the particle is maximum at the mean position.
 (a) Both true, R explains A
 (b) Both true, R does not explain A
 (c) A false, R true
 (d) Both false
- 43** **NEET pattern** A wave is given by $y = 0.05\sin(40\pi t - 0.5\pi x)$ in SI units. Its speed and wavelength are
 (a) 80 m/s , 2 m (b) 20 m/s , 4 m
 (c) 40 m/s , 0.5 m (d) 80 m/s , 4 m

44. **NEET pattern** A pipe closed at one end is 34 cm long. If the speed of sound is 340 m/s, the frequency of its **first overtone** is
- (a) 1000 Hz (b) 750 Hz (c) 250 Hz (d) 500 Hz

45. **JEE Main pattern** Tuning fork A produces 4 beats/s with a 256 Hz fork. When A is loaded with a little wax, the beat frequency becomes 6 beats/s. The original frequency of A is
- (a) 260 Hz (b) 252 Hz (c) 262 Hz (d) 250 Hz

✓ Answer Key & Quick Solutions

- 1(a) 2(a) 3(b) 4(c) 5(d) 6(a) 7(d) 8(b) 9(c) 10(a) 11(b) 12(d) 13(d) 14(d) 15(c)
 16(c) 17(c) 18(c) 19(a) 20(b) 21(b) 22(a) 23(a) 24(c) 25(d) 26(a) 27(d) 28(b) 29(a)
 30(a) 31(b) 32(c) 33(a) 34(d) 35(a) 36(d) 37(b) 38(d) 39(c) 40(b) 41(a) 42(c) 43(d)
 44(b) 45(b)

1. (a) $[ML^2T^{-1}] / ([M][LT^{-1}]) = [L]$ – a length (the Compton wavelength).
2. (a) $\frac{\Delta A}{A} = \frac{0.1}{5.0} + \frac{0.1}{2.0} = 0.07 \Rightarrow \Delta A = 0.7 \text{ cm}^2$.
3. (b) $t = \sqrt{2h/g} = 4 \text{ s}$. $s_4 - s_3 = 80 - 45 = 35 \text{ m}$.
4. (c) Area = $\frac{1}{2}(4)(20) + 4(20) + \frac{1}{2}(2)(20) = 40 + 80 + 20 = 140 \text{ m}$; $140/10 = 14 \text{ m/s}$.
5. (d) $v = 3(t-1)(t-3)$ turns at $t = 1 \text{ s}$ and 3 s . $x: 0 \rightarrow 4 \rightarrow 0 \rightarrow 4 \text{ m}$, so distance = $4 + 4 + 4 = 12 \text{ m}$ (displacement only 4 m).
6. (a) $\frac{H}{R} = \frac{\tan \theta}{4} = 1 \Rightarrow \tan \theta = 4$.
7. (d) Horizontal component unchanged; vertical goes from +10 to -10 m/s. $\Delta \vec{v} = 2u \sin \theta = 20 \text{ m/s}$ downward ($= \vec{g}T$).
8. (b) Rain relative to cyclist: 30 m/s down and 10 m/s west, so it comes from the front: $\tan \theta = 10/30$, tilt forward (east).
9. (c) $a_c = v^2/r = 8$, $a_t = 3 \Rightarrow a = \sqrt{64+9} = \sqrt{73} \approx 8.5 \text{ m/s}^2$.
10. (a) $a = 10/5 = 2 \text{ m/s}^2$. Contact force pushes only the 3 kg block: $3 \times 2 = 6 \text{ N}$.
11. (b) $T = \frac{2m_1m_2g}{m_1+m_2} = \frac{2(3)(5)(10)}{8} = 37.5 \text{ N}$.
12. (d) Limiting friction = $0.4 \times 50 = 20 \text{ N} > 15 \text{ N}$, so the block stays at rest; static friction adjusts to 15 N.
13. (d) $\Delta p = 0.5(10+8) = 9 \text{ kg m/s}$ (direction reverses, so speeds add). $F = 9/0.1 = 90 \text{ N}$.
14. (d) $W = \int_1^3 (2x+3) dx = [x^2+3x]_1^3 = 18-4 = 14 \text{ J}$.
15. (c) $K = p^2/2m \Rightarrow K' = (1.2)^2 K = 1.44K$ – an increase of 44%.
16. (c) $\frac{4m_1m_2}{(m_1+m_2)^2} = \frac{8}{9}$. (Check: $v_2 = 4 \text{ m/s}$, $K_2 = 16 \text{ J}$ of 18 J.)
17. (c) $T_{\text{bottom}} - T_{\text{top}} = 6mg = 6(0.2)(10) = 12 \text{ N}$, independent of speed and radius.
18. (c) $x = \frac{2 \times 2}{6} = \frac{2}{3} \text{ m}$, $y = \frac{3 \times 3}{6} = \frac{3}{2} \text{ m}$.
19. (a) Diameter: $MR^2/4$ (perpendicular axes). Parallel axes: $\frac{MR^2}{4} + MR^2 = \frac{5}{4}MR^2$.
20. (b) $K_{\text{rot}} = \frac{1}{2} \cdot \frac{2}{5}mv^2$, $K_{\text{tot}} = \frac{7}{10}mv^2 \Rightarrow$ ratio 2/7.
21. (b) Torques about 40 cm: $mg(30) = 2g(10) \Rightarrow m = 2/3 \approx 0.67 \text{ kg}$.
22. (a) L conserved; $K = L^2/2I$ with $I' = 0.6I \Rightarrow K' = K/0.6 = \frac{5}{3}K$ (extra energy from her muscles).
23. (a) Depth: $g(1 - \frac{1}{2}) = \frac{g}{2}$. Height: $\frac{g}{(3/2)^2} = \frac{4g}{9}$. Ratio = $\frac{1/2}{4/9} = \frac{9}{8}$.
24. (c) $T \propto r^{3/2} \Rightarrow T' = 2 \times 4^{3/2} = 2 \times 8 = 16 \text{ h}$.
25. (d) $v_e = \sqrt{2GM/R}$ depends only on the planet – neither on direction (energy is a scalar) nor on the mass of the body. Both false.

26. (a) $\Delta L = \frac{FL}{AY} = \frac{100 \times 2}{10^{-6} \times 2 \times 10^{11}} = 10^{-3} \text{ m} = 1 \text{ mm}.$

27. (d) $\Delta L \propto L / r^2; \frac{L / 4r^2}{2L / r^2} = \frac{1}{8}.$

28. (b) Volume doubles $\Rightarrow R = 2^{1/3}r; v_t \propto r^2 \Rightarrow v = 5 \times 2^{2/3} \approx 7.9 \text{ cm/s}.$

29. (a) Continuity: $v_2 = 2 \times 4 = 8 \text{ m/s}.$ $\Delta P = \frac{1}{2}\rho(v_2^2 - v_1^2) = 500(64 - 4) = 3 \times 10^4 \text{ Pa}.$

30. (a) $h = \frac{2Scos\theta}{\rho g r}$; $cos\theta > 0$ for water (rise), < 0 for mercury (fall). R is the correct explanation.

31. (b) $\gamma = 3\alpha = 6 \times 10^{-5}; \Delta V / V = 6 \times 10^{-5} \times 50 = 3 \times 10^{-3} = 0.3\%.$

32. (c) Water can give $100 \times 60 = 6000 \text{ cal}$; this melts $6000/80 = 75 \text{ g}.$ Left: $25 \text{ g},$ final temperature $0^\circ\text{C}.$

33. (a) Use kelvin: $300 \text{ K} \rightarrow 600 \text{ K}.$ $P \propto T^4 \Rightarrow T^4 = 16 \text{ times}.$

34. (d) $\Delta U = Q - W_{by} = 400 - (-150) = 550 \text{ J}.$

35. (a) $Q = 0 \Rightarrow \Delta U = -W;$ $W > 0$ in expansion $\Rightarrow U$ and hence T fall. R explains A.

36. (d) $\eta = 1 - \frac{400}{800} = 0.5$ (kelvin!) $\Rightarrow W = 500 \text{ J}.$ Using $^\circ\text{C}$ wrongly gives $\approx 759 \text{ J}.$

37. (b) Area = $(2 \times 10^5 \text{ Pa})(2 \times 10^{-3} \text{ m}^3) = 400 \text{ J};$ clockwise on a P–V diagram $\Rightarrow$ positive work by the gas.

38. (d) $v_{rms} \propto \sqrt{T/M}; \frac{T_H}{2} = \frac{400}{32} \Rightarrow T_H = 25 \text{ K}.$

39. (c) $C_v = \frac{\frac{3}{2}R + \frac{5}{2}R}{2} = 2R, C_p = 3R \Rightarrow \gamma = 1.5.$

40. (b) $\omega = 2\pi / T = \pi \text{ rad/s}; v = \omega\sqrt{A^2 - x^2} = \pi\sqrt{25 - 9} = 4\pi \text{ cm/s}.$

41. (a) Half the length $\Rightarrow k' = 2k; T \propto 1 / \sqrt{k} \Rightarrow T / \sqrt{2}.$

42. (c) $a = -\omega^2 x$ is zero at the mean position and maximum at the extremes, so A is false; R is true.

43. (d) $v = \omega / k = 40\pi / 0.5\pi = 80 \text{ m/s}; \lambda = 2\pi / k = 4 \text{ m}.$

44. (b) Fundamental = $v / 4L = 250 \text{ Hz}.$ A closed pipe has only odd harmonics, so the first overtone = $3 \times 250 = 750 \text{ Hz}.$

45. (b) A = 260 or 252 Hz. Wax lowers the frequency; beats rise to 6 only if A moves further below 256: $252 \rightarrow 250.$ So A = 252 Hz.

Mock analysis (Day 28)

Write **every** wrong, guessed or skipped question below. Be honest about the type of mistake – that decides the fix.

🎯 How to analyse (Day 28) and re-attempt (Day 29)

- **Concept** mistake $\rightarrow$ re-read that section and its formula box in the chapter; solve 5 questions of that topic.
- **Calculation** mistake $\rightarrow$ redo the question on paper without the solution; practise approximations ($\sqrt{2} \approx 1.41, \pi^2 \approx 10$).
- **Silly** mistake ($^\circ\text{C}$ instead of K, "on" vs "by", distance vs displacement) $\rightarrow$ add it to the Traps list and read it before every test.
- **Time** problem $\rightarrow$ note which question type ate time; in the exam, skip it in Round 1.
- **Day 29:** cover the solutions and re-attempt every question in the table below. Target: 100% correct. Anything still wrong goes into the Error Log at the end of the book.

📊 Score bands

For this mock: **150+** excellent – keep revising traps; **110–150** good – fix calculation speed; **< 110** – go back to the formula sheets (Days 25–26) before the next test.

THE FINAL SPRINT

DAYS
25–30**Last-Week Toolkit**

Day 25–26 • Formula sheets

Top 60 traps

10 graphs at a glance

Exam-day strategy

Error log

Everything you need in the last six days on a few pages: every NEET formula of Class 11, the traps where marks are lost, the graphs NEET loves, a minute-by-minute exam plan and your personal error log.

A • Formula Sheets (Days 25–26)

How to use: Day 25 – Chapters 1–7 (mechanics); Day 26 – Chapters 8–14. Cover the right-hand meaning and say each formula aloud with its condition. Tick the ones you had to look up and revise them again at night.

Day 25 – Mechanics **Ch 1 • Units & Measurement**

$\Delta Z = \Delta A + \Delta B$	$Z = A \pm B$: absolute errors add	$\frac{\Delta Z}{Z} = \frac{\Delta A}{A} + \frac{\Delta B}{B}$	$Z = AB$ or A/B : relative errors add
$\frac{\Delta Z}{Z} = p\frac{\Delta A}{A} + q\frac{\Delta B}{B} + r\frac{\Delta C}{C}$	for $Z = A^p B^q / C^r$	$\bar{\Delta a} = \frac{\sum \Delta a_i }{n}$	Mean absolute error; % error = $\frac{\bar{\Delta a}}{a} \times 100$
$LC = 1 \text{ MSD} - 1 \text{ VSD}$	Vernier calipers least count	$LC = \frac{\text{pitch}}{N}$	Screw gauge (N circular divisions)
$n_2 = n_1 \left[\frac{M_1}{M_2} \right]^a \left[\frac{L_1}{L_2} \right]^b \left[\frac{T_1}{T_2} \right]^c$	Converting a value between unit systems	Reading = MSR + $n \times LC$ – e	
		e = zero error, subtracted with sign	

 Must-know dimensions

$[MLT^{-2}]$	Force, weight, thrust	$[ML^2T^{-2}]$	Work, energy, torque
$[ML^2T^{-3}]$	Power	$[MLT^{-1}]$	Momentum, impulse
$[ML^{-1}T^{-2}]$	Pressure, stress, Y, B, energy density	$[MT^{-2}]$	Surface tension, spring constant
$[ML^{-1}T^{-1}]$	Coefficient of viscosity η	$[M^{-1}L^3T^{-2}]$	Gravitational constant G
$[ML^2T^{-1}]$	Planck constant, angular momentum	$[ML^2]$	Moment of inertia

 Constants to keep at your fingertips

$g \approx 9.8$ (use 10) m s^{-2} · $G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$ · $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1}$ · $k_B = 1.38 \times 10^{-23} \text{ J K}^{-1}$ ·
 $N_A = 6.02 \times 10^{23} \text{ mol}^{-1}$ · $\sigma = 5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$ · $b = 2.9 \times 10^{-3} \text{ m K}$ · $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa}$ · $1 \text{ cal} = 4.186 \text{ J}$ · $R_E = 6.4 \times 10^6 \text{ m}$ · speed of sound in air $\approx 331 \text{ m/s}$ at 0°C .

**Ch 2 • Motion in a Straight Line**

$v = u + at$	Uniform acceleration only	$s = ut + \frac{1}{2}at^2$	Displacement in time t
$v^2 = u^2 + 2as$	No time involved	$s_n = u + \frac{a}{2}(2n - 1)$	Distance in the n -th second
$v = \frac{dx}{dt}, a = \frac{dv}{dt} = v \frac{dv}{dx}$	Calculus definitions	$H = \frac{u^2}{2g}, T = \frac{2u}{g}$	Thrown up: max height, total time
$t = \sqrt{\frac{2h}{g}}, v = \sqrt{2gh}$	Dropped from height h	$s_1:s_2:s_3 = 1:3:5$	Equal successive intervals, from rest
$\bar{v} = \frac{2v_1v_2}{v_1 + v_2}$	Average speed, equal distances	$\bar{v} = \frac{v_1 + v_2}{2}$	Average speed, equal time intervals
$s_{stop} = \frac{u^2}{2a}$	Stopping distance $\propto u^2$	$\vec{v}_{AB} = \vec{v}_A - \vec{v}_B$	Velocity of A relative to B

**Ch 3 • Motion in a Plane**

$R = \sqrt{A^2 + B^2 + 2AB\cos\theta}$	Resultant of two vectors	$\tan\alpha = \frac{B\sin\theta}{A + B\cos\theta}$	Angle of resultant with A
$\vec{A} \cdot \vec{B} = AB\cos\theta$	Dot product (zero if $\perp$)	$T = \frac{2u\sin\theta}{g}$	Time of flight
$H = \frac{u^2\sin^2\theta}{2g}$	Maximum height	$R = \frac{u^2\sin 2\theta}{g}$	Range; max u^2 / g at 45°
$\frac{H}{R} = \frac{\tan\theta}{4}$	Height-to-range ratio	$y = x\tan\theta - \frac{gx^2}{2u^2\cos^2\theta}$	Trajectory (parabola)
$R = u\sqrt{\frac{2h}{g}}$	Horizontal projection from height h	$R(\theta) = R(90^\circ - \theta)$	Same range for complementary angles
$a_c = \frac{v^2}{r} = \omega^2 r$	Centripetal acceleration	$v = \omega r, \omega = \frac{2\pi}{T} = 2\pi\nu$	Linear-angular speed
$a = \sqrt{a_c^2 + a_t^2}$	Non-uniform circular motion	$t_{min} = \frac{d}{v_b}, x = \frac{v_r d}{v_b}$	River: shortest time, drift
$\sin\theta = \frac{v_r}{v_b}$	River: shortest path (upstream from normal)	$\tan\theta = \frac{v_{man}}{v_{rain}}$	Umbrella tilt from vertical

**Projectile shortcuts**

- $R = 4H\cot\theta$ and $T^2 = \frac{8H}{g}$; at $45^\circ, R = 4H$.
- Complementary angles θ and $90^\circ - \theta$: $T_1 T_2 = \frac{2R}{g}$ and $H_1 + H_2 = \frac{u^2}{2g}$.
- Horizontal velocity never changes; the vertical velocity changes by g every second.

Ch 4 • Laws of Motion

$\vec{F} = \frac{d\vec{p}}{dt} = m\vec{a}$	Newton's second law	$\vec{J} = \vec{F}\Delta t = \Delta\vec{p}$	Impulse = area under F–t graph
$m_1\vec{u}_1 + m_2\vec{u}_2 = m_1\vec{v}_1 + m_2\vec{v}_2$	Momentum conservation	$V = -\frac{mv}{M}$	Recoil of gun
$N = m(g \pm a)$	Lift accelerating up (+) / down (–)	$f_s \leq \mu_s N, f_k = \mu_k N$	Static (self-adjusting), kinetic
$\tan\theta_r = \mu_s$	Angle of repose	$a = g(\sin\theta - \mu_k \cos\theta)$	Sliding down a rough incline
$a = \frac{(m_1 - m_2)g}{m_1 + m_2}$	Atwood machine	$T = \frac{2m_1m_2g}{m_1 + m_2}$	Atwood tension
$v_{max} = \sqrt{\mu_s rg}$	Level curved road	$v_0 = \sqrt{rg \tan\theta}$	Banked road, no friction needed
$v_{max} = \sqrt{rg \frac{\mu_s + \tan\theta}{1 - \mu_s \tan\theta}}$	Banked road with friction	$F_c = \frac{mv^2}{r}$	Centripetal force needed

Ch 5 • Work, Energy and Power

$W = \vec{F} \cdot \vec{d} = Fd\cos\theta$	Work by a constant force	$W = \int F dx$	Variable force = area under F–x
$W_{net} = \Delta K$	Work–energy theorem	$K = \frac{p^2}{2m}$	KE–momentum relation
$U = mgh, U = \frac{1}{2}kx^2$	Gravitational, spring PE	$F = -\frac{dU}{dx}$	Conservative force from U
$P = \frac{dW}{dt} = \vec{F} \cdot \vec{v}$	Instantaneous power	$e = \frac{v_2 - v_1}{u_1 - u_2}$	e = 1 elastic, e = 0 perfectly inelastic
$v_1 = \frac{(m_1 - m_2)u_1 + 2m_2u_2}{m_1 + m_2}$	1-D elastic (swap 1 ↔ 2 for v_2)	$U = \frac{m_1u_1 + m_2u_2}{m_1 + m_2}$	Bodies stick together
$\Delta K = \frac{m_1m_2(u_1 - u_2)^2}{2(m_1 + m_2)}$	KE lost when they stick	$\frac{4m_1m_2}{(m_1 + m_2)^2}$	KE fraction transferred (elastic, target at rest)
$v_{bot} \geq \sqrt{5gl}, v_{top} \geq \sqrt{gl}$	Vertical circle on a string	$T_{bot} - T_{top} = 6mg$	Tension difference in vertical circle

Collision shortcuts

- Equal masses, elastic, head-on: velocities are **exchanged**.
- Very heavy body hits a light one at rest: light one moves off at $\approx 2u$.
- Light body hits a very heavy one at rest: rebounds with $\approx -u$.
- Ball dropped from h, coefficient e: rebound height e^2h .

Power shortcuts

- Lifting at constant speed: $P = mgv$.
- Pump raising water: $P = \frac{m}{t}gh$ (add $\frac{1}{2}\frac{m}{t}v^2$ if it leaves with speed v).
- 1 hp = 746 W; 1 kWh = 3.6×10^6 J.



Ch 6 • System of Particles & Rotational Motion

$$\vec{r}_{cm} = \frac{\sum m_i \vec{r}_i}{\sum m_i} \quad \text{Centre of mass}$$

$$\vec{\tau} = \vec{r} \times \vec{F}, \tau = rF \sin \theta \quad \text{Torque}$$

$$\vec{L} = \vec{r} \times \vec{p}, L = I\omega \quad \text{Angular momentum}$$

$$\tau = I\alpha = \frac{dL}{dt} \quad \text{Rotational second law}$$

$$I = \sum m_i r_i^2 = Mk^2 \quad k = \text{radius of gyration}$$

$$I = I_{cm} + Md^2 \quad \text{Parallel axes theorem}$$

$$I_z = I_x + I_y \quad \text{Perpendicular axes (plane body)}$$

$$I_1 \omega_1 = I_2 \omega_2 \quad \text{No external torque}$$

$$K = \frac{1}{2} I \omega^2, P = \tau \omega \quad \text{Rotational KE, power}$$

$$K = \frac{1}{2} m v^2 \left(1 + \frac{k^2}{R^2} \right) \quad \text{Rolling KE } (v = R\omega)$$

$$a = \frac{g \sin \theta}{1 + k^2 / R^2} \quad \text{Rolling down an incline}$$

$$v = \sqrt{\frac{2gh}{1 + k^2 / R^2}} \quad \text{Speed at bottom (rolling)}$$

$$\omega = \omega_0 + \alpha t \quad \text{Rotational kinematics (constant } \alpha)$$

$$\omega^2 = \omega_0^2 + 2\alpha\theta \quad \text{Also } \theta = \omega_0 t + \frac{1}{2} \alpha t^2$$

Body (mass M)	Axis	I	k^2 / R^2
Ring / thin hollow cylinder	central, $\perp$ plane	MR^2	1
Disc / solid cylinder	central, $\perp$ plane	$\frac{1}{2} MR^2$	1/2
Solid sphere	diameter	$\frac{2}{5} MR^2$	2/5
Hollow sphere (shell)	diameter	$\frac{2}{3} MR^2$	2/3
Ring / disc	diameter	$\frac{1}{2} MR^2 / \frac{1}{4} MR^2$	–
Thin rod, length L	centre / end, $\perp$ rod	$\frac{ML^2}{12} / \frac{ML^2}{3}$	–



Ch 7 • Gravitation

$$F = \frac{Gm_1 m_2}{r^2} \quad G = 6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$$

$$g = \frac{GM}{R^2} \quad g \text{ at the surface}$$

$$g_h = \frac{g}{(1 + h/R)^2} \approx g \left(1 - \frac{2h}{R} \right) \quad \text{Height } h \text{ (approx. for } h \ll R)$$

$$g_d = g \left(1 - \frac{d}{R} \right) \quad \text{Depth } d; \text{ zero at centre}$$

$$U = -\frac{GMm}{r}, V = -\frac{GM}{r} \quad \text{Potential energy, potential}$$

$$v_e = \sqrt{\frac{2GM}{R}} = \sqrt{2gR} \quad \text{Escape speed } \approx 11.2 \text{ km/s}$$

$$v_o = \sqrt{\frac{GM}{r}} \quad \text{Orbital speed } (\approx 7.9 \text{ km/s near surface})$$

$$v_e = \sqrt{2} v_o \quad \text{Near the surface}$$

$$T^2 = \frac{4\pi^2}{GM} r^3 \quad \text{Kepler's 3rd law } (T^2 \propto a^3)$$

$$\frac{dA}{dt} = \frac{L}{2m} \quad \text{Areal velocity constant (2nd law)}$$

$$K = \frac{GMm}{2r}, E = -\frac{GMm}{2r} \quad \text{Satellite: KE, total energy}$$

$$\Delta U = \frac{mgh}{1 + h/R} \quad \text{Energy to lift } m \text{ to height } h$$

 NCERT Line to Remember

Geostationary satellite: period 24 h, orbit in the equatorial plane, height $\approx 35\,800$ km (orbit radius $\approx 4.22 \times 10^4$ km). Polar satellites: height ≈ 500 – 800 km, period ≈ 100 min.

Day 26 – Properties of Matter, Heat, Oscillations & Waves

 Ch 8 • Mechanical Properties of Solids

$$\text{stress} = \frac{F}{A}, \text{ strain} = \frac{\Delta L}{L} \quad \text{Longitudinal}$$

$$Y = \frac{FL}{A\Delta L} \quad \text{Young's modulus}$$

$$\Delta L = \frac{FL}{AY} = \frac{4FL}{\pi d^2 Y} \quad \text{Extension of a wire (d = diameter)}$$

$$B = -\frac{p}{\Delta V/V} \quad \text{Bulk modulus; compressibility} = 1/B$$

$$G = \frac{F/A}{\theta} \quad \text{Shear modulus (\theta = shear strain), } G \approx Y/3$$

$$\sigma = \frac{\Delta d/d}{\Delta L/L} \quad \text{Poisson's ratio}$$

$$U = \frac{1}{2}F\Delta L \quad \text{Elastic PE stored in a wire}$$

$$u = \frac{1}{2} \text{stress} \times \text{strain} \quad \text{Energy per unit volume}$$

 Ch 9 • Mechanical Properties of Fluids

$$P = P_a + \rho gh \quad \text{Absolute pressure at depth h}$$

$$P - P_a = \rho gh \quad \text{Gauge pressure}$$

$$\frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \text{Hydraulic lift (Pascal's law)}$$

$$F_B = \rho_l V_{sub} g \quad \text{Buoyant force (Archimedes)}$$

$$\frac{V_{sub}}{V} = \frac{\rho_b}{\rho_l} \quad \text{Floating body}$$

$$A_1 v_1 = A_2 v_2 \quad \text{Equation of continuity}$$

$$P + \frac{1}{2}\rho v^2 + \rho gh = \text{const} \quad \text{Bernoulli's principle}$$

$$v = \sqrt{2gh} \quad \text{Torricelli's law (speed of efflux)}$$

$$F = \eta A \frac{dv}{dx} \quad \text{Viscous force}$$

$$F = 6\pi\eta r v \quad \text{Stokes' law}$$

$$v_t = \frac{2r^2(\rho - \sigma)g}{9\eta} \quad \text{Terminal velocity}$$

$$R_e = \frac{\rho v d}{\eta} \quad \begin{array}{l} < 1000 \text{ streamline, } > 2000 \\ & \text{turbulent} \end{array}$$

$$S = \frac{F}{l}, W = S\Delta A \quad \text{Surface tension, surface energy}$$

$$\Delta P = \frac{2S}{r} \quad \text{Drop; air bubble inside liquid}$$

$$\Delta P = \frac{4S}{r} \quad \text{Soap bubble in air (two surfaces)}$$

$$h = \frac{2S\cos\theta}{\rho g r} \quad \text{Capillary rise}$$

 Fluids in one glance

- Floating: weight = buoyant force; fraction outside $= 1 - \frac{\rho_b}{\rho_l}$.
- Capillary: $hr = \text{constant}$ – half the radius, double the rise.
- Surface tension and viscosity of liquids **decrease** with temperature.

 NCERT Line to Remember

- Viscosity of gases **increases** with temperature.
- Water: $\rho = 1000 \text{ kg/m}^3$; mercury: $13.6 \times 10^3 \text{ kg/m}^3$.
- Bernoulli needs steady, incompressible, non-viscous flow along a streamline.

Ch 10 • Thermal Properties of Matter

$T_F = \frac{9}{5}T_C + 32$	Fahrenheit–Celsius	$T = T_C + 273.15$	Kelvin
$\Delta L = \alpha_l L \Delta T$	Linear expansion	$\alpha_A = 2\alpha_l, \alpha_V = 3\alpha_l$	Area, volume expansion
$Q = mc \Delta T$	Specific heat	$Q = mL$	Latent heat (no temperature change)
$m_1 c_1 (T_1 - T) = m_2 c_2 (T - T_2)$	Calorimetry: heat lost = heat gained	$H = \frac{KA(T_1 - T_2)}{L}$	Rate of conduction
$R = \frac{L}{KA}$	Thermal resistance (series: add)	$H = e\sigma AT^4$	Stefan–Boltzmann; $\sigma = 5.67 \times 10^{-8}$ SI
$H_{net} = e\sigma A(T^4 - T_s^4)$	Net radiation to surroundings	$\lambda_m T = b$	Wien's law, $b = 2.9 \times 10^{-3}$ m K
$\frac{dT}{dt} = -k(T - T_s)$	Newton's law of cooling	$\frac{T_1 - T_2}{t} = k \left(\frac{T_1 + T_2}{2} - T_s \right)$	Average form (for MCQs)

Ch 11 • Thermodynamics

$\Delta Q = \Delta U + \Delta W$	First law (W = work by gas)	$W = \int P dV$	Area under P–V curve
$\Delta U = nC_v \Delta T$	Ideal gas, any process	$C_p - C_v = R, \gamma = \frac{C_p}{C_v}$	Mayer's relation
$W = nRT \ln \frac{V_2}{V_1}$	Isothermal ($\Delta U = 0, Q = W$)	$W = P \Delta V$	Isobaric
$W = 0, Q = \Delta U$	Isochoric	$PV^\gamma = \text{const}, TV^{\gamma-1} = \text{const}$	Adiabatic ($Q = 0$)
$W = \frac{nR(T_1 - T_2)}{\gamma - 1}$	Adiabatic work = $\frac{P_1 V_1 - P_2 V_2}{\gamma - 1}$	$\text{slope}_{adi} = \gamma \times \text{slope}_{iso}$	Adiabatic is steeper
$\eta = \frac{W}{Q_1} = 1 - \frac{Q_2}{Q_1}$	Heat engine efficiency	$\eta = 1 - \frac{T_2}{T_1}$	Carnot (T in kelvin!)
$\alpha = \frac{Q_2}{W}$	Refrigerator: coefficient of performance	$\alpha = \frac{T_2}{T_1 - T_2}$	Carnot refrigerator

Ch 12 • Kinetic Theory

$PV = nRT = Nk_B T$	Ideal gas equation	$P = \frac{1}{3}\rho \overline{v^2}$	Kinetic pressure
$v_{rms} = \sqrt{\frac{3RT}{M}} = \sqrt{\frac{3k_B T}{m}}$	M = molar mass in kg/mol	$\bar{E} = \frac{3}{2}k_B T$	Average translational KE per molecule
$\frac{1}{2}k_B T$	Energy per degree of freedom (equipartition)	$C_v = \frac{f}{2}R, \gamma = 1 + \frac{2}{f}$	f = degrees of freedom
$f = 3, 5, 6$	Mono, rigid di-, rigid non-linear poly-atomic	$\gamma = \frac{5}{3}, \frac{7}{5}, \frac{4}{3}$	Same order as above
$C_{v,mix} = \frac{n_1 C_{v1} + n_2 C_{v2}}{n_1 + n_2}$	Mixture of gases	$\lambda = \frac{1}{\sqrt{2} n \pi d^2}$	Mean free path (n = number density)
$P = P_1 + P_2 + \dots$	Dalton's law of partial pressures	$k_B = \frac{R}{N_A} = 1.38 \times 10^{-23}$ J/K; $N_A = 6.02 \times 10^{23}$	

Ch 13 • Oscillations

$$x = A \cos(\omega t + \phi) \quad \text{Displacement in SHM}$$

$$v = \omega \sqrt{A^2 - x^2} \quad \text{Speed at displacement } x$$

$$a = -\omega^2 x \quad \text{Defining condition of SHM}$$

$$v_{\max} = A\omega, \quad a_{\max} = A\omega^2 \quad \text{At mean / extreme position}$$

$$\omega = \frac{2\pi}{T} = 2\pi\nu \quad \text{Angular frequency}$$

$$T = 2\pi \sqrt{\frac{m}{k}} \quad \text{Spring-mass system}$$

$$T = 2\pi \sqrt{\frac{L}{g_{\text{eff}}}} \quad \text{Simple pendulum (small angle)}$$

$$E = \frac{1}{2}kA^2 = \frac{1}{2}m\omega^2 A^2 \quad \text{Total energy (constant)}$$

$$K = \frac{1}{2}k(A^2 - x^2), \quad U = \frac{1}{2}kx^2 \quad \text{KE and PE at } x$$

$$K = U \text{ at } x = \frac{A}{\sqrt{2}} \quad \text{KE, PE vary with period } T/2$$

$$k_s = \frac{k_1 k_2}{k_1 + k_2}, \quad k_p = k_1 + k_2 \quad \text{Springs in series, parallel}$$

$$k' = nk \quad \text{Spring cut into } n \text{ equal parts}$$

Ch 14 • Waves

$$y = a \sin(kx - \omega t + \phi) \quad \text{Wave moving along } +x$$

$$v = \frac{\omega}{k} = \nu \lambda, \quad k = \frac{2\pi}{\lambda} \quad \text{Wave speed, wave number}$$

$$v = \sqrt{\frac{T}{\mu}} \quad \text{Transverse wave on a string}$$

$$v = \sqrt{\frac{B}{\rho}} \quad \text{Longitudinal wave in a fluid}$$

$$v = \sqrt{\frac{\gamma P}{\rho}} \quad \text{Laplace (air } \approx 331 \text{ m/s at } 0^\circ\text{C)}$$

$$v \propto \sqrt{T} \quad \text{Sound speed vs temperature (K)}$$

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta x \quad \text{Phase vs path difference}$$

$$y = 2a \sin kx \cos \omega t \quad \text{Standing wave; nodes } \lambda/2 \text{ apart}$$

$$\nu_n = \frac{nv}{2L} \quad \text{String fixed at both ends / open pipe}$$

$$\nu_n = \frac{(2n-1)v}{4L} \quad \text{Closed pipe: odd harmonics only}$$

$$\nu_{\text{beat}} = |\nu_1 - \nu_2| \quad \text{Beat frequency}$$

$$\frac{\lambda}{4} \quad \text{Node to nearest antinode}$$

B • Top 60 Traps where students lose marks

Read this list on Day 30 and again on the morning of the exam. Each line is a mark saved.

! Ch 1 • Units & Measurement

- Errors never subtract: for $g = 4\pi^2 l / T^2$, % error = %l + 2(%T).
- 0.0500 has 3 SF; changing units (5.00 m = 500 cm) never changes SF – write 5.00×10^2 cm.
- Zero error is subtracted with its sign: a negative zero error is **added**.
- Dimensionally correct $\neq$ physically correct; constants like 2π , $1/2$ cannot be found.

! Ch 2 • Straight-line motion

- Distance $\neq$ displacement: if v changes sign, split the interval at $v = 0$.
- Equal distances $\rightarrow$ average speed is $\frac{2v_1 v_2}{v_1 + v_2}$, not the simple mean.
- At the top of a vertical throw $v = 0$ but $a = g$ (not zero).
- Negative a does not always mean slowing down – only when a and v are opposite.
- Area under v - t = displacement (take signs); area under a - t = change in velocity.

! Ch 3 • Motion in a plane

10. At the highest point speed = $u \cos \theta$, not zero; only there is $\vec{v} \perp \vec{a}$.
11. θ and $90^\circ - \theta$ give the same range but different H and T.
12. Object dropped from a moving plane/train keeps the vehicle's horizontal velocity.
13. Uniform circular motion: speed constant, but velocity and acceleration vectors keep changing.
14. Umbrella tilts **forward** (towards motion); angle is measured from the vertical.

! Ch 4 • Laws of motion

15. Static friction self-adjusts: $f = \text{applied force}$ until it reaches $\mu_s N$.
16. $N \neq mg$ always: incline $mg \cos \theta$, lift $m(g \pm a)$, pull at angle $mg - F \sin \theta$.
17. Action and reaction act on different bodies – they never cancel each other.
18. Rebound: $\Delta p = m(v + u)$ because the direction reverses.
19. Mass of the whole system goes in $a = F / M$; for the contact force use only the mass ahead.

! Ch 5 • Work, energy, power

20. Friction does not always do negative work (box on an accelerating truck).
21. Centripetal force does zero work in uniform circular motion.
22. Momentum is conserved in every collision; KE only in elastic ones.
23. Same $p \rightarrow$ lighter body has more KE ($K = p^2 / 2m$).
24. Spring: work to stretch $0 \rightarrow x$ and $x \rightarrow 2x$ are in ratio 1 : 3.

! Ch 6 • Rotation

25. Always state the axis: thin rod $ML^2 / 12$ about centre but $ML^2 / 3$ about an end.
26. Parallel-axis theorem works only from the axis through the COM.
27. Pure rolling: friction does no work; on a level surface with no push, friction = 0.
28. Skater pulls arms in: L conserved, ω increases and KE **increases**.
29. COM may lie outside the body (ring, boomerang).

! Ch 7 • Gravitation

30. g decreases both above and below the surface; $g = 0$ at the centre.
31. $g(1 - 2h / R)$ only for $h \ll R$; otherwise use $gR^2 / (R + h)^2$.
32. Escape speed is independent of the body's mass and of the direction of projection.
33. Satellite: $E < 0$ and $KE = -E$; orbital speed does not depend on satellite mass.
34. Astronauts feel weightless because they are in free fall, not because $g = 0$.

! Ch 8 • Solids

35. Y depends only on the material – length and area change the extension, not Y .
36. Steel is **more** elastic than rubber (larger Y , less strain for same stress).
37. Use the **original** length for strain; for liquids only B exists (no Y , no G).

 Ch 9 • Fluids

- 38.** Pressure depends on depth, not on the shape of the vessel.
- 39.** Soap bubble: $4S / r$ (two surfaces); drop or air bubble in liquid: $2S / r$.
- 40.** Terminal velocity $\propto r^2$ – when n drops merge, $v' = n^{2/3}v$.
- 41.** A capillary of insufficient length never overflows – the meniscus radius adjusts.
- 42.** Faster flow $\Rightarrow$ lower pressure (Bernoulli) – that is why roofs blow off in storms.

 Ch 10 • Thermal properties

- 43.** Use kelvin in T^4 , Carnot and gas laws; a change of $1^\circ\text{C} =$ a change of 1 K .
- 44.** A hole in a plate **expands** on heating, like the material around it.
- 45.** Ice + water: first check if all ice melts – the final temperature may stay 0°C .
- 46.** Stefan: $P \propto AT^4$ (area matters); Wien: $\lambda_m \propto 1 / T$ (hotter $\rightarrow$ bluer).

 Ch 11 • Thermodynamics

- 47.** Signs: work **by** gas +, work **on** gas –; heat absorbed +.
- 48.** Isothermal: $\Delta U = 0$ but $Q \neq 0$. Adiabatic: $Q = 0$ but $\Delta T \neq 0$.
- 49.** The adiabatic curve is γ times steeper than the isothermal through the same point.
- 50.** Cyclic process: $\Delta U = 0$; clockwise on P – $V \Rightarrow$ net work by gas is positive.
- 51.** No engine, not even Carnot, reaches 100% efficiency (second law).

 Ch 12 • Kinetic theory

- 52.** Doubling T makes v_{rms} only $\sqrt{2}$ times; use M in kg/mol .
- 53.** Average KE per molecule depends only on T – same for H_2 and O_2 at the same T .
- 54.** Diatomic gas at room temperature: $f = 5$ (7 only if vibration is counted).

 Ch 13 • Oscillations

- 55.** Pendulum T is independent of mass; in a lift use $g \pm a$.
- 56.** KE and PE oscillate with period $T/2$ (frequency 2ν).
- 57.** Cutting a spring in half **doubles** k ; springs in series are softer.

 Ch 14 • Waves

- 58.** Closed pipe: first overtone = **3rd** harmonic (only odd harmonics).
- 59.** Frequency is set by the source – it does not change when a wave enters a new medium.
- 60.** Wax on a tuning fork lowers its frequency; filing raises it.

 Make this list yours

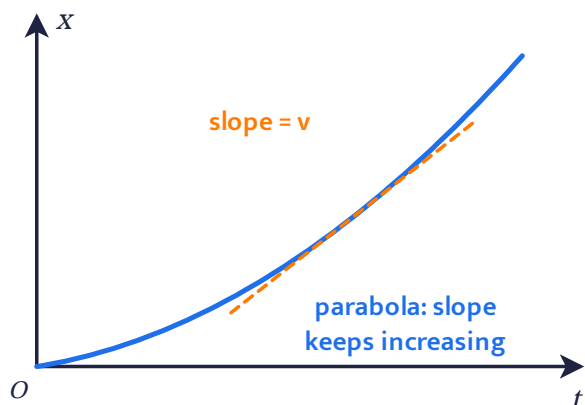
- Put a ✓ next to every trap you have actually fallen into in a test.
- Before each test, read only the ticked traps – it takes 3 minutes.
- Add your own traps from the Error Log at the end.

 The 5 silent killers

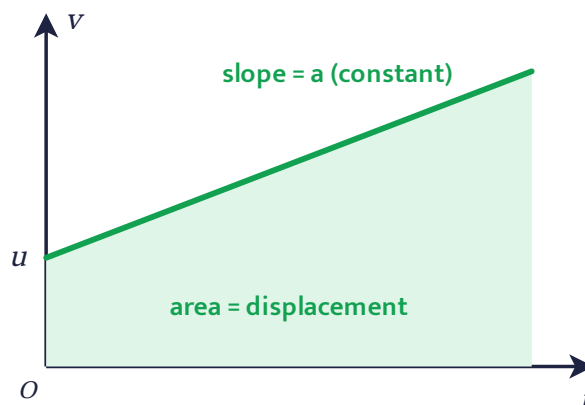
- $^\circ\text{C}$ used instead of K
- "work on" read as "work by"
- distance vs displacement
- radius vs diameter
- "incorrect statement" read as "correct"

C • Important graphs at a glance

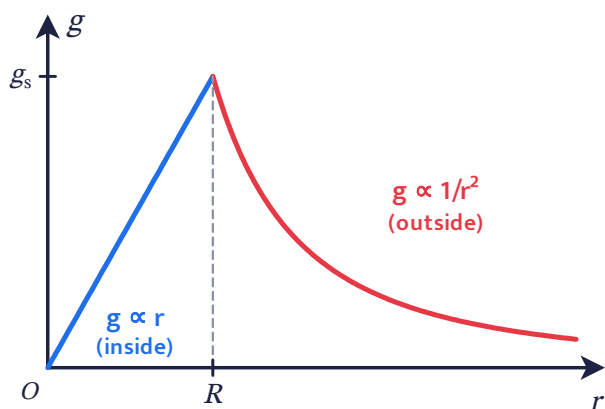
NEET asks at least 2–3 graph-based questions every year. Sketch each of these from memory once on Day 30.



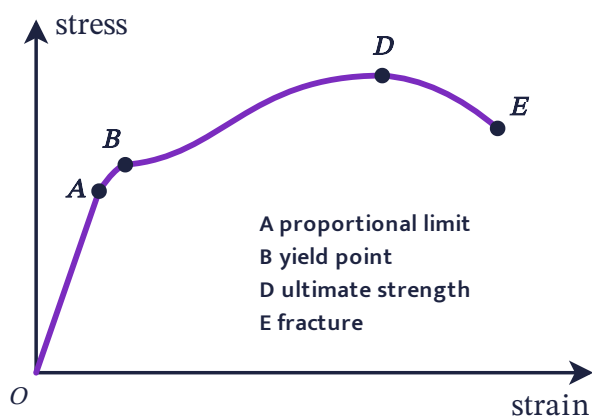
$x-t$, uniform acceleration: parabola



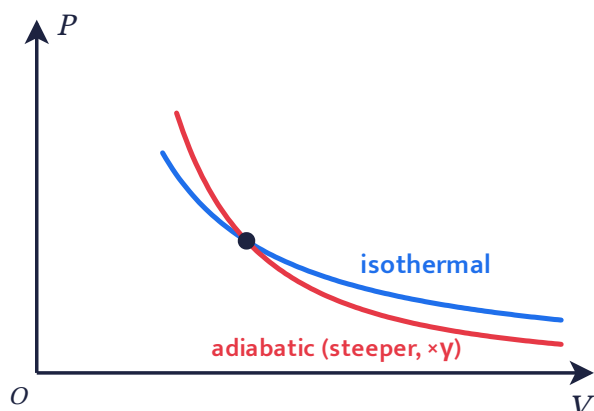
$v-t$, uniform acceleration: straight line



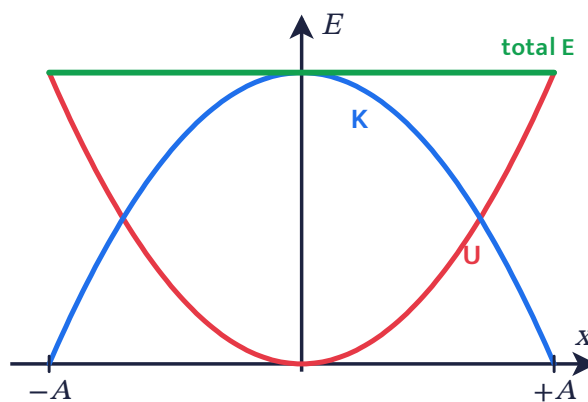
g vs r : maximum at the surface



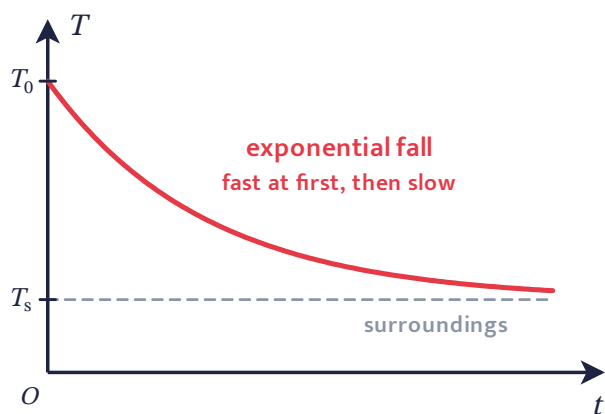
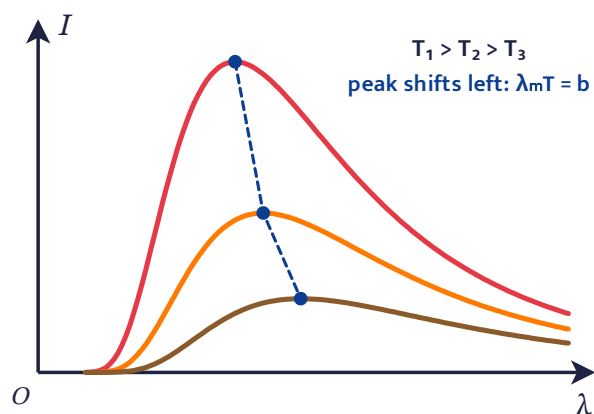
Stress-strain curve of a metal wire



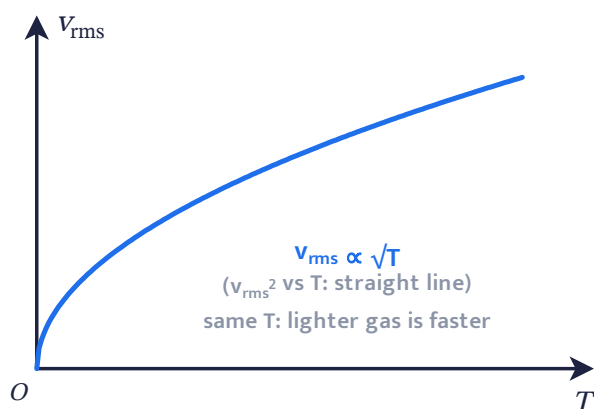
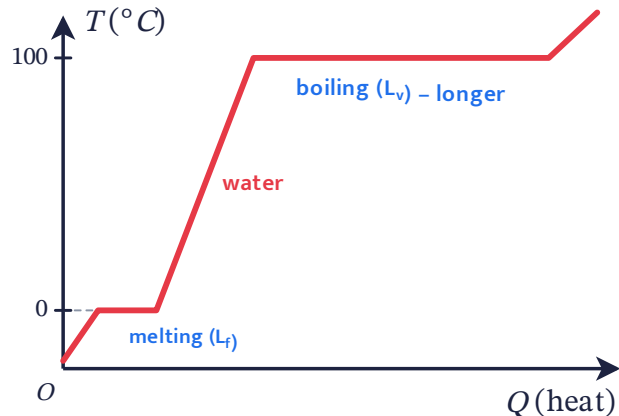
Isothermal vs adiabatic through one point



SHM energy vs x : $K + U = \text{constant}$

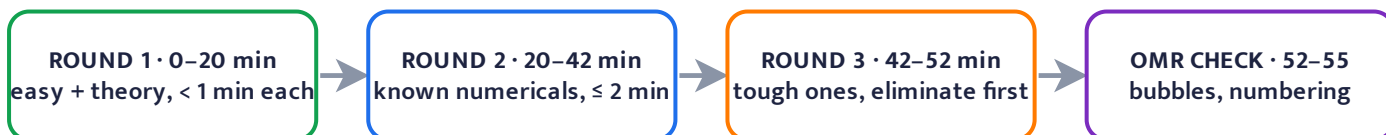
Newton's cooling: T approaches T_s 

Black-body spectrum at three temperatures

 v_{rms} vs T for a given gasHeating curve of ice ($-20^\circ\text{C} \rightarrow \text{steam}$)

D • Exam-day strategy for NEET & JEE Physics

Physics has **45 compulsory questions (180 marks)**. Budget about **50–55 minutes** for them, so that Chemistry and Biology get their share. The 3-round method below stops one hard question from eating ten easy ones.



The 3-round method – never get stuck on the first pass

Round	Time	What to do	Target
1 • Sweep	0–20 min	NCERT facts, graphs, one-formula questions, AR. If a question needs more than 1 minute – mark "?" and move on.	25–30 done
2 • Numericals	20–42 min	Return to "?" questions whose method you know. Hard cap: 2 minutes each. Approximate ($\pi^2 \approx 10, g = 10$).	+10–12
3 • Tough	42–52 min	Eliminate options using units, limiting cases and dimensions. Mark only if 2 options are eliminated.	+3–5
4 • Check	52–55 min	Question number = bubble number; no bubble half-filled; no stray marks.	✓

 OMR tips

- Fill the OMR **after every 10 questions** – never leave all 45 to the end.
- Use only the pen given/allowed; darken the bubble fully and neatly.
- A marked answer cannot be changed – decide first, then bubble.
- Every 10 questions, re-check that Q-number and bubble row match.
- Fill roll number, test-booklet code and signature first – calmly.

**Guessing – the maths**

- Blind guess (1 of 4): expected gain = $\frac{1}{4}(4) - \frac{3}{4}(1) = +0.25$ – tiny, and risky.
- After eliminating 2 options: $\frac{1}{2}(4) - \frac{1}{2}(1) = +1.5$ – worth it.
- Rule: **guess only when you have eliminated two options.**

**Food, sleep & the last 24 hours**

- Sleep 7–8 hours on each of the last three nights – do not stay up the night before.
- Eat a light, familiar meal 2–3 hours before the exam; no new or oily food.
- Drink water, but not a litre just before entering.
- Last day: only formula sheets, traps and your error log – **no new topic.**
- Keep admit card, photo ID and items listed by NTA ready the night before; reach the centre early.

**Calm-down breathing (1 minute)**

- Breathe **in** through the nose – count 4.
- **Hold** – count 4.
- Breathe **out** slowly through the mouth – count 6.
- Repeat 4–5 times: before the paper starts and whenever you feel stuck.
- Then say: "Next question." A hard question is hard for everyone.

Final checklist – tick before you sleep on the last night

- ☐ Formula sheets A revised (Days 25–26)
- ☐ Mock test done and analysed (Days 27–28)
- ☐ All wrong questions re-attempted (Day 29)
- ☐ 60 traps + graphs + error log read (Day 30)
- ☐ Admit card, ID, photo, route and reporting time checked

 First 2 minutes in the hall

- Fill the OMR details, then take three slow breaths.
- Start Round 1 at Q1 and move steadily – do not hunt for "your" chapter.
- Write the start time on the question paper; check it at Q15 and Q30.

**If the paper feels hard**

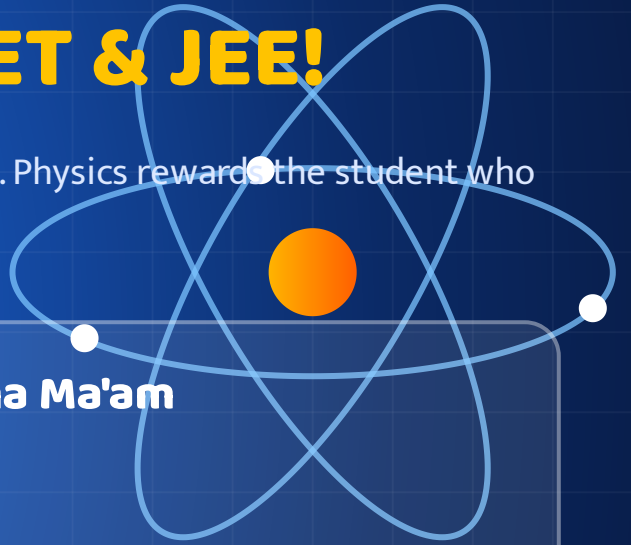
- It is hard for every candidate – ranks are relative.
- Wild guesses cost marks: 38 right + 2 wrong = 150, but 38 right + 7 wrong = 145.
- Never carry a bad question into the next one.

$$v^2 = u^2 + 2as$$

All the best for NEET & JEE!

$$F = ma$$

Revise daily, test weekly, analyse every mistake. Physics rewards the student who practises smartly.



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All books: vcakota20-web.github.io/ai-mastery-with-vs-sir/books/

$$E = \frac{1}{2}mv^2$$

$$\tau = I\alpha$$

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$$PV = nRT$$

$$T = 2\pi\sqrt{l/g}$$

